

DRAWING
TITLE=K22
ABBREV=DRAWING
LAST_MODIFIED=

87654321

DRAWING TITLE

SCH, D8, MLB

Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED.

DRAWING NUMBER

051-9504

REVISION

5.3.0

BRANCH

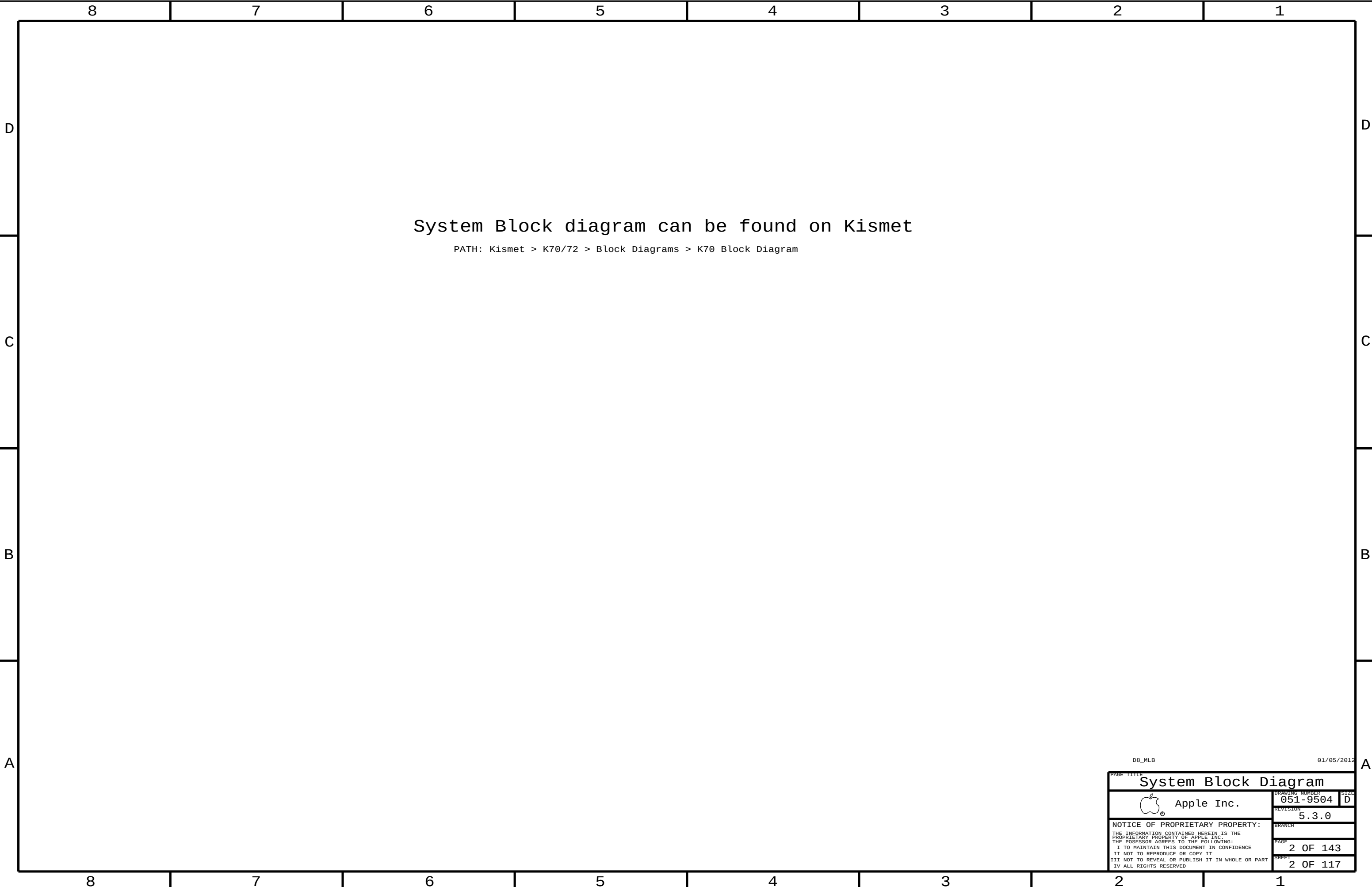
PAGE

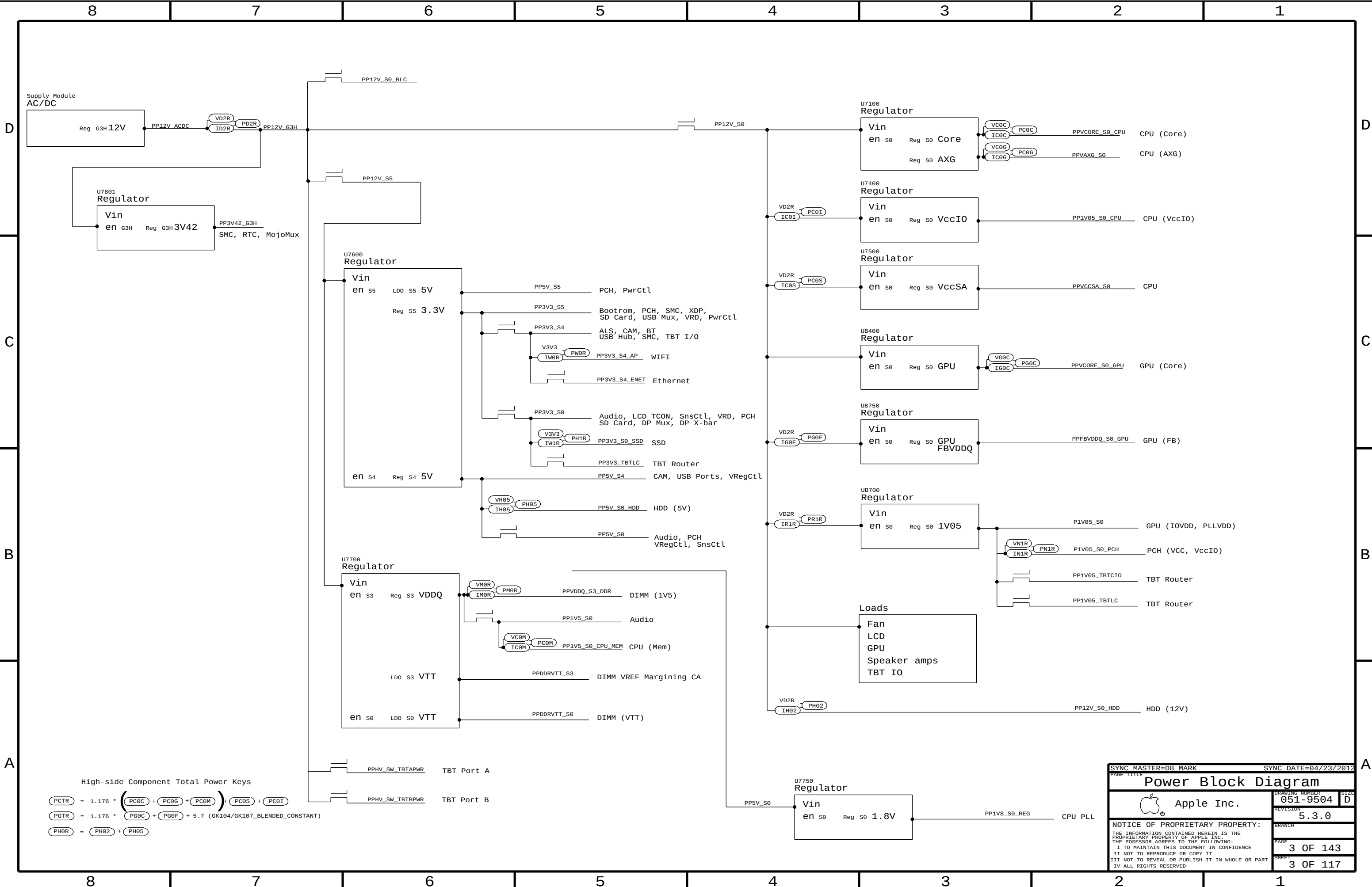
1 OF 143

SHEET

1 OF 117

SIZE
D





SYNC MASTER=D8 MARK

SYNC DATE=04/23/2012

Power Block Diagram

Apple Inc.

DRAWING NUMBER
051-9504

REVISION
5.3.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

PAGE
3 OF 143

SHEET
3 OF 117

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-3816	PCBA, MLB, 2.9G, 4C, GK107, SAM, D8	D8_COMMON, D8, CPU:4C_2P9GHZ, GPU:107EGE, FB:80TH_SAMSUNG, EEEE:F2FR
639-3952	PCBA, MLB, 3.1G, 4C, GK107, SAM, D8	D8_COMMON, D8, CPU:4C_3P1GHZ, GPU:107EGE, FB:80TH_SAMSUNG, EEEE:F49T
639-4092	PCBA, MLB, 2.9G, 4C, GK107, HYN.D8	D8_COMMON, D8, CPU:4C_2P9GHZ, GPU:107EGE, FB:80TH_HYNIX, EEEE:F653
639-4093	PCBA, MLB, 3.1G, 4C, GK107, HYN.D8	D8_COMMON, D8, CPU:4C_3P1GHZ, GPU:107EGE, FB:80TH_HYNIX, EEEE:F654
085-4433	PCBA, MLB, DEV, D8	DEVELOPMENT, D8_DEVEL

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_DHNM	CRITICAL	EEEE:DHNM
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F2FR	CRITICAL	EEEE:F2FR
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F49V	CRITICAL	EEEE:F49V
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F49T	CRITICAL	EEEE:F49T
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F653	CRITICAL	EEEE:F653
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F654	CRITICAL	EEEE:F654

BOM Groups

BOM GROUP	BOM OPTIONS
D8_COMMON	COMMON, ALTERNATE, D8_COMMON1, D8_PROGPARTS
D8_COMMON1	XDP, RSMRST:GATE, SPEAKERID, VREF:CPU, TBTHV:P12V, PRE_BOOST:Y, FBA, FBB
D8_PROGPARTS	SMC:PROG, BOOTROM:PROG, TBTRON:PROG, CIVROM:PROG, CAMROM:PROG, BLCMCU:PROG
D8_DEVEL	XDP_CONN, LCPPLUS, VREFMRGN:EXT, BKLT_PWM, DEVEL_AUDIO, TEMPSNSDEV

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337SA355	1	IVB, SR0TA, PRQ, N1, 2, 9, 65W, 4+1, 1, 1, 6H, LGA	CPU	CRITICAL	CPU: 4C_2P9GHZ
337SA372	1	IVB, SR0T9, PRQ, N0, 3, 1, 65W, 4+1, 1, 1, 6H, LGA	CPU	CRITICAL	CPU: 4C_3P1GHZ

ASICs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33754277	1	IC, PANTHER POINT, C1, SLJC7, PRQ, DB0277	U1800	CRITICAL	
33851133	1	IC, TBT, CR - 4C, B1, PRQ, 28B FC6BA, 12X32MM	U3600	CRITICAL	
33754280	1	IC, GPU, VN GK107 - GE - PS - A2	U8000	CRITICAL	GPU:107EGE
343S0616	1	IC, BCM57766A1, ENET&SD, 8X8	U3900	CRITICAL	

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S3640	1	IC, EEPROM, CR, V1.1 (B1), D8	U3690	CRITICAL	TBTROM:PROG
335S0805	1	IC, EEPROM, SERIAL, 8KB, MLP8	U3690	CRITICAL	TBTROM:BLANK
341S3641	1	IC, PROGRMD, EFI ROM, PIB, D7/D8	U5110	CRITICAL	BOOTROM:PROG
335S0807	1	IC, 64 MBIT SPI SERIAL FLASH	U5110	CRITICAL	BOOTROM:BLANK
341S3394	1	IC, PROGRMD, SMC, A3, V2, 2A32, D8	U4900	CRITICAL	SMC:PROG
338S1098	1	IC, SMC, LX4FS1AH5BBCIGA3	U4900	CRITICAL	SMC:BLANK
341S3646	1	IC, CAMERA FLASH, V7226, D7/D8	U4202	CRITICAL	CAMROM:PROG
335S0852	1	IC, FLASH, SPI, 1MBIT, 3V3	U4202	CRITICAL	CAMROM:BLANK
341S3645	1	IC, ENET 1MBIT, SPI, ROM, V1.13 D8	U3990	CRITICAL	CIVROM:PROG
335S0862	1	IC, SERIAL FLASH, 2MBIT, 2.7V, REF F	U3990	CRITICAL	CIVROM:BLANK
341S3627	1	IC, BLC, MCU, PPROGRAMMED, V9203, D8	U9700	CRITICAL	BLCMCU:PROG
337S3978	1	IC, BLC MCU LPC2132FBD64/01, LQFP64	U9700	CRITICAL	BLCMCU:BLANK

ALTERNATE: 335S0812

ALTERNATE: 335S0854

CPU SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0073	1	SOCKET, MOLEX, LGA1155, CPU- LF	U1000	CRITICAL	

CPU SOCKET ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
511S0071	511S0073		ALL	TYCO SOCKET
511S0072	511S0073		ALL	FOXCONN SOCKET

D8 SCHEMATIC / PCB #'S

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-9504	1	SCH, MLB, D8	SCH1	CRITICAL	D8
820-3298	1	PCBF, MLB, D8	PCB1	CRITICAL	D8

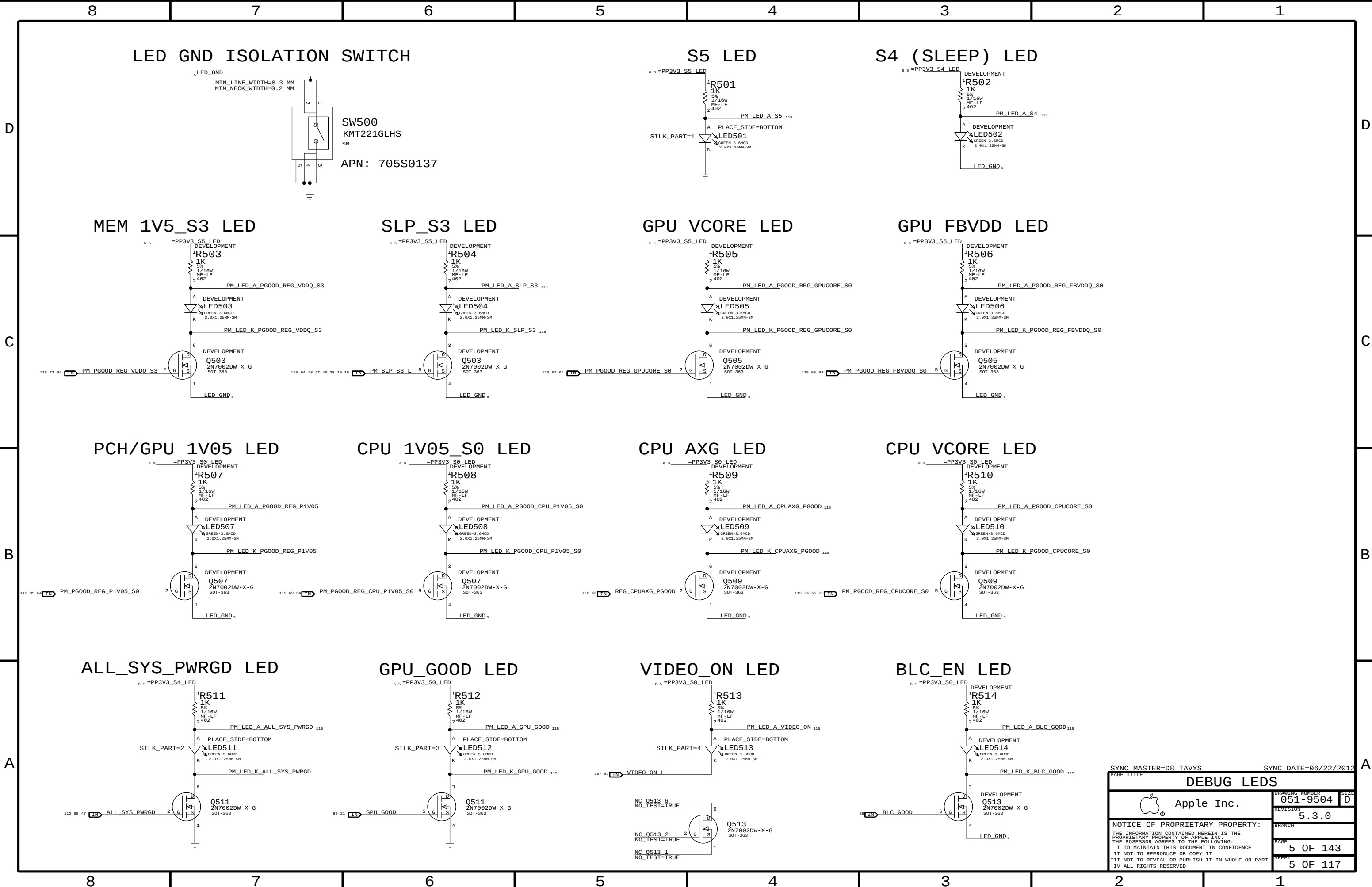
D8 ALTERNATES

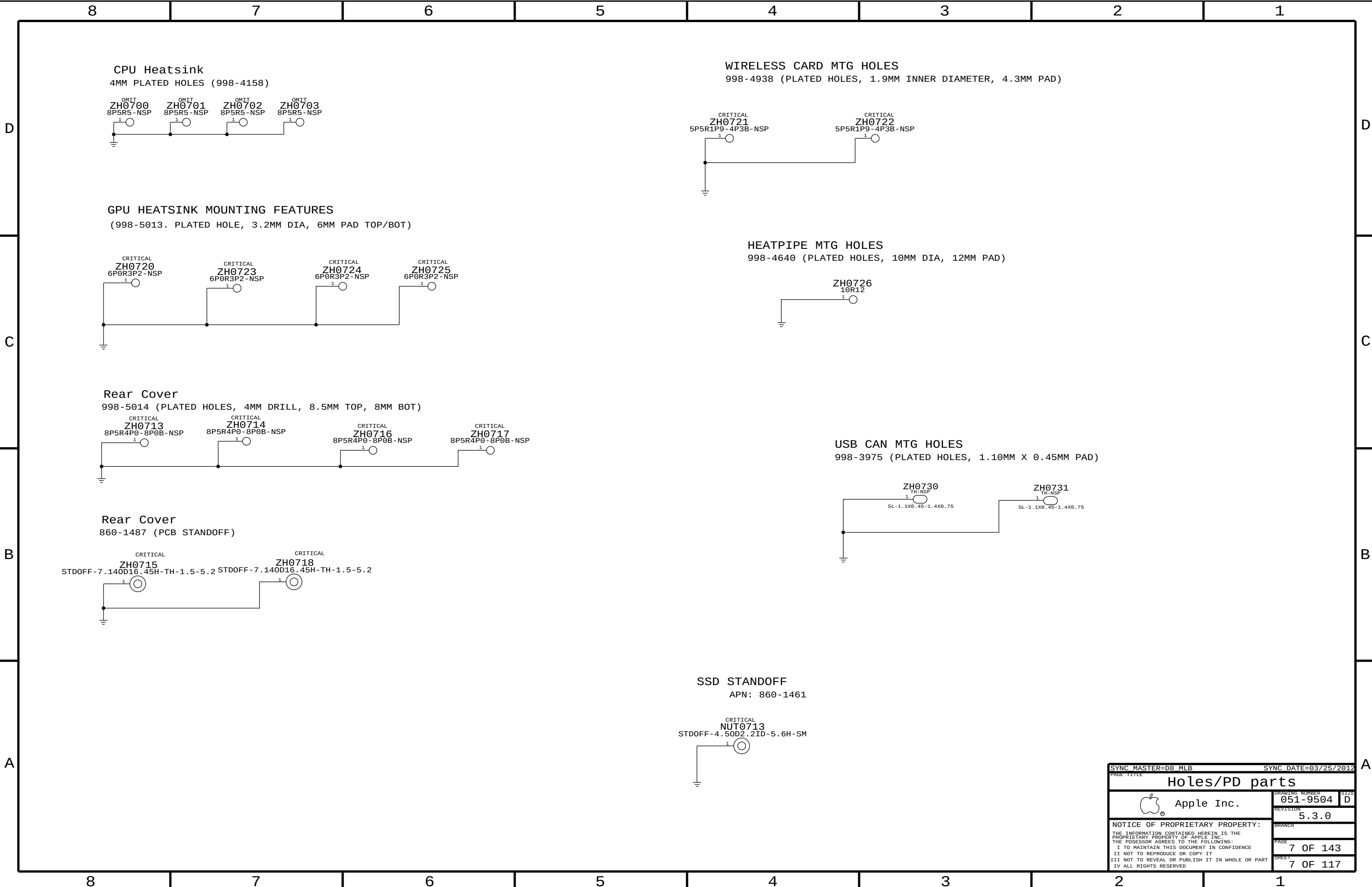
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0147	377S0126		ALL	USB diodes
157S0084	157S0058		ALL	Enet Magnetics
341S3644	341S3645		U3990	CIVROM
376S0975	376S1081		ALL	P/NCH DUAL FET
152S1668	152S1591		ALL	33UH BLC INDUCTOR
128S0365	128S0368		ALL	150UF CAPS BLK

VRAM MODULE PARTS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
333S0619	4	IC, SGRAM, GDDR5, 32MX32, 1.5GHZ, G-DIE, HF	U8400, U8450, U8500, U8550	CRITICAL	FB: BOTH_SAMSUNG
333S0620	4	IC, GDDR5, 32MX32, 1.5GHZ, VEGA 44NM, B-DIE	U8400, U8450, U8500, U8550	CRITICAL	FB: BOTH_HYNIX

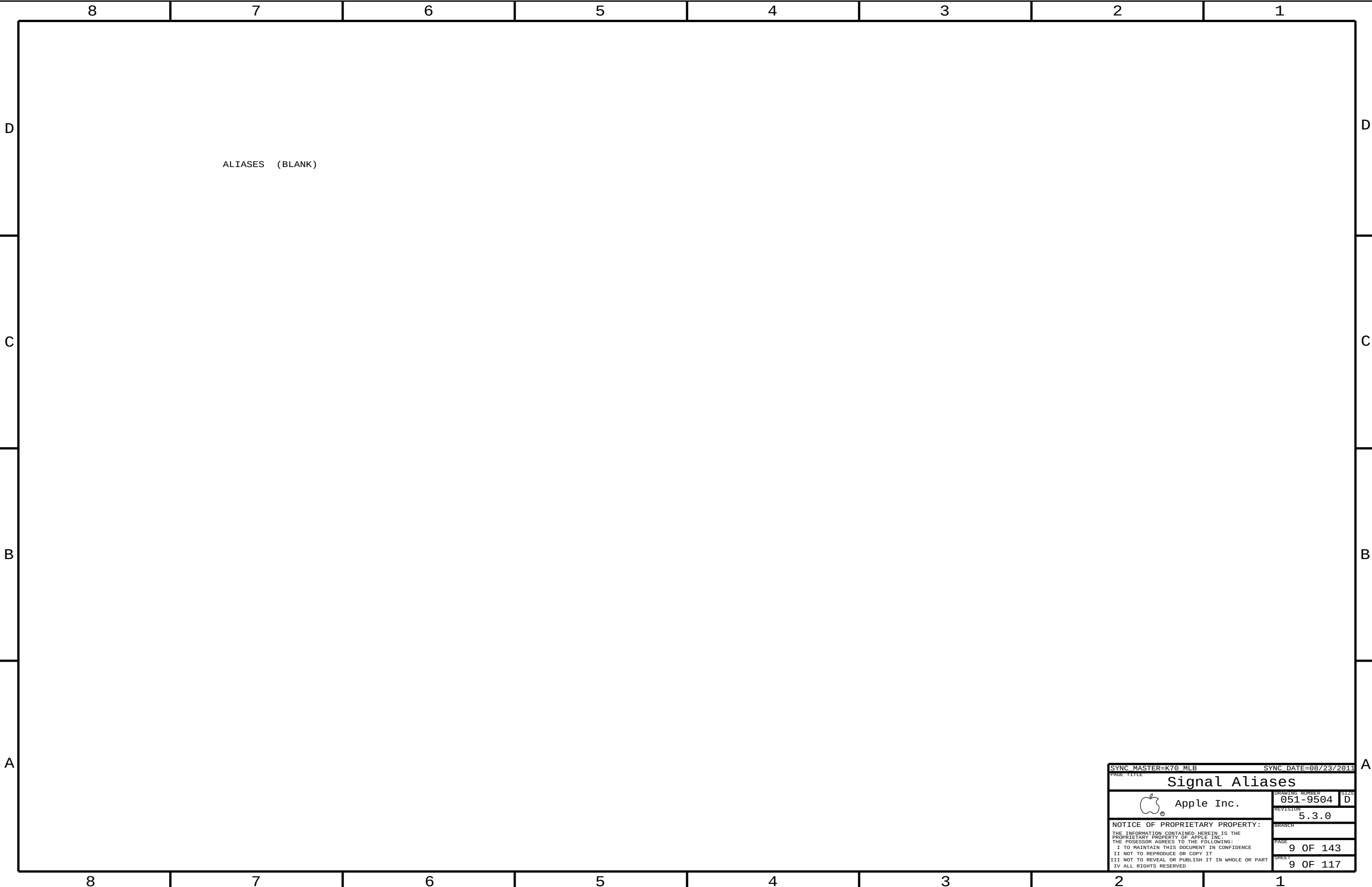
SYNC MASTER=D8 MLB		SYNC DATE=12/19/2011	
PAGE TITLE			
BOM Configuration			
 Apple Inc.		DRAWING NUMBER 051-9504	
		SIZE D	
		REVISION 5.3.0	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			
BRANCH		PAGE	
		4 OF 143	
SHEET		4 OF 117	





SYNC MASTER=D8_MLB		SYNC DATE=03/25/2012	
PAGE TITLE			
Holes/PD parts			
 Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	7 OF 143
		SHEET	7 OF 117

8	7	6	5	4	3	2	1		
CPU Reserved		PCH PCIE		PCH Unused Display		PCH SATA		PCH Test Points	
10 TP_CPU_RSVD<16..1> == NC_CPU_RSVD<16..1> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE1_D2RN == NC_PCIE1_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_RED == NC_CRT_IG_RED MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_R2D_CN == NC_SATA_C_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP1 == NC_PCH_TP1 MAKE_BASE=TRUE NO_TEST=TRUE	
10 TP_CPU_RSVD<46..19> == NC_CPU_RSVD<46..19> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE1_D2RP == NC_PCIE1_D2RP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_GREEN == NC_CRT_IG_GREEN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_R2D_CP == NC_SATA_C_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP2 == NC_PCH_TP2 MAKE_BASE=TRUE NO_TEST=TRUE	
10 CPU_CFG<15..12> == TP_CPU_CFG<15..12> MAKE_BASE=TRUE		10 TP_PCIE1_R2D_CN == NC_PCIE1_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_BLUE == NC_CRT_IG_BLUE MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_D2RN == NC_SATA_C_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP3 == NC_PCH_TP3 MAKE_BASE=TRUE NO_TEST=TRUE	
CPU Memory		10 TP_PCIE1_R2D_CP == NC_PCIE1_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_HSYNC == NC_CRT_IG_HSYNC MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_D2RP == NC_SATA_C_D2RP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP4 == NC_PCH_TP4 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_A_DQ_CB<7..0> == NC_MEM_A_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE2_D2RN == NC_PCIE2_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_VSYNC == NC_CRT_IG_VSYNC MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_D_R2D_CN == NC_SATA_D_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP5 == NC_PCH_TP5 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_A_DQS_N<8> == NC_MEM_A_DQS_N<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE2_D2RP == NC_PCIE2_D2RP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_DDC_CLK == NC_CRT_IG_DDC_CLK MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_D_R2D_CP == NC_SATA_D_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP6 == NC_PCH_TP6 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_A_DQS_P<8> == NC_MEM_A_DQSPX<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE2_R2D_CN == NC_PCIE2_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_DDC_DATA == NC_CRT_IG_DDC_DATA MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_D_D2RN == NC_SATA_D_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP7 == NC_PCH_TP7 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_B_DQ_CB<7..0> == NC_MEM_B_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE2_R2D_CP == NC_PCIE2_R2D_PNX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_MLN<3..0> == NC_DP_IG_B_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_D_D2RP == NC_SATA_D_D2RPX MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP8 == NC_PCH_TP8 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_B_DQS_N<8> == NC_MEM_B_DQSNX<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 DMI_MIDBUS_CLK100M_N == NC_DMI_MIDBUS_CLK100NX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_MLP<3..0> == NC_DP_IG_B_MLPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_R2D_CN == NC_SATA_E_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP9 == NC_PCH_TP9 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP_MEM_B_DQS_P<8> == NC_MEM_B_DQSPX<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 DMI_MIDBUS_CLK100M_P == NC_DMI_MIDBUS_CLK100PX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_AUX_N == NC_DP_IG_B_AUXNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_R2D_CP == NC_SATA_E_R2D_CPX MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP10 == NC_PCH_TP10 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE0N == NC_PCIE_CLK100M_PE0NX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_AUX_P == NC_DP_IG_B_AUXPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_D2RN == NC_SATA_E_D2RNX MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP11 == NC_PCH_TP11 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE0P == NC_PCIE_CLK100M_PE0PX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_HPD == NC_DP_IG_B_HPD MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_D2RP == NC_SATA_E_D2RPX MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP12 == NC_PCH_TP12 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE4N == NC_PCIE_CLK100M_PE4NX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_DDC_CLK == NC_DP_IG_B_DDC_CLK MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP13 == NC_PCH_TP13 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE4P == NC_PCIE_CLK100M_PE4PX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_DDC_DATA == NC_DP_IG_B_DDC_DATA MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP14 == NC_PCH_TP14 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE5N == NC_PCIE_CLK100M_PESNX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_MLN<3..0> == NC_DP_IG_C_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP15 == NC_PCH_TP15 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCIE_CLK100M_PE5P == NC_PCIE_CLK100M_PESPX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_MLP<3..0> == NC_DP_IG_C_MLPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP16 == NC_PCH_TP16 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP_PCIE_CLK100M_PE6N == NC_PCIE_CLK100M_PEGNX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_AUX_N == NC_DP_IG_C_AUXNX MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP17 == NC_PCH_TP17 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP_PCIE_CLK100M_PE6P == NC_PCIE_CLK100M_PEGPX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_AUX_P == NC_DP_IG_C_AUXPX MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP18 == NC_PCH_TP18 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP_PCIE_CLK100M_PE7N == NC_PCIE_CLK100M_PETNX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_HPD == NC_DP_IG_C_HPD MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP19 == NC_PCH_TP19 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP_PCIE_CLK100M_PE7P == NC_PCIE_CLK100M_PETPX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_CTRL_CLK == NC_DP_IG_C_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP20 == NC_PCH_TP20 MAKE_BASE=TRUE NO_TEST=TRUE	
				10 DP_IG_C_CTRL_DATA == NC_DP_IG_C_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE					
		10 TP_PE_TX_N<3..0> == NC_PE_TNX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_D_MLN<3..0> == NC_DP_IG_D_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		PCH Reserved		PCH PCI	
		10 TP_PE_TX_P<3..0> == NC_PE_TPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_D_MLP<3..0> == NC_DP_IG_D_MLPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_0 == NC_PCH_RESERVE_0 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCI_AD<31..0> == NC_PCI_AD<31..0> MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PE_RX_N<3..0> == NC_PE_RNX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_D_AUXN == NC_DP_IG_D_AUXNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_1 == NC_PCH_RESERVE_1 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCI_C_BE_L<3..0> == NC_PCI_C_BE_L<3..0> MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PE_RX_P<3..0> == NC_PE_RPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_D_AUXP == NC_DP_IG_D_AUXPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_2 == NC_PCH_RESERVE_2 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCI_PAR == NC_PCI_PAR MAKE_BASE=TRUE NO_TEST=TRUE	
				10 DP_IG_D_HPD == NC_DP_IG_D_HPD MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_3 == NC_PCH_RESERVE_3 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCI_RESET_L == NC_PCI_RESET_L MAKE_BASE=TRUE NO_TEST=TRUE	
				10 DP_IG_D_CTRL_CLK == NC_DP_IG_D_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_4 == NC_PCH_RESERVE_4 MAKE_BASE=TRUE NO_TEST=TRUE		20 TP_PCH_PCI_GNT0_L == NC_PCI_GNT0_L MAKE_BASE=TRUE NO_TEST=TRUE	
				10 DP_IG_D_CTRL_DATA == NC_DP_IG_D_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_5 == NC_PCH_RESERVE_5 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_INIT3V3_L == NC_PCH_INIT3V3_L MAKE_BASE=TRUE NO_TEST=TRUE	
		PCH USB		10 TP_SDVO_TVCLKINN == NC_SDVO_TVCLKINNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_6 == NC_PCH_RESERVE_6 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_LPC_DREQ0_L == NC_LPC_DREQ0_L MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_4_N == NC_USB_PCH_4NX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_TVCLKINP == NC_SDVO_TVCLKINPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_7 == NC_PCH_RESERVE_7 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_4_P == NC_USB_PCH_4PX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_STALLN == NC_SDVO_STALLNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_8 == NC_PCH_RESERVE_8 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_5_N == NC_USB_PCH_5NX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_STALLP == NC_SDVO_STALLPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_9 == NC_PCH_RESERVE_9 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_5_P == NC_USB_PCH_5PX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_INTN == NC_SDVO_INTNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_10 == NC_PCH_RESERVE_10 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_6_N == NC_USB_PCH_6NX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_INTP == NC_SDVO_INTPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_RESERVE_11 == NC_PCH_RESERVE_11 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_6_P == NC_USB_PCH_6PX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_12 == NC_PCH_RESERVE_12 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_11_N == NC_USB_PCH_11NX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_13 == NC_PCH_RESERVE_13 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_11_P == NC_USB_PCH_11PX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_14 == NC_PCH_RESERVE_14 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_12_N == NC_USB_PCH_12NX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_15 == NC_PCH_RESERVE_15 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_12_P == NC_USB_PCH_12PX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_16 == NC_PCH_RESERVE_16 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_13_N == NC_USB_PCH_13NX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_17 == NC_PCH_RESERVE_17 MAKE_BASE=TRUE NO_TEST=TRUE			
		20 USB_PCH_13_P == NC_USB_PCH_13PX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_18 == NC_PCH_RESERVE_18 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_19 == NC_PCH_RESERVE_19 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_20 == NC_PCH_RESERVE_20 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_21 == NC_PCH_RESERVE_21 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_22 == NC_PCH_RESERVE_22 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_23 == NC_PCH_RESERVE_23 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_24 == NC_PCH_RESERVE_24 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_25 == NC_PCH_RESERVE_25 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_26 == NC_PCH_RESERVE_26 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_27 == NC_PCH_RESERVE_27 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_28 == NC_PCH_RESERVE_28 MAKE_BASE=TRUE NO_TEST=TRUE			
						PCH and CPU FDI			
						10 TP_CPU_FDI_TX_N<7..0> == NC_CPU_FDI_TNX<7..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_CPU_FDI_TX_P<7..0> == NC_CPU_FDI_TPX<7..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_RX_N<7..0> == NC_PCH_FDI_RNX<7..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_RX_P<7..0> == NC_PCH_FDI_RPX<7..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_CPU_FDI_FSYNC<1..0> == NC_CPU_FDI_FSYNC<1..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_CPU_FDI_LSYNC<1..0> == NC_CPU_FDI_LSYNC<1..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_CPU_FDI_INT == NC_CPU_FDI_INT MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_FSYNC<1..0> == NC_PCH_FDI_FSYNC<1..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_LSYNC<1..0> == NC_PCH_FDI_LSYNC<1..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_INT == NC_PCH_FDI_INT MAKE_BASE=TRUE NO_TEST=TRUE			



SYNC_MASTER=K70_MLB

SYNC_DATE=08/23/2011

PAGE_TITLE

Signal Aliases

 Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING_NUMBER

051-9504

REVISION

5.3.0

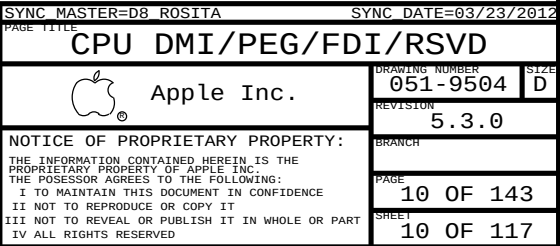
BRANCH

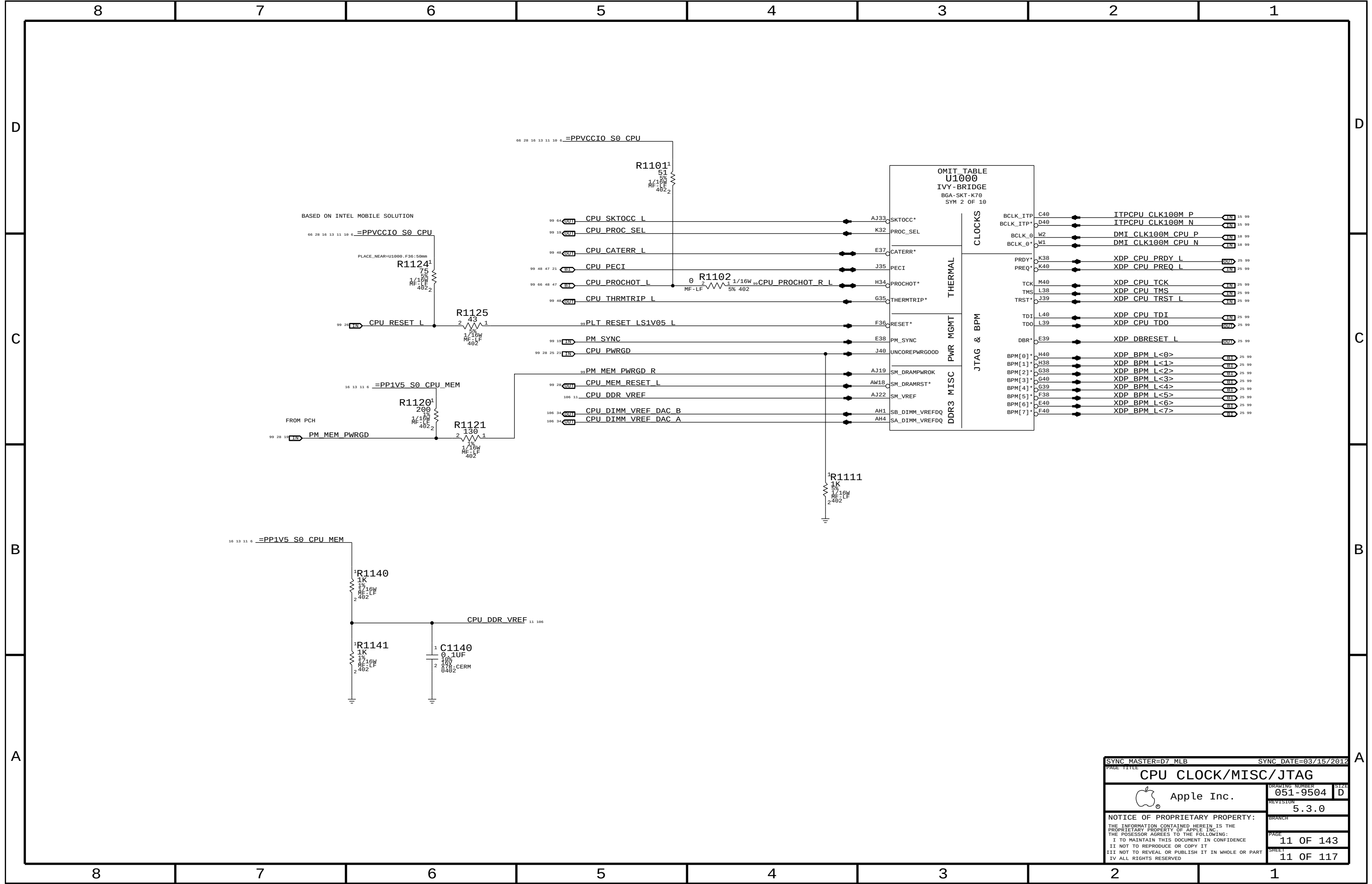
PAGE

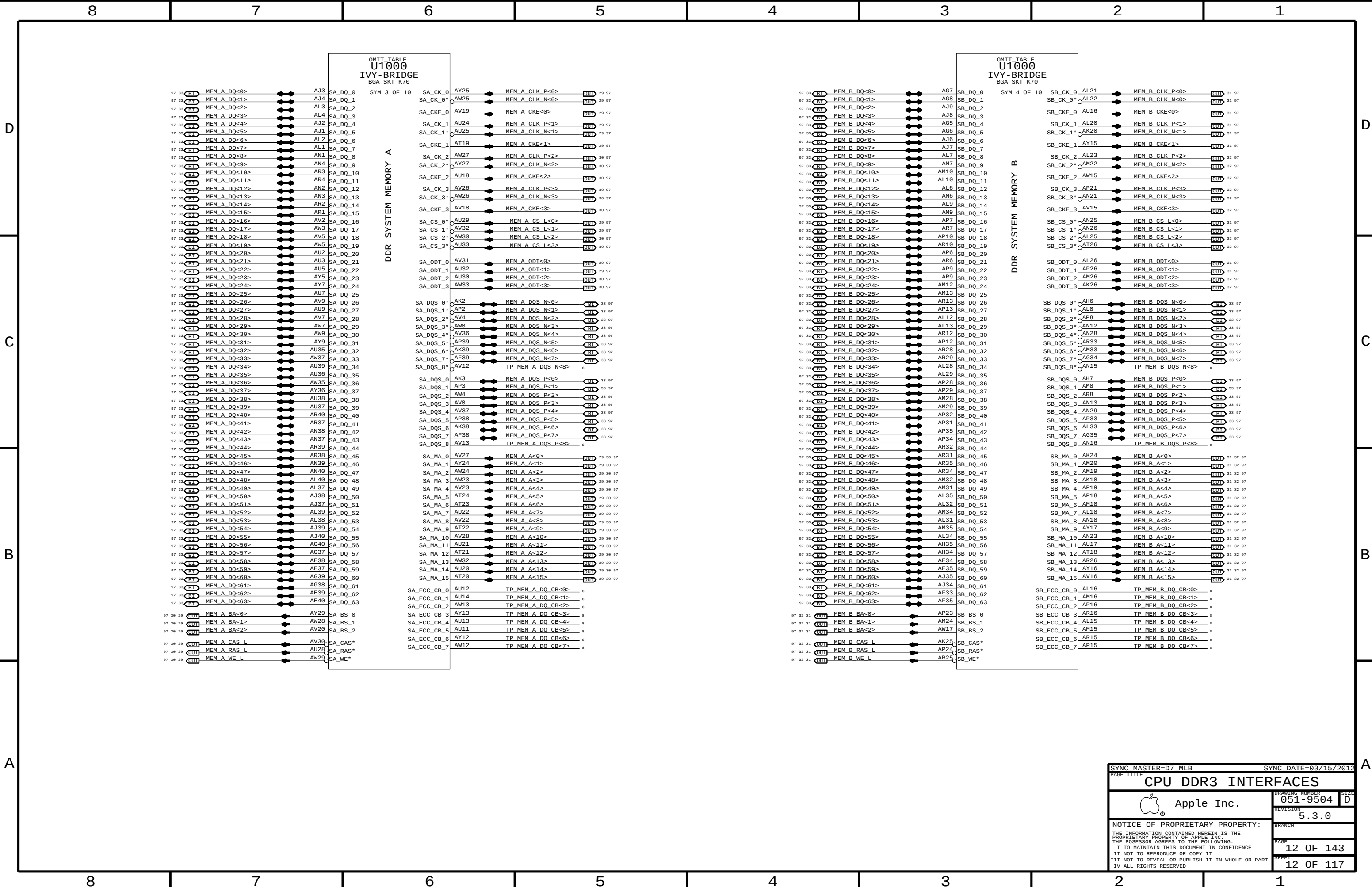
9 OF 143

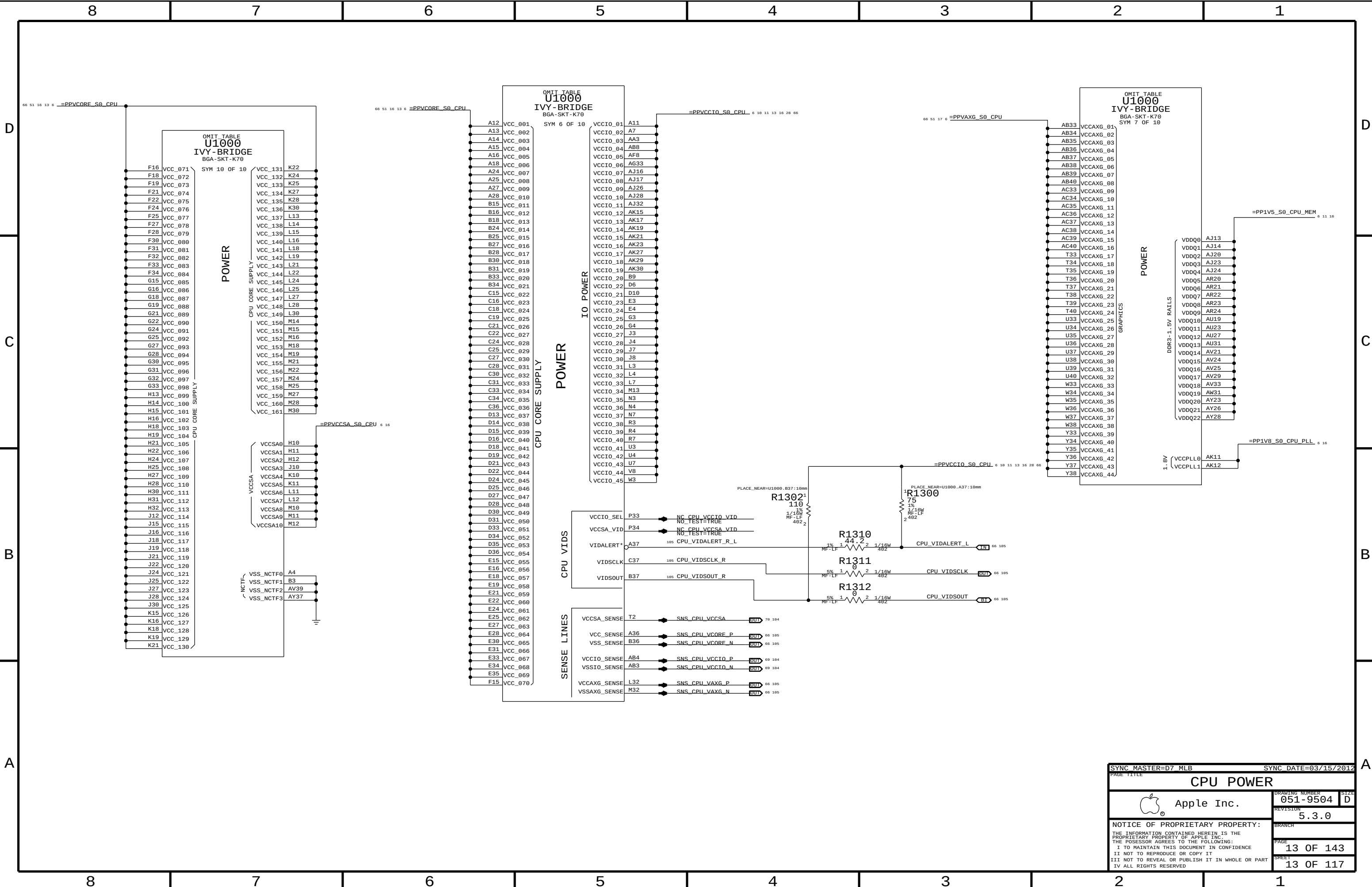
SHEET

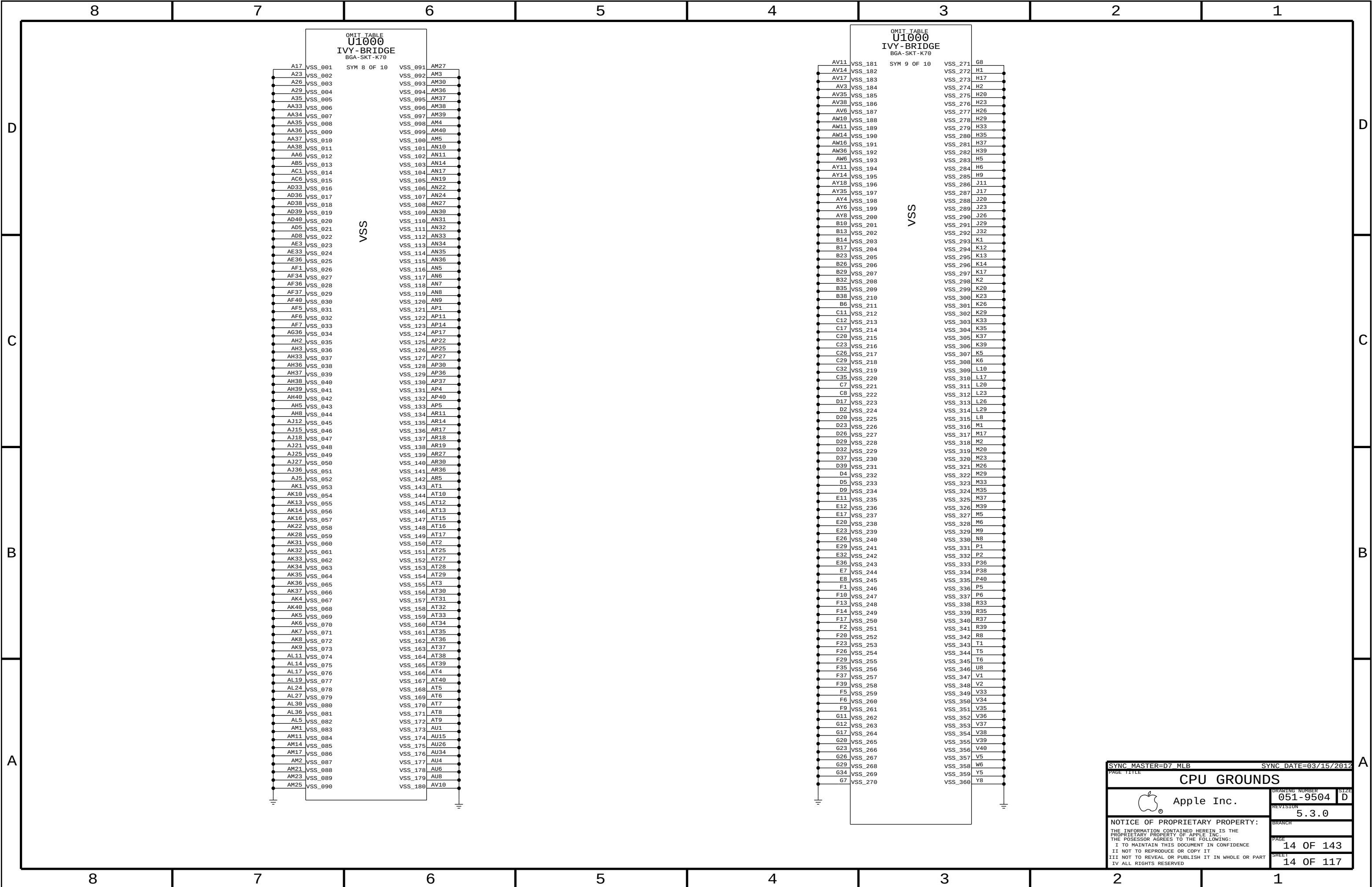
9 OF 117







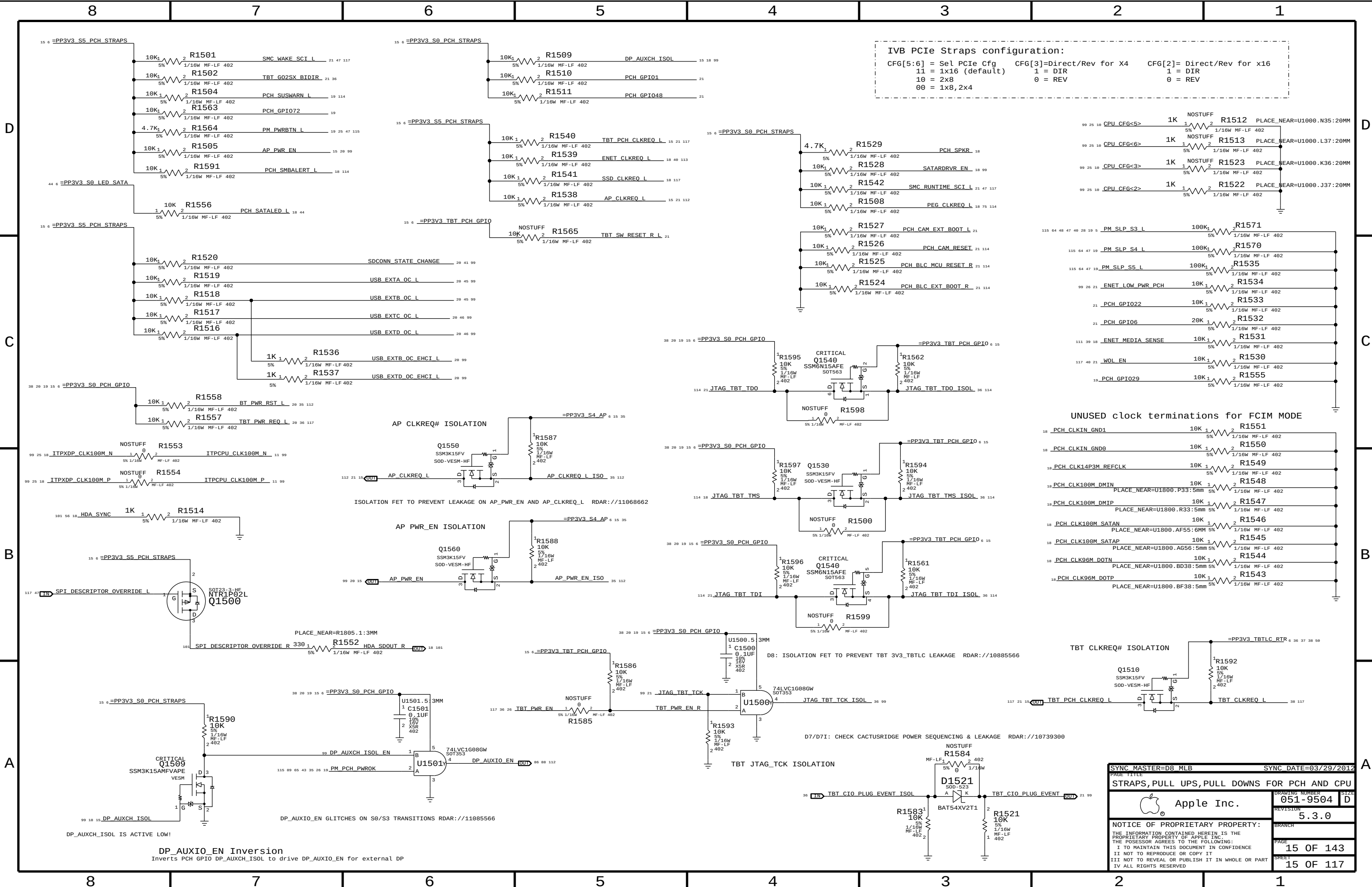




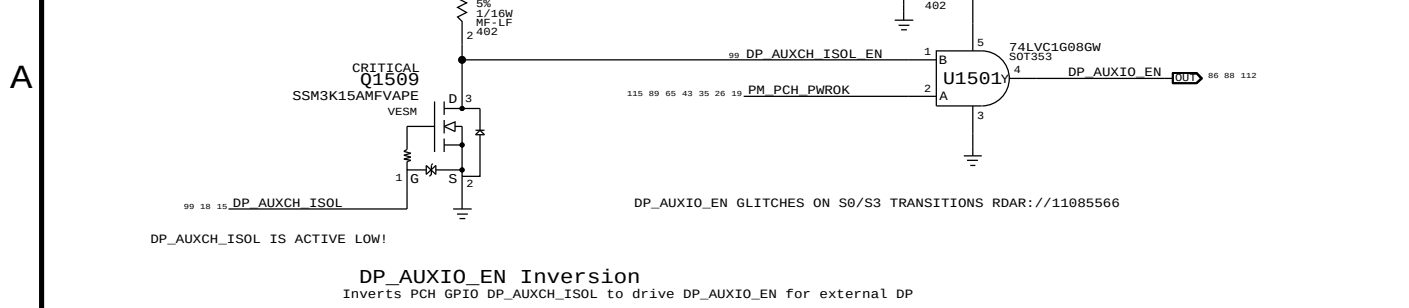
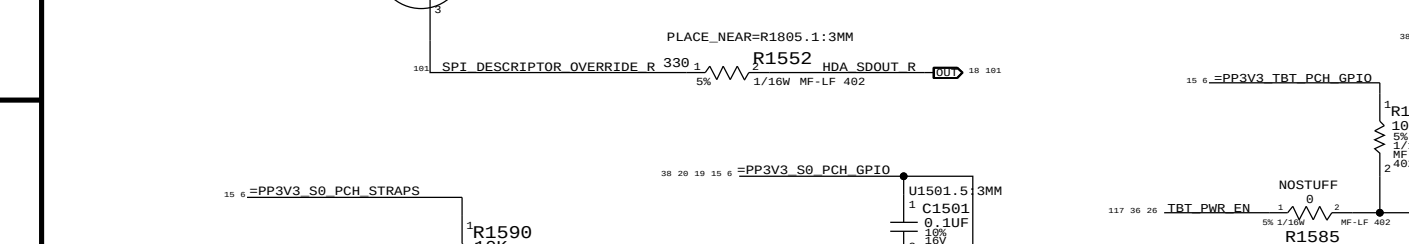
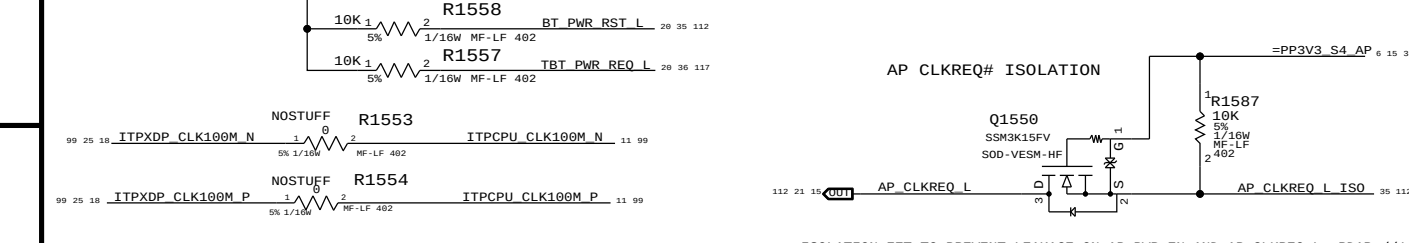
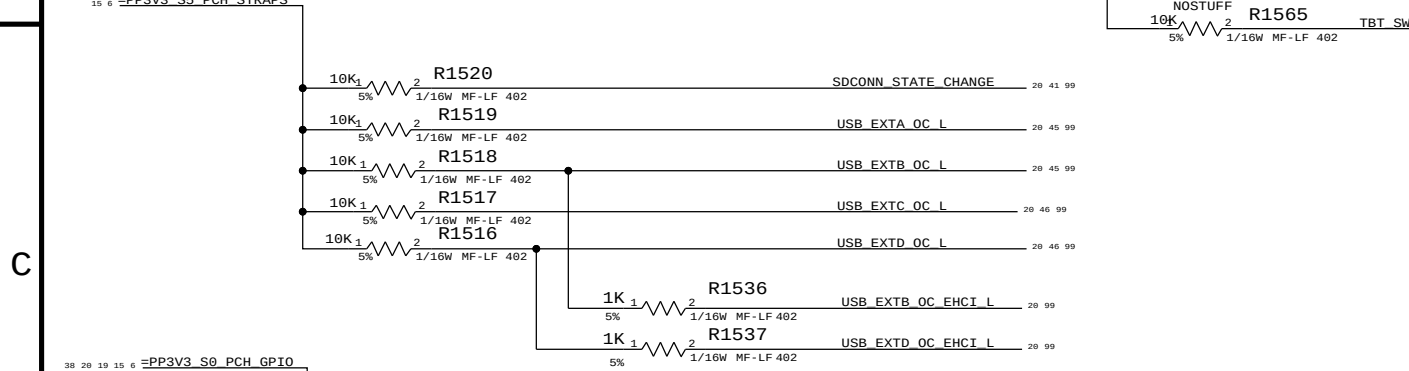
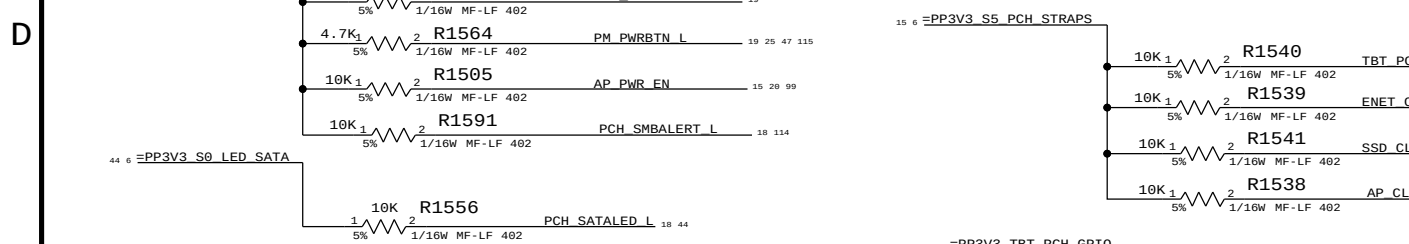
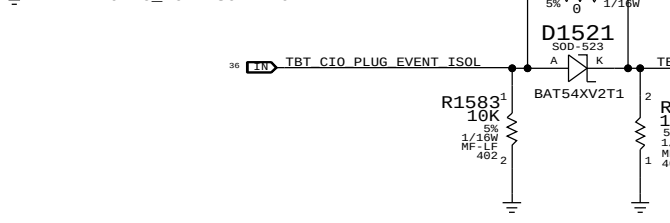
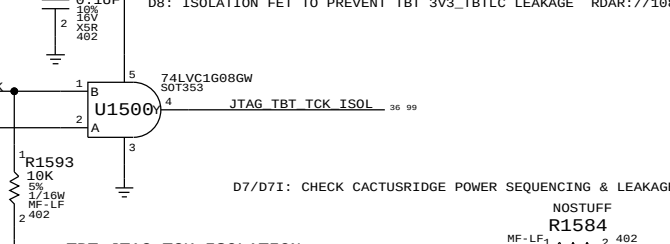
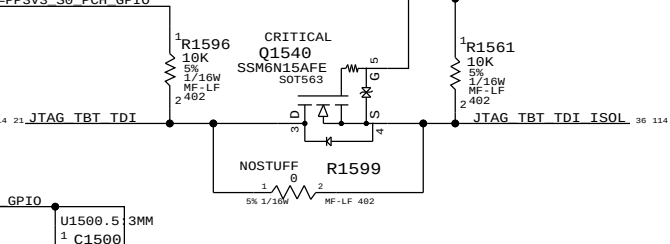
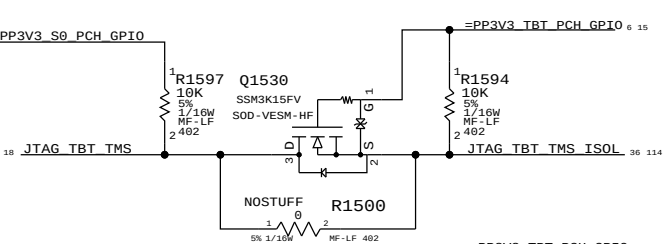
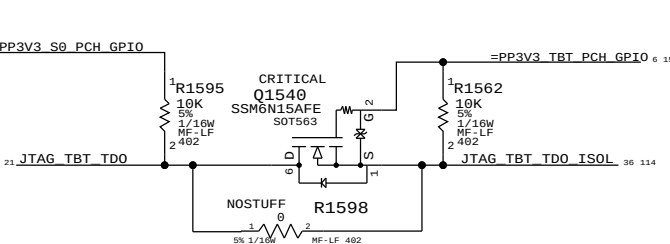
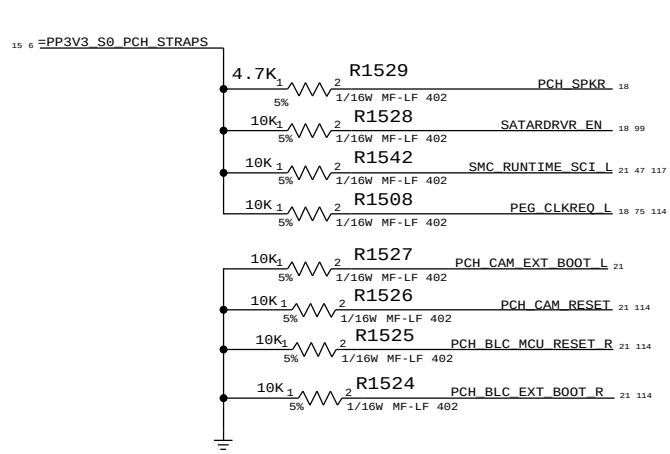
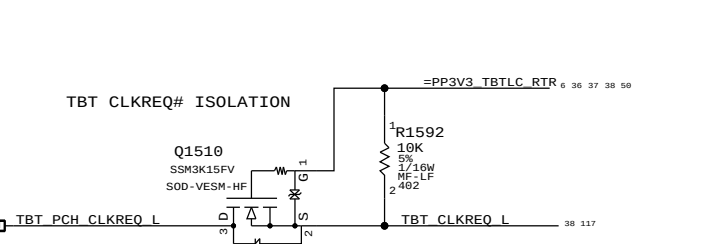
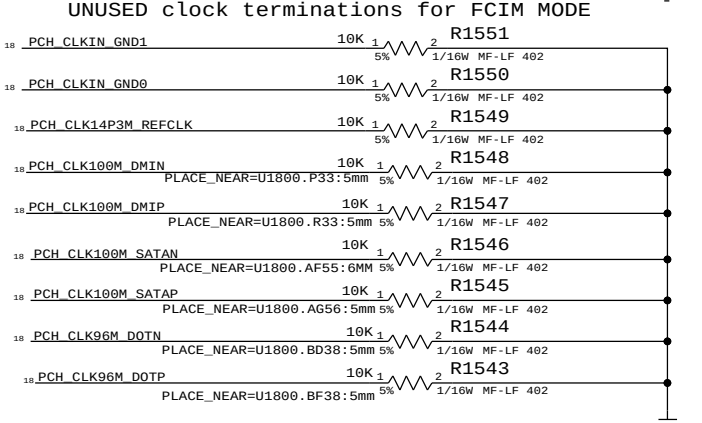
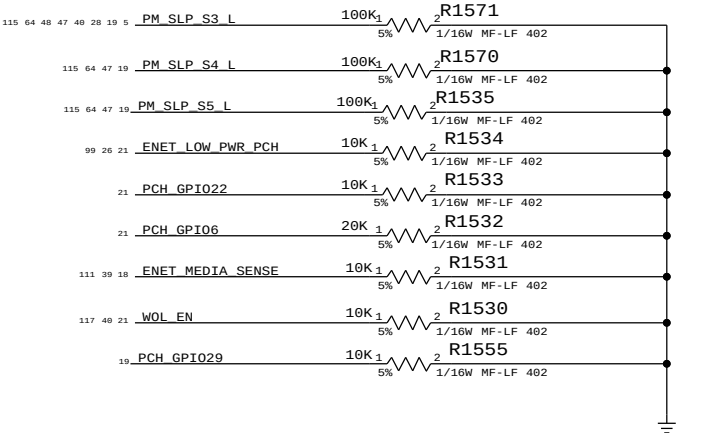
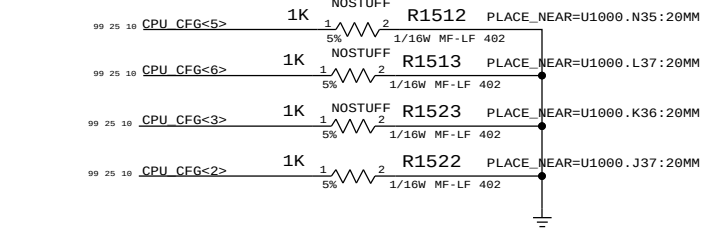
SYNC MASTER=D7 MLB

SYNC DATE=03/15/2012

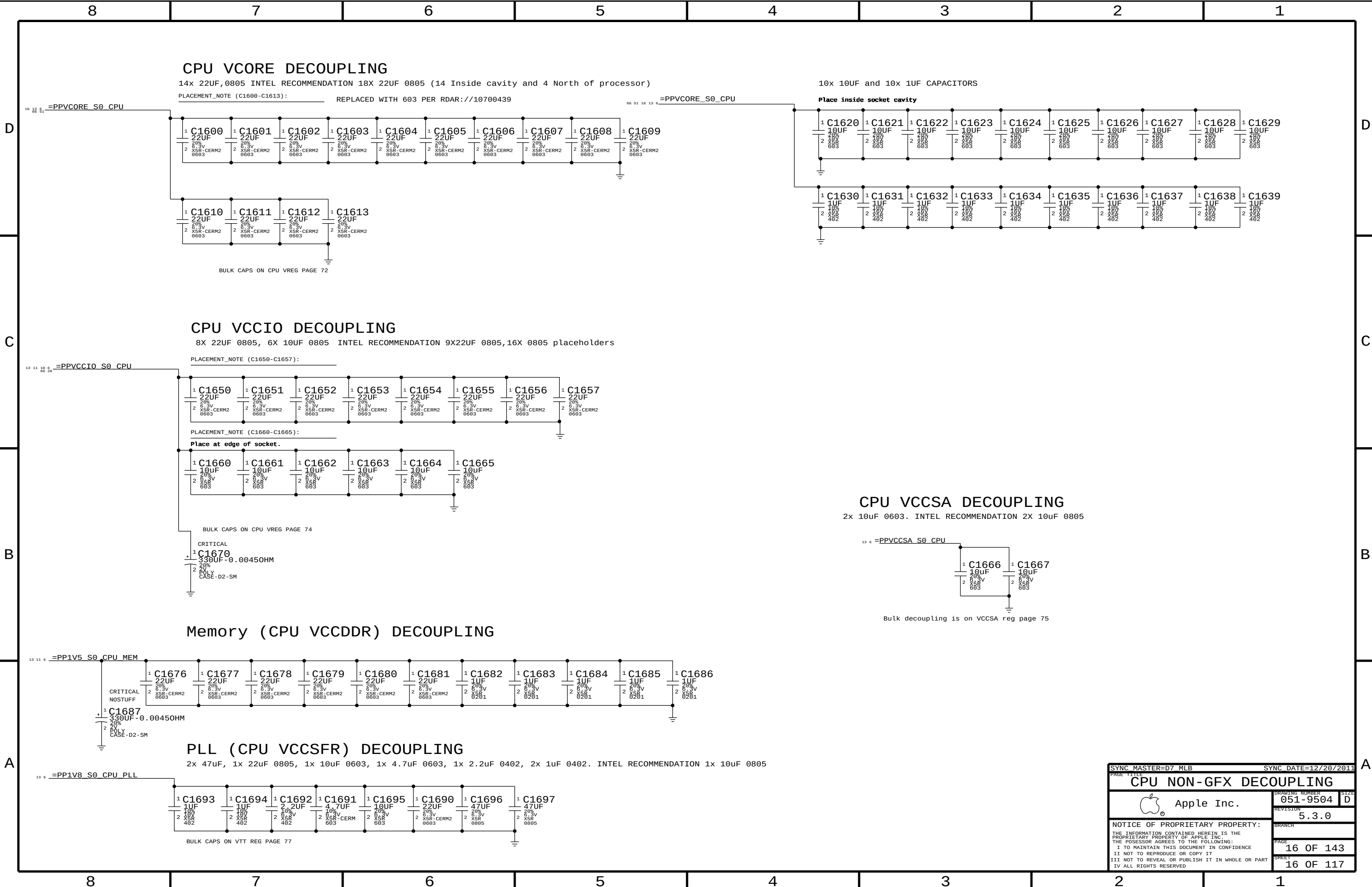
PAGE TITLE		CPU GROUNDS	
 Apple Inc.	DRAWING NUMBER	051-9504	SIZE
	REVISION	5.3.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	14 OF 143
		SHEET	14 OF 117



IVB PCIe Straps configuration:
CFG[5:6] = Sel PCIe Cfg CFG[3]=Direct/Rev for X4 CFG[2]= Direct/Rev for x16
11 = 1x16 (default) 1 = DIR 1 = DIR
10 = 2x8 0 = REV 0 = REV
00 = 1x8,2x4



SYNC MASTER=D8_MLB		SYNC DATE=03/29/2012	
PAGE TITLE		STRAPS,PULL UPS,PULL DOWNS FOR PCH AND CPU	
		DRAWING NUMBER	051-9504
Apple Inc.		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	15 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	15 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



CPU Vcore DECOUPLING

14x 22uF,0805 INTEL RECOMMENDATION 18X 22uF 0805 (14 Inside cavity and 4 North of processor)

PLACEMENT_NOTE (C1600-C1613): REPLACED WITH 603 PER RDAR://10700439

10x 10uF and 10x 1uF CAPACITORS

Place inside socket cavity

BULK CAPS ON CPU VREG PAGE 72

CPU VCCIO DECOUPLING

8X 22uF 0805, 6X 10uF 0805 INTEL RECOMMENDATION 9X22uF 0805,16X 0805 placeholders

PLACEMENT_NOTE (C1650-C1657):

PLACEMENT_NOTE (C1660-C1665):

Place at edge of socket.

BULK CAPS ON CPU VREG PAGE 74

CRITICAL
C1670
330uF-0.00450HM
20%
POLY
CASE-D2-SM

CPU VCCSA DECOUPLING

2x 10uF 0603. INTEL RECOMMENDATION 2X 10uF 0805

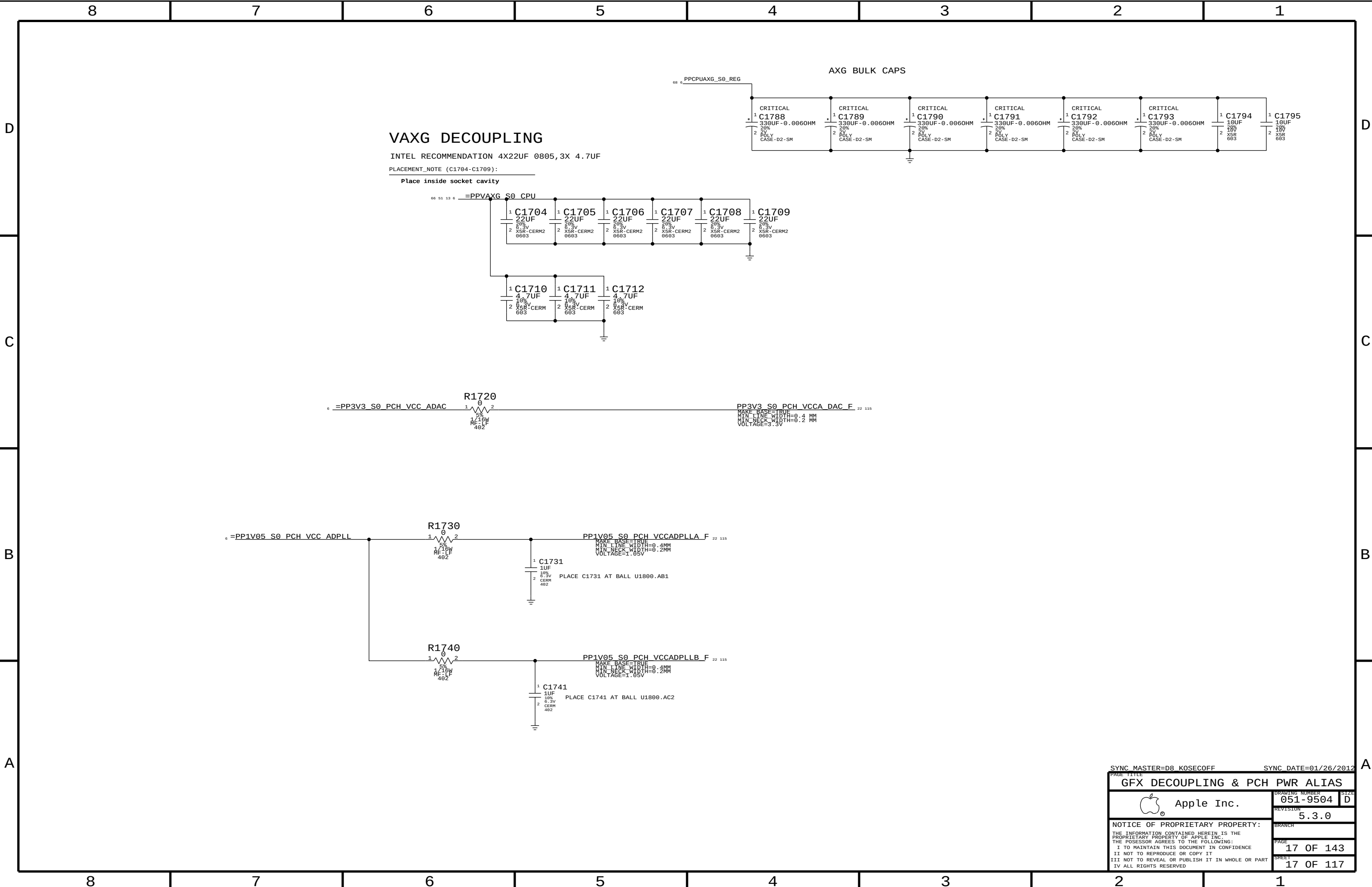
Bulk decoupling is on VCCSA reg page 75

Memory (CPU VCCDDR) DECOUPLING

PLL (CPU VCCSFR) DECOUPLING

2x 47uF, 1x 22uF 0805, 1x 10uF 0603, 1x 4.7uF 0603, 1x 2.2uF 0402, 2x 1uF 0402. INTEL RECOMMENDATION 1x 10uF 0805

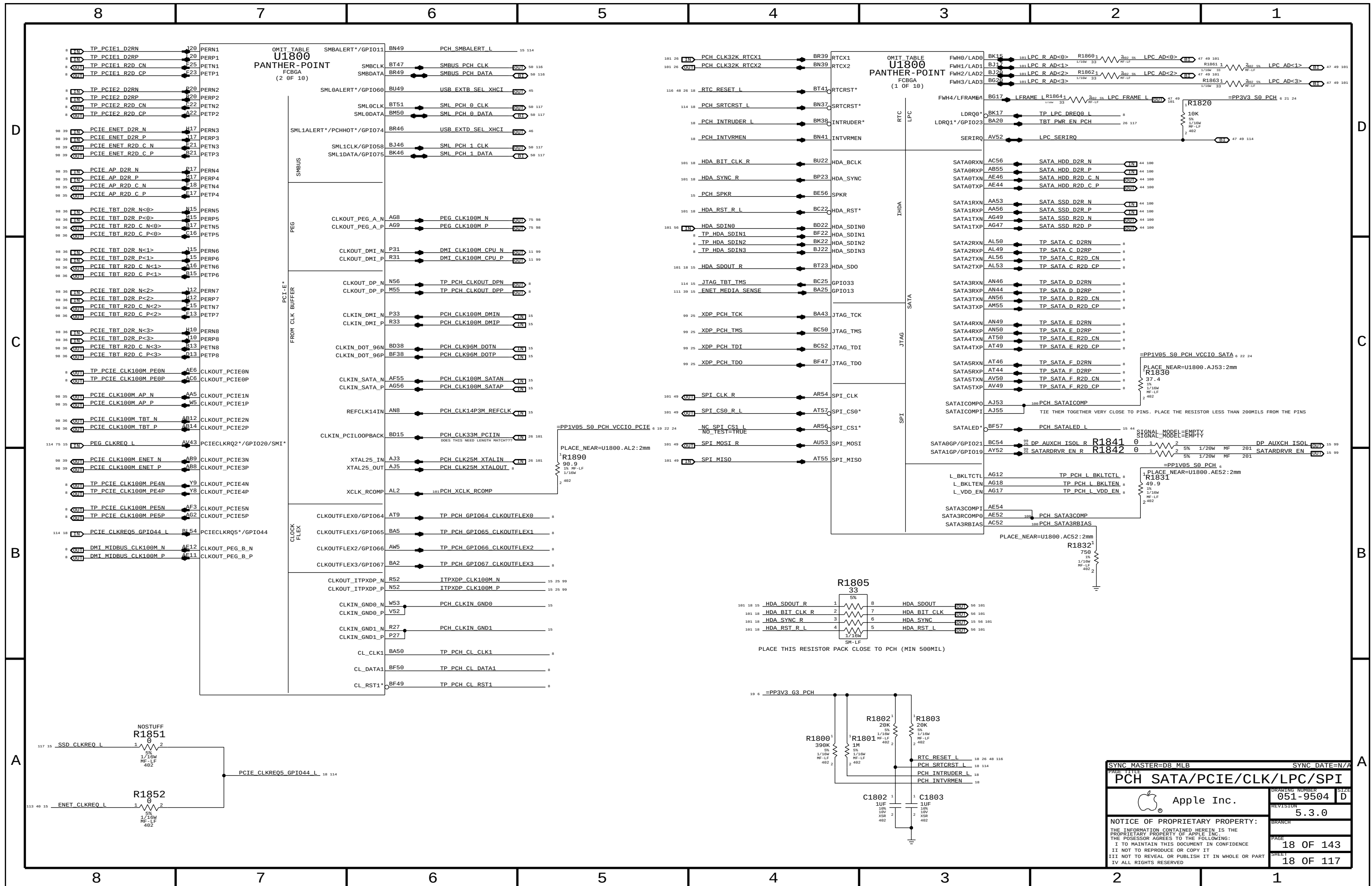
SYNC MASTER=D7_MLB		SYNC DATE=12/20/2011	
PAGE TITLE			
CPU NON-GFX DECOUPLING			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-9504		D
		REVISION	
		5.3.0	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.			
THE POSSESSOR AGREES TO THE FOLLOWING:			
I I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
		BRANCH	
		PAGE	16 OF 143
		SHEET	16 OF 117

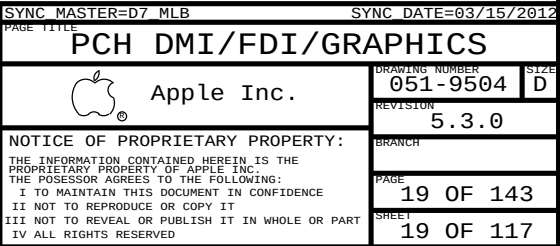


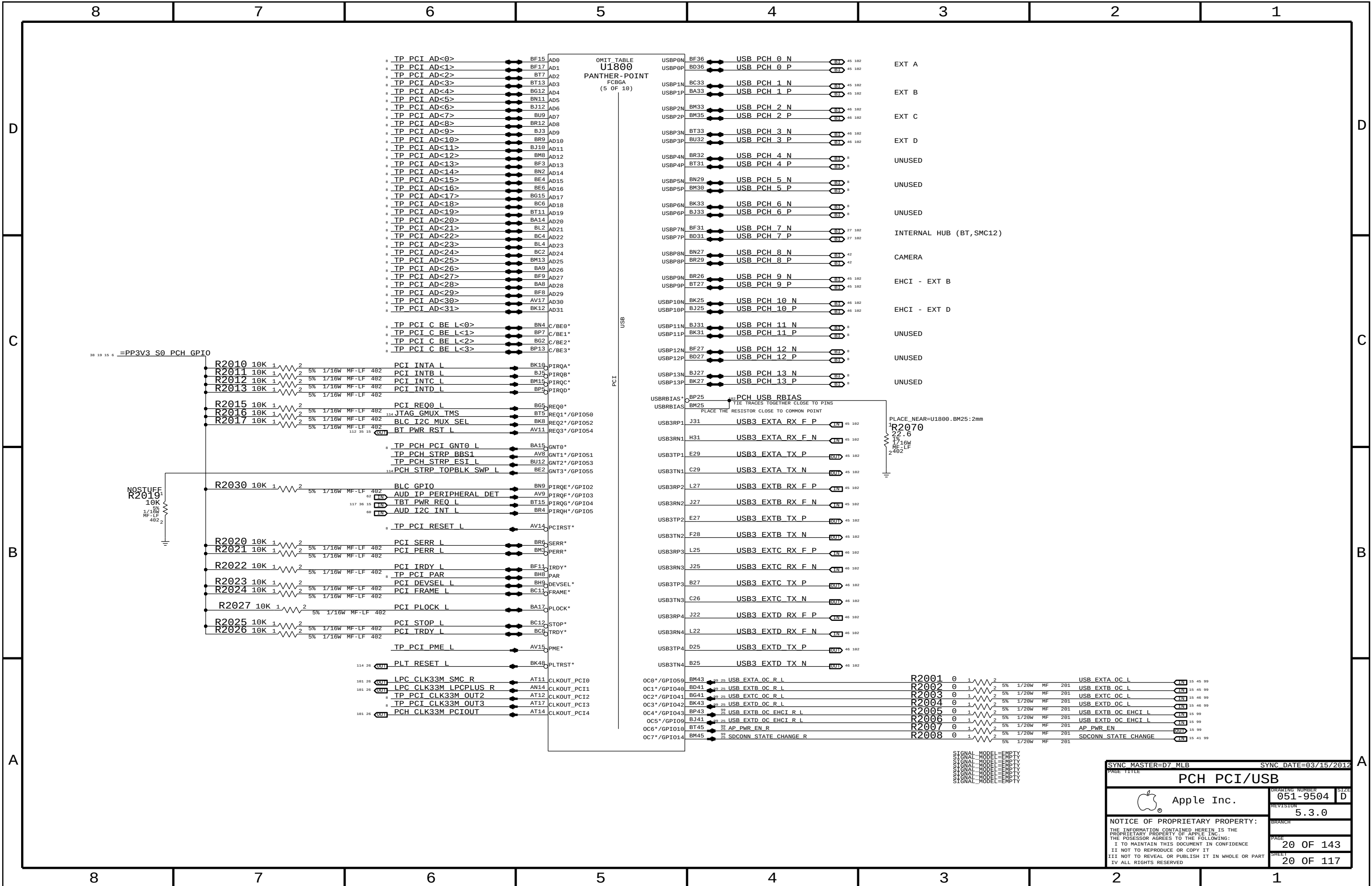
SYNC MASTER=D8 KOSECOFF

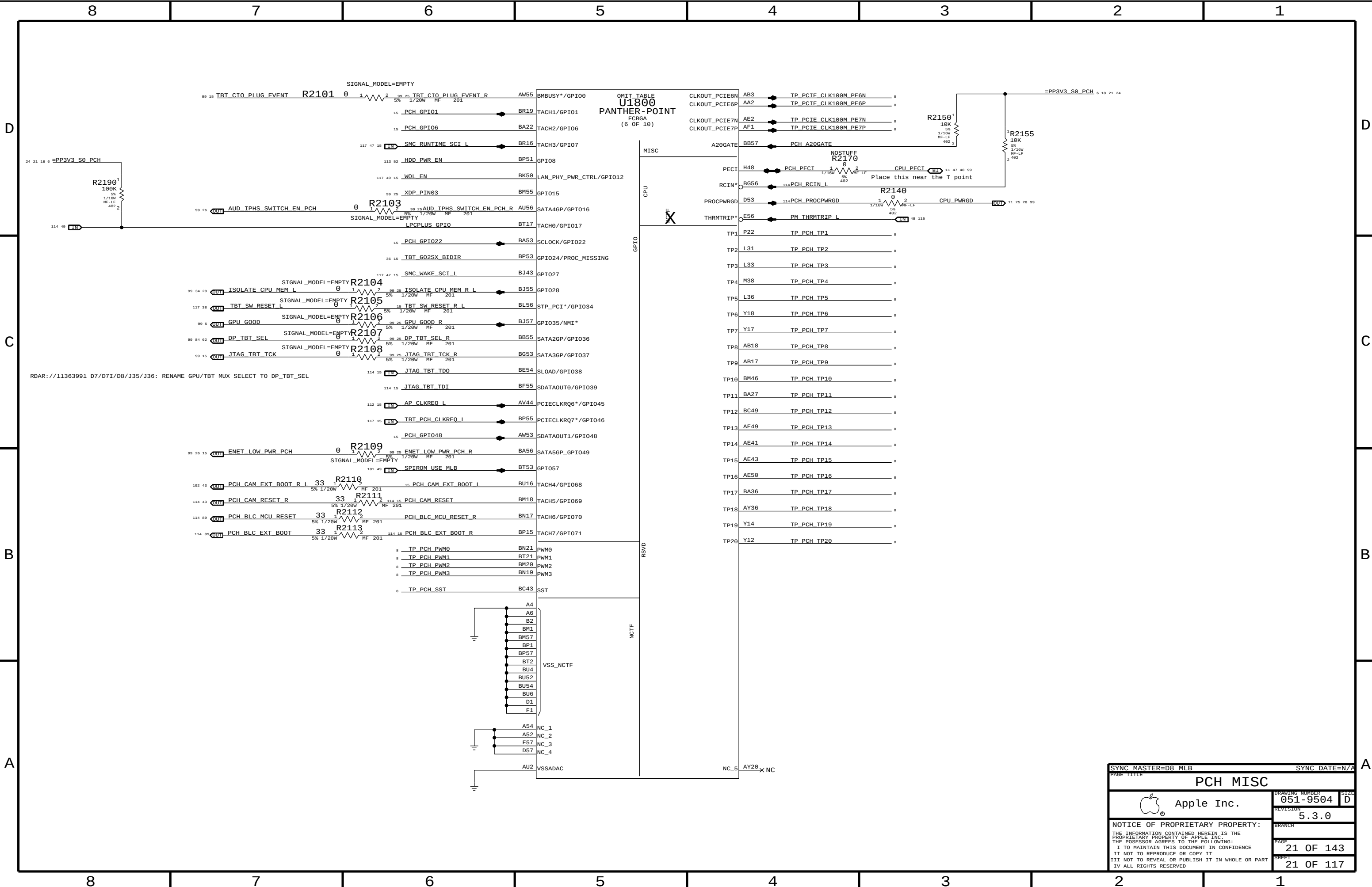
SYNC DATE=01/26/2012

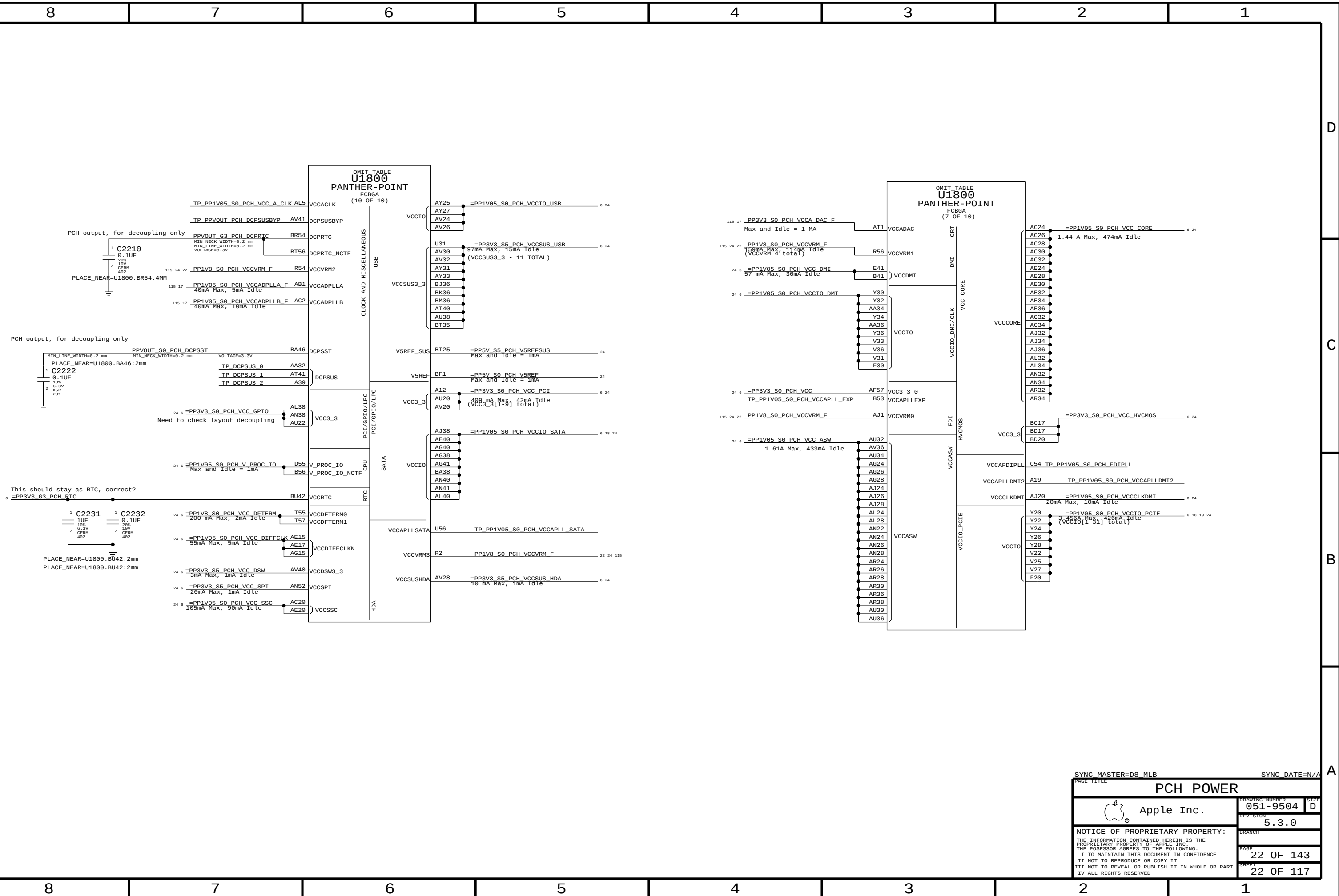
PAGE TITLE		
GFX DECOUPLING & PCH PWR ALIAS		
 Apple Inc.	DRAWING NUMBER	051-9504
	REVISION	5.3.0
	BRANCH	
NOTICE OF PROPRIETARY PROPERTY:		PAGE
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		17 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET
II NOT TO REPRODUCE OR COPY IT		
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		17 OF 117
IV ALL RIGHTS RESERVED		

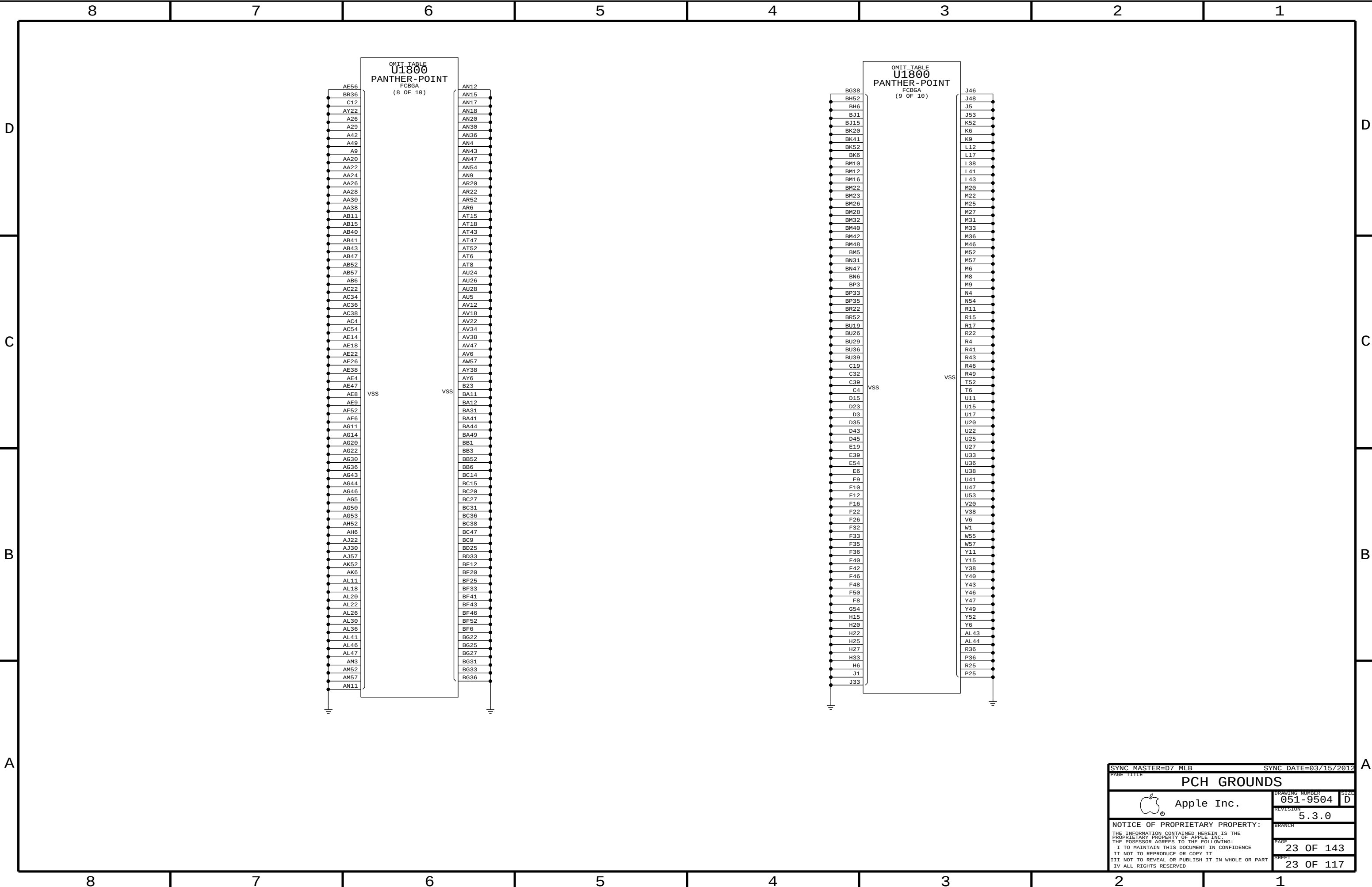








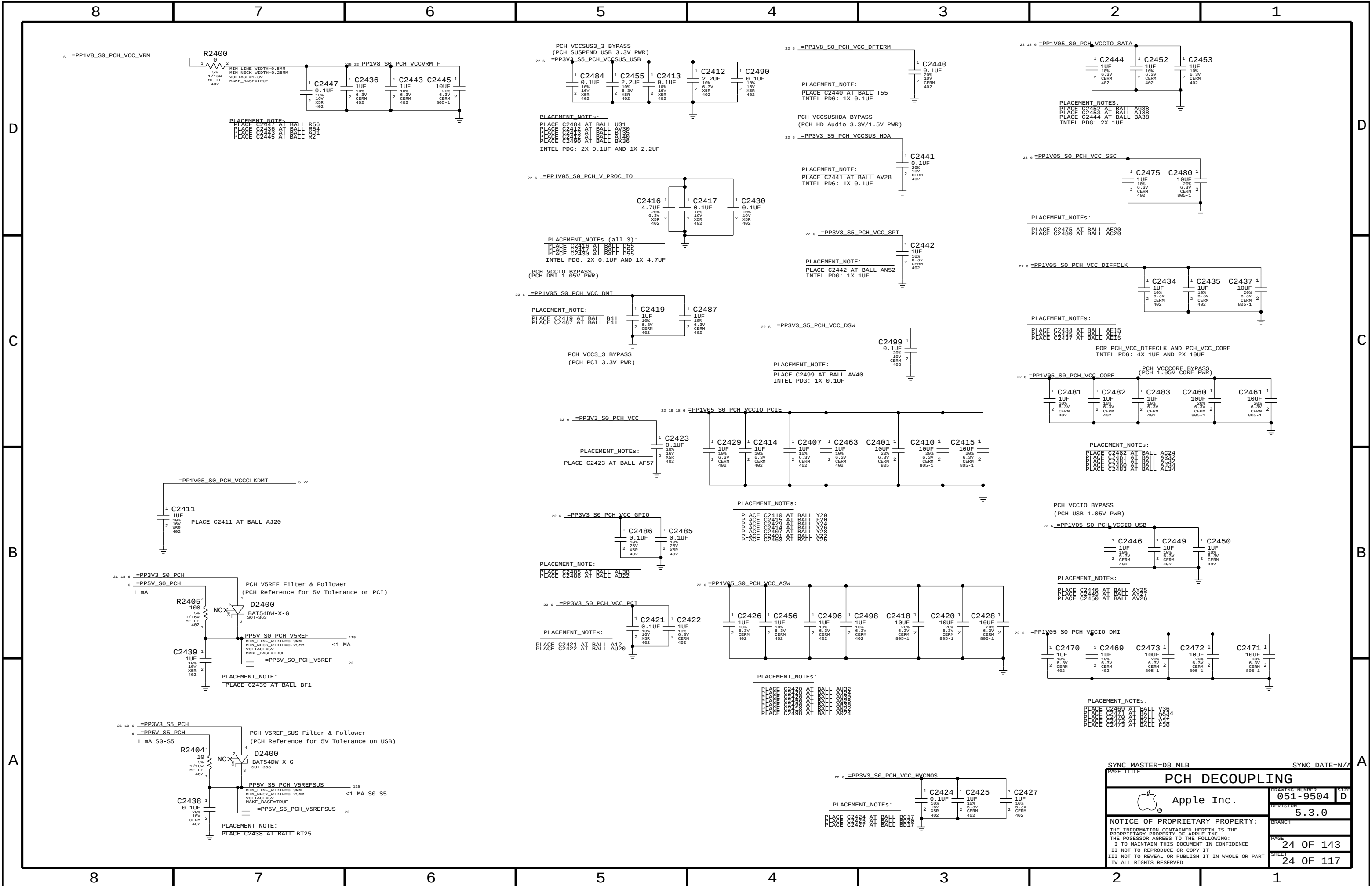




SYNC MASTER=D7_MLB

SYNC DATE=03/15/2012

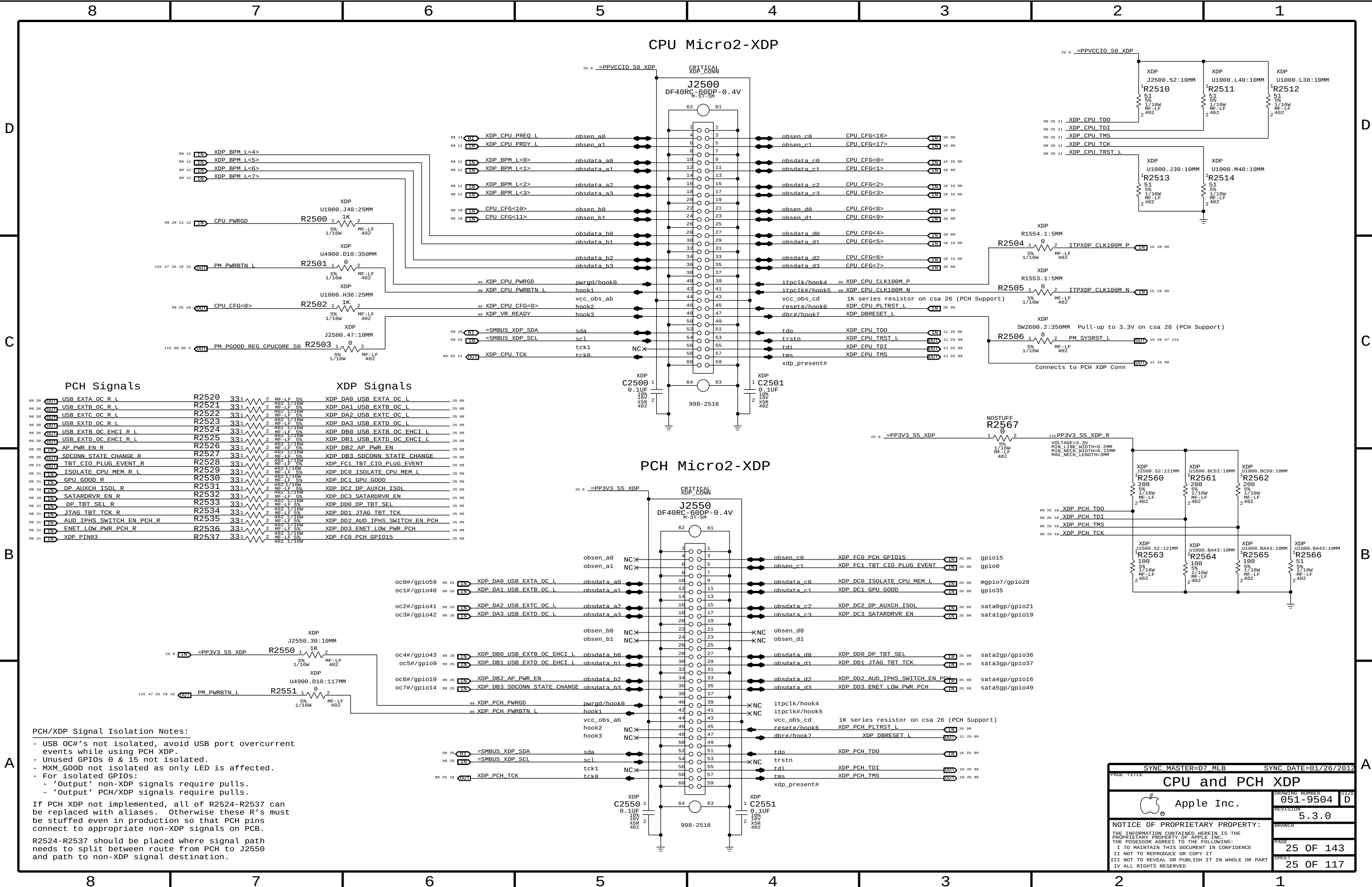
PAGE TITLE		PCH GROUNDS	
 Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	23 OF 143
		SHEET	23 OF 117



SYNC MASTER=D8_MLB

SYNC DATE=N/A

PAGE TITLE		PCH DECOUPLING	
 Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	24 OF 143
		SHEET	24 OF 117



PCH/XDP Signal Isolation Notes:

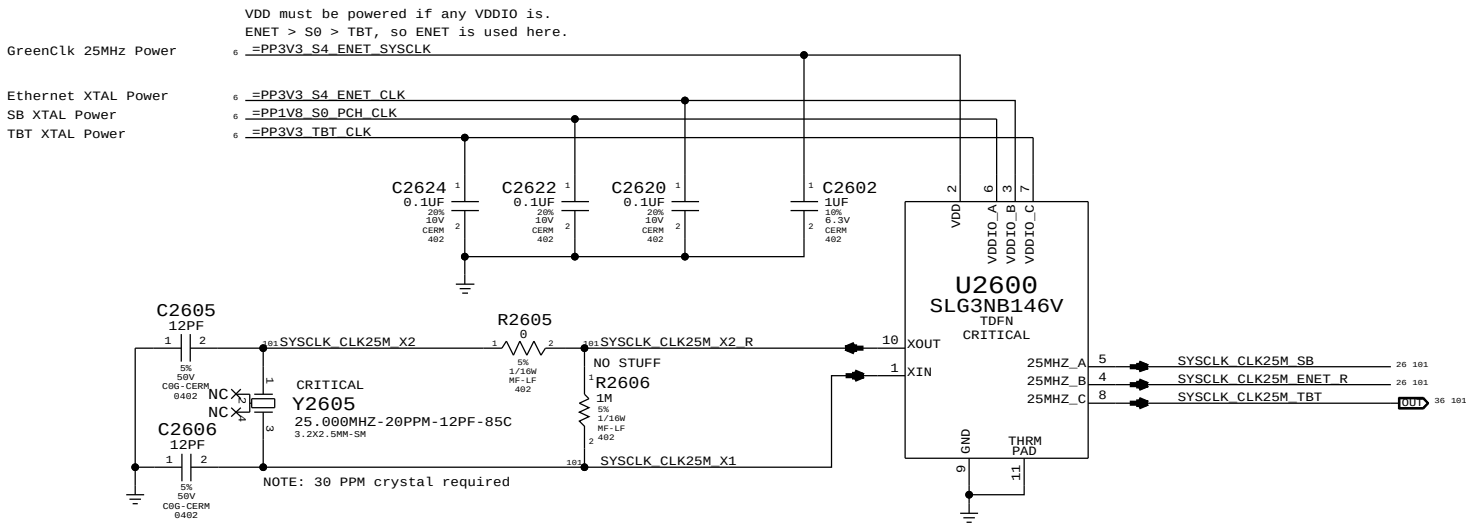
- USB OC#'s not isolated, avoid USB port overcurrent events while using PCH XDP.
- Unused GPIOs 0 & 15 not isolated.
- MXM_GOOD not isolated as only LED is affected.
- For isolated GPIOs:
 - 'Output' non-XDP signals require pulls.
 - 'Output' PCH/XDP signals require pulls.

If PCH XDP not implemented, all of R2524-R2537 can be replaced with aliases. Otherwise these R's must be stuffed even in production so that PCH pins connect to appropriate non-XDP signals on PCB.

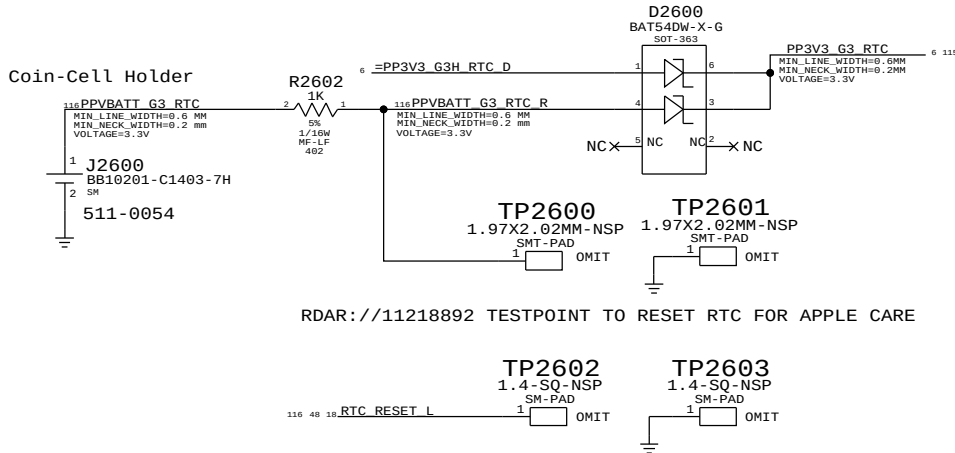
R2524-R2537 should be placed where signal path needs to split between route from PCH to J2550 and path to non-XDP signal destination.

SYNC MASTER=D7 MLB		SYNC DATE=01/26/2012	
CPU and PCH XDP			
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	25 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	25 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

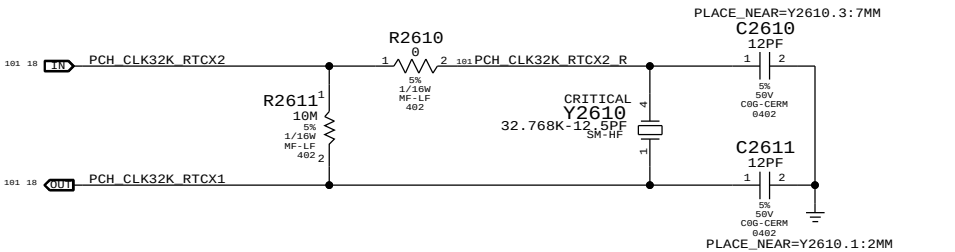
System 25MHz Clock Generator



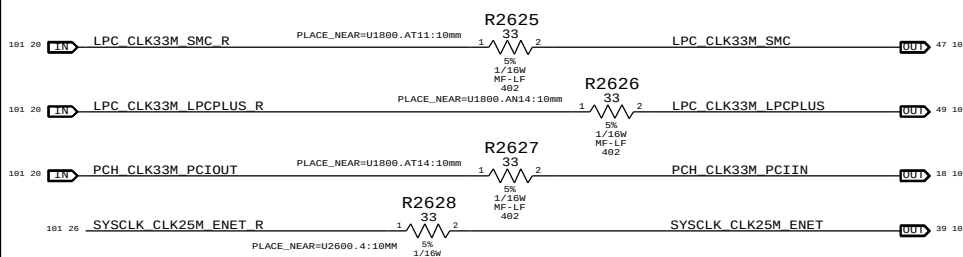
RTC Power Sources



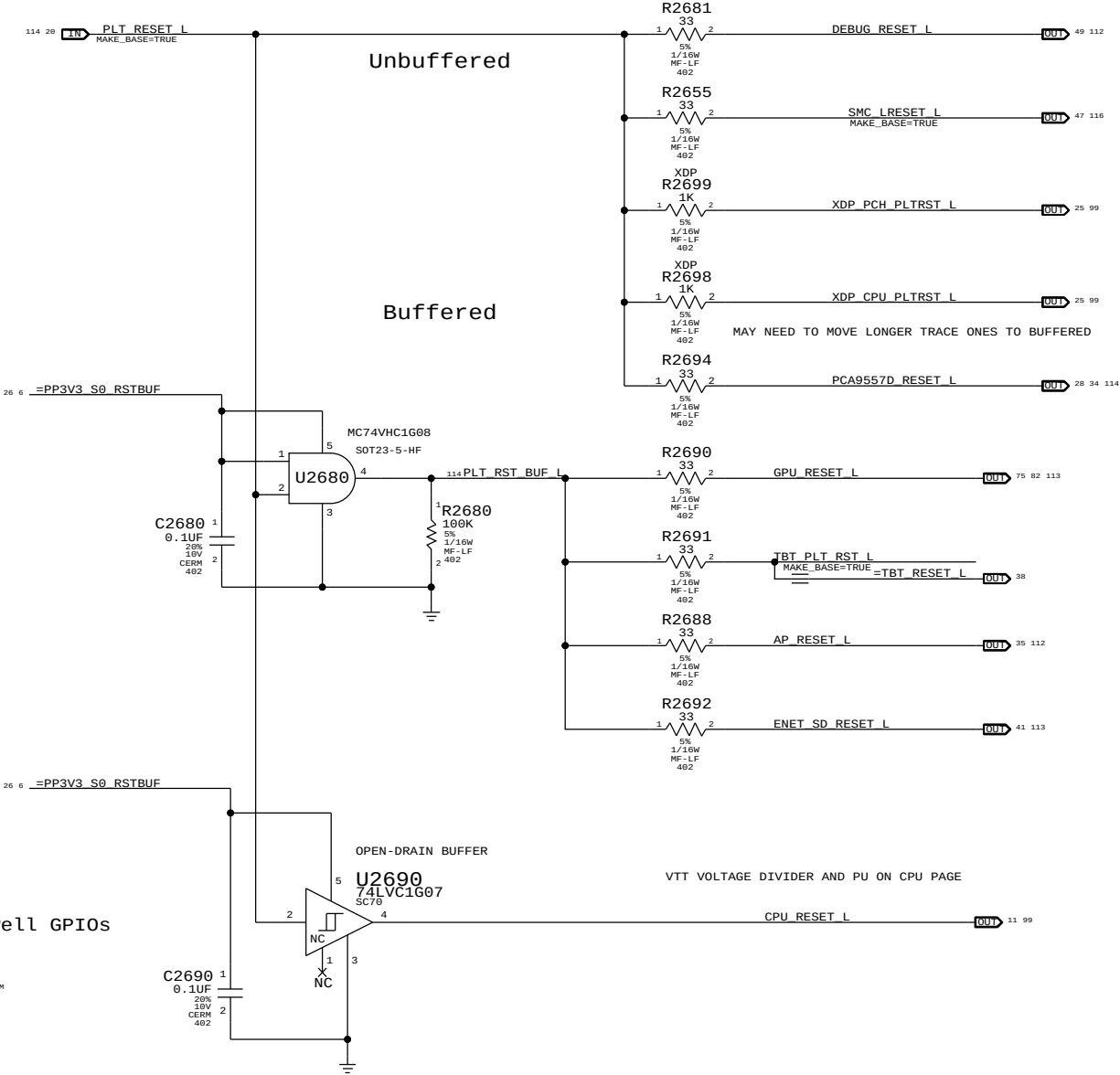
PCH RTC Crystal



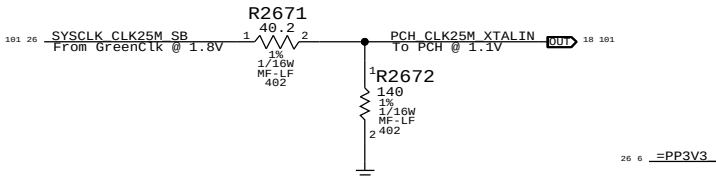
Clock series termination



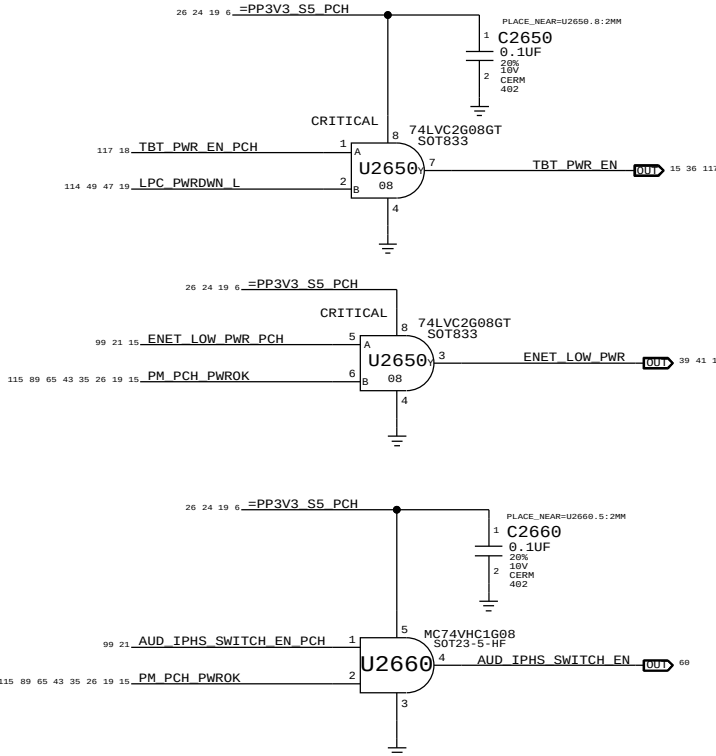
Platform Reset Connections



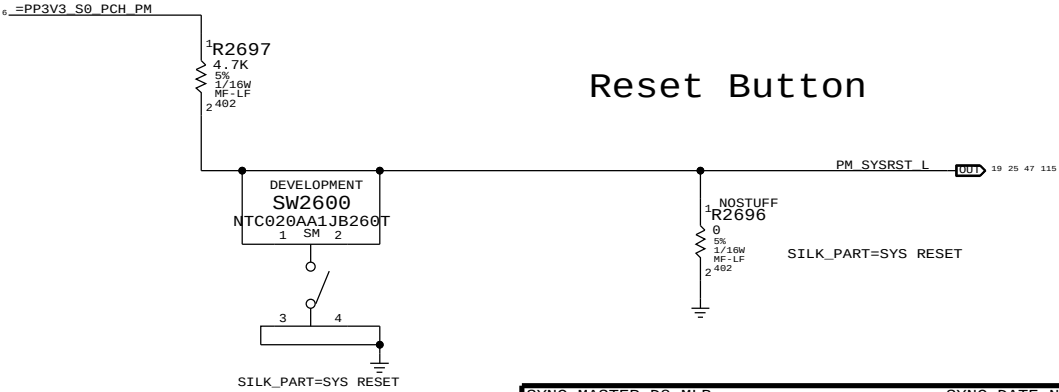
PCH 25MHZ CLOCK



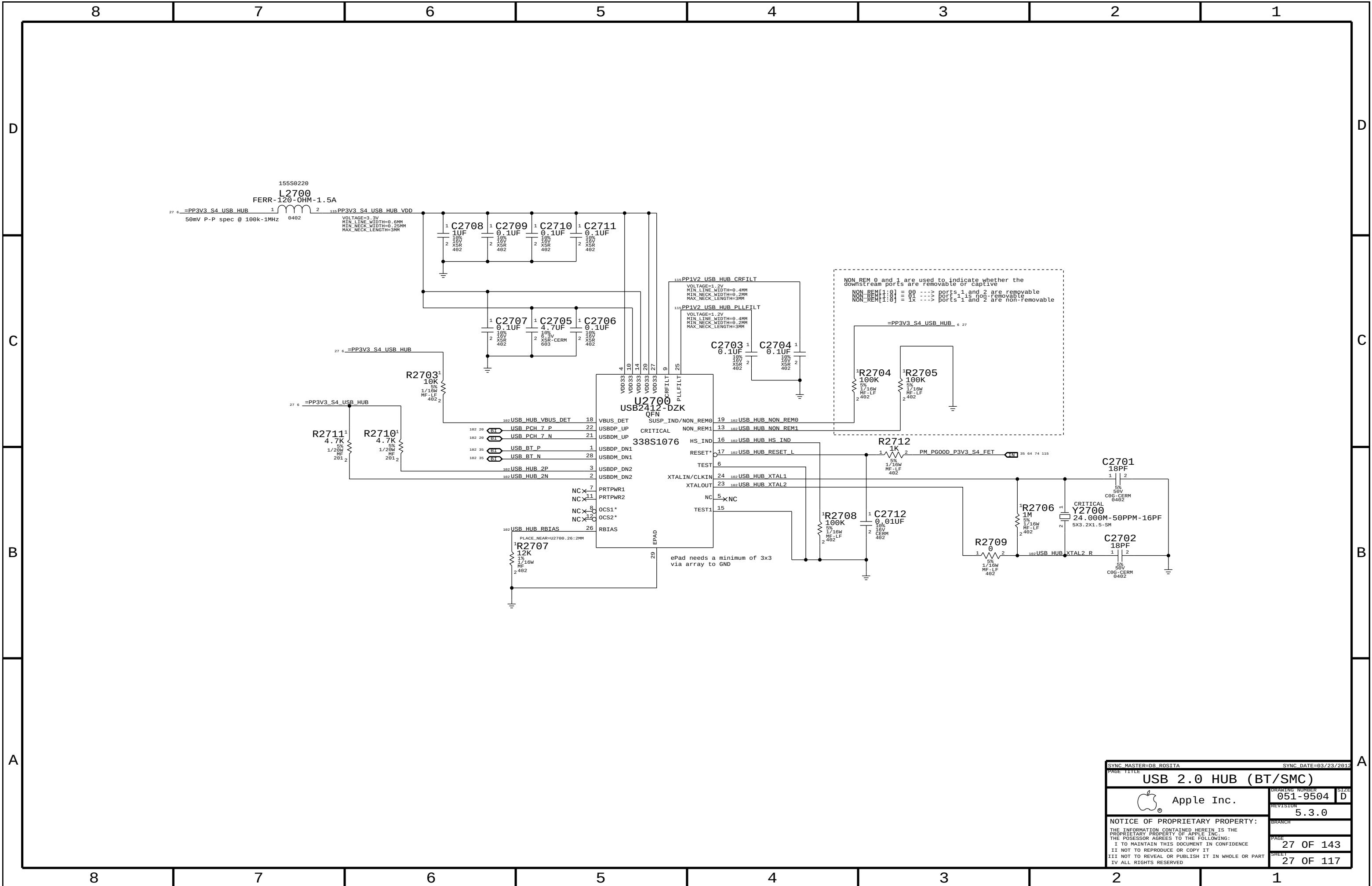
GPIO Isolation to prevent glitches on critical core well GPIOs



Reset Button



SYNC MASTER=D8_MLB		SYNC DATE=N/A	
PAGE TITLE		CHIPSET SUPPORT	
 Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	26 OF 143
		SHEET	26 OF 117



SYNC MASTER=D8 ROSITA		SYNC DATE=03/23/2012	
PAGE TITLE			
USB 2.0 HUB (BT/SMC)			
 Apple Inc.		DRAWING NUMBER	051-9504
		SIZE	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	5.3.0
		BRANCH	
		PAGE	27 OF 143
		SHEET	27 OF 117

MEM_RESET_L Generator

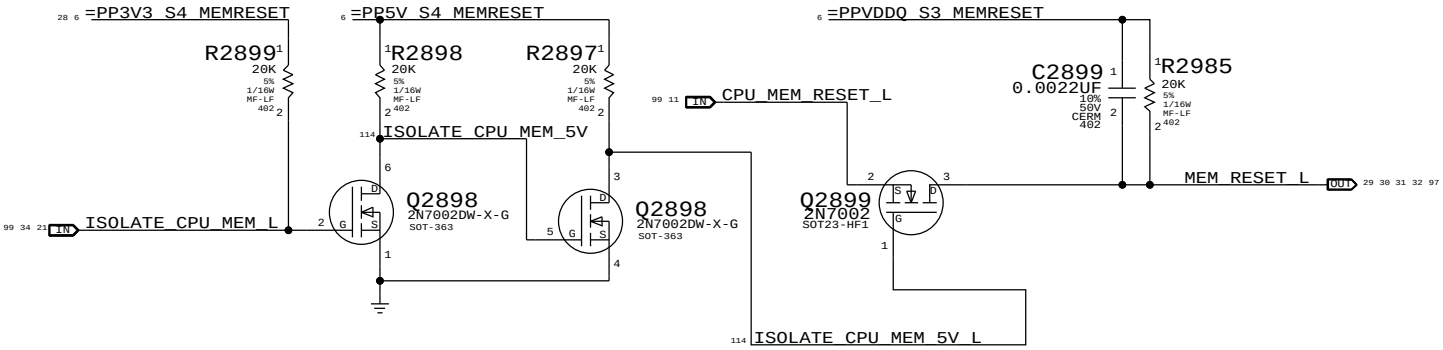
The circuits below handle MEMVTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behaviour of signals.

WHEN HIGH: MEM_RESET_L NOT ISOLATED.

WHEN LOW: MEM_RESET_L IS ISOLATED.

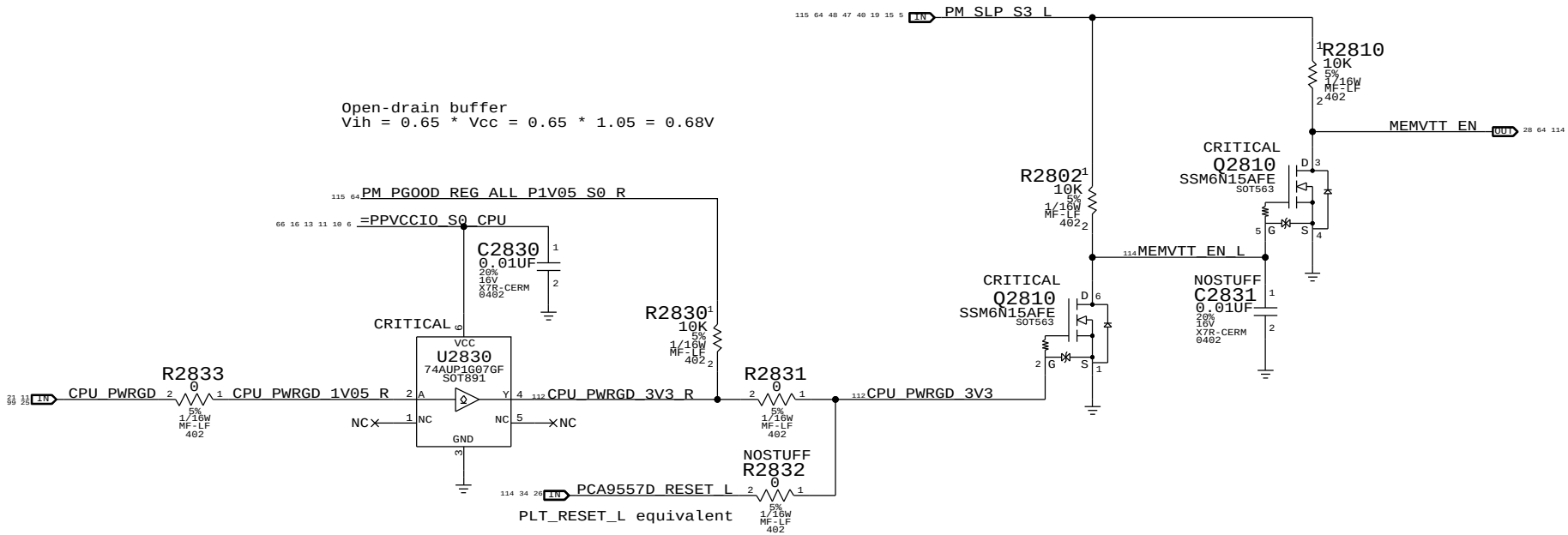
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L (Block CPU from driving MEM_RESET_L in S3)



MEMVTT_EN Generator

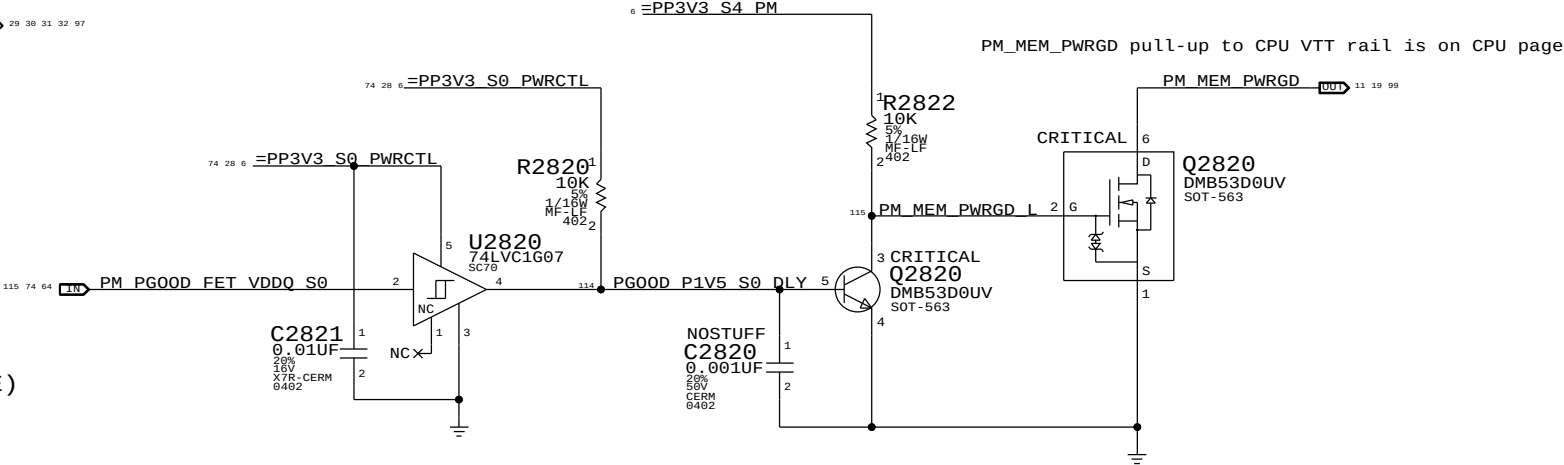
rdar://11117167 Enables MEMVTT when PCH drives CPU PWRGD.

CPU does not drive MEM_CKE until VCCORE activated but CPU 1V5 (VDDQ) leaks into it. Clamping MEMVTT will keep the MEM_CKE low until CPU actively controls it. MEMVTT Clamp actively holds MEMVTT rail low until MEMVTT is enabled. MEMVTT_EN = CPU_PWRGD * PM_SLP_S3_L (VTT is enabled when PCH tells CPU to enable VCCORE)



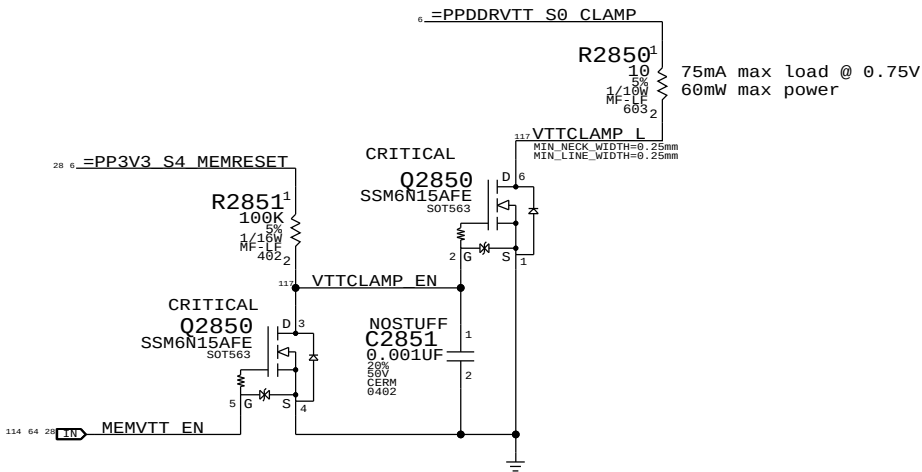
1V5 S0 "PGOOD" for CPU

With optional delay from 1V5 S0 PG00D



MEMVTT Clamp

Ensures CKE signals are held low in S3 and in S0 before CPU PWRGD

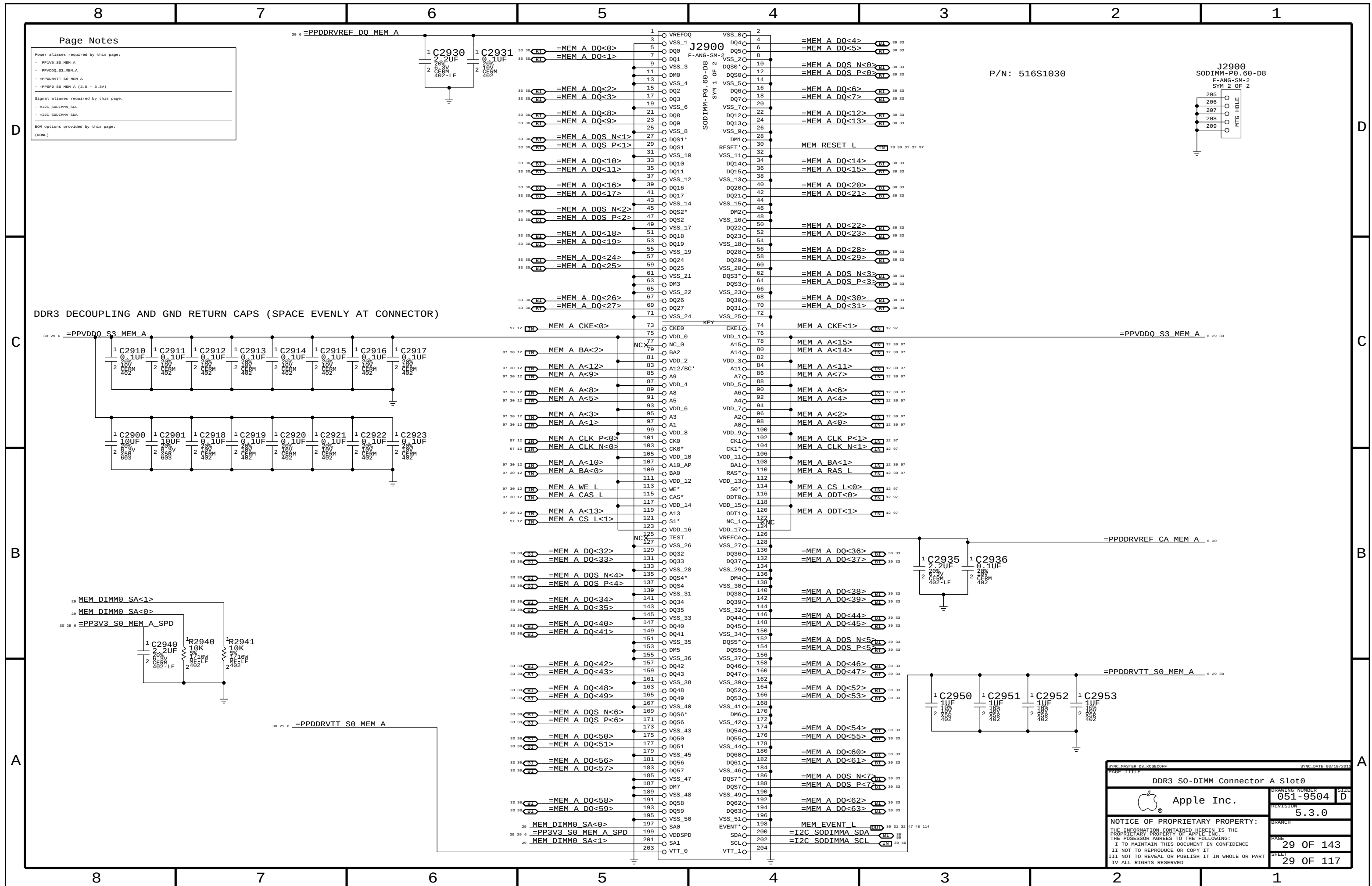


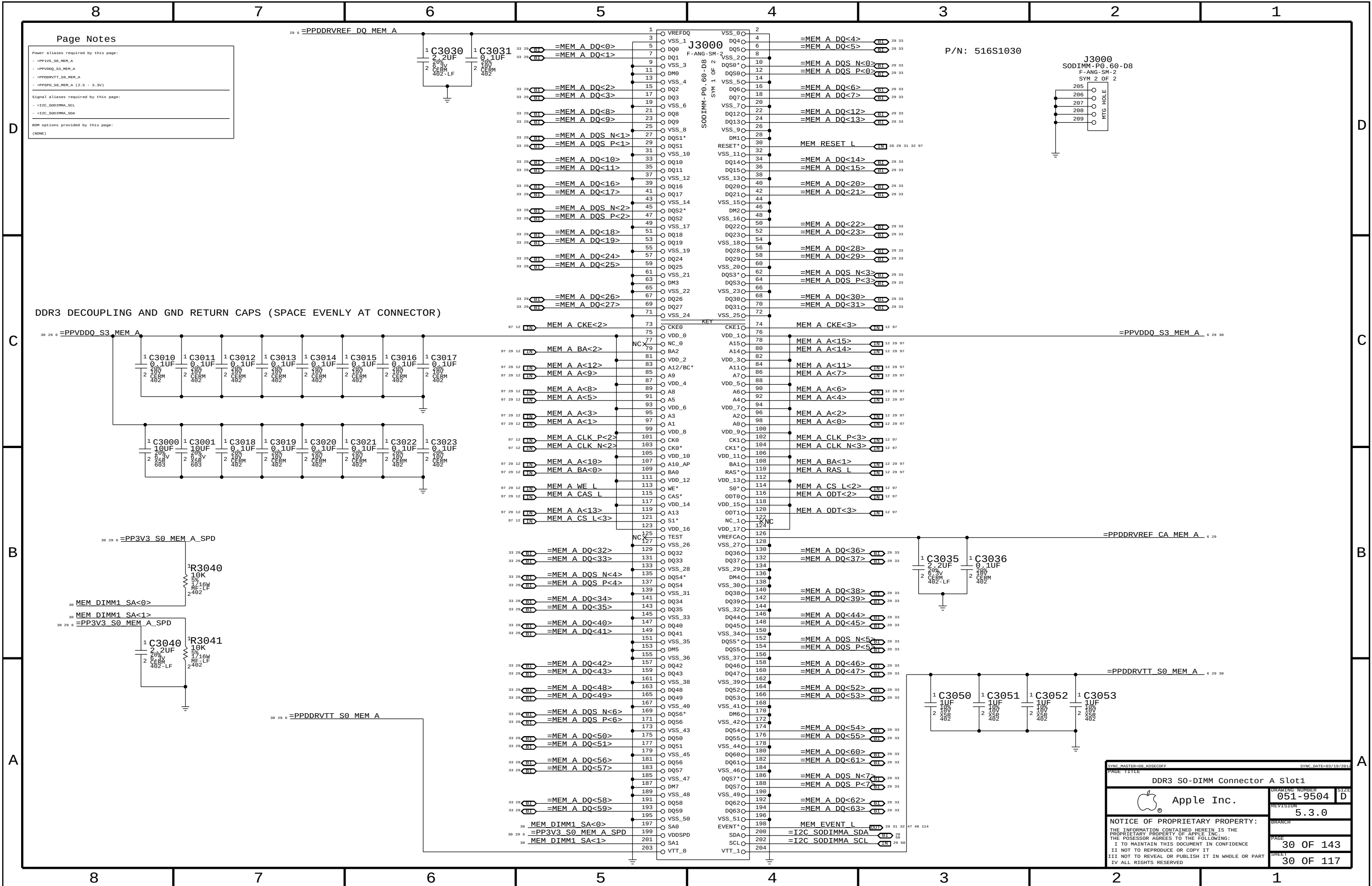
Step	ISOLATE_CPU_MEM_L	PM_SLP_S3_L	CPU_PWRGD	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	CPU_MEM_RESET_L	1	1
1	0	1	1	1	1	1
2	0	1	1	1	1	1
3	0	0	0	X	1	0
4	0	0	0	X	1	0
5	0	1	1	0 (*)	1	1
6	0	1	1	1	1	1
S0	7	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: On a S5->S0 transition, ISOLATE_CPU_MEM_L will default low. Rails will power-up as if from S3, but MEM_RESET_L now needs to be asserted in S0. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC MASTER=D8 MARK		SYNC DATE=04/23/2012	
PAGE TITLE		CPU Memory S3 Support	
DRAWING NUMBER		051-9504	SIZE D
REVISION		5.3.0	BRANCH
NOTICE OF PROPRIETARY PROPERTY:		PAGE 28 OF 143	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		SHEET 28 OF 117	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			





Page Notes

Power aliases required by this page:

- =PP3V3_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVT_S0_MEM_A
- =PPDDRREF_S0_MEM_A (2.5 - 3.3V)

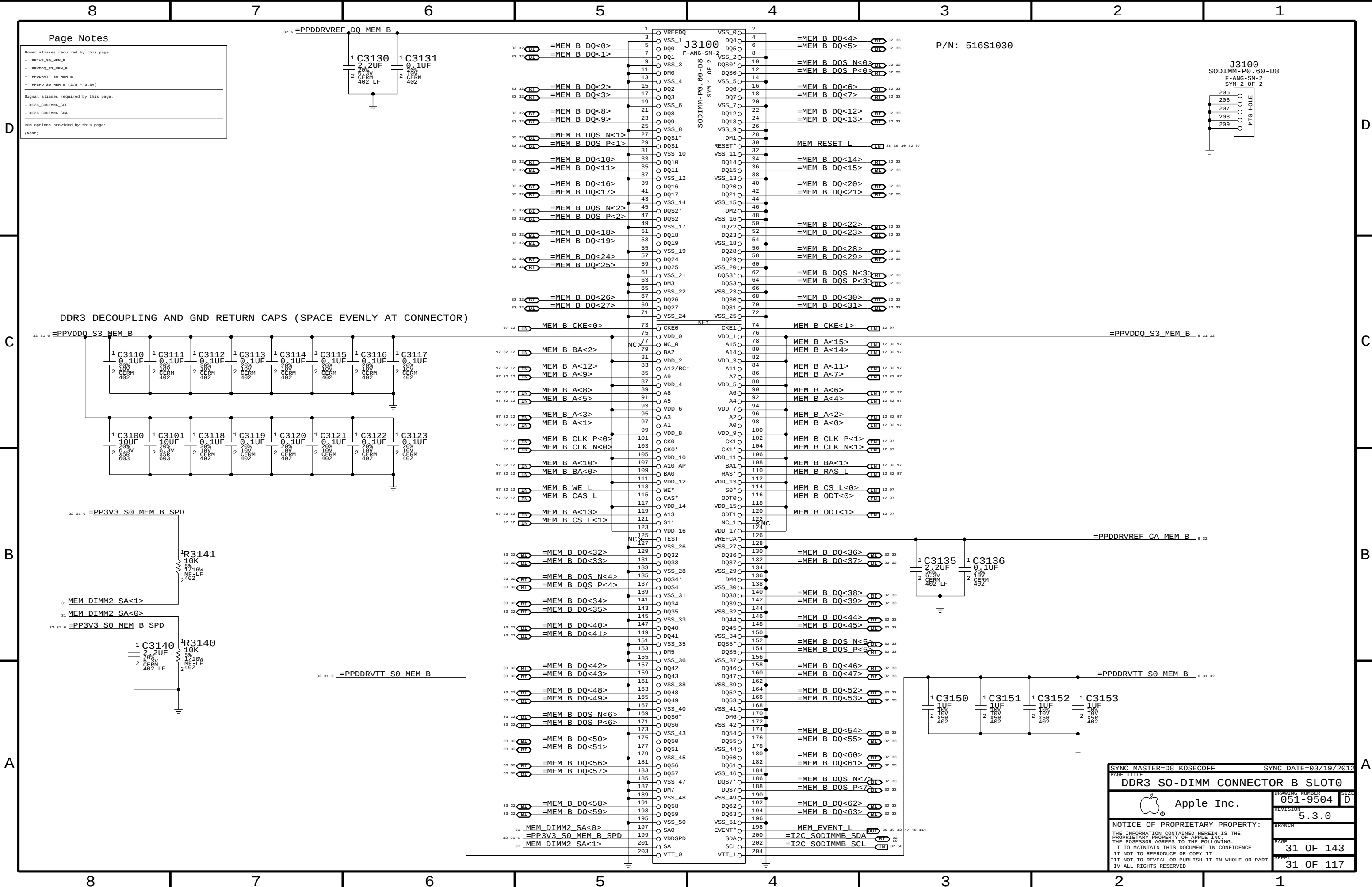
Signal aliases required by this page:

- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)

SYNC MASTER=09_K05EC0FF		SYNC DATE=03/19/2015	
PAGE TITLE			
DDR3 S0-DIMM Connector A Slot1		DRAWING NUMBER	SIZE
	Apple Inc.	051-9504	D
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
PAGE	30 OF 143		
SHEET	30 OF 117		



Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVTT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

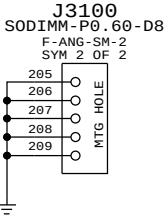
- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA

BOM options provided by this page:

(NONE)

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

P/N: 516S1030



SYNC MASTER=D8 KOSECOFF		SYNC DATE=03/19/2012	
PAGE TITLE		DRAWING NUMBER	
DDR3 SO-DIMM CONNECTOR B SLOT0		051-9504	
Apple Inc.		REVISION	
		5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		31 OF 143	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		31 OF 117	
IV ALL RIGHTS RESERVED			

8		7		6		5		4		3		2		1	
D	97 12	MEM A DQS N<0>	==	=MEM A DQS N<0>	29 30	97 12	MEM B DQS N<0>	==	=MEM B DQS N<0>	31 32	D				
	97 12	MEM A DQS P<0>	==	=MEM A DQS P<0>	29 30	97 12	MEM B DQS P<0>	==	=MEM B DQS P<0>	31 32					
	97 12	MEM A DQ<7>	==	=MEM A DQ<7>	29 30	97 12	MEM B DQ<7>	==	=MEM B DQ<3>	31 32					
	97 12	MEM A DQ<6>	==	=MEM A DQ<6>	29 30	97 12	MEM B DQ<6>	==	=MEM B DQ<6>	31 32					
	97 12	MEM A DQ<5>	==	=MEM A DQ<4>	29 30	97 12	MEM B DQ<5>	==	=MEM B DQ<5>	31 32					
	97 12	MEM A DQ<4>	==	=MEM A DQ<1>	29 30	97 12	MEM B DQ<4>	==	=MEM B DQ<4>	31 32					
	97 12	MEM A DQ<3>	==	=MEM A DQ<3>	29 30	97 12	MEM B DQ<3>	==	=MEM B DQ<7>	31 32					
	97 12	MEM A DQ<2>	==	=MEM A DQ<2>	29 30	97 12	MEM B DQ<2>	==	=MEM B DQ<2>	31 32					
	97 12	MEM A DQ<1>	==	=MEM A DQ<5>	29 30	97 12	MEM B DQ<1>	==	=MEM B DQ<1>	31 32					
	97 12	MEM A DQ<0>	==	=MEM A DQ<0>	29 30	97 12	MEM B DQ<0>	==	=MEM B DQ<0>	31 32					
C	97 12	MEM A DQS N<1>	==	=MEM A DQS N<1>	29 30	97 12	MEM B DQS N<1>	==	=MEM B DQS N<1>	31 32	C				
	97 12	MEM A DQS P<1>	==	=MEM A DQS P<1>	29 30	97 12	MEM B DQS P<1>	==	=MEM B DQS P<1>	31 32					
	97 12	MEM A DQ<15>	==	=MEM A DQ<15>	29 30	97 12	MEM B DQ<15>	==	=MEM B DQ<11>	31 32					
	97 12	MEM A DQ<14>	==	=MEM A DQ<14>	29 30	97 12	MEM B DQ<14>	==	=MEM B DQ<10>	31 32					
	97 12	MEM A DQ<13>	==	=MEM A DQ<13>	29 30	97 12	MEM B DQ<13>	==	=MEM B DQ<13>	31 32					
	97 12	MEM A DQ<12>	==	=MEM A DQ<12>	29 30	97 12	MEM B DQ<12>	==	=MEM B DQ<12>	31 32					
	97 12	MEM A DQ<11>	==	=MEM A DQ<11>	29 30	97 12	MEM B DQ<11>	==	=MEM B DQ<15>	31 32					
	97 12	MEM A DQ<10>	==	=MEM A DQ<10>	29 30	97 12	MEM B DQ<10>	==	=MEM B DQ<14>	31 32					
	97 12	MEM A DQ<9>	==	=MEM A DQ<9>	29 30	97 12	MEM B DQ<9>	==	=MEM B DQ<8>	31 32					
	97 12	MEM A DQ<8>	==	=MEM A DQ<8>	29 30	97 12	MEM B DQ<8>	==	=MEM B DQ<9>	31 32					
B	97 12	MEM A DQS N<2>	==	=MEM A DQS N<2>	29 30	97 12	MEM B DQS N<2>	==	=MEM B DQS N<2>	31 32	B				
	97 12	MEM A DQS P<2>	==	=MEM A DQS P<2>	29 30	97 12	MEM B DQS P<2>	==	=MEM B DQS P<2>	31 32					
	97 12	MEM A DQ<23>	==	=MEM A DQ<23>	29 30	97 12	MEM B DQ<23>	==	=MEM B DQ<22>	31 32					
	97 12	MEM A DQ<22>	==	=MEM A DQ<22>	29 30	97 12	MEM B DQ<22>	==	=MEM B DQ<18>	31 32					
	97 12	MEM A DQ<21>	==	=MEM A DQ<21>	29 30	97 12	MEM B DQ<21>	==	=MEM B DQ<21>	31 32					
	97 12	MEM A DQ<20>	==	=MEM A DQ<20>	29 30	97 12	MEM B DQ<20>	==	=MEM B DQ<20>	31 32					
	97 12	MEM A DQ<19>	==	=MEM A DQ<18>	29 30	97 12	MEM B DQ<19>	==	=MEM B DQ<23>	31 32					
	97 12	MEM A DQ<18>	==	=MEM A DQ<19>	29 30	97 12	MEM B DQ<18>	==	=MEM B DQ<19>	31 32					
	97 12	MEM A DQ<17>	==	=MEM A DQ<17>	29 30	97 12	MEM B DQ<17>	==	=MEM B DQ<17>	31 32					
	97 12	MEM A DQ<16>	==	=MEM A DQ<16>	29 30	97 12	MEM B DQ<16>	==	=MEM B DQ<16>	31 32					
A	97 12	MEM A DQS N<3>	==	=MEM A DQS N<3>	29 30	97 12	MEM B DQS N<3>	==	=MEM B DQS N<3>	31 32	A				
	97 12	MEM A DQS P<3>	==	=MEM A DQS P<3>	29 30	97 12	MEM B DQS P<3>	==	=MEM B DQS P<3>	31 32					
	97 12	MEM A DQ<31>	==	=MEM A DQ<31>	29 30	97 12	MEM B DQ<31>	==	=MEM B DQ<31>	31 32					
	97 12	MEM A DQ<30>	==	=MEM A DQ<30>	29 30	97 12	MEM B DQ<30>	==	=MEM B DQ<30>	31 32					
	97 12	MEM A DQ<29>	==	=MEM A DQ<29>	29 30	97 12	MEM B DQ<29>	==	=MEM B DQ<29>	31 32					
	97 12	MEM A DQ<28>	==	=MEM A DQ<28>	29 30	97 12	MEM B DQ<28>	==	=MEM B DQ<28>	31 32					
	97 12	MEM A DQ<27>	==	=MEM A DQ<27>	29 30	97 12	MEM B DQ<27>	==	=MEM B DQ<27>	31 32					
	97 12	MEM A DQ<26>	==	=MEM A DQ<26>	29 30	97 12	MEM B DQ<26>	==	=MEM B DQ<26>	31 32					
	97 12	MEM A DQ<25>	==	=MEM A DQ<25>	29 30	97 12	MEM B DQ<25>	==	=MEM B DQ<25>	31 32					
	97 12	MEM A DQ<24>	==	=MEM A DQ<24>	29 30	97 12	MEM B DQ<24>	==	=MEM B DQ<24>	31 32					

SYNC MASTER=D8 KOSECOFF

SYNC DATE=03/19/2012

PAGE TITLE

DDR3 ALIASES AND BITSWAPS

 Apple Inc.

DRAWING NUMBER
051-9504

SIZE
D

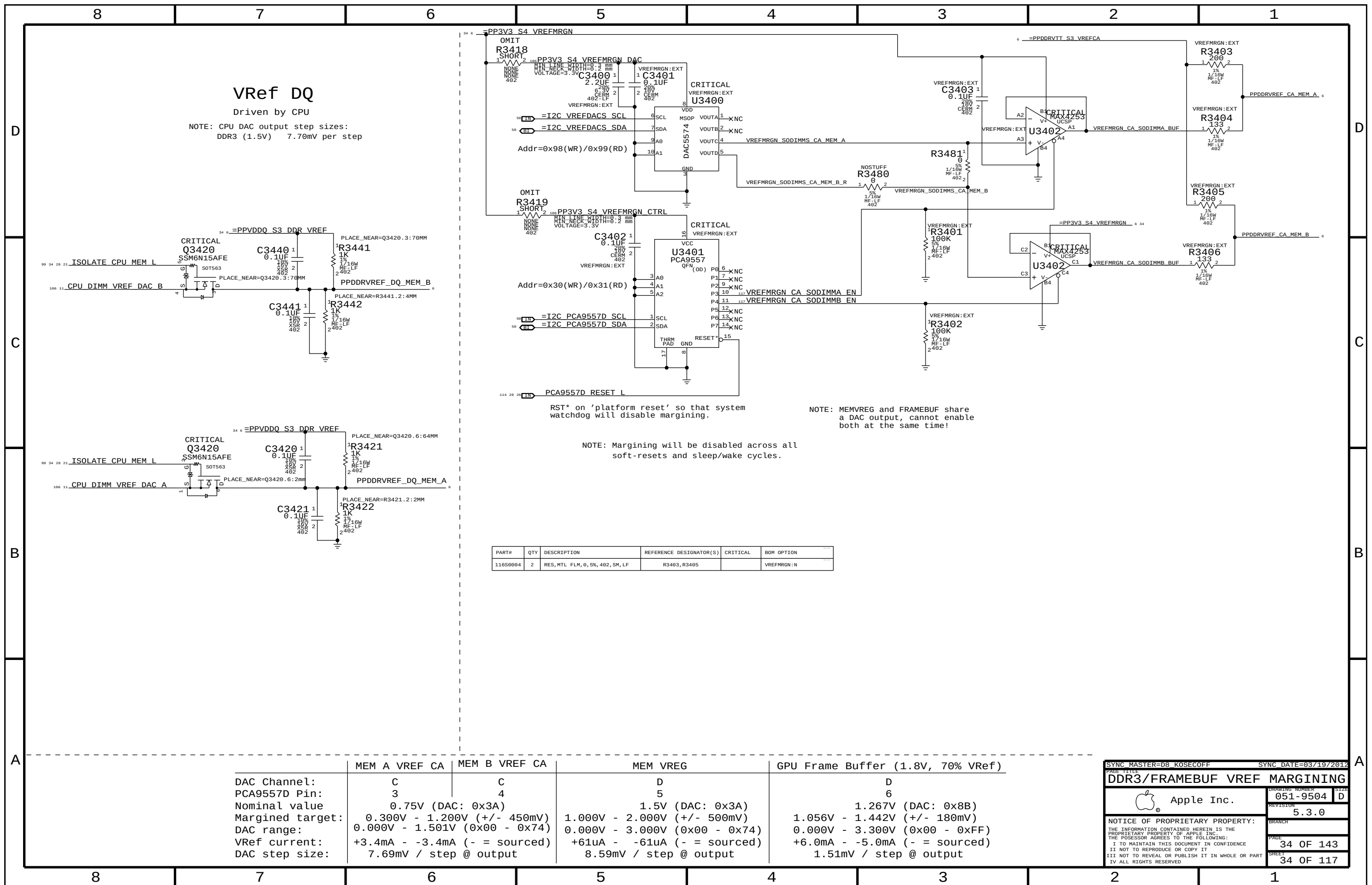
REVISION
5.3.0

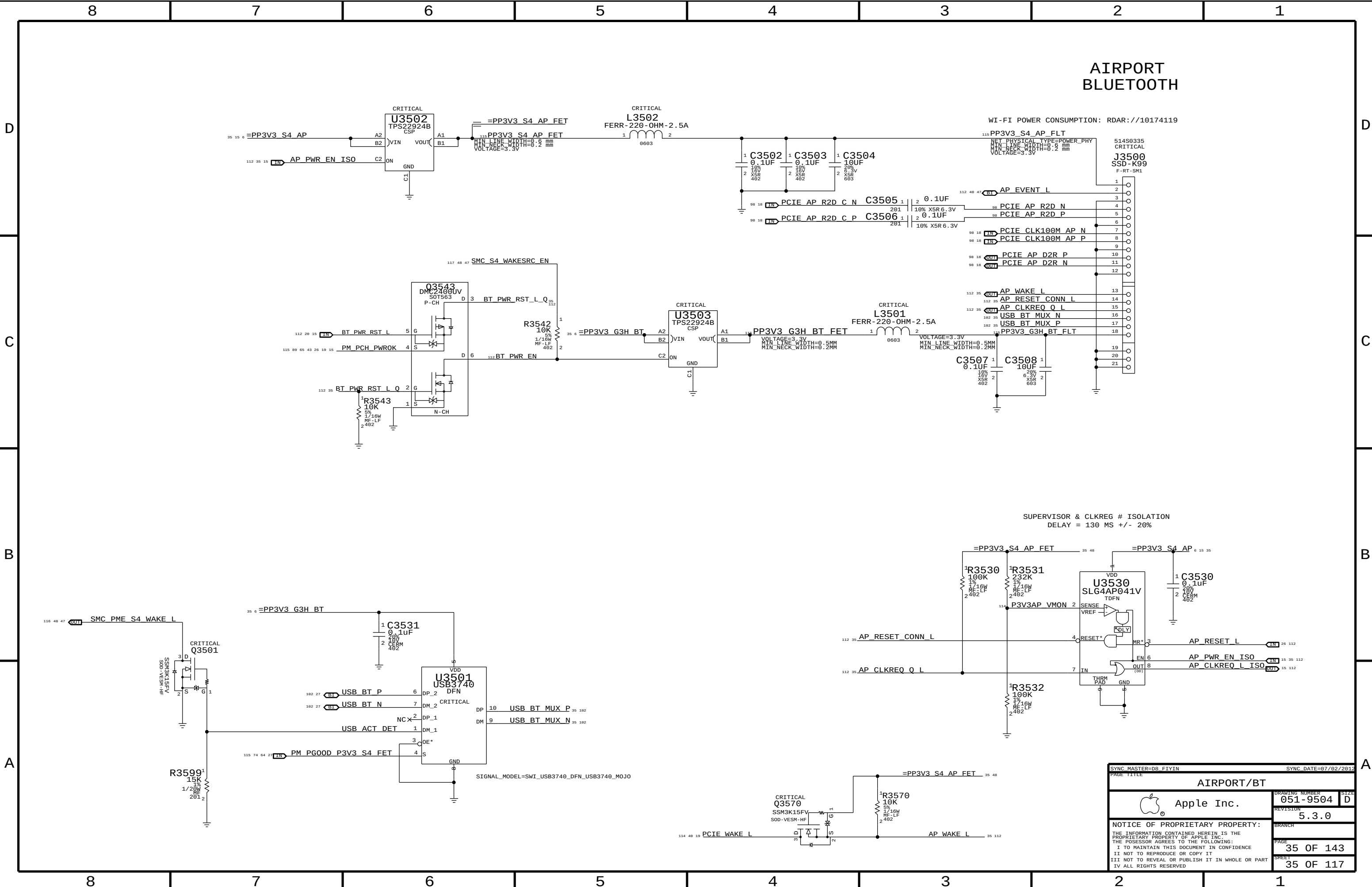
BRANCH

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

PAGE
33 OF 143

SHEET
33 OF 117



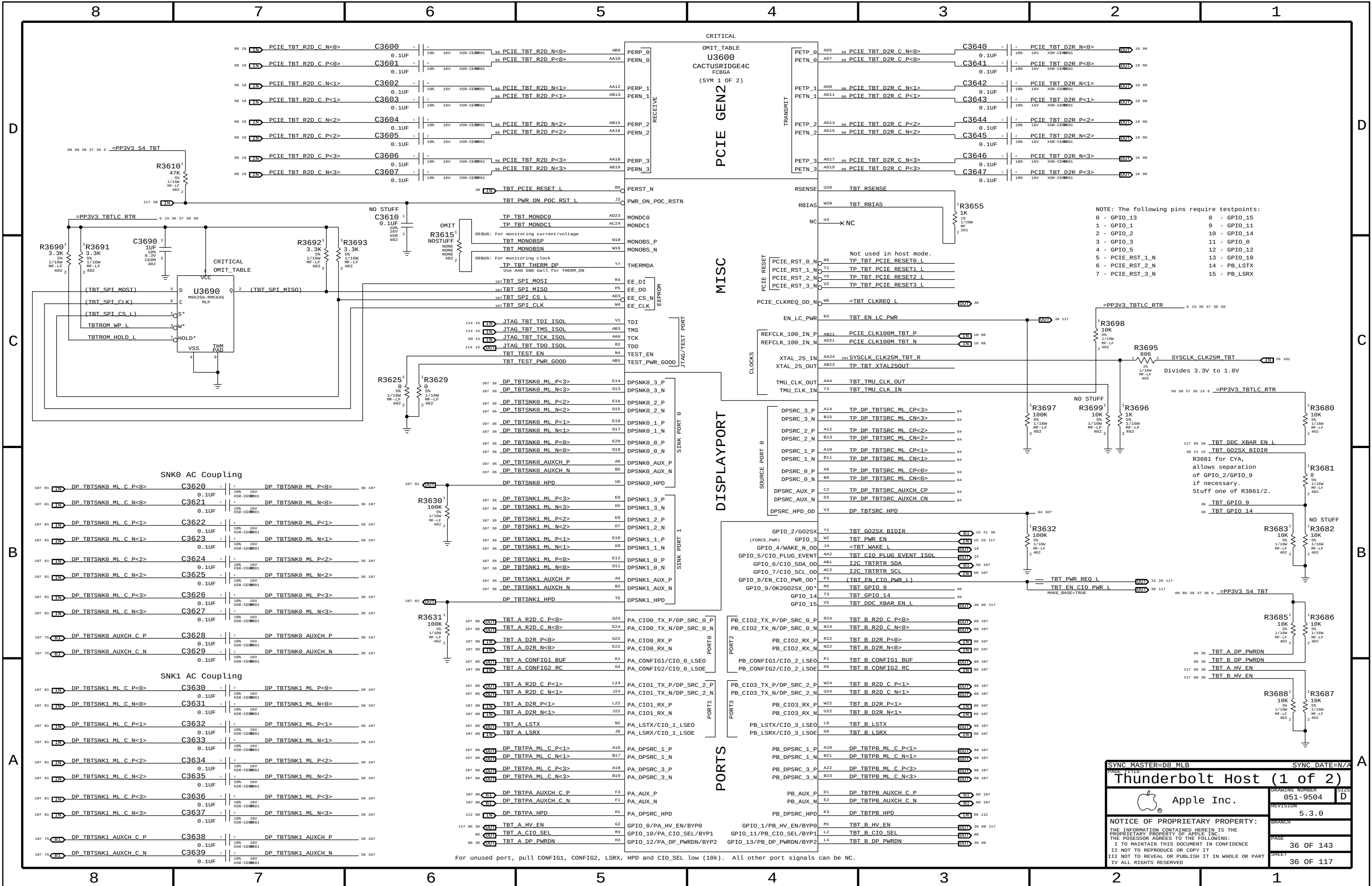


AIRPORT
BLUETOOTH

WI-FI POWER CONSUMPTION: RDAR://10174119

SUPERVISOR & CLKREG # ISOLATION
DELAY = 130 MS +/- 20%

SYNC MASTER=D8 FIYIN		SYNC DATE=07/02/2012	
PAGE TITLE			
AIRPORT/BT			
	Apple Inc.	DRAWING NUMBER	051-9504
		SIZE	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	5.3.0
		BRANCH	
		PAGE	35 OF 143
		SHEET	35 OF 117



SYNC MASTER=D8_MLB

SYNC DATE=N/A

PAGE TITLE

Thunderbolt Host (1 of 2)

 Apple Inc.

DRAWING NUMBER
051-9504

SIZE
D

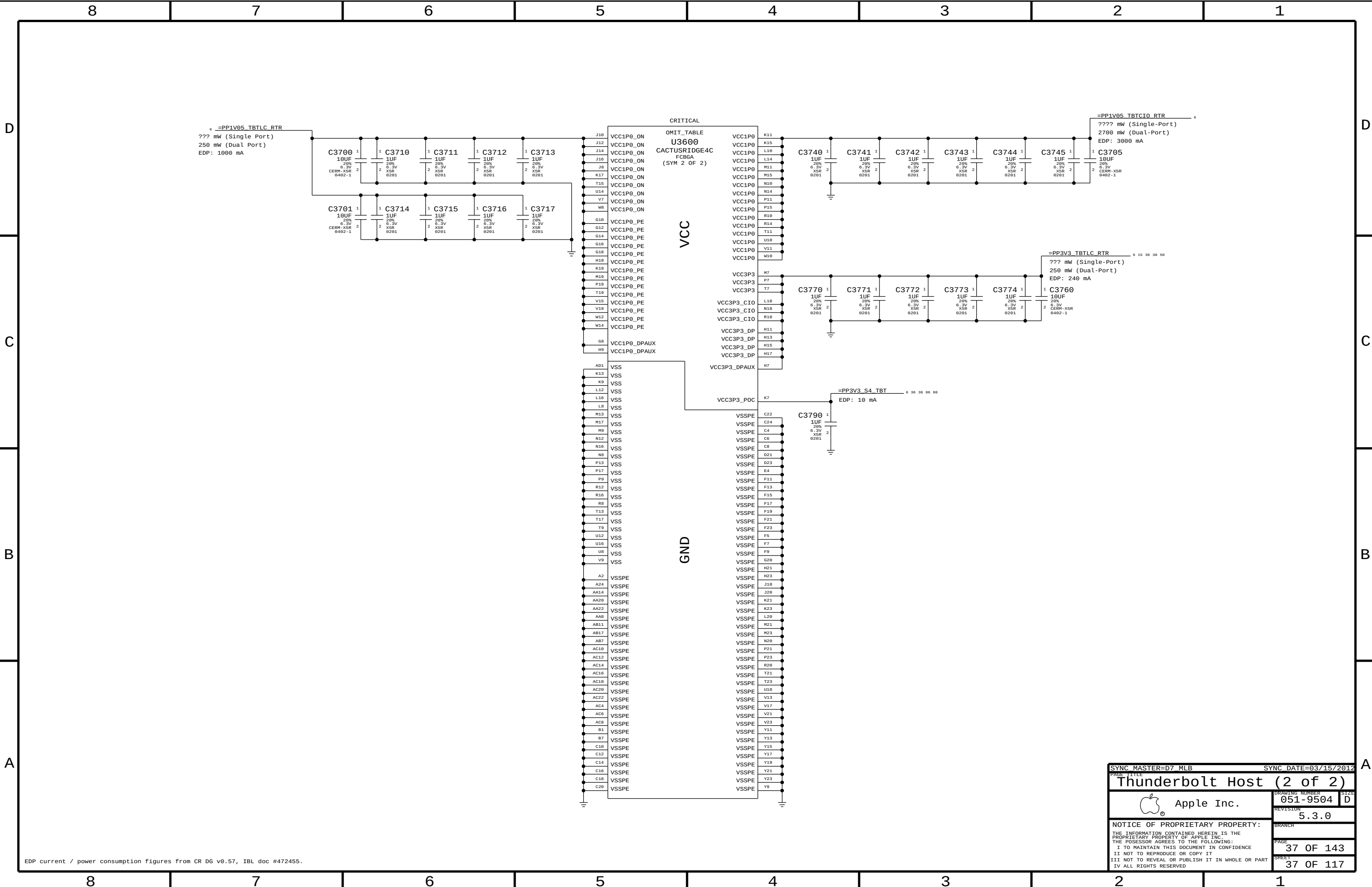
REVISION
5.3.0

BRANCH

PAGE
36 OF 143

SHEET
36 OF 117

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED



EDP current / power consumption figures from CR DG v0.57, IBL doc #472455.

PAGE TITLE		SYNC DATE=03/15/2012	
Thunderbolt Host (2 of 2)		DRAWING NUMBER	
Apple Inc.		051-9504	
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		5.3.0	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	
II NOT TO REPRODUCE OR COPY IT		37 OF 143	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		37 OF 117	

[illegible]

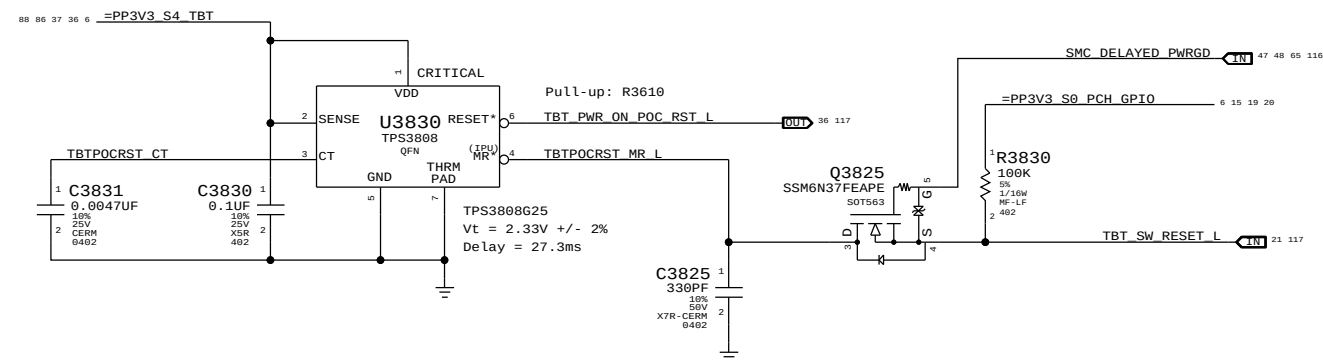
BOM options provided by this page:
TBTBST:Y - Stuffs 15V boost circuitry.

The schematic diagram illustrates the internal circuitry of the TBT module, centered around the U3800 SLG4AP016V logic device. The device is configured with several pins and components:

- Power and Ground:**
 - VDD:** Connected to the PP3V3_S0_TBT_PWRCTL signal.
 - TDFN:** Connected to the PP1V05_TBTLC signal.
 - GND:** Connected to the TBT_EN_LC_ISO_L signal.
 - THRM_PAD:** Connected to the TBT_CLKREQ_ISO_L signal.
- Logic and Timing:**
 - SENSE:** Connected to the TBT_PCIE_RESET_L signal.
 - MR*:** Connected to the TBT_EN_LC_ISO_L signal.
 - RESET*:** Connected to the TBT_CLKREQ_ISO_L signal.
 - IN:** Connected to the TBT_CLKREQ_ISO_L signal.
 - OUT:** Connected to the TBT_CLKREQ_ISO_L signal.
 - EN:** Connected to the TBT_CLKREQ_ISO_L signal.
 - OUT (00):** Connected to the TBT_CLKREQ_ISO_L signal.
- Components:**
 - Q3840 SSM3K15AMFVAPE:** A MOSFET used for signal switching, controlled by the TBT_EN_LC_PWR signal.
 - R3840:** A 10K resistor connected to the VDD and GND lines.
 - C3800:** A 0.1uF capacitor connected to the VDD and GND lines.
 - R3807:** A 100K resistor connected to the TBT_PCIE_RESET_L signal.
- Labels and Notes:**
 - CRITICAL:** A label indicating the importance of the timing and signal integrity.
 - DLY = 60 ms +/- 20%:** A note indicating the delay time for the TBT_PCIE_RESET_L signal.
 - Pull-up provided by SB page.** A note indicating that the pull-up for the TBT_CLKREQ_ISO_L signal is provided by the system board (SB) page.

Schematic diagram of a power supply circuit. The circuit includes a TPS22924 CSP (A1, A2, B1, B2) with VIN, VOUT, and GND pins. A resistor R3815 (10k, 6.3V, CSRR, 482) is connected between the input and the VIN pin. A resistor R3816 (5%, 1/16W, NF=LF, 482) is connected between the input and the VOUT pin. A capacitor C3815 (10F, 6.3V, CSRR, 482) is connected between the input and ground. A capacitor C3816 (10F, 6.3V, CSRR, 482) is connected between the VOUT pin and ground. The output is connected to a P1V05_TBTLC_FET (A1, B1) with a maximum current of 2A (85C).

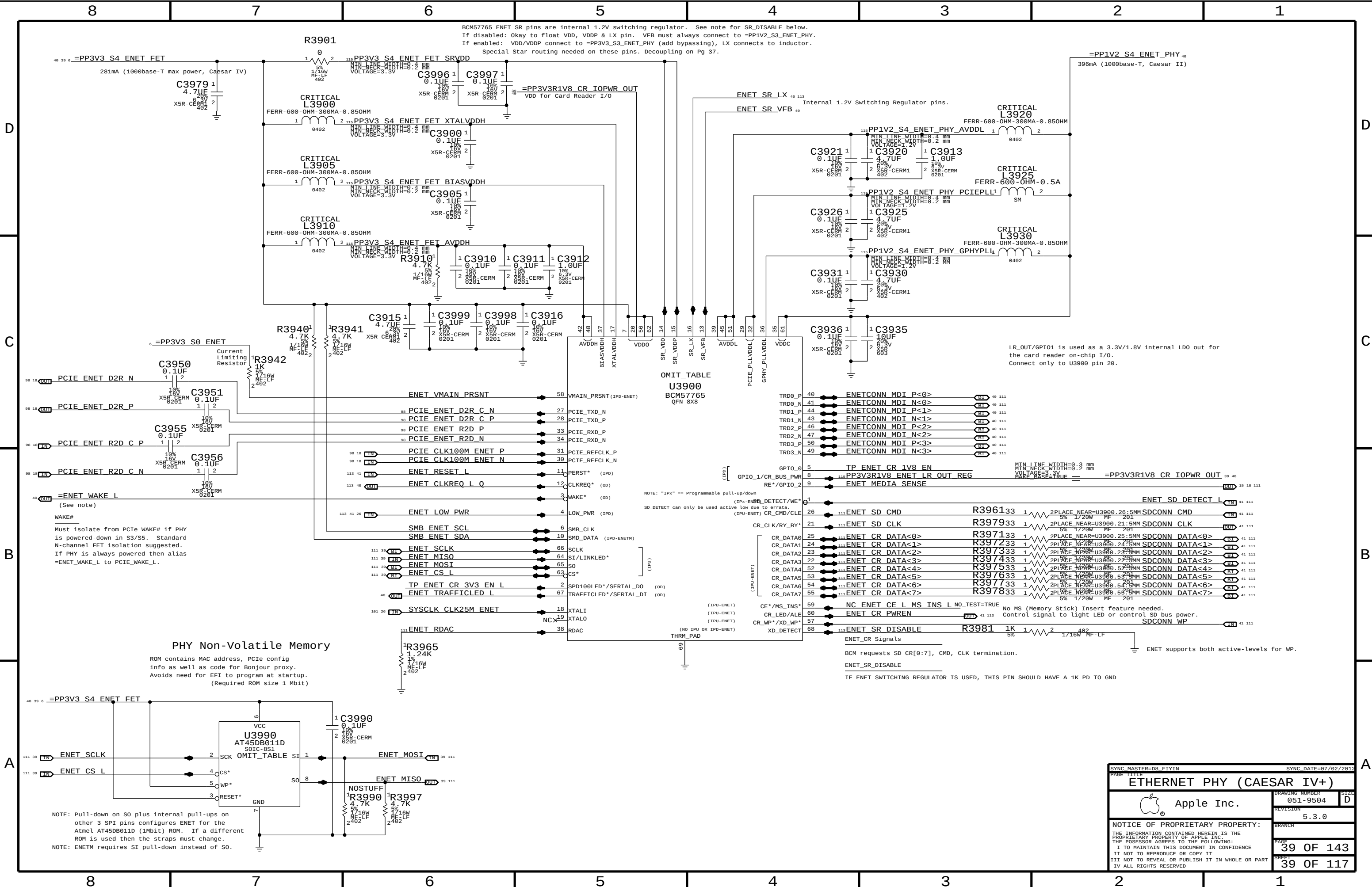
Intel investigating whether RC is sufficient.

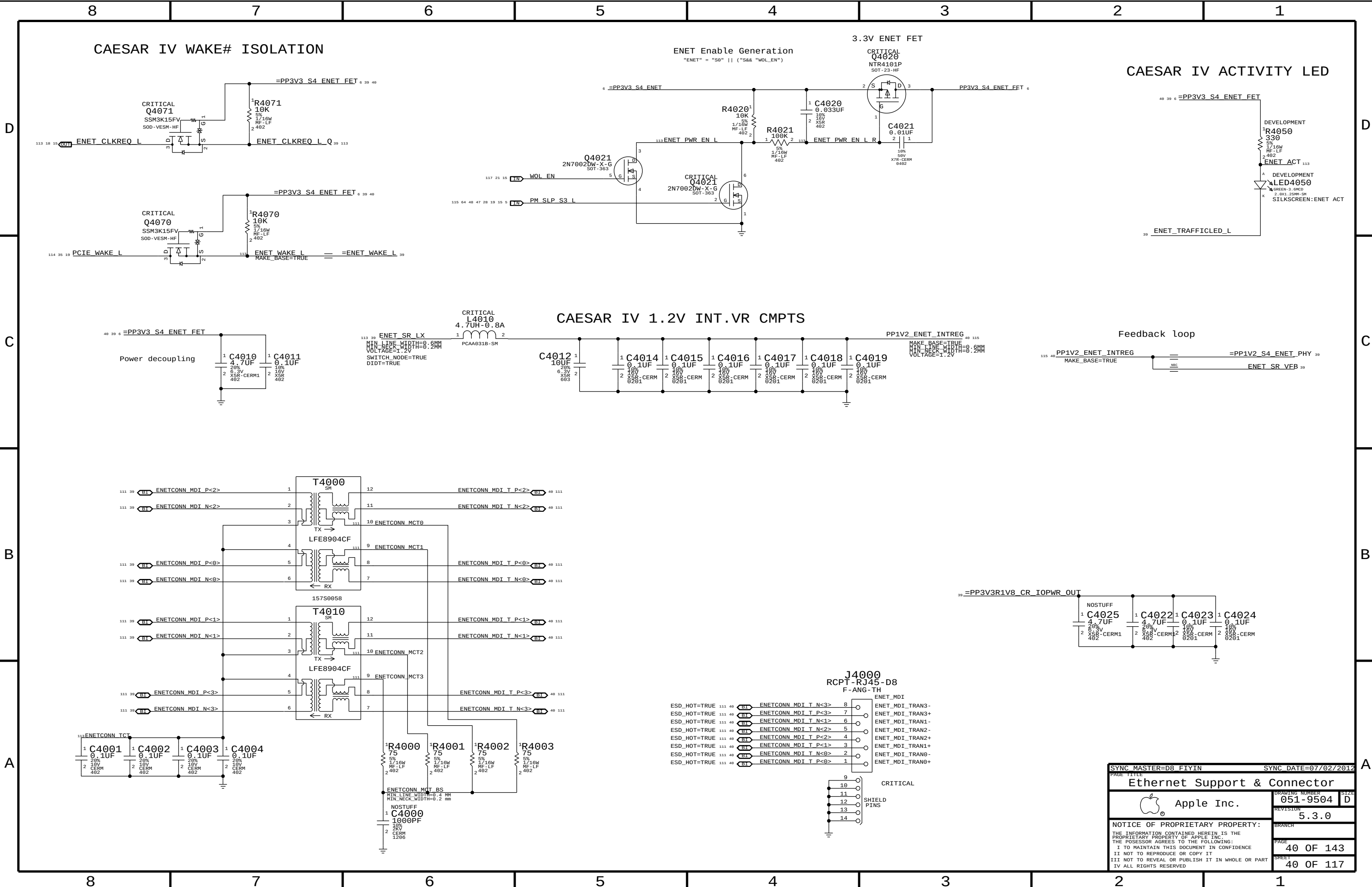
[illegible]

U3815	
Part	TPS22924C
Type	Load Switch
R(on)	20.3 mOhm Typ
@ 1.0V	28.6 mOhm Max

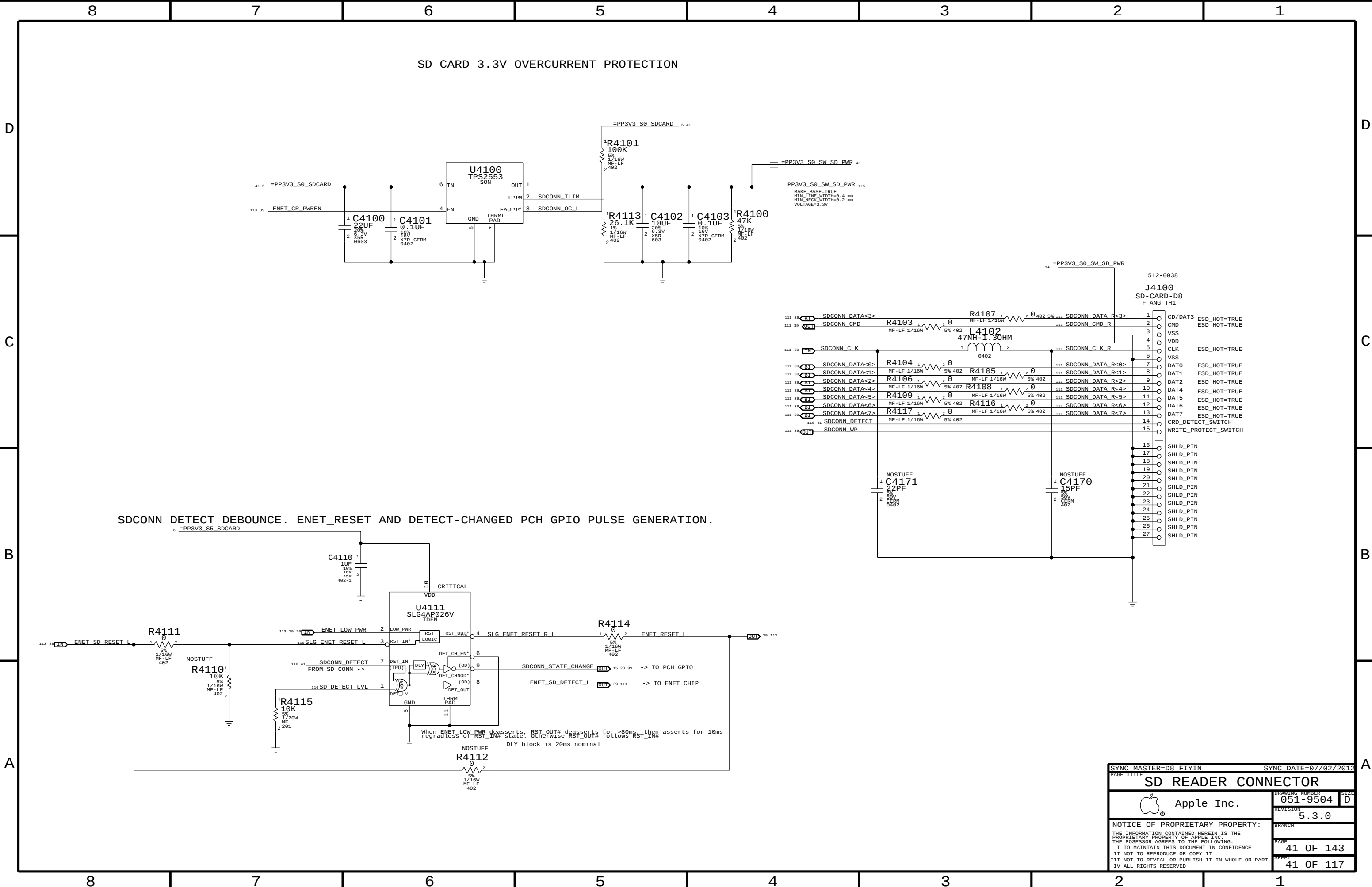
U3820	
Part	TPS22920
Type	Load Switch
R(on)	8 mOhm Typ
@ 1.05V	11.5 mOhm Max

SYNCH MASTER=D7 MLB		SYNCH DATE=03/15/2012	
PAGE 1 FILE			
Thunderbolt Power Support			
	Apple Inc.		DRAWING NUMBER 051-9504
			SIZE D
		REVISION 5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I. TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II. NOT TO REPRODUCE OR COPY IT III. NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV. ALL RIGHTS RESERVED		PAGE 38 OF 143	
		SHEET 38 OF 117	





PAGE TITLE		SYNC DATE=07/02/2012	
Ethernet Support & Connector		DRAWING NUMBER	051-9504
Apple Inc.		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	40 OF 143
		SHEET	40 OF 117



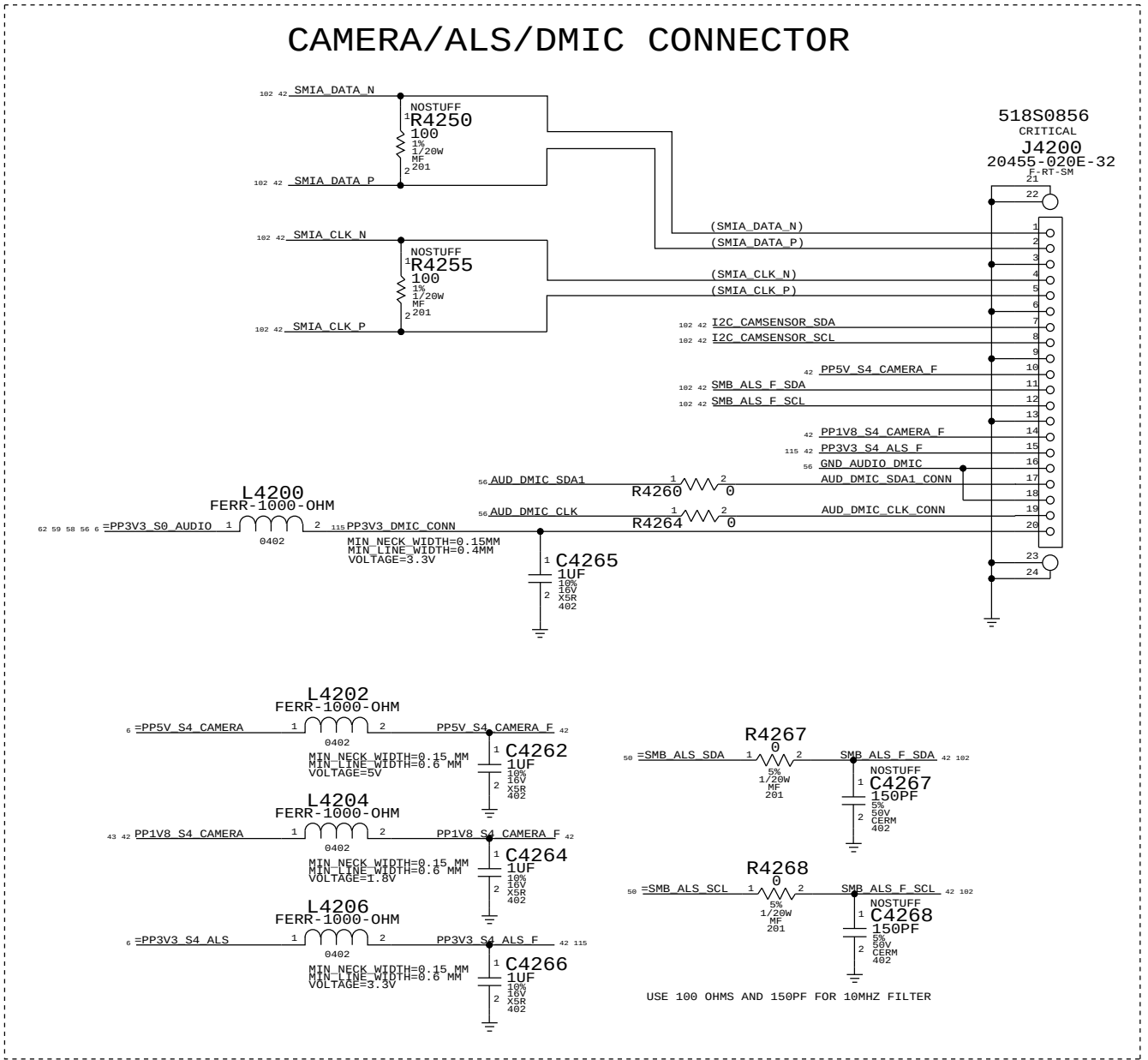
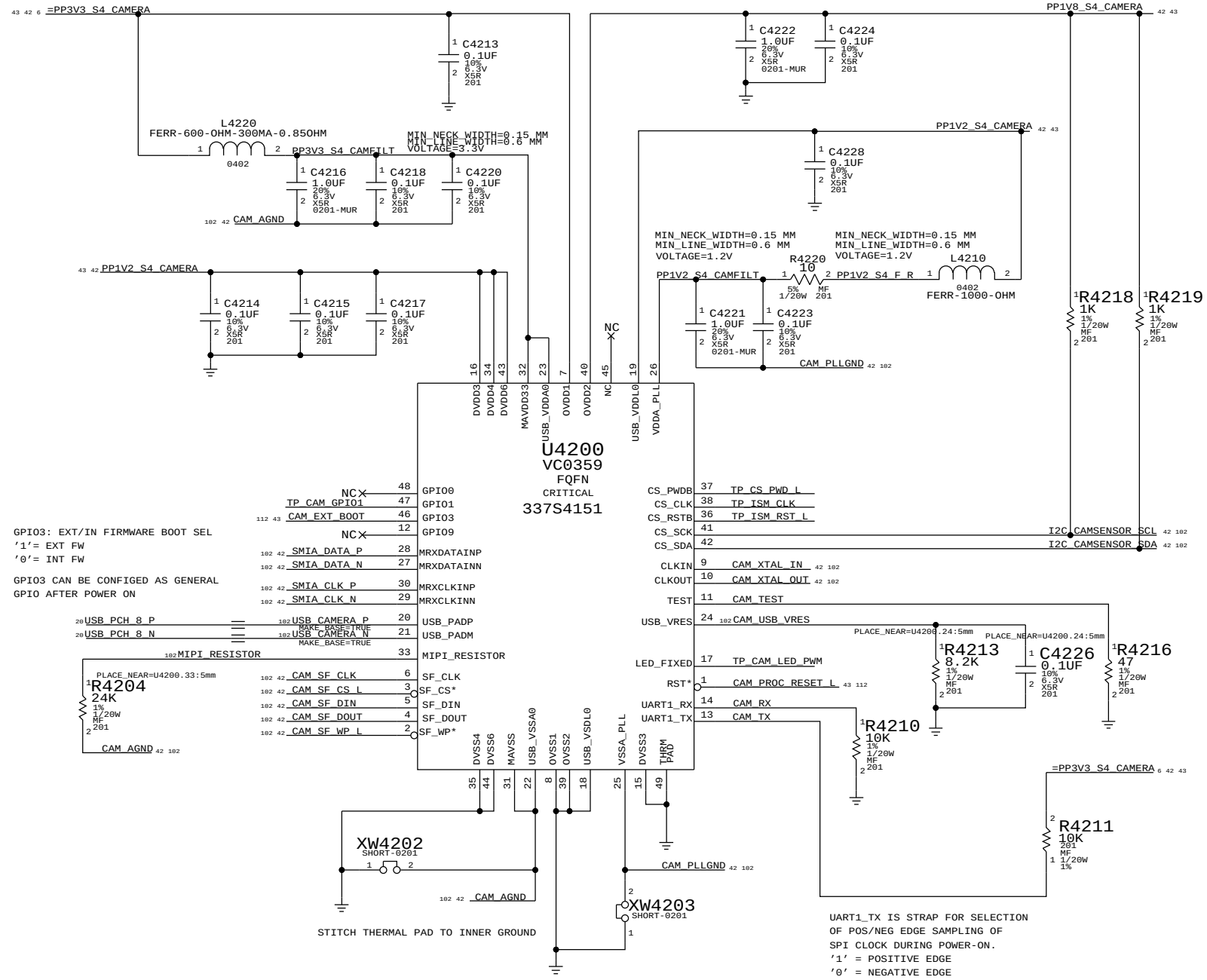
USB CAMERA CONTROLLER

D

C

B

A

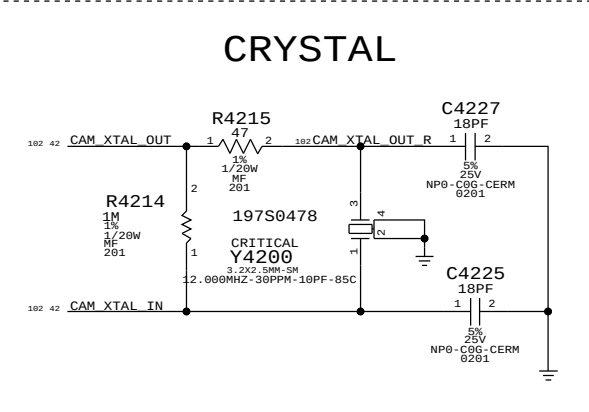
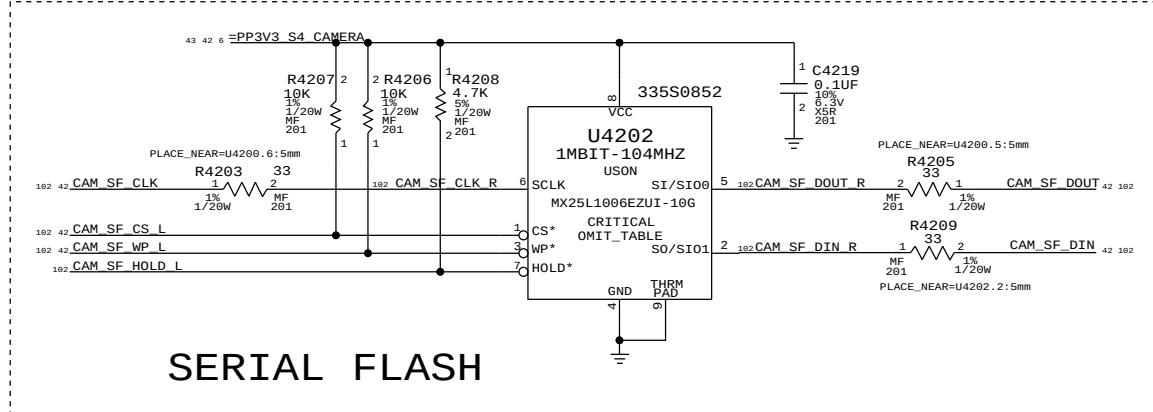


D

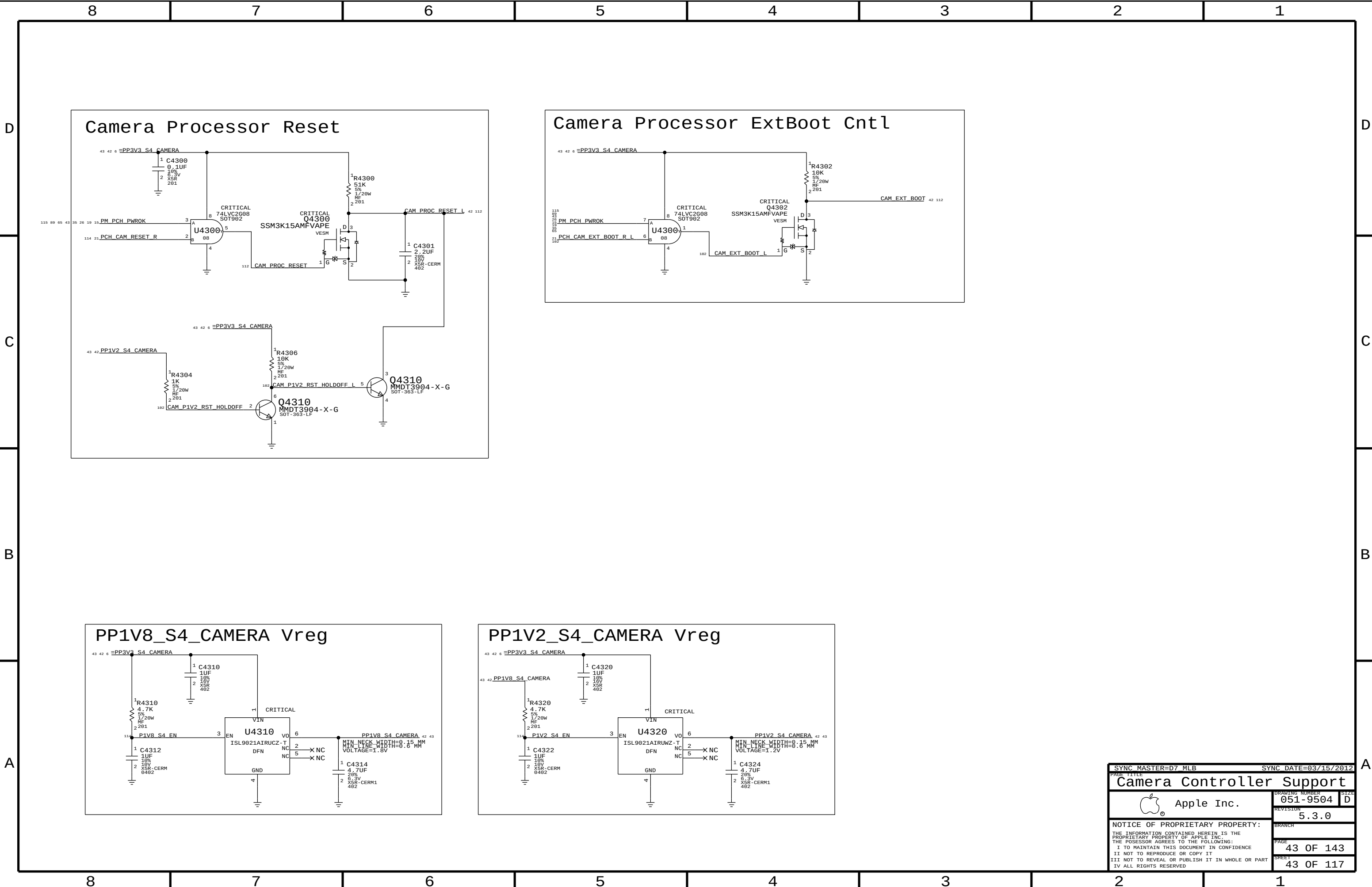
C

B

A



PAGE TITLE		SYNC DATE=03/23/2012	
Camera Controller		DRAWING NUMBER	
Apple Inc.		051-9504	
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		5.3.0	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		BRANCH	
II NOT TO REPRODUCE OR COPY IT		PAGE	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		42 OF 143	
IV ALL RIGHTS RESERVED		SHEET	
		42 OF 117	



Camera Processor Reset

Camera Processor ExtBoot Cntl

PP1V8_S4_CAMERA Vreg

PP1V2_S4_CAMERA Vreg

D

C

B

A

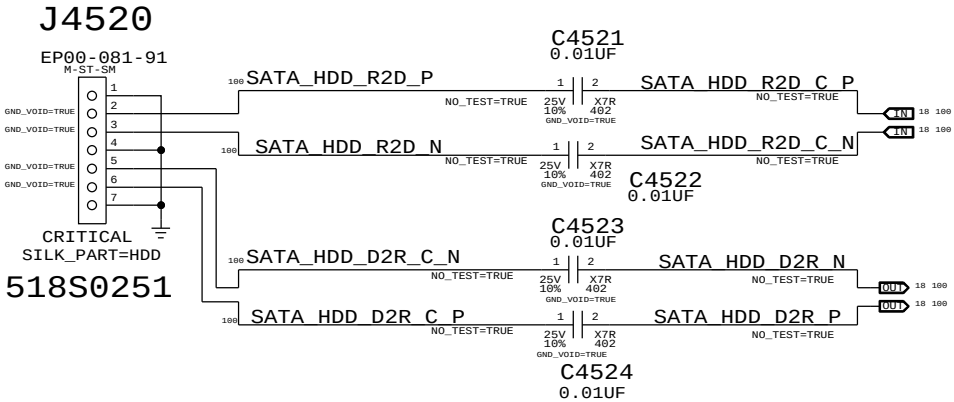
D

C

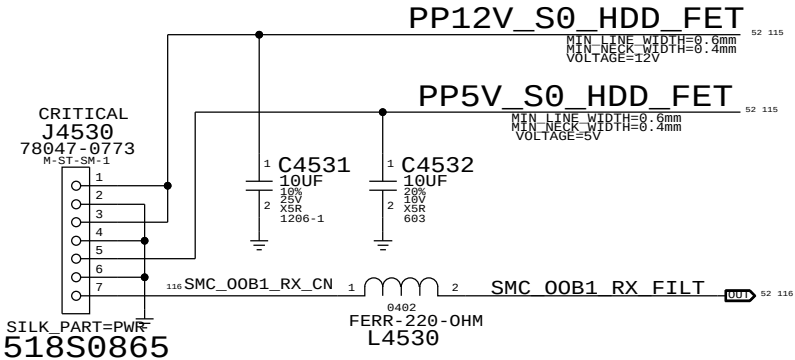
B

A

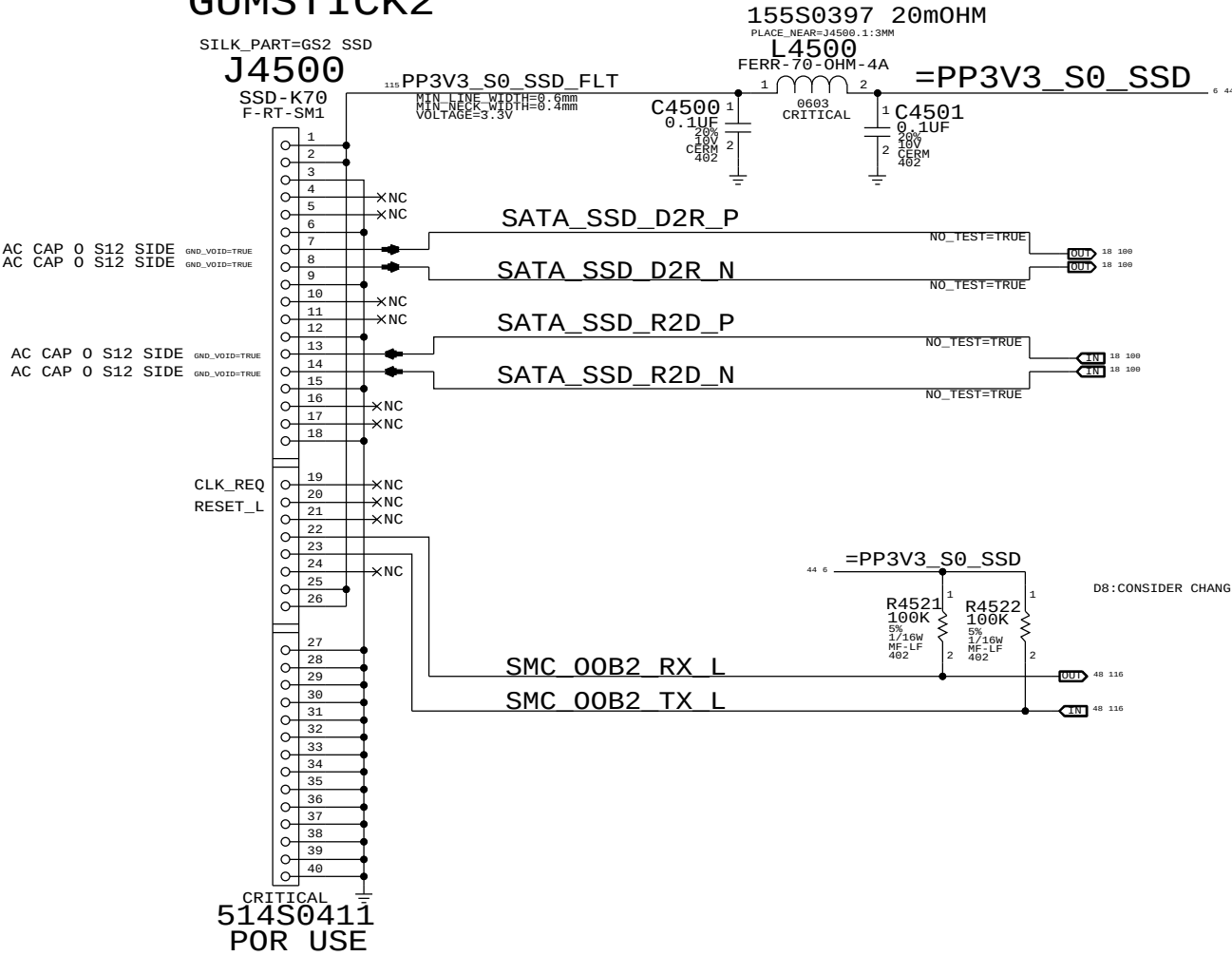
HDD DATA



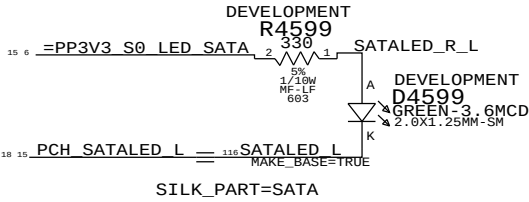
HDD POWER



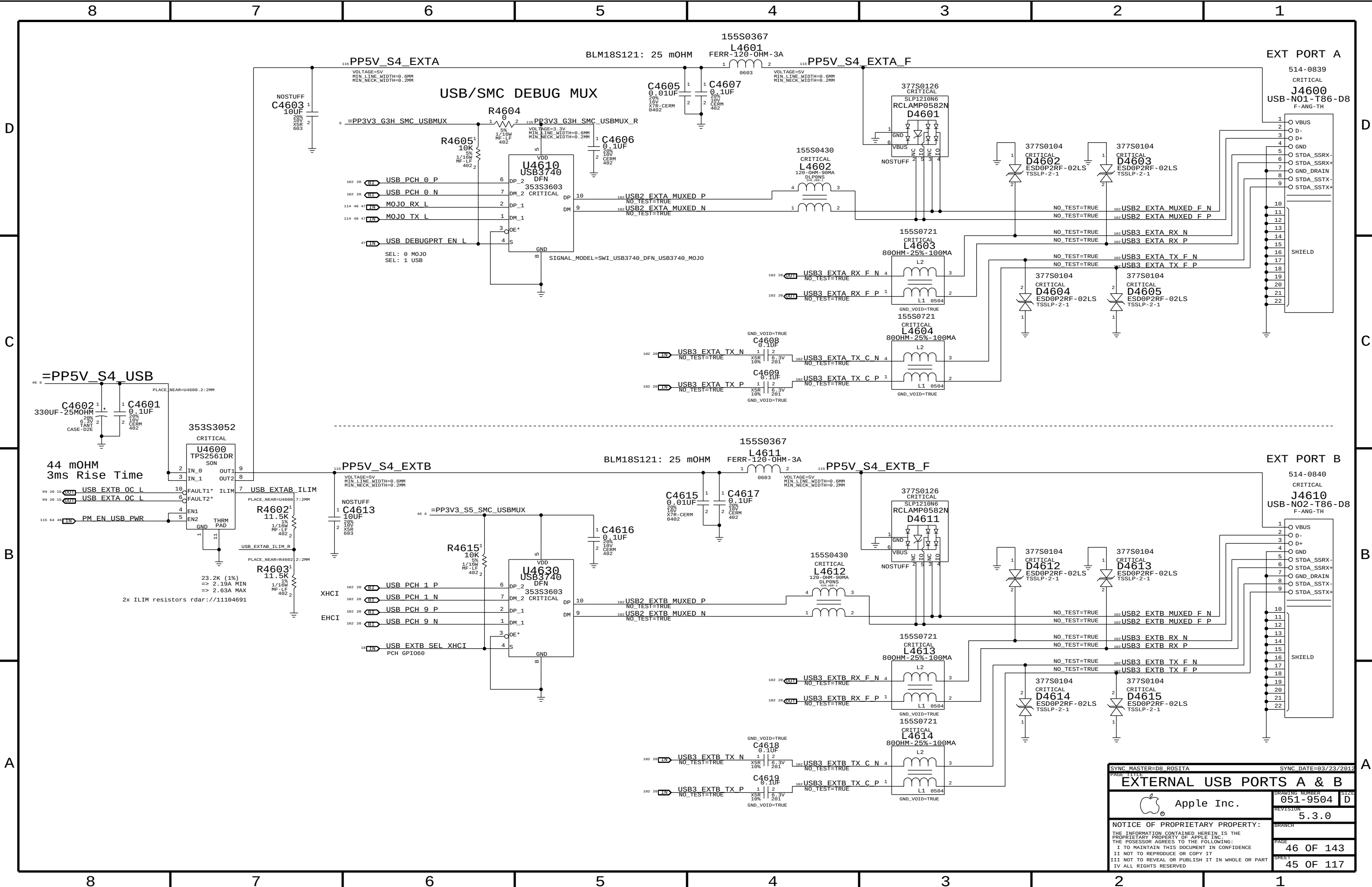
GUMSTICK2



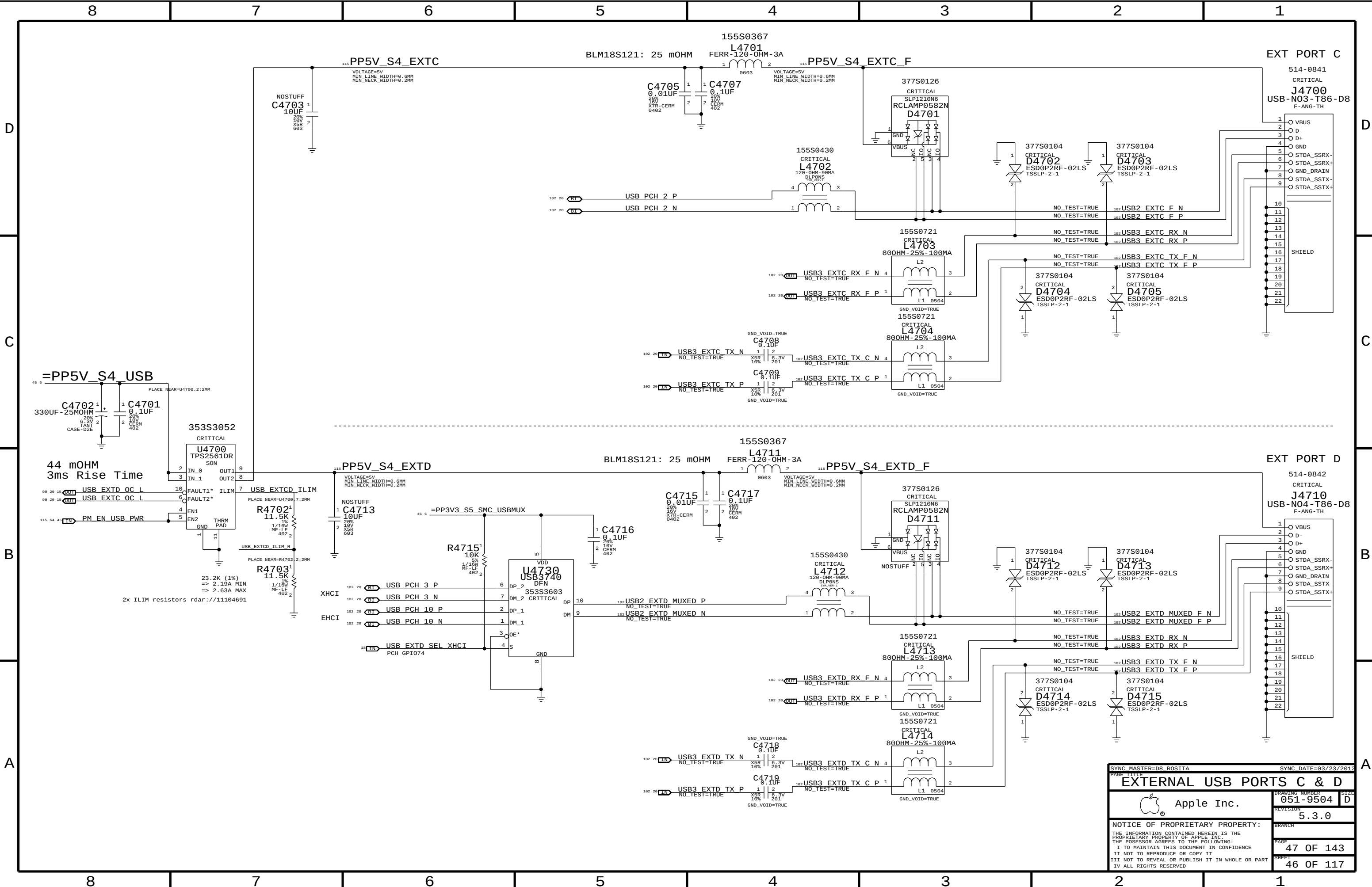
SATA Activity LED



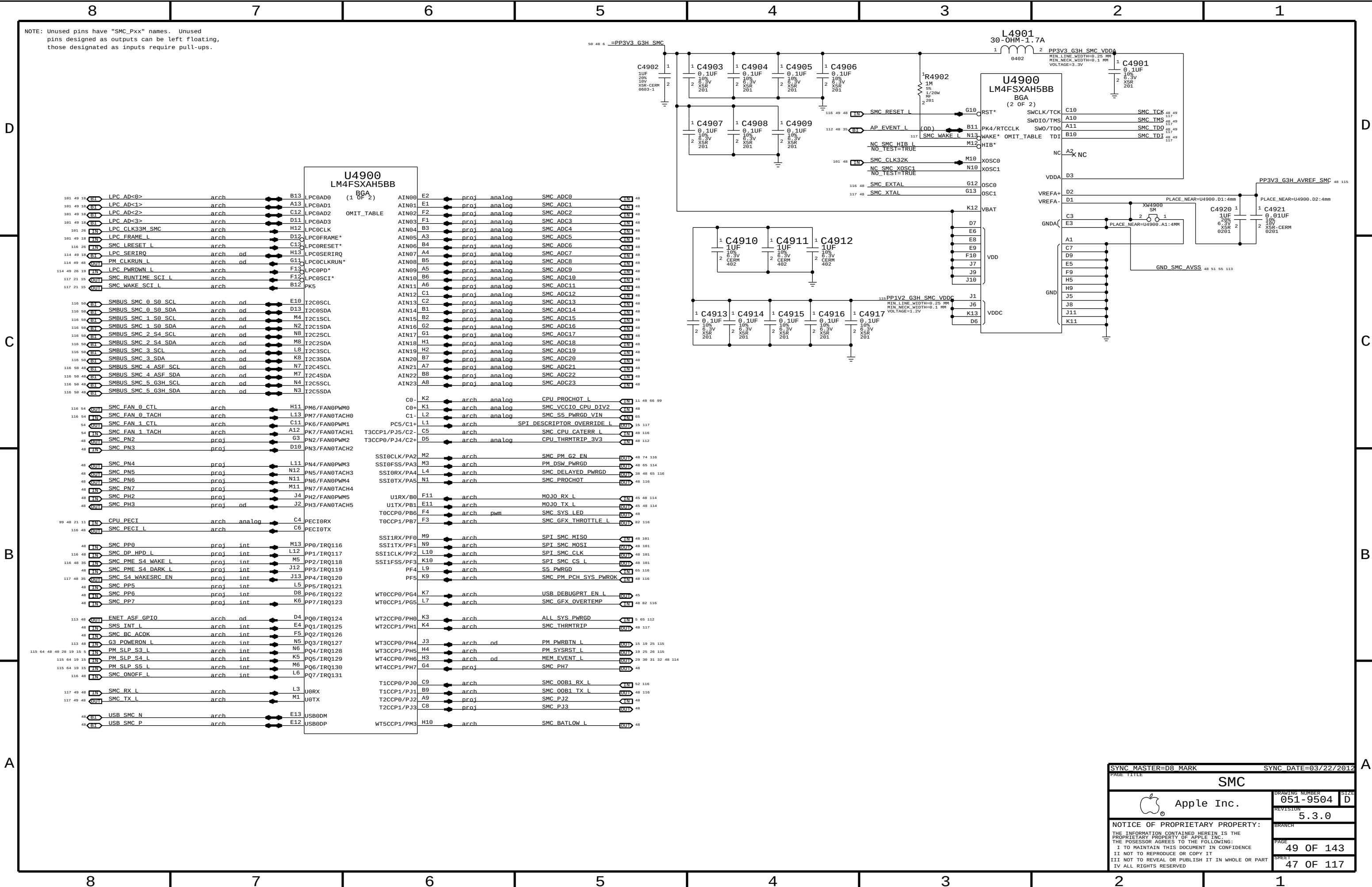
SYNC MASTER=D8_JERRY		SYNC DATE=01/31/2012	
PAGE TITLE		SATA Connectors	
DRAWING NUMBER		051-9504	SIZE D
REVISION		5.3.0	BRANCH
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE 45 OF 143	SHEET 44 OF 117

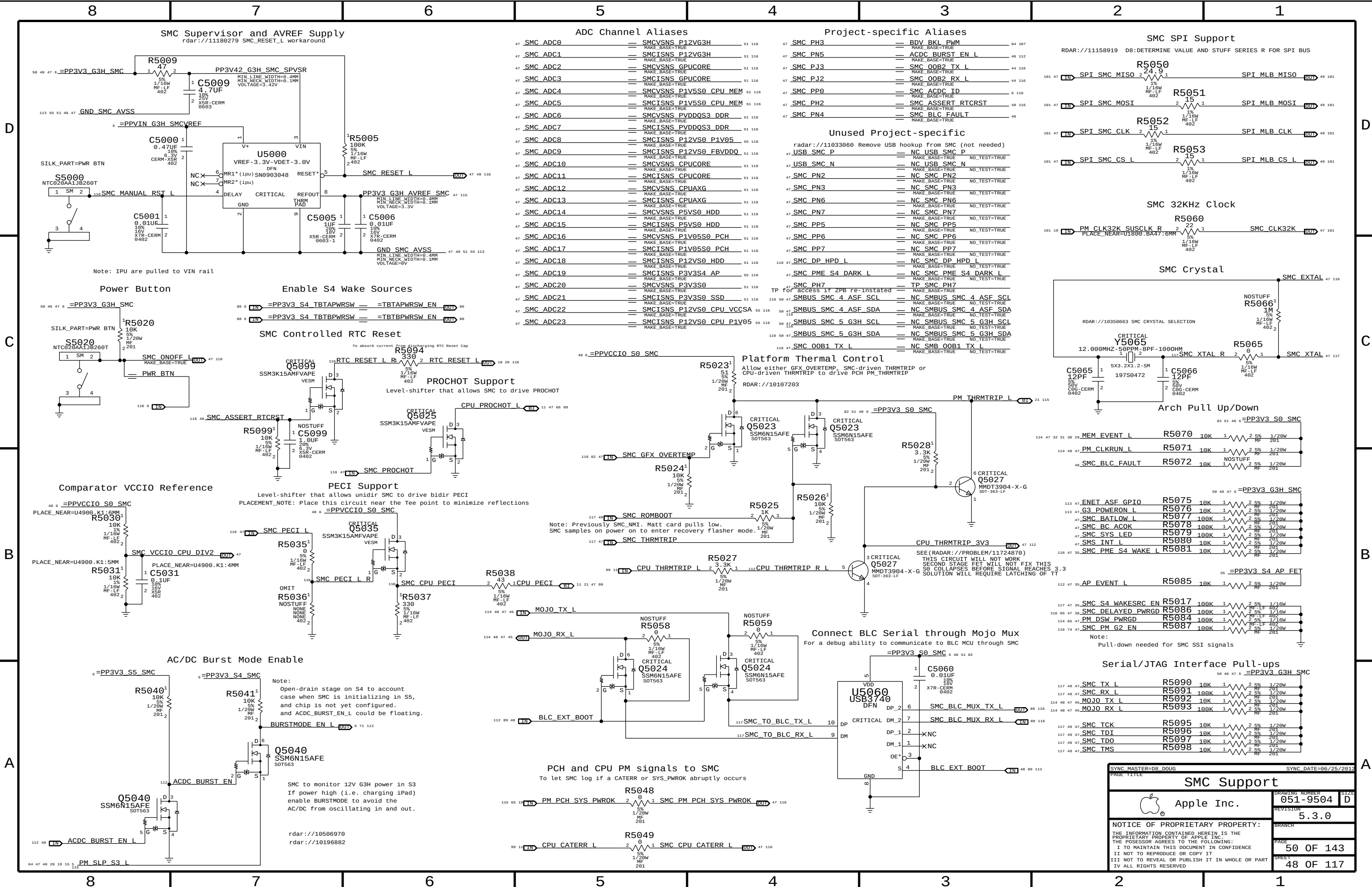


SYNC MASTER=D8 ROSITA		SYNC DATE=03/23/2012	
EXTERNAL USB PORTS A & B		DRAWING NUMBER	
Apple Inc.		051-9504	
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		5.3.0	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		BRANCH	
II NOT TO REPRODUCE OR COPY IT		PAGE	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		46 OF 143	
IV ALL RIGHTS RESERVED		SHEET	
		45 OF 117	



SYNC MASTER=D8 ROSITA		SYNC DATE=03/23/2012	
PAGE TITLE		DRAWING NUMBER	
EXTERNAL USB PORTS C & D		051-9504	
Apple Inc.		REVISION	
		5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		47 OF 143	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		46 OF 117	
IV ALL RIGHTS RESERVED			





SMC Supervisor and AVREF Supply

rdar://11180279 SMC_RESET_L workaround

ADC Channel Aliases

Project-specific Aliases

SMC SPI Support

RDAR://11158919 D8:DETERMINE VALUE AND STUFF SERIES R FOR SPI BUS

Unused Project-specific

radar://11033060 Remove USB hookup from SMC (not needed)

SMC 32KHz Clock

SMC Crystal

RDAR://10350663 SMC CRYSTAL SELECTION

Platform Thermal Control

Allow either GFX OVERTEMP, SMC-driven THRMTrip or CPU-driven THRMTrip to drive PCH PM_THRMTrip

RDAR://10107203

Connect BLC Serial through Mojo Mux

For a debug ability to communicate to BLC MCU through SMC

Serial/JTAG Interface Pull-ups

PCH and CPU PM signals to SMC

To let SMC log if a CATERR or SYS_PWROK abruptly occurs

SYNC MASTER=D8 DOUG

SYNC DATE=06/25/2012

PAGE TITLE

SMC Support

Apple Inc.

DRAWING NUMBER

051-9504

REVISION

5.3.0

NOTICE OF PROPRIETARY PROPERTY:

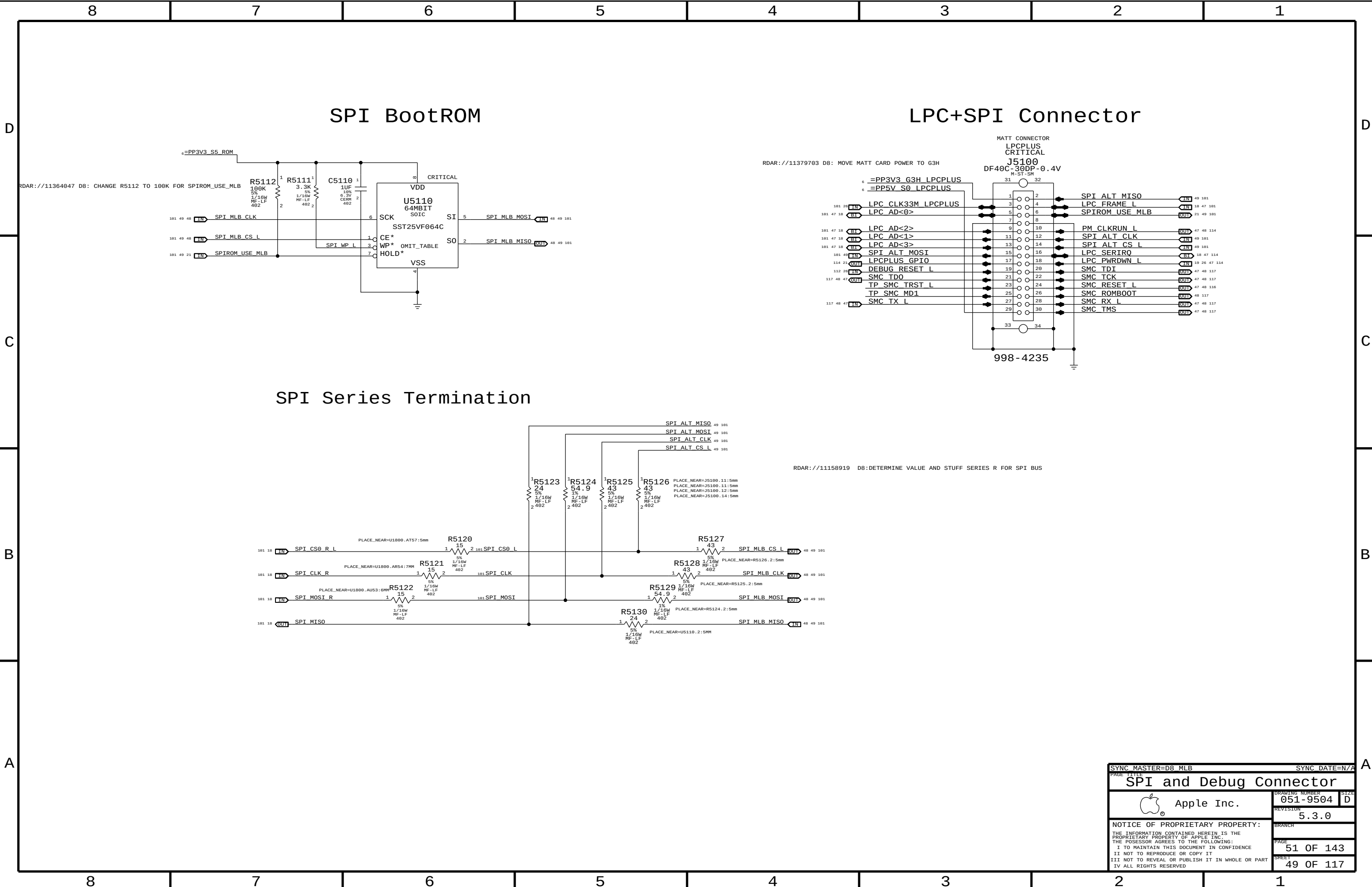
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

PAGE

50 OF 143

SHEET

48 OF 117

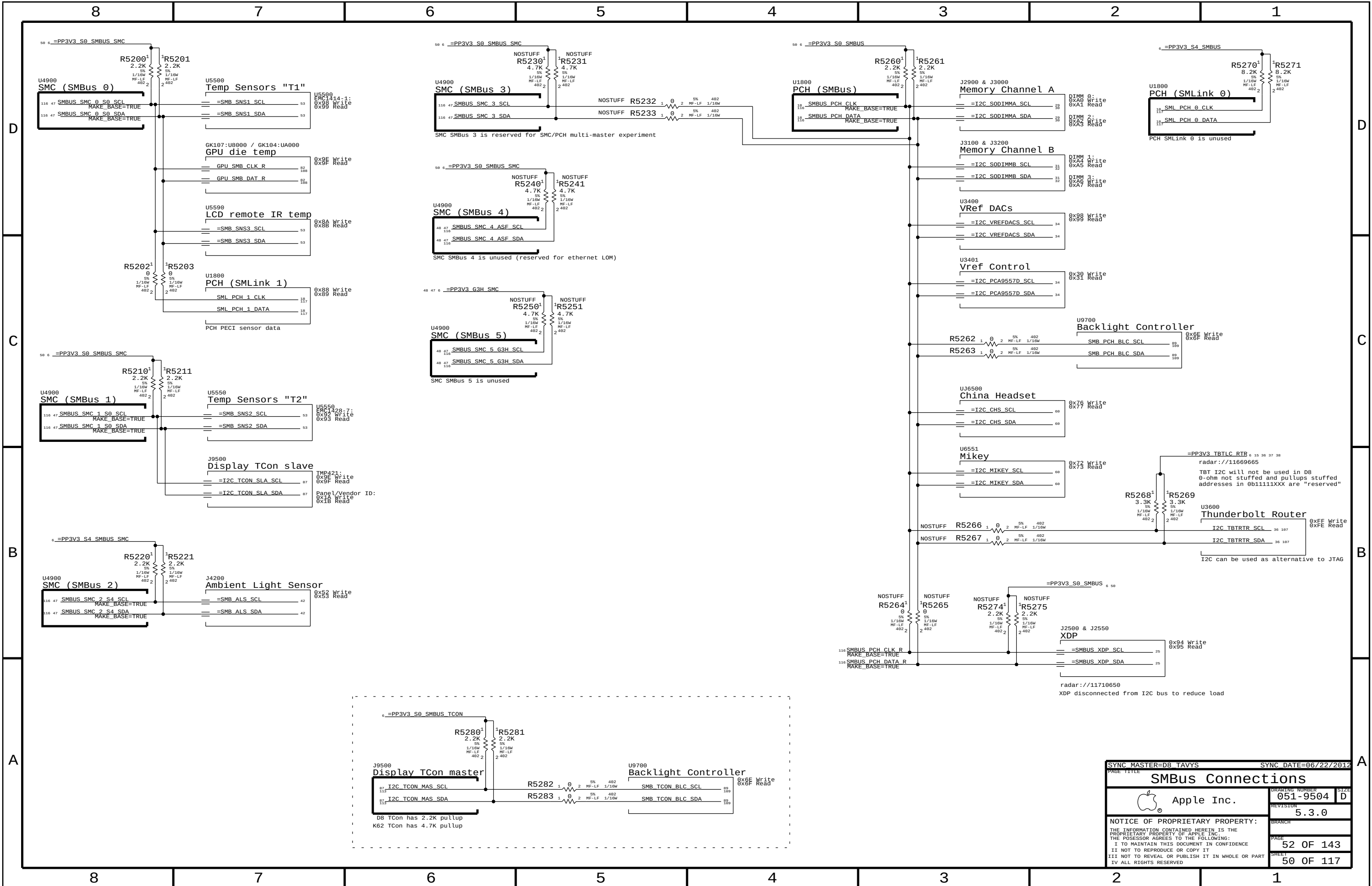


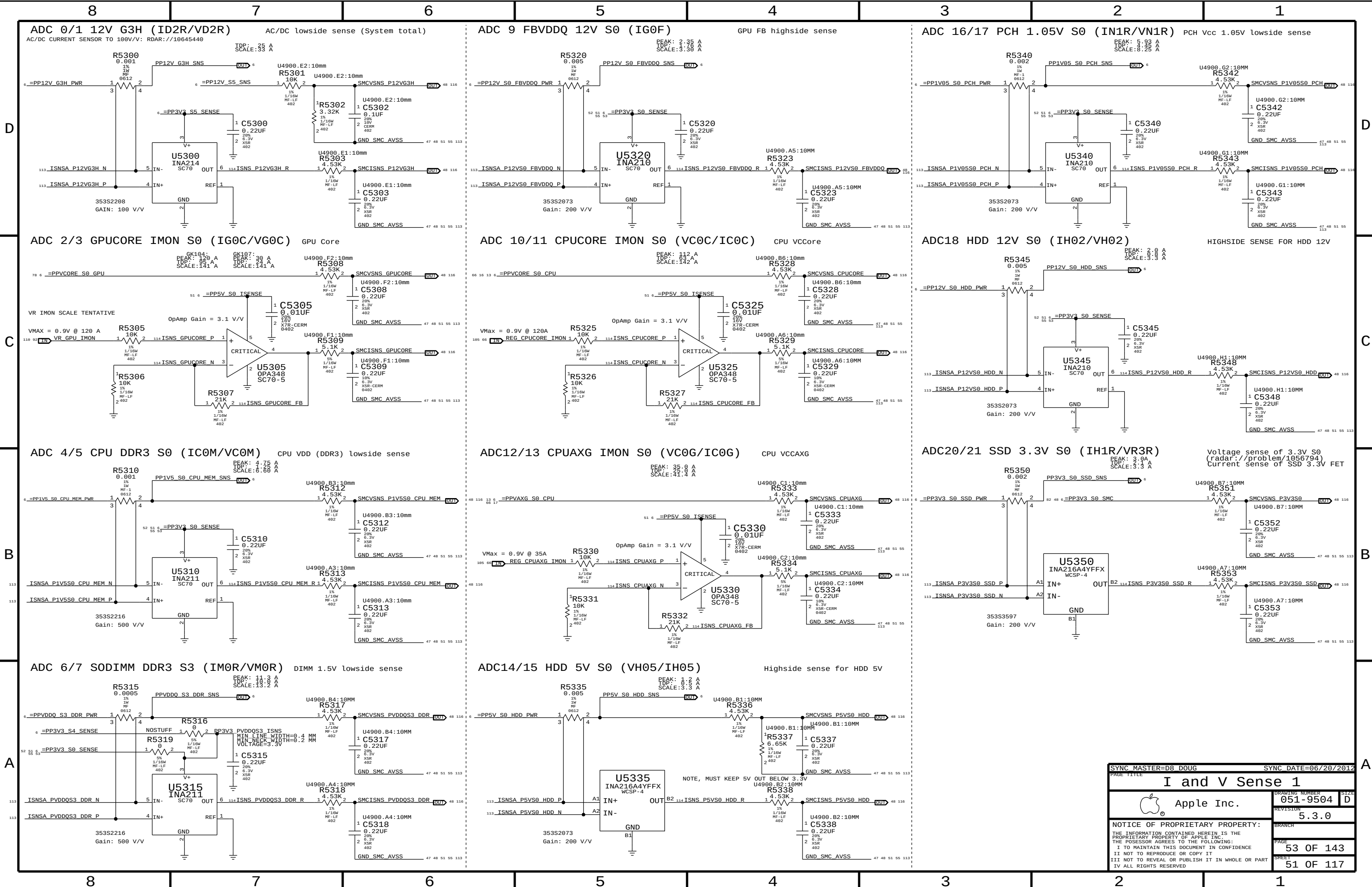
SPI BootROM

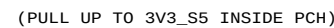
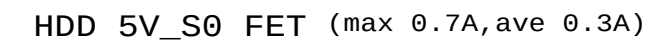
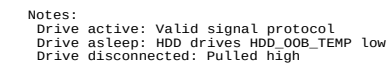
LPC+SPI Connector

SPI Series Termination

PAGE TITLE		SYNC DATE=N/A	
SPI and Debug Connector		DRAWING NUMBER	
Apple Inc.		051-9504	D
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		5.3.0	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		BRANCH	
II NOT TO REPRODUCE OR COPY IT		PAGE	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		51 OF 143	
IV ALL RIGHTS RESERVED		SHEET	
		49 OF 117	







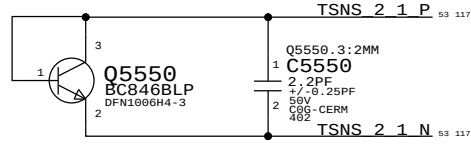
- SYNC-MASTER=D8- JERRY- - - - - SYNC-DATE=02/25/2012

PAGE TITLE		HDD/SSD Temp Sense	
	Apple Inc.	DRAWING NUMBER	051-9504
		SIZE	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: 1 TO MAINTAIN THIS DOCUMENT IN CONFIDENCE IT NOT TO REPRODUCE OR COPY IT IT NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IN ALL RIGHTS RESERVED		REVISION	5.3.0
		BRANCH	
		PAGE	54 OF 143
		SHEET	52 OF 117

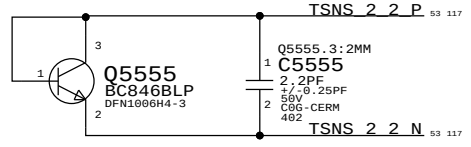
Temperature Sensor T2 EMC1428: Near GPU VR

SNS T2: TEMP SENSOR IC

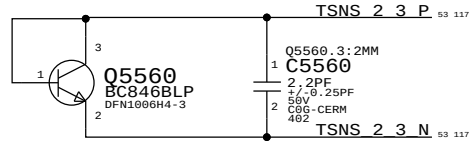
GPU Proximity



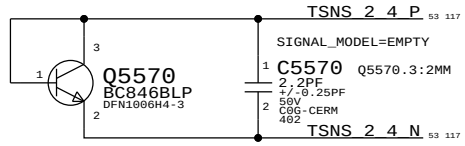
Ambient



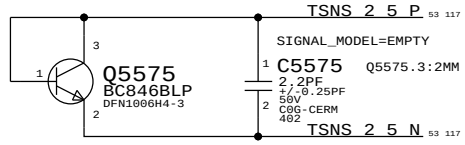
BLC Proximity



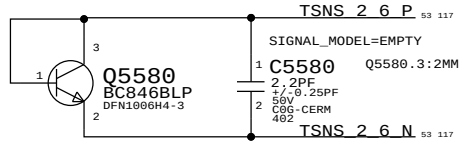
SO-DIMM Proximity 1



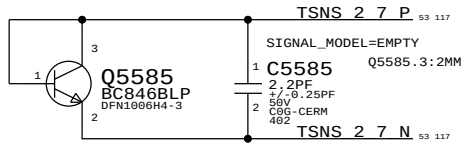
SO-DIMM Proximity 2



SO-DIMM Proximity 3



SO-DIMM Proximity 4



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

GPU Prox (T60p) TSNS_2_1_P
TSNS_2_1_N

Ambient (TA0p) TSNS_2_2_P
TSNS_2_2_N

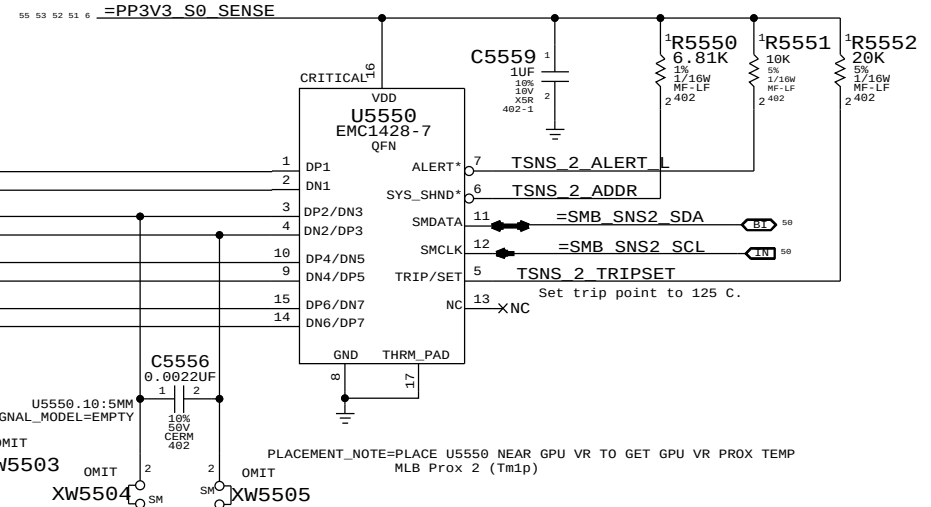
SoDIMM Prox 1 (TM0p) TSNS_2_4_P
TSNS_2_4_N

SoDIMM Prox 3 (TM2p) TSNS_2_6_P
TSNS_2_6_N

SoDIMM Prox 4 (TM3p) TSNS_2_7_P
TSNS_2_7_N

SoDIMM Prox 2 (TM1p) TSNS_2_5_P
TSNS_2_5_N

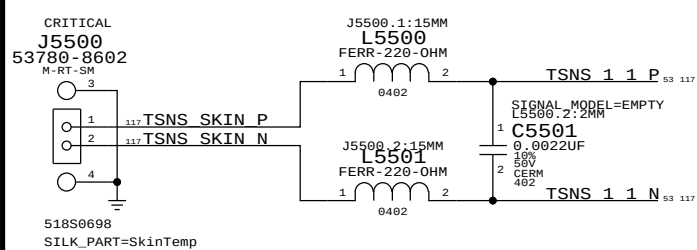
BLC PROX (Tb0p) TSNS_2_3_P
TSNS_2_3_N



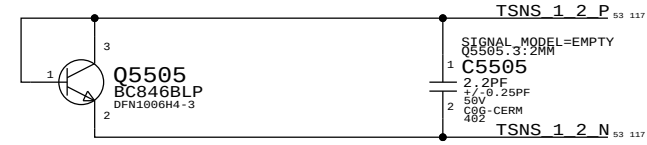
Temperature Sensor T1 EMC1414: Near PSU Conn

Temperature Sensor T3: LCD Remote Sensor (Dev4Now)

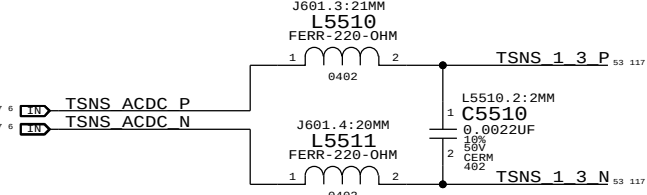
Skin



CPU Proximity



AC/DC



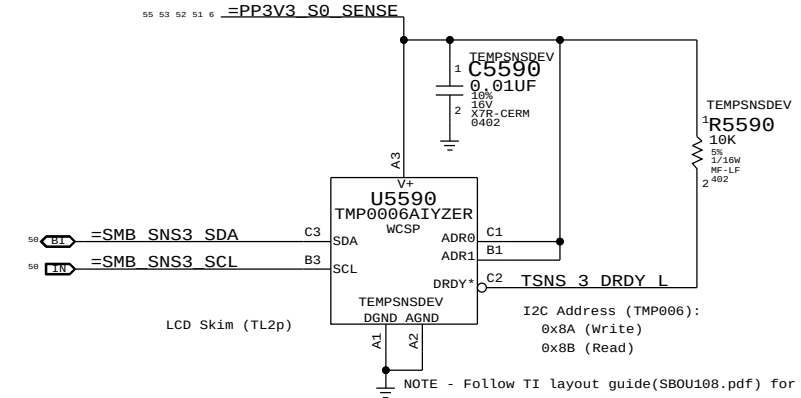
Skin Temp (TS0p)

CPU Prox (TC0p)

AC/DC (Tp2p)

Note:
Internal sensor of the EMC 1414 will be used as MLB sensor.
MLB Prox 1 (TM0p)

PLACEMENT_NOTE=PLACE U5500 NEAR PSU CONNECTOR



SYNC MASTER=D8 DOUG		SYNC DATE=06/07/2012	
PAGE TITLE		Temperature Sensors	
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	55 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	53 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

D

C

B

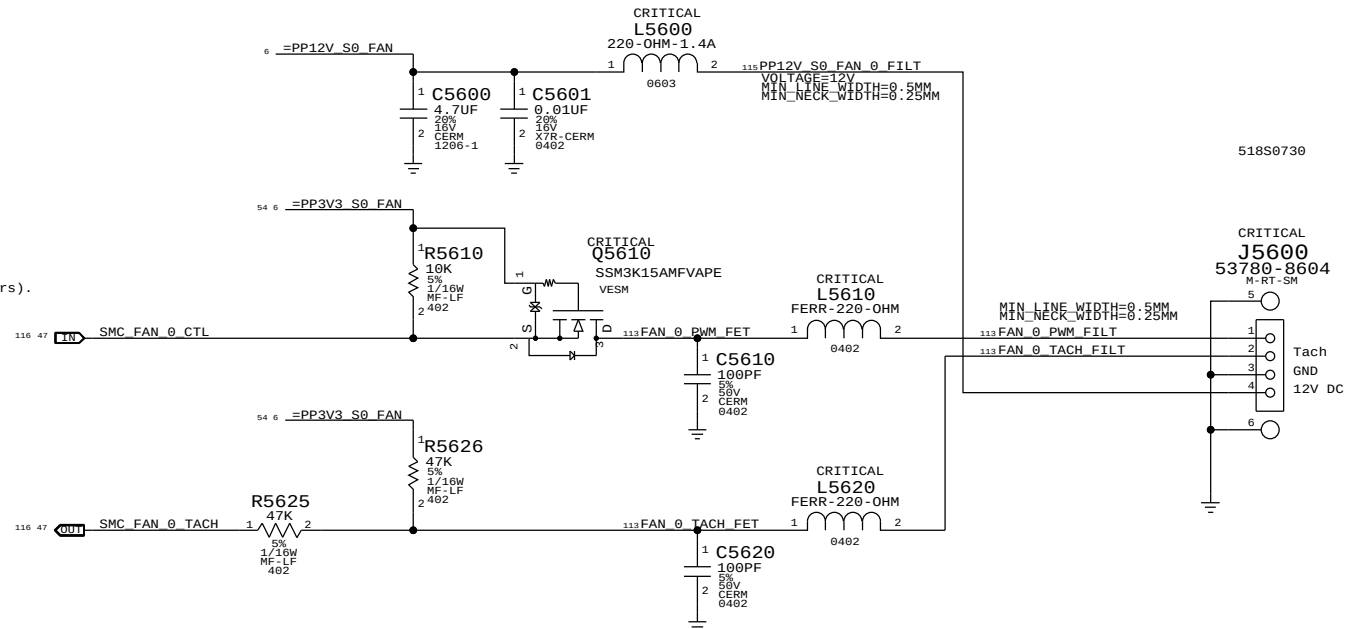
A

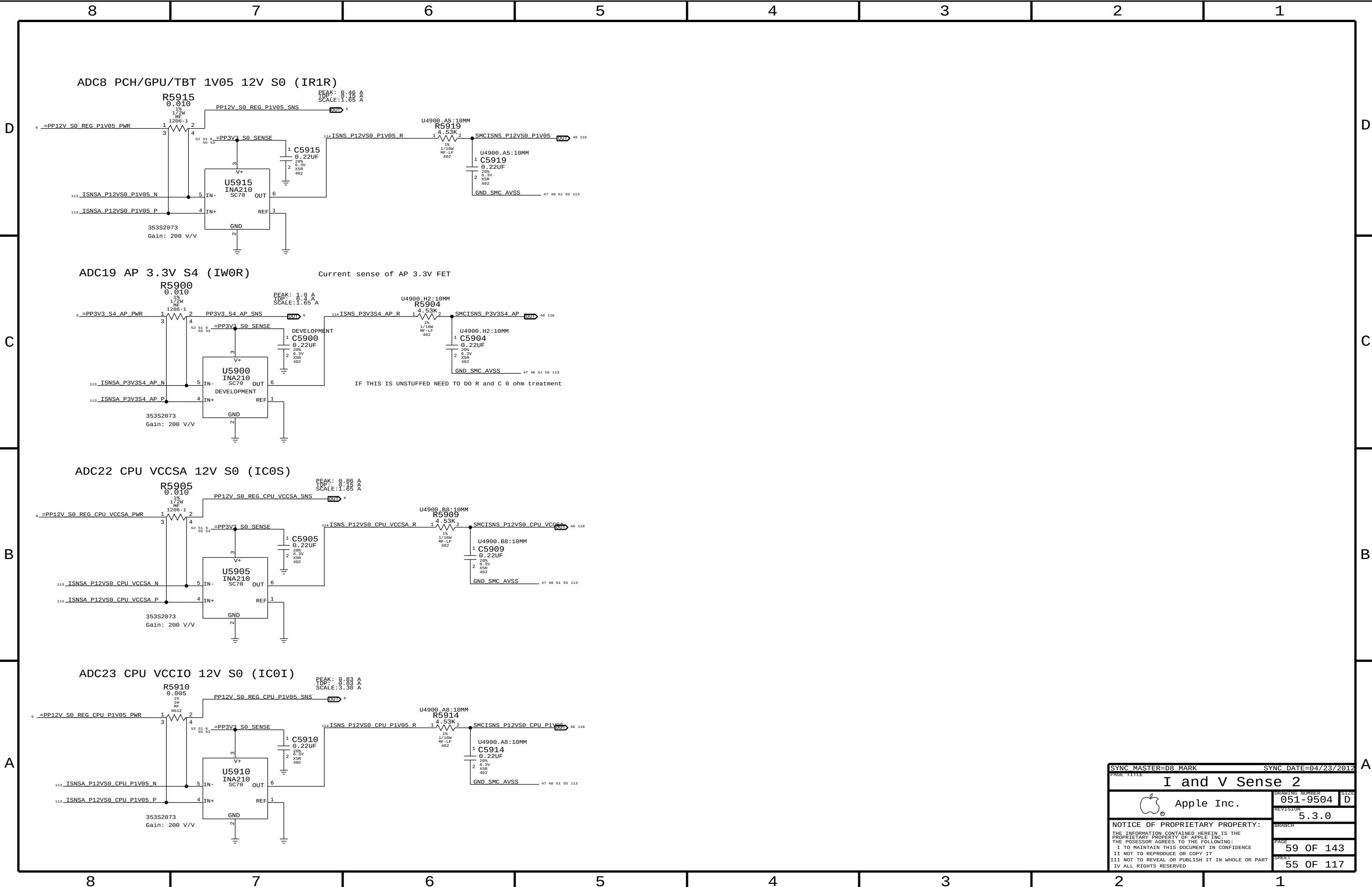
```

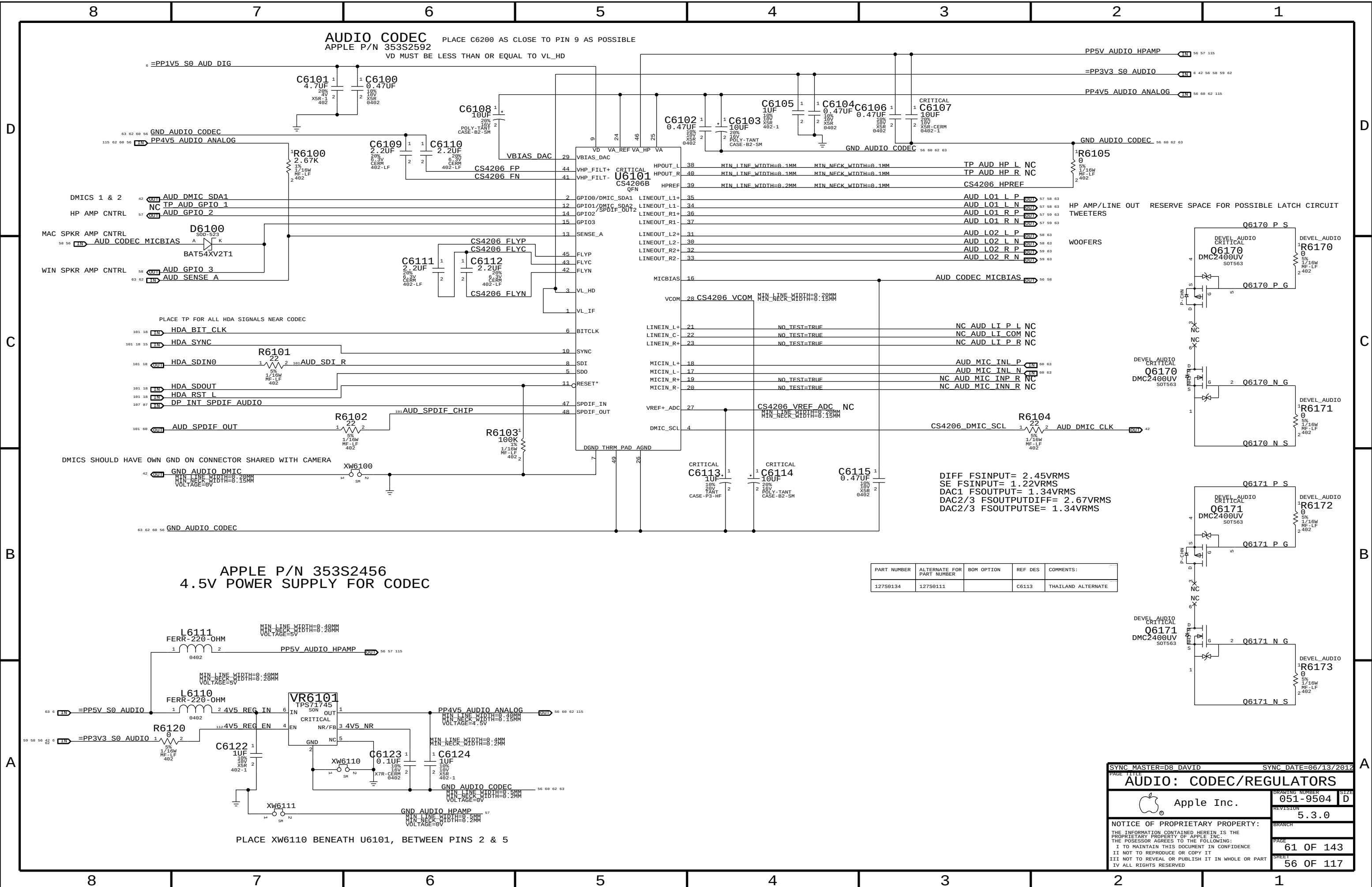
47 [IN] SMC_FAN_1_CTL == NC_SMC_FAN_1_CTL
== MAKE_BASE=TRUE NO_TEST=TRUE

47 [OUT] SMC_FAN_1_TACH == NC_SMC_FAN_1_TACH
== MAKE_BASE=TRUE NO_TEST=TRUE

```







PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
127S0134	127S0111		C6113	THAILAND ALTERNATE

SYNC MASTER=D8 DAVID

SYNC DATE=06/13/2012

AUDIO: CODEC/REGULATORS

Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

DRAWING NUMBER

051-9504

REVISION

5.3.0

PAGE

61 OF 143

SHEET

56 OF 117

LEFT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +9 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)

D

D

C

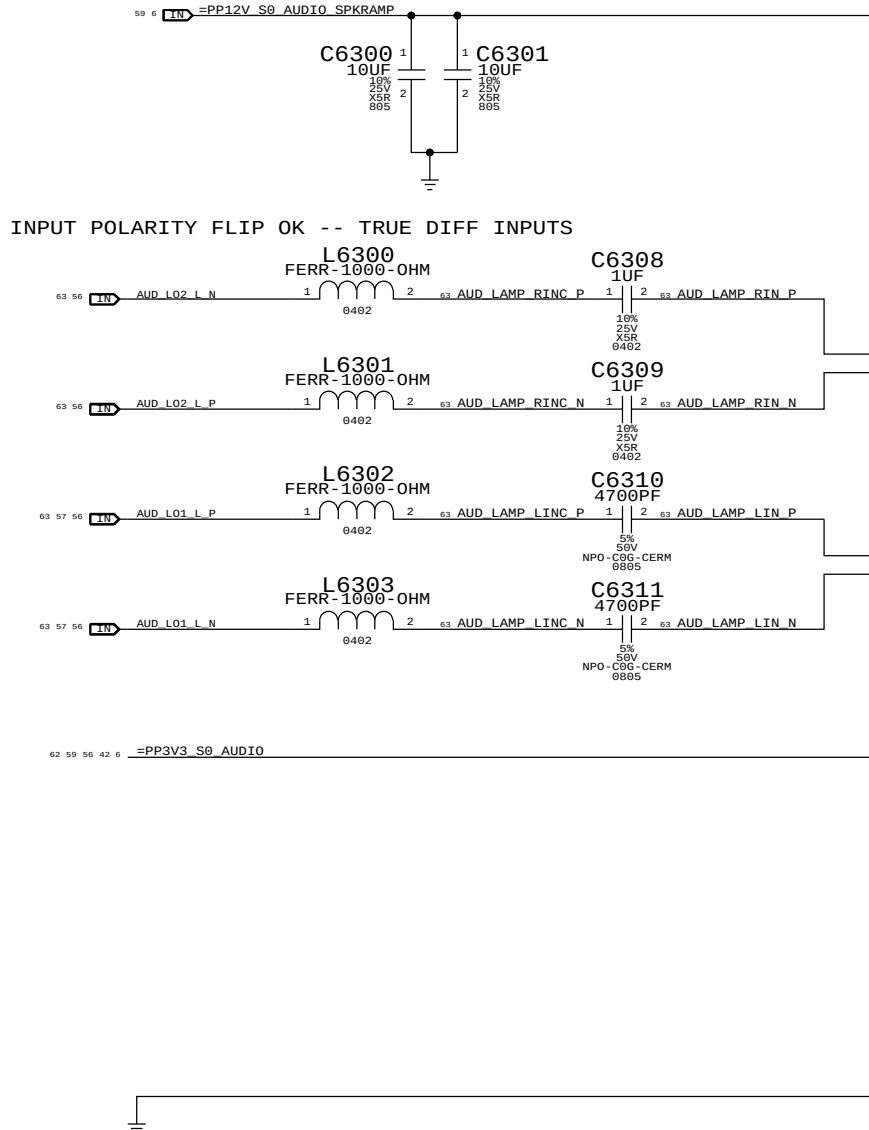
C

B

B

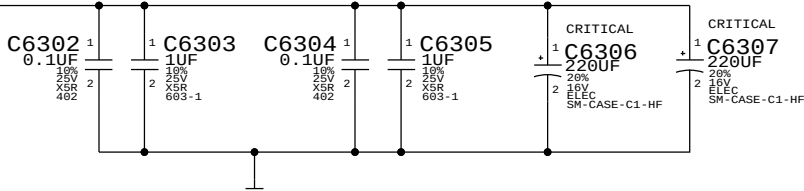
A

A

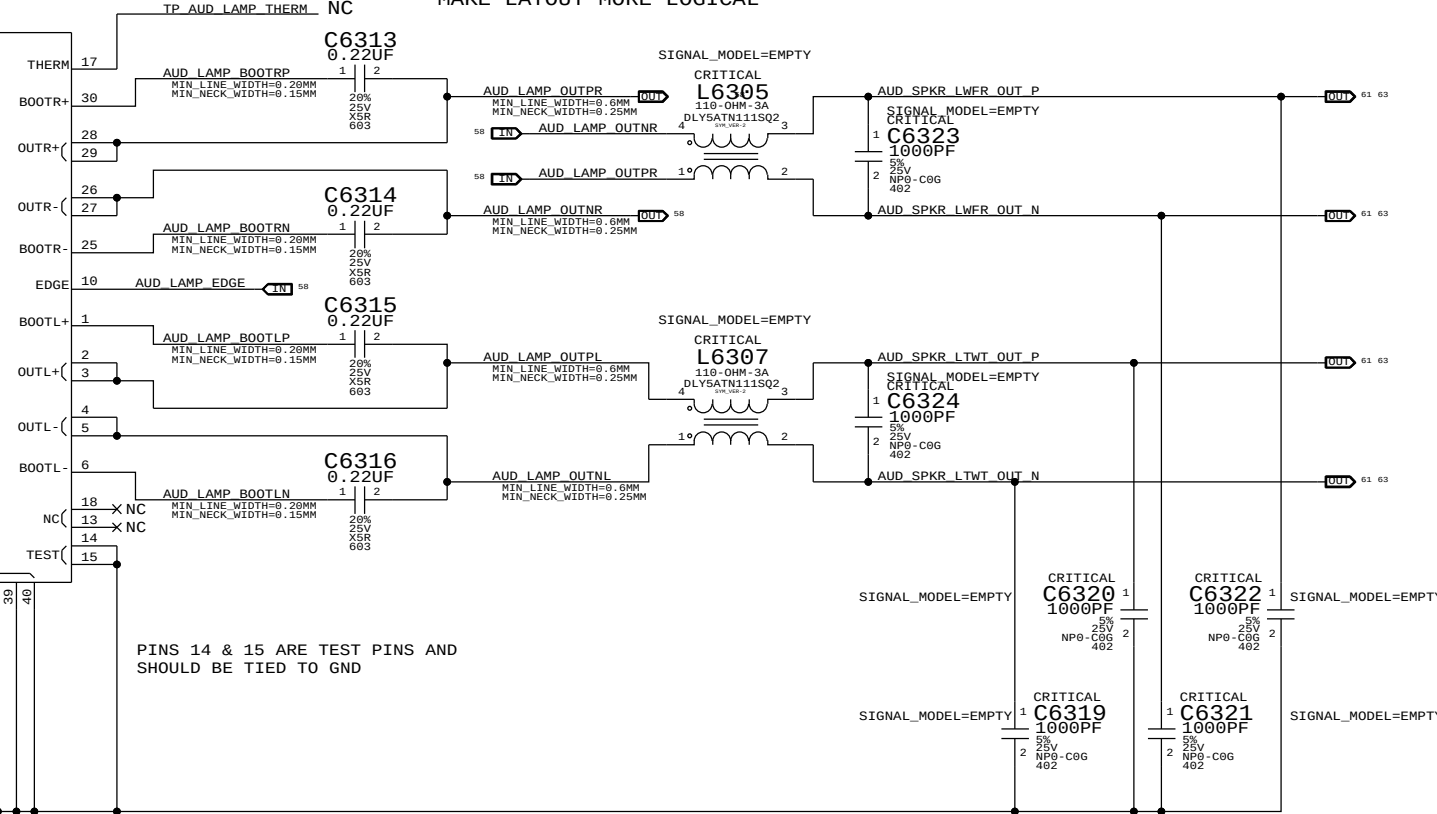


ONLY WOOFERS ON UNDER WINDOWS
WOOFERS & TWEETERS ON UNDER MAC OS

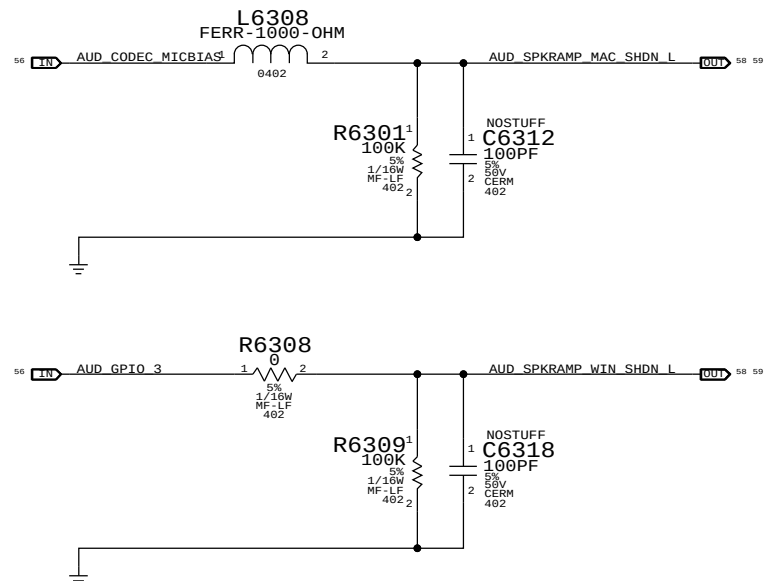
CRITICAL
U6300
SSM3302
LFCSP



OUTPUT POLARITY FLIP TO
MAKE LAYOUT MORE LOGICAL



PINS 14 & 15 ARE TEST PINS AND
SHOULD BE TIED TO GND



EDGE RATE
CONTROL
ON
OFF

R6304
0 OHM
NOSTUFF

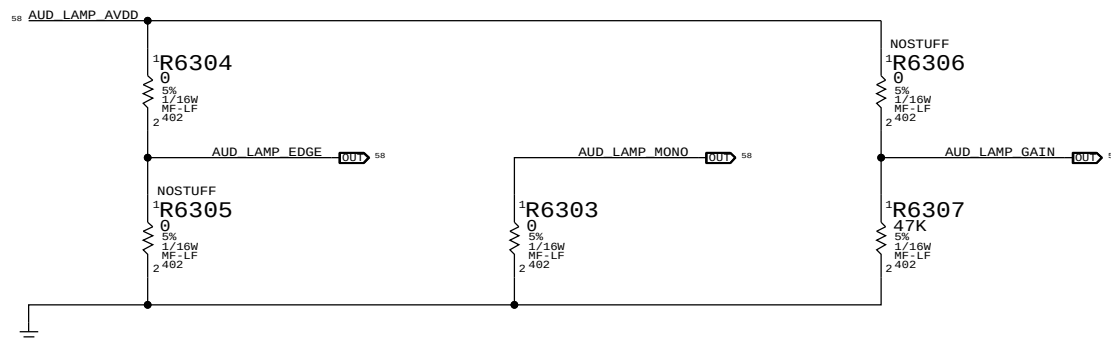
R6305
0 OHM
NOSTUFF

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN
+9 DB
+12 DB
+15 DB
+18 DB
+24 DB

R6306
NOSTUFF
NOSTUFF
NOSTUFF
NOSTUFF
NOSTUFF

R6307
0 OHM
47 KOHM
NOSTUFF
NOSTUFF
NOSTUFF



SYNC MASTER=D8 DAVID		SYNC DATE=06/13/2012	
PAGE TITLE		AUDIO: LEFT SPKR AMP	
Apple Inc.		DRAWING NUMBER	051-9504
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	5.3.0
		BRANCH	
		PAGE	63 OF 143
		SHEET	58 OF 117

RIGHT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +9 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)

D

D

C

C

B

B

A

A

INPUT POLARITY FLIP OK -- TRUE DIFF INPUTS

OUTPUT POLARITY FLIP TO
MAKE LAYOUT MORE LOGICAL

PINS 14 & 15 ARE TEST PINS AND
SHOULD BE TIED TO GND

EDGE RATE
CONTROL
ON
OFF

R6404
0 OHM
NOSTUFF

R6405
NOSTUFF
0 OHM

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN
+9 DB
+12 DB
+15 DB
+18 DB
+24 DB

R6406
NOSTUFF
NOSTUFF
47 KOHM
0 OHM

R6407
0 OHM
NOSTUFF
NOSTUFF
NOSTUFF
NOSTUFF

SYNC MASTER=08 DAVID		SYNC DATE=06/13/2012	
PAGE TITLE			
AUDIO: RIGHT SPKR AMP			
 Apple Inc.		DRAWING NUMBER	051-9504
		SIZE	D
		REVISION	5.3.0
		BRANCH	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
		PAGE	64 OF 143
		SHEET	59 OF 117

MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S3231

I2C ADDRESSES

MIKEY	U6551	READ	0111 0011	0X73
MIKEY	U6551	WRITE	0111 0010	0X72

D

D

C

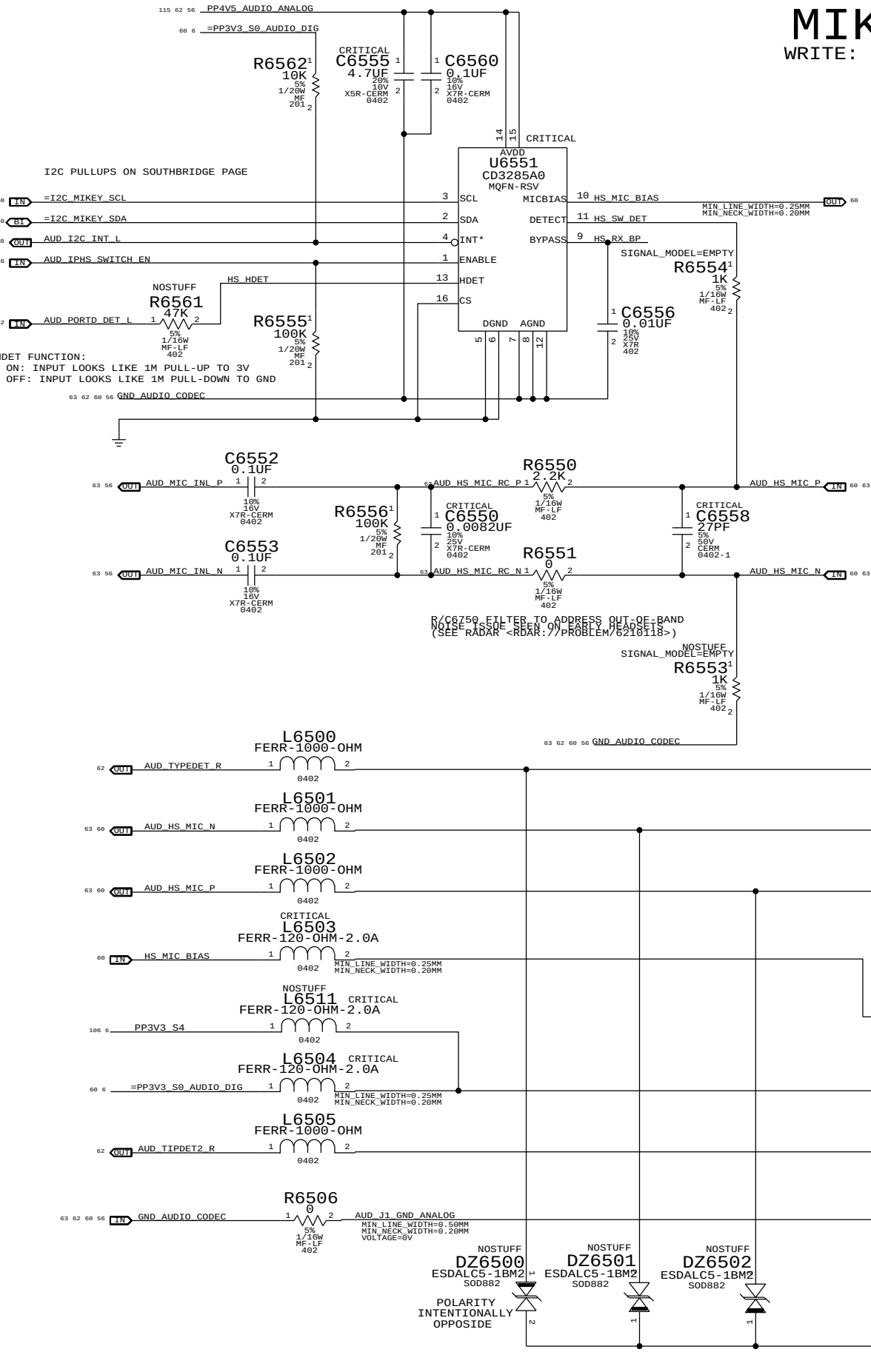
C

B

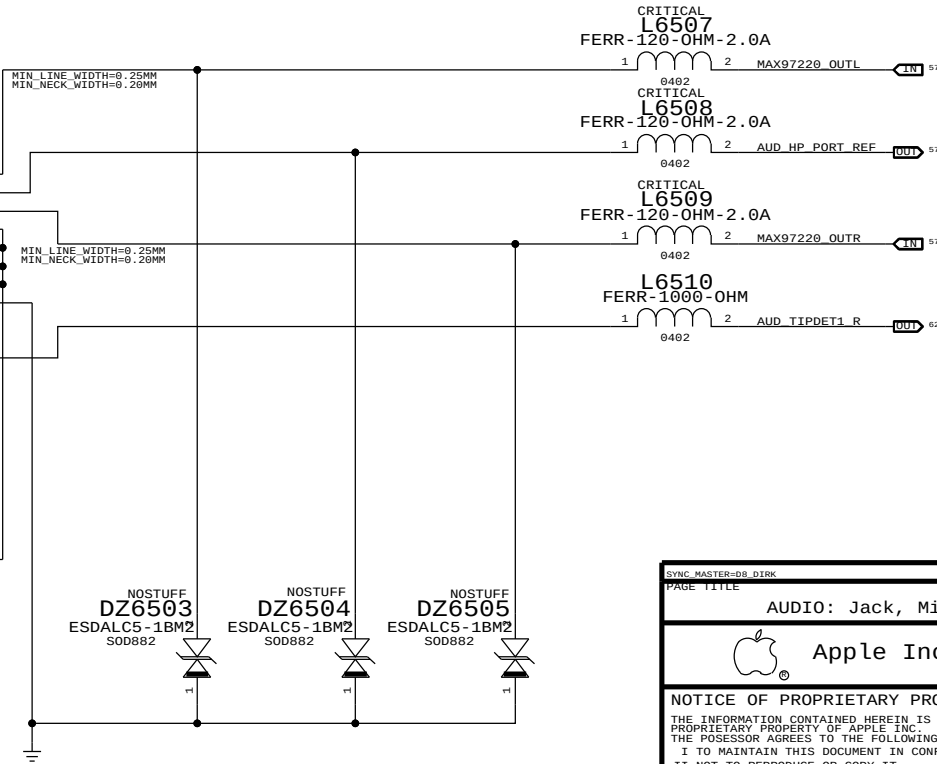
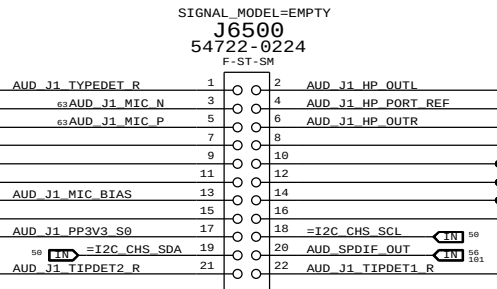
B

A

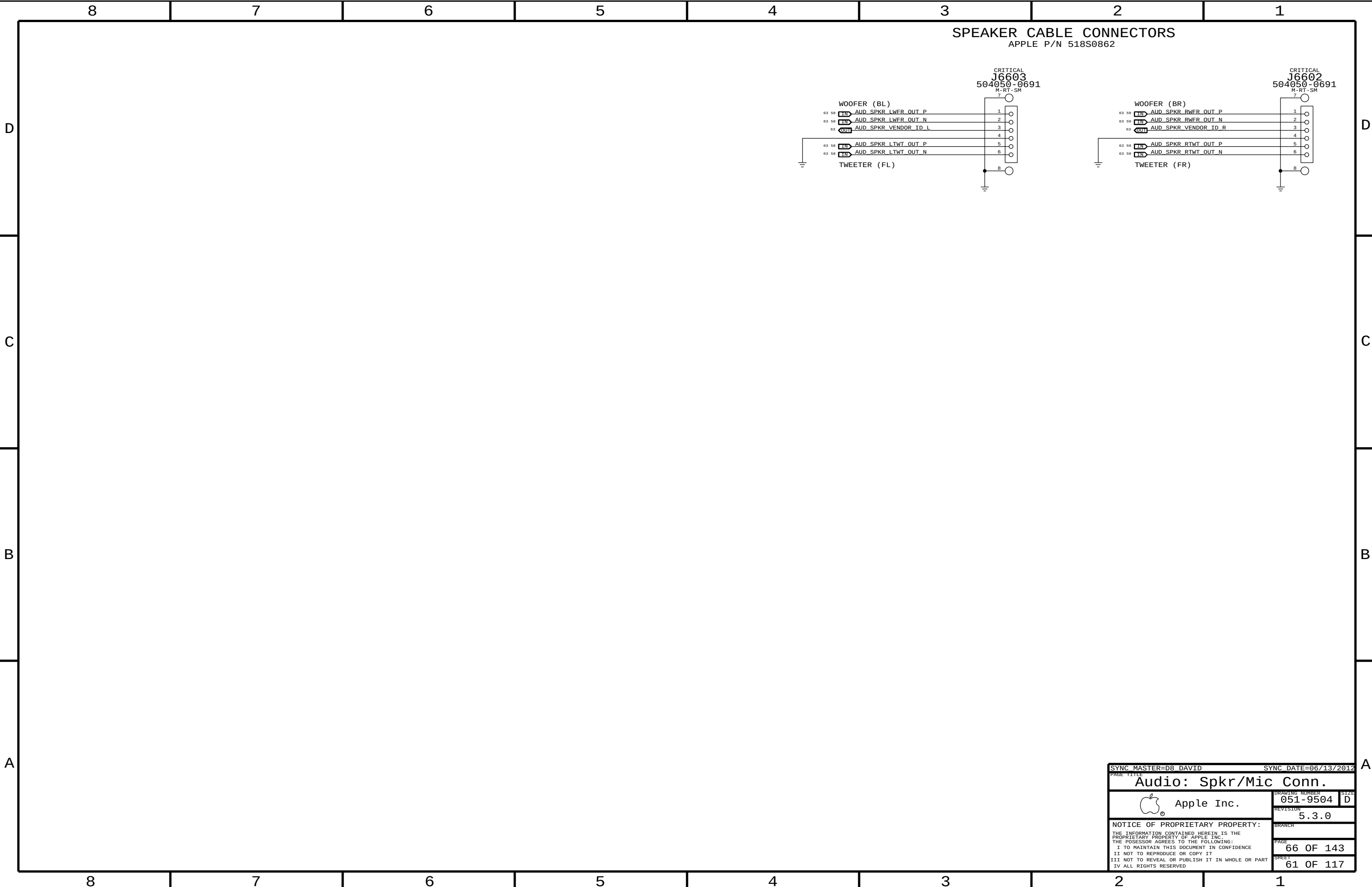
A



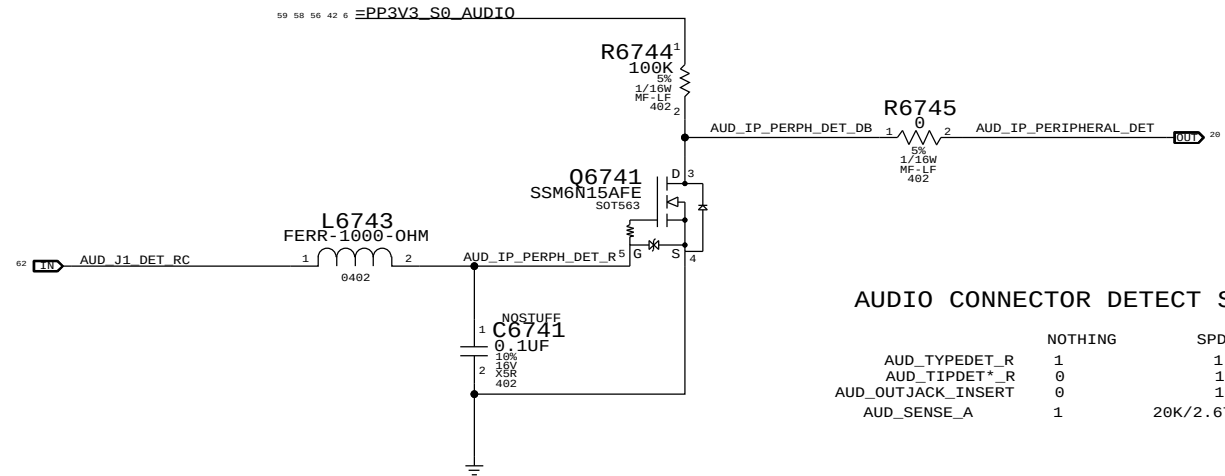
APPLE P/N 518S0687



PAGE TITLE		PAGE TITLE	
AUDIO: Jack, Mikey, CHS Switch		DRAWING NUMBER	
Apple Inc.		051-9504	
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		5.3.0	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	
II NOT TO REPRODUCE OR COPY IT		65 OF 143	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		60 OF 117	

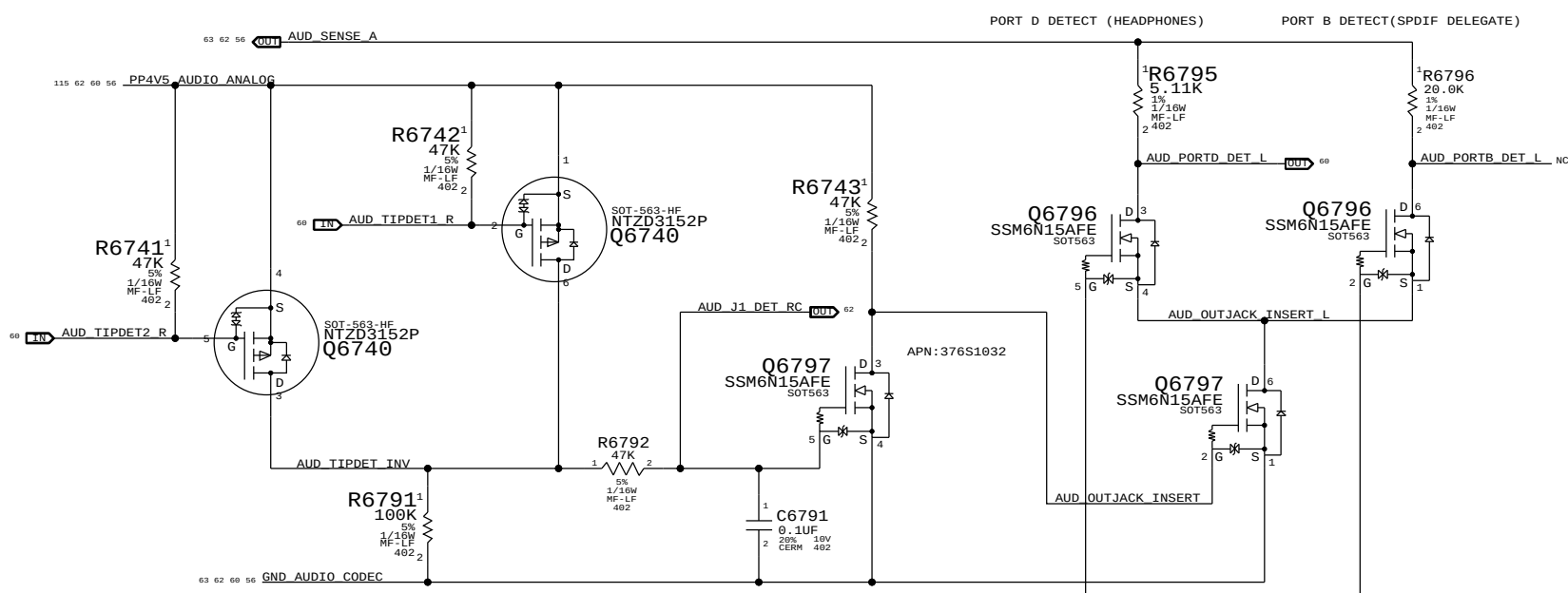
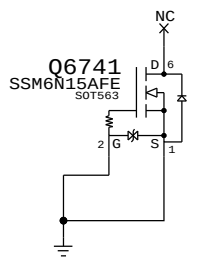


IPHS HS Detect Debounce CKT

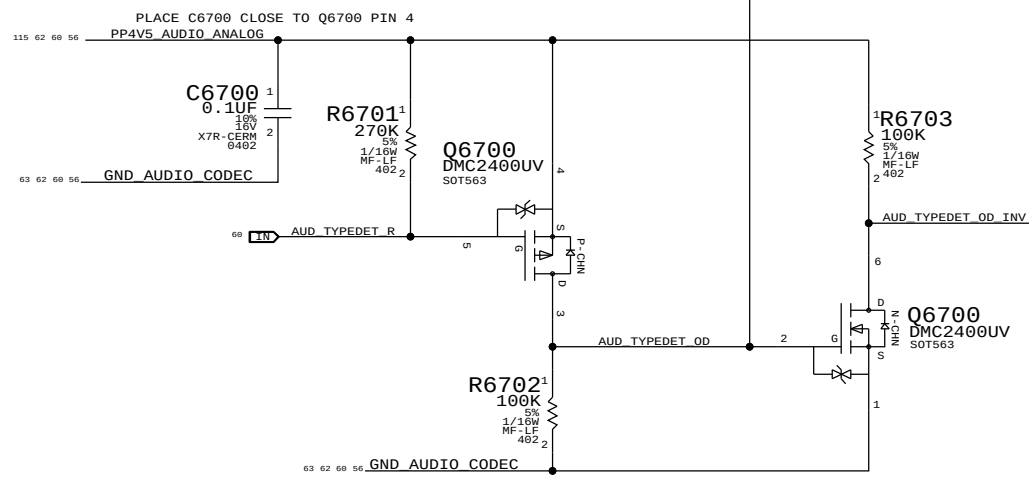
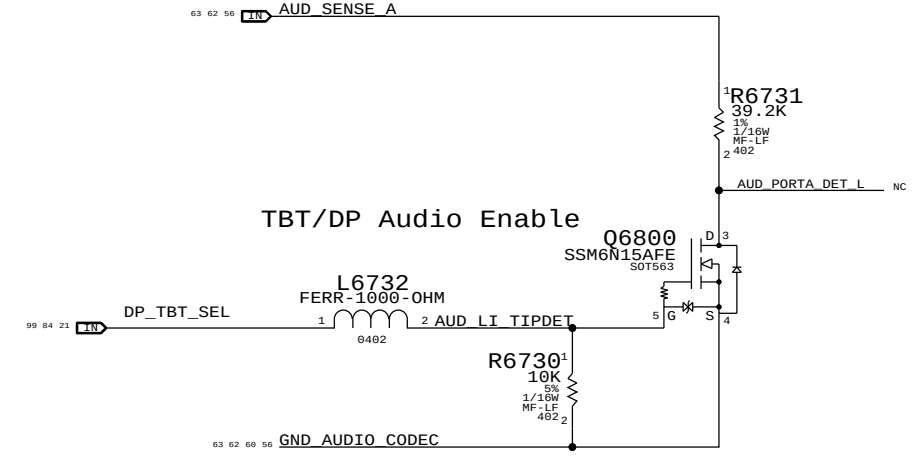


AUDIO CONNECTOR DETECT STATES

	NOTHING	SPDIF	HEADPHONE
AUD_TYPEDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_A	1	20K/2.67K RDIV	5.11K/2.67K RDIV



LI Insert Detect (DETECT A)



SYNC MASTER=D8 DAVID

SYNC DATE=06/13/2012

AUDIO: Detects/Grounding

Apple Inc.

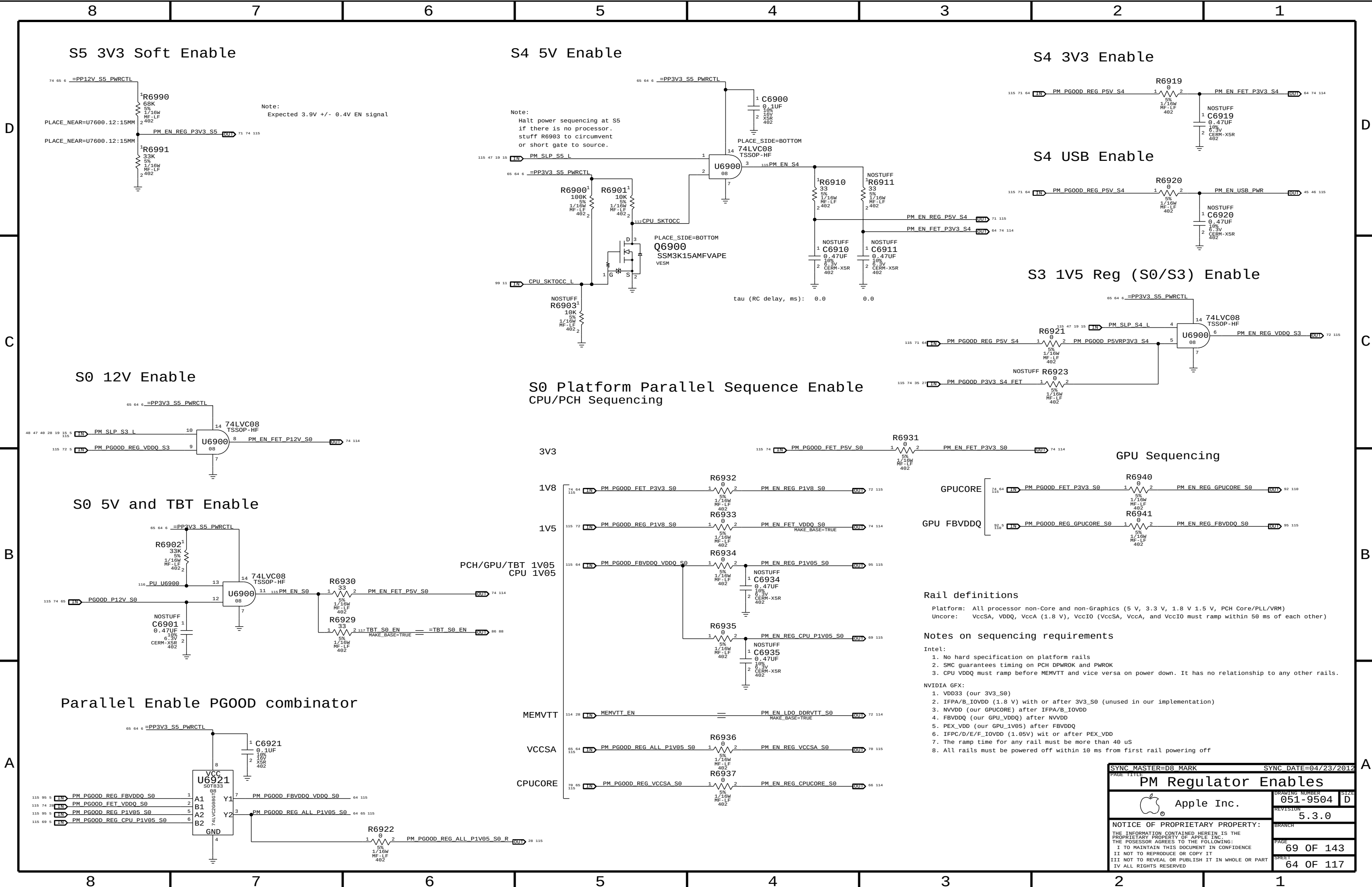
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER
051-9504

REVISION
5.3.0

PAGE
67 OF 143

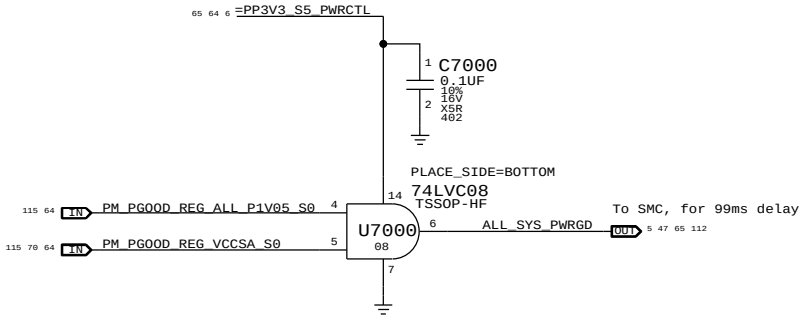
SHEET
62 OF 117



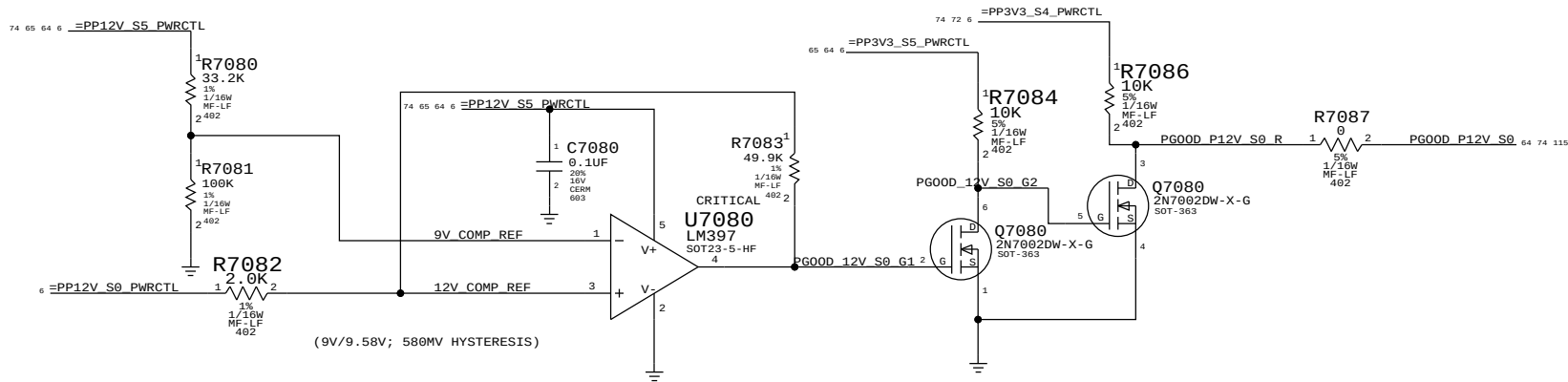
SYNC MASTER=D8 MARK		SYNC DATE=04/23/2012	
PAGE TITLE		PM Regulator Enables	
 Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	69 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	64 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

Platform Power Good derive SMC ALL_SYS_PWRGD

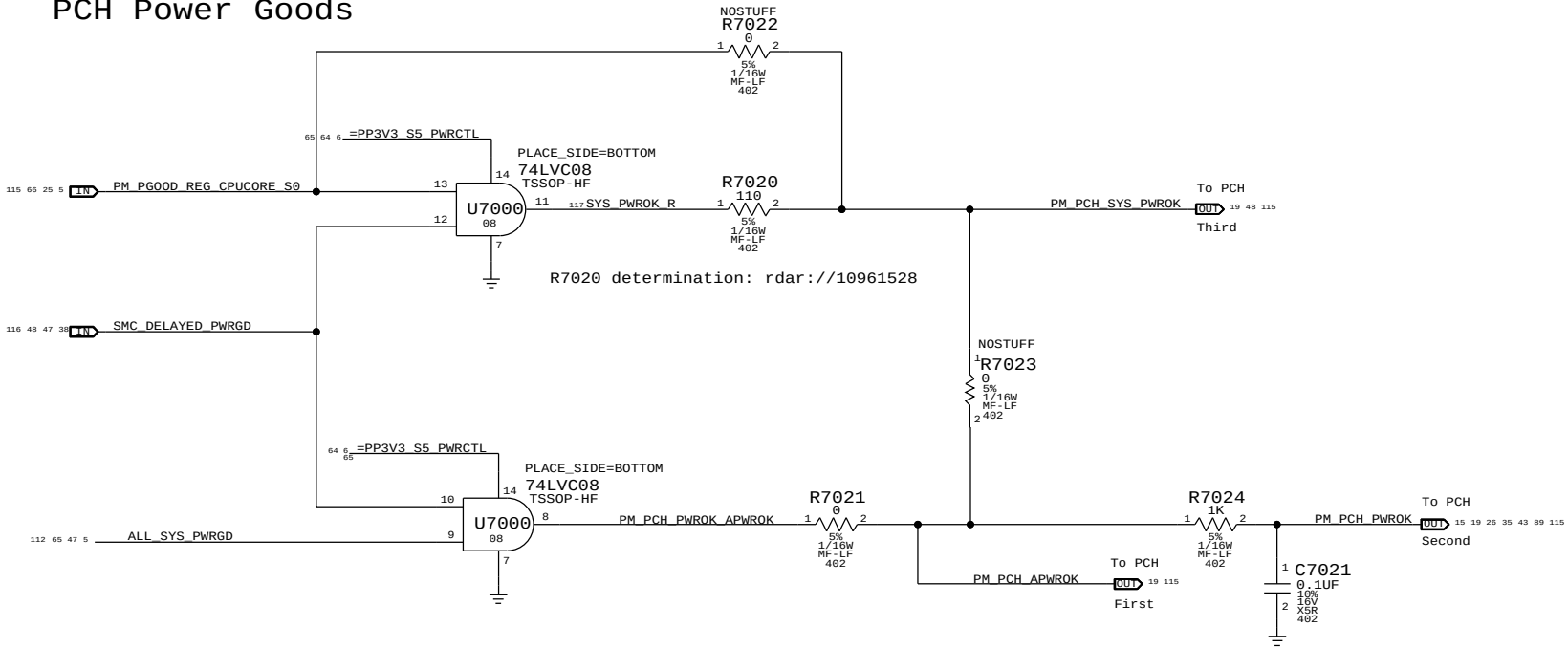
The end of the power sequence for S0 rails except CPU CORE.



PGOOD COMPARATORS FOR PP12V_S0



PCH Power Goods



radar://11043352 Need AND Gate to deassert PM_PCH_PWROK to PCH when unexpected power loss happens

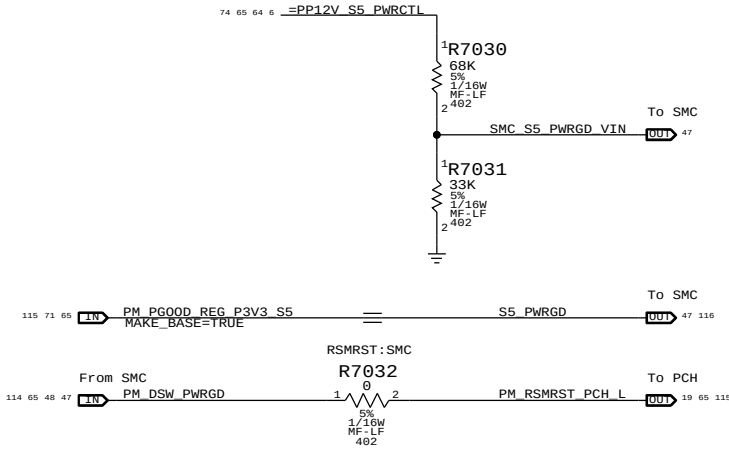
Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

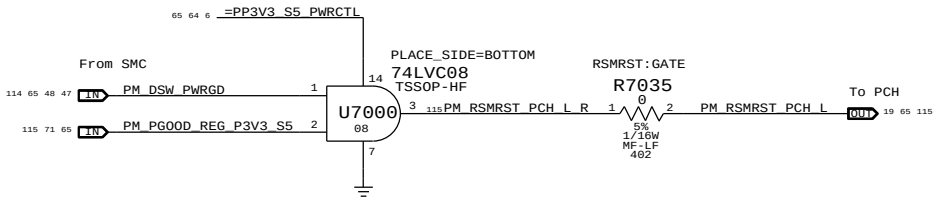
Note:
The iMac K70K72 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
to allow PCH to switch suspend well to battery without excessive loading

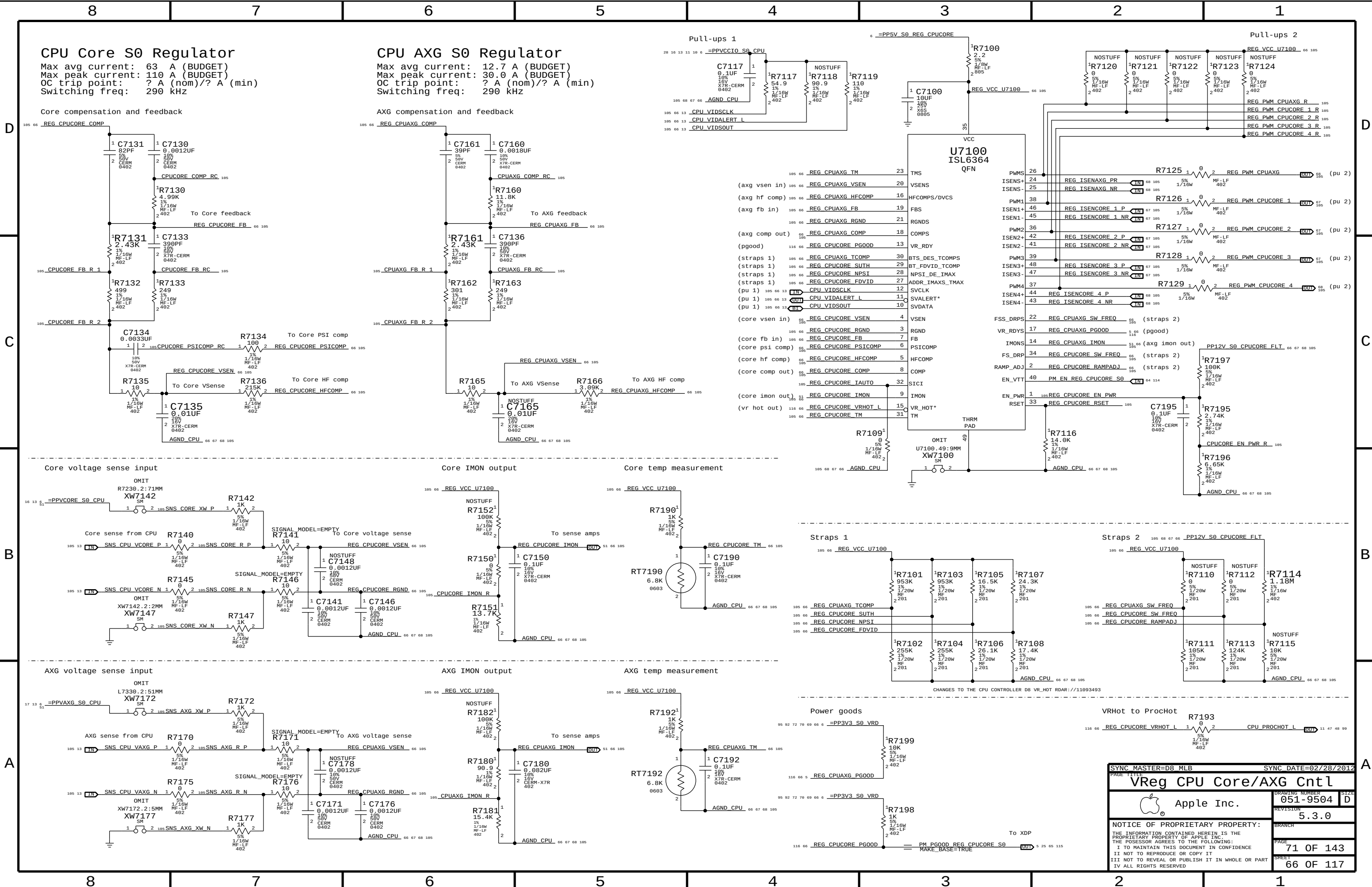
Primary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation.
SMC de-asserts RSMRST# (PM_DSW_PWRGD) when S5_PWRGD input is asserted and SMC_S5_PWRGD_VIN input is above comparator input level of 1.5 V.
SMC asserts RSMRST# (PM_DSW_PWRGD) when SMC_S5_PWRGD_VIN input drops from 1.8 V to 1.5 V (as implemented) when 12 V S5 rail drops to 10 V.



Secondary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.
RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



SYNC MASTER=D8 MARK		SYNC DATE=04/23/2012	
PAGE TITLE		PM Power Good	
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	70 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	65 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



CPU Core S0 Regulator
Max avg current: 63 A (BUDGET)
Max peak current: 110 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: 290 kHz

CPU AXG S0 Regulator
Max avg current: 12.7 A (BUDGET)
Max peak current: 30.0 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: 290 kHz

Core compensation and feedback

AXG compensation and feedback

Core voltage sense input

Core IMON output

Core temp measurement

AXG voltage sense input

AXG IMON output

AXG temp measurement

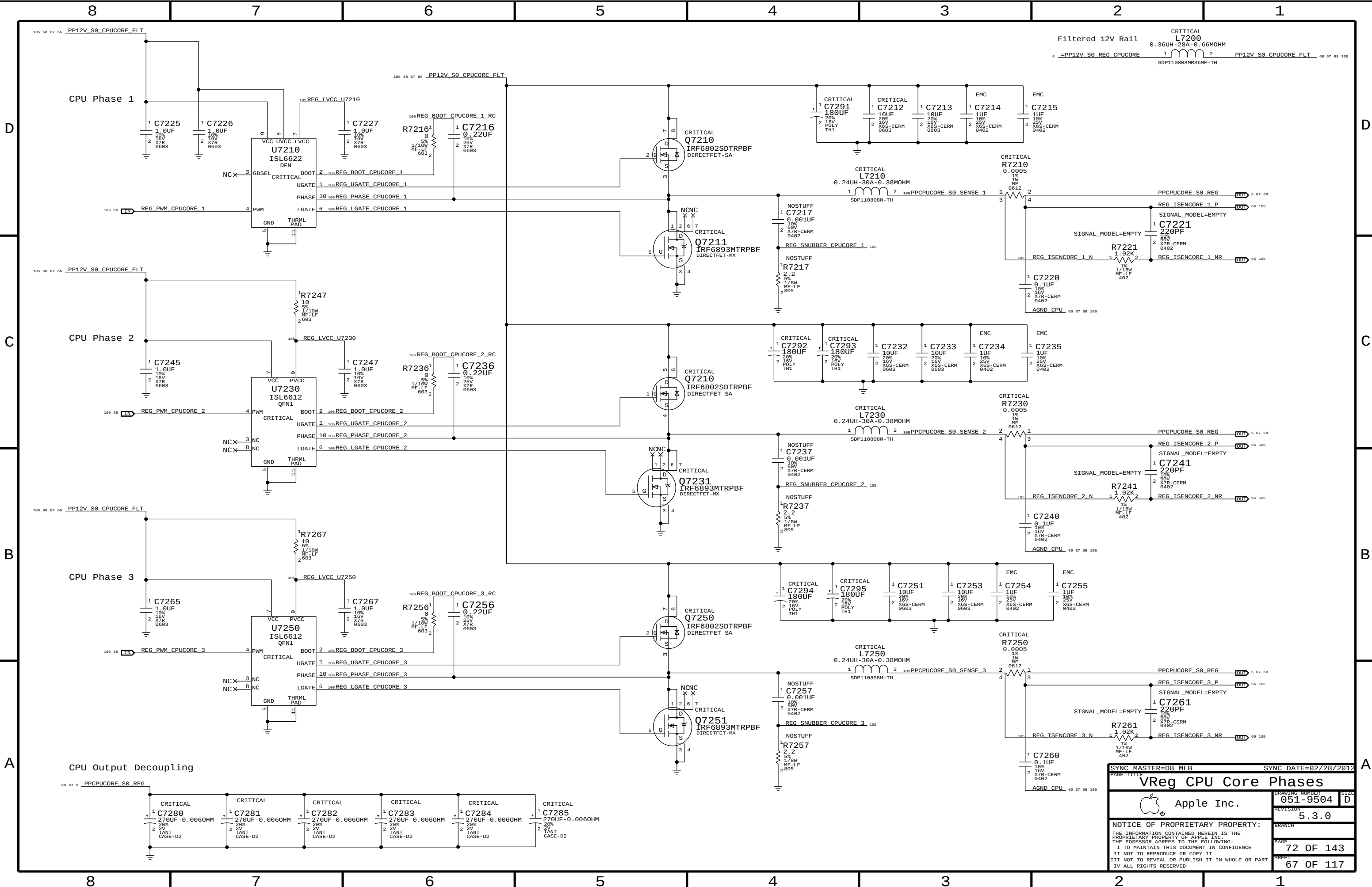
Straps 1

Straps 2

Power goods

VRHot to Prochot

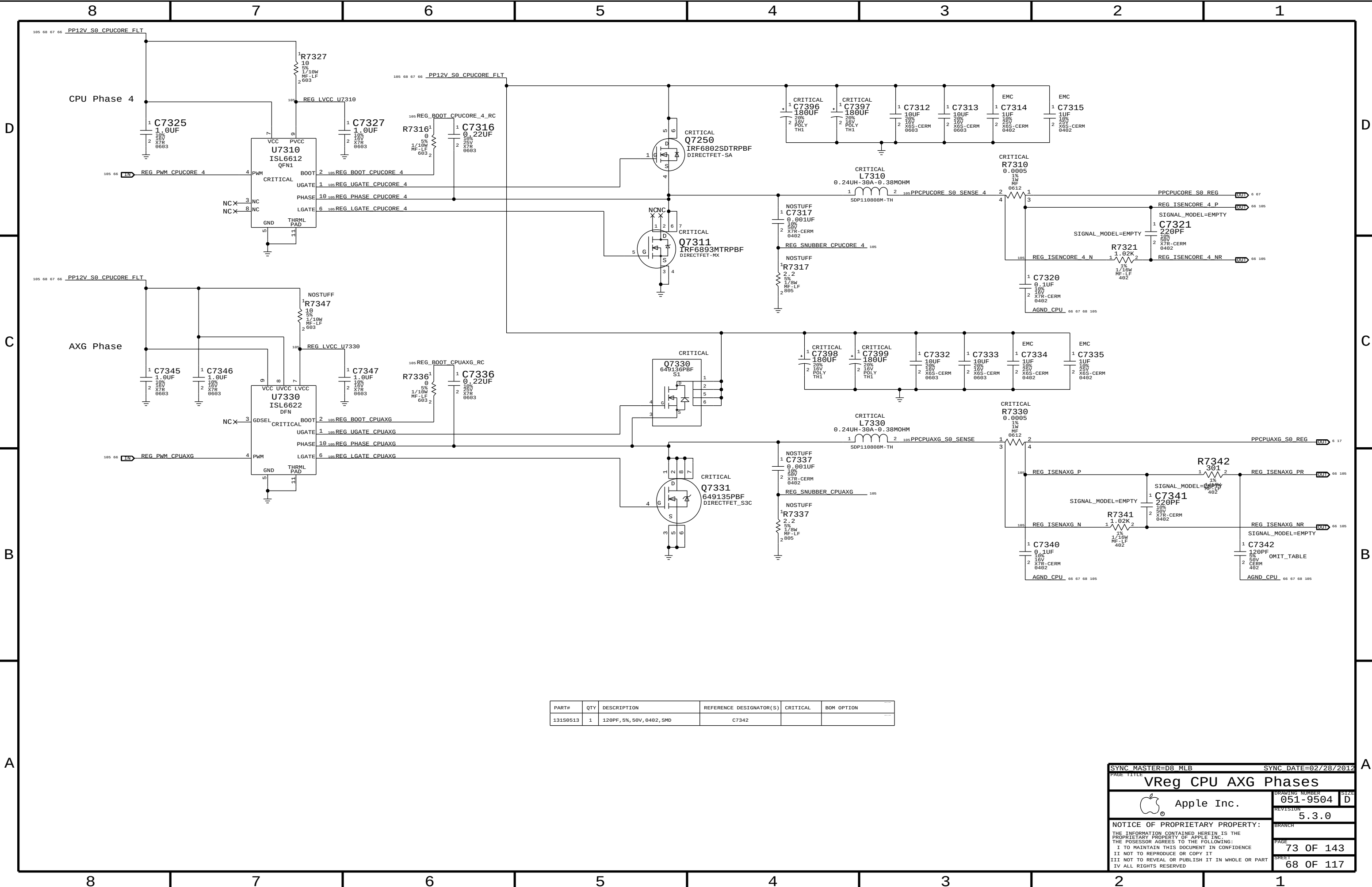
SYNC MASTER=D8 MLB		SYNC DATE=02/28/2012	
PAGE TITLE		VReg CPU Core/AXG Cntl	
Apple Inc.		DRAWING NUMBER	051-9504
NOTICE OF PROPRIETARY PROPERTY:		REVISION	5.3.0
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	71 OF 143
II NOT TO REPRODUCE OR COPY IT		SHEET	66 OF 117
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



SYNC MASTER=D8_MLB

SYNC DATE=02/28/2012

VReg CPU Core Phases	
	Apple Inc.
DRAWING NUMBER	051-9504
REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE	
II NOT TO REPRODUCE OR COPY IT	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART	
IV ALL RIGHTS RESERVED	
PAGE	72 OF 143
SHEET	67 OF 117



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S0513	1	120PF, 5%, 50V, 0402, SMD	C7342		

SYNC MASTER=D8_MLB

SYNC DATE=02/28/2012

VReg CPU AXG Phases

Apple Inc.

051-9504

5.3.0

NOTICE OF PROPRIETARY PROPERTY:

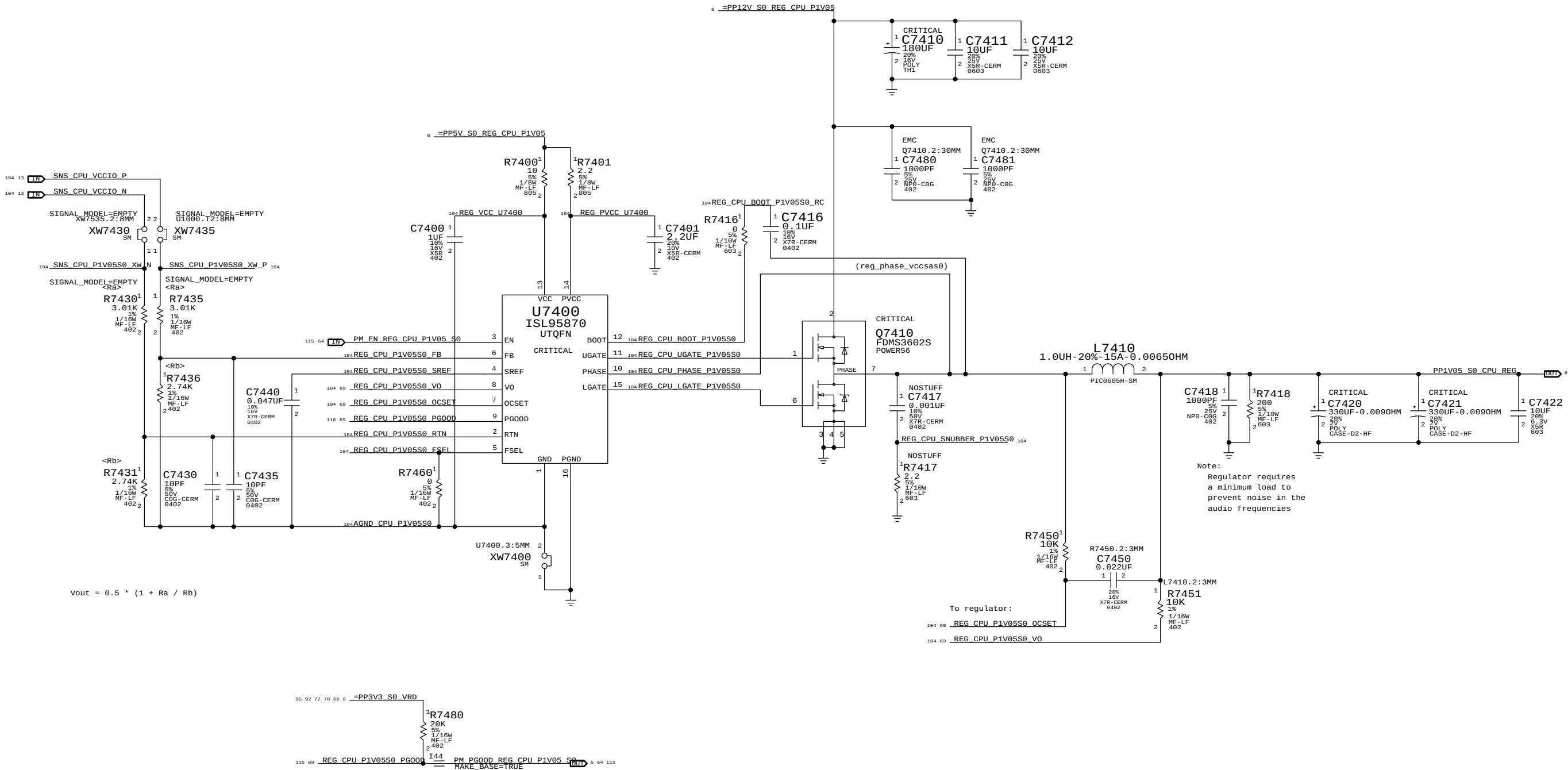
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

73 OF 143

68 OF 117

CPU VccIO (1.05V) S0 Regulator

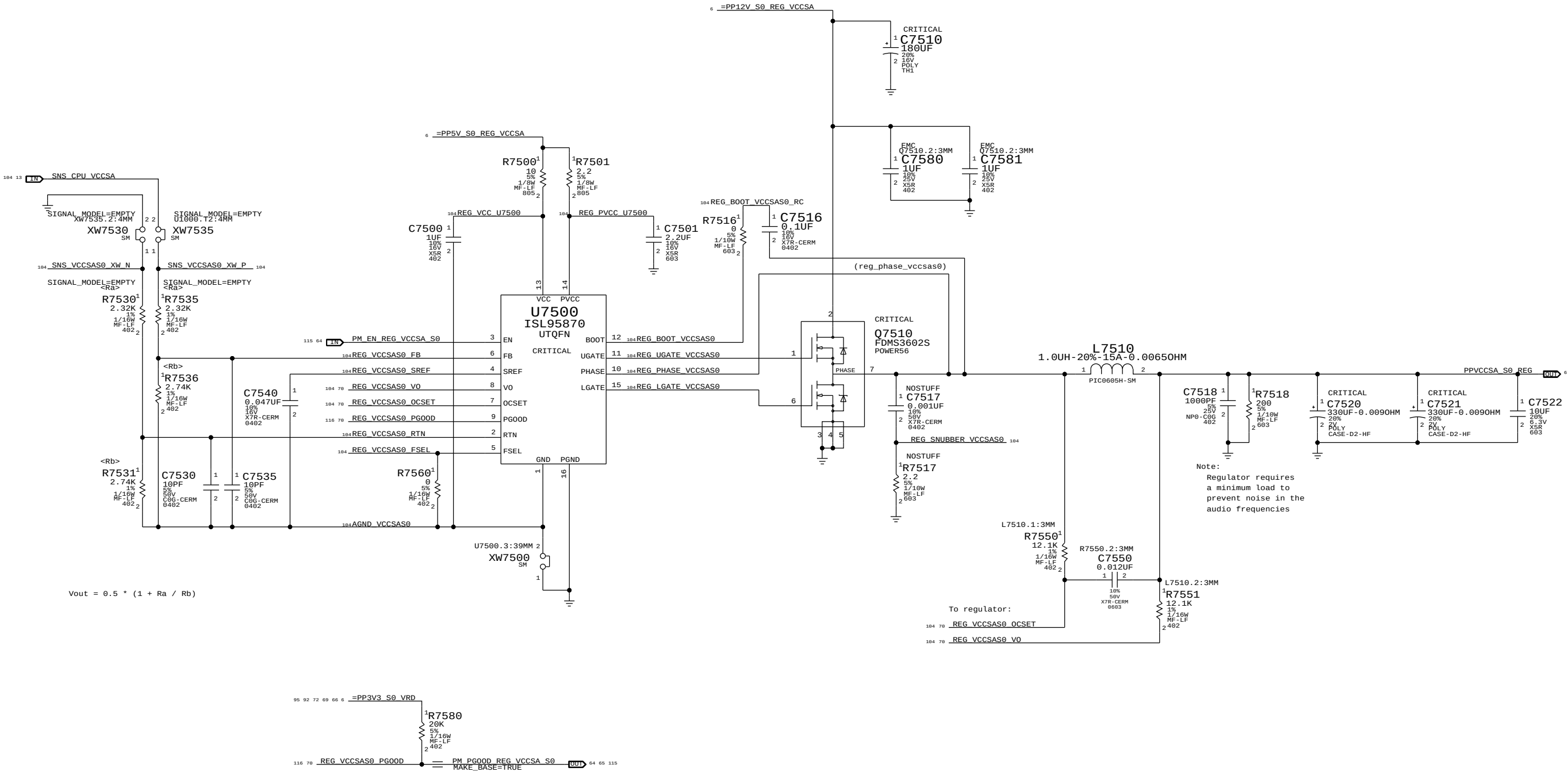
Max avg current: 8.10 A (BUDGET)
Max peak current: 8.50 A (BUDGET)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz



SYNC MASTER=D8 KOSECOFF		SYNC DATE=02/25/2012	
PAGE TITLE		VReg CPU 1.05V S0	
 Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	74 OF 143
		SHEET	69 OF 117

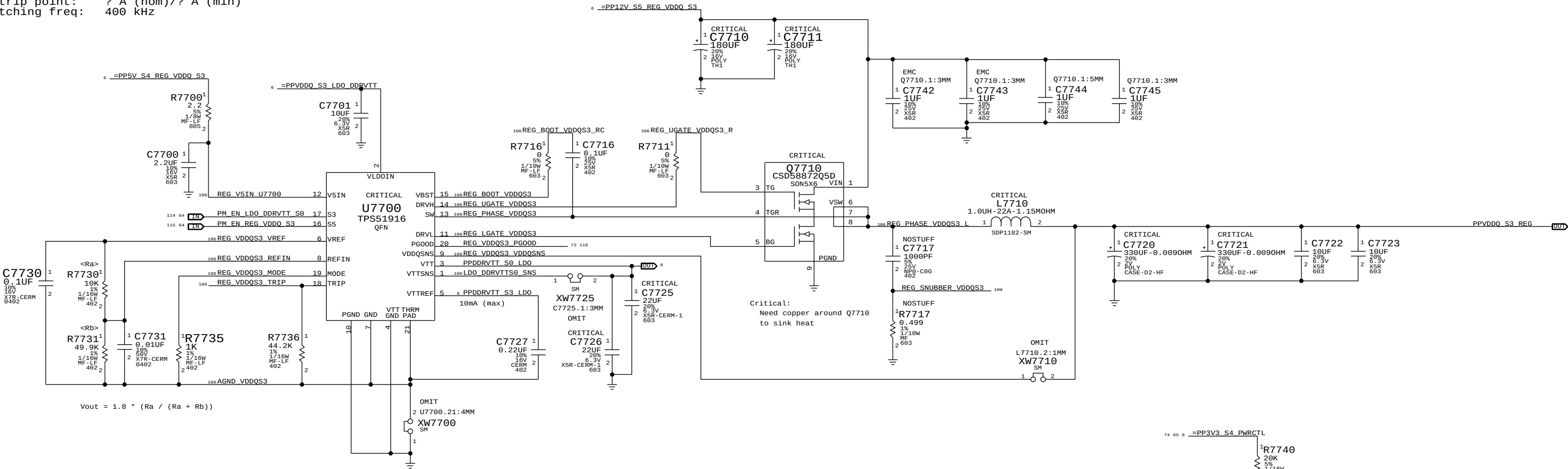
CPU VccSA (0.925V) S0 Regulator

Max avg current: 12.07 A (BUDGET)
Max peak current: 30 A (BUDGET)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 KHz



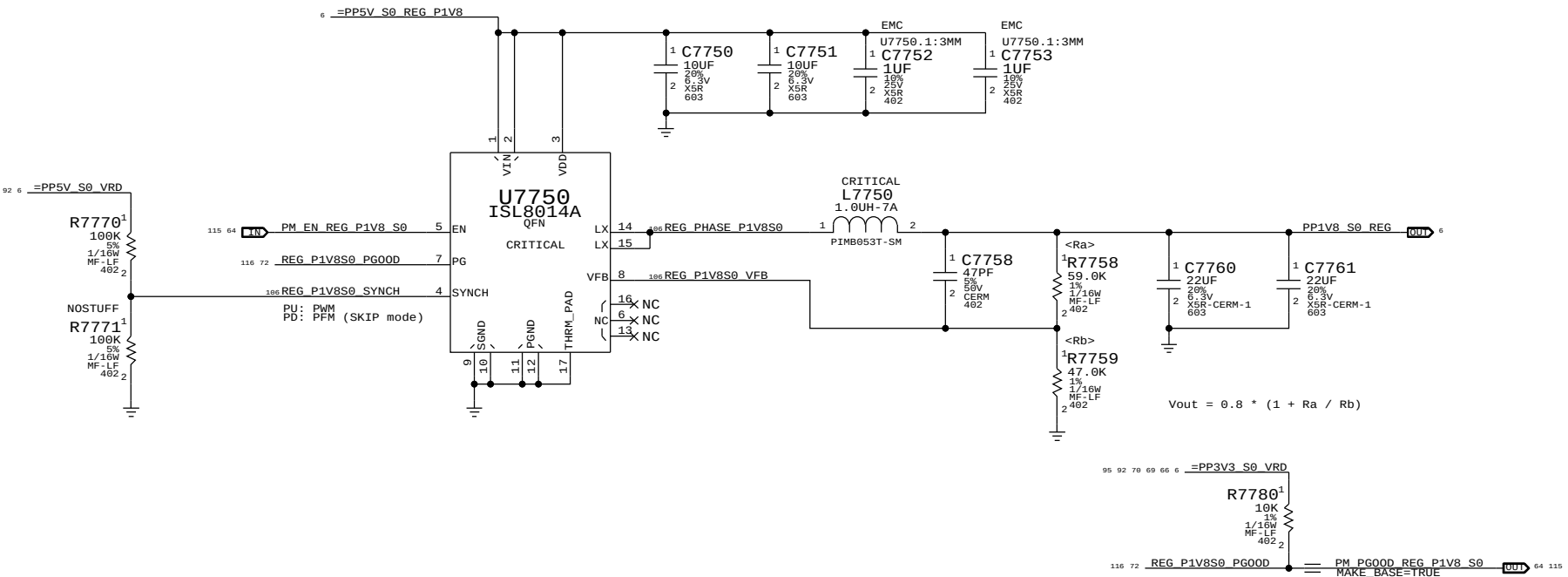
VDDQ (1.5V) S3 Regulator

Max avg current: 9.0 A (BUDGET)
Max peak current: 11.3 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: 400 kHz

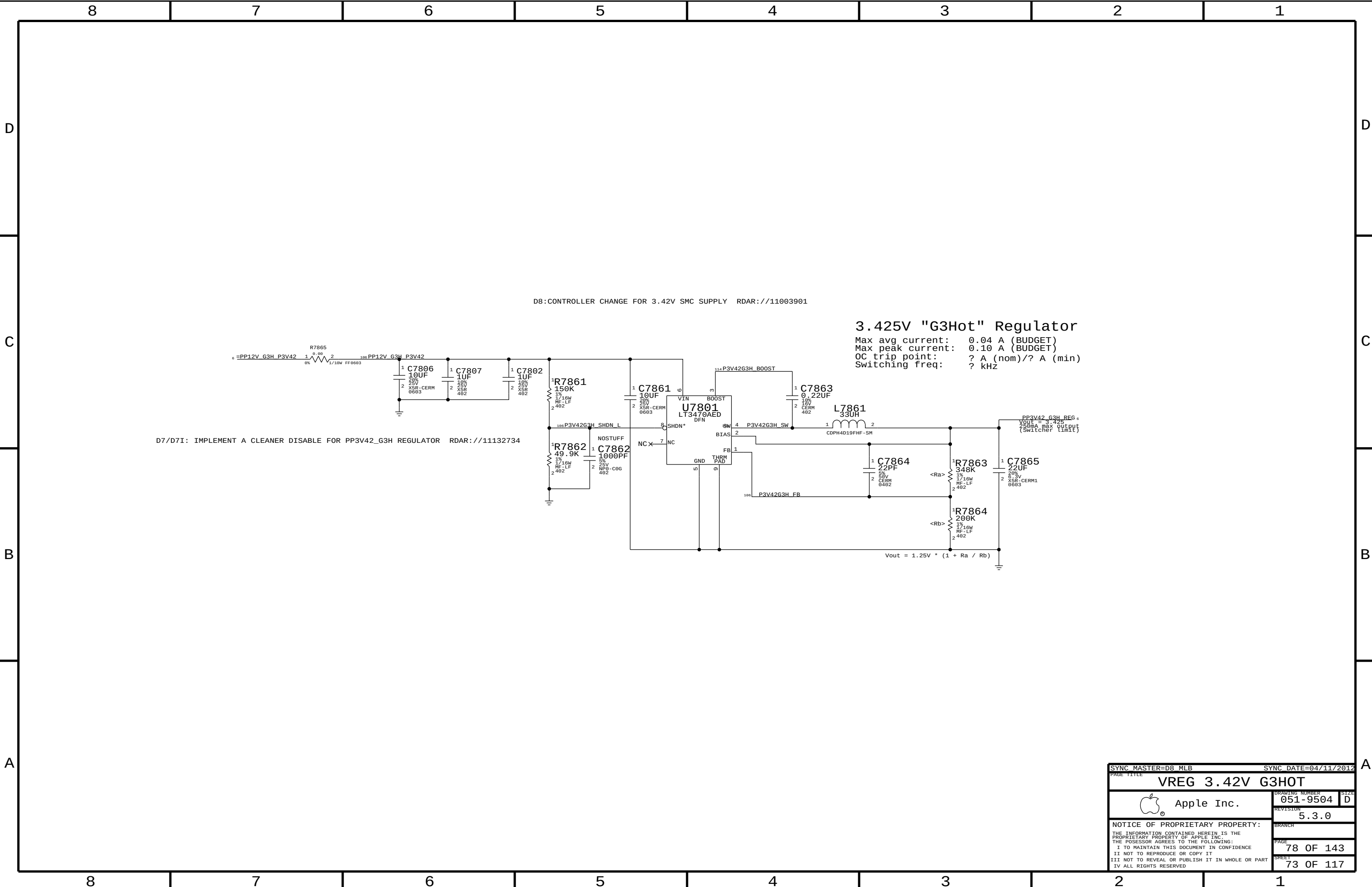


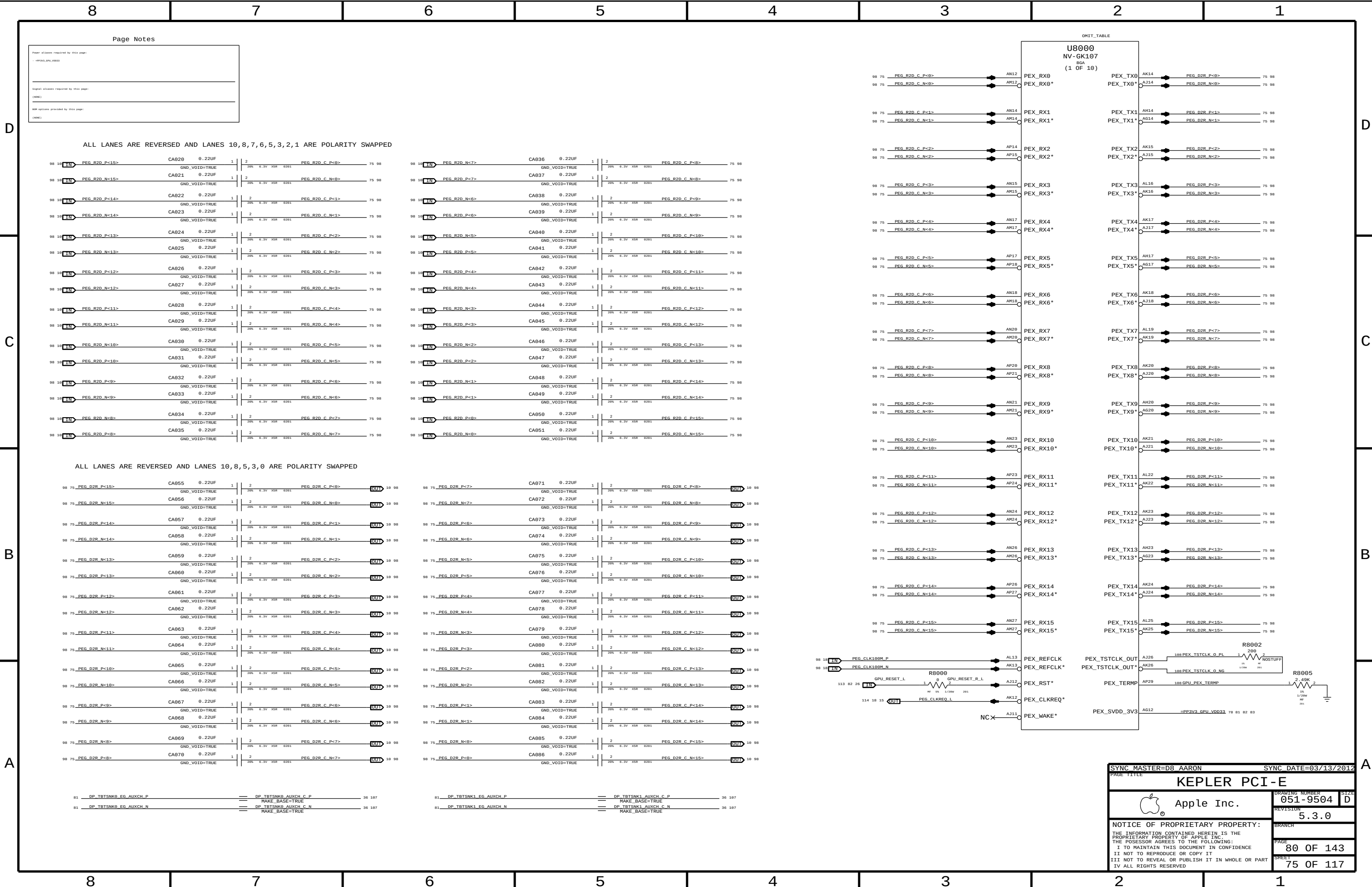
1.8V S0 Regulator

Max avg current: 0.6 A (BUDGET)
Max peak current: 1.7 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: ? kHz



PAGE TITLE		PAGE TITLE	
VReg VDDQ and 1.8V S0		VReg VDDQ and 1.8V S0	
Apple Inc.		Apple Inc.	
DRAWING NUMBER		DRAWING NUMBER	
051-9504		051-9504	
REVISION		REVISION	
5.3.0		5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		NOTICE OF PROPRIETARY PROPERTY:	
THE INFORMATION CONTAINED HEREIN IS THE		THE INFORMATION CONTAINED HEREIN IS THE	
PROPRIETARY PROPERTY OF APPLE INC.		PROPRIETARY PROPERTY OF APPLE INC.	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE	
II NOT TO REPRODUCE OR COPY IT		II NOT TO REPRODUCE OR COPY IT	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART	
IV ALL RIGHTS RESERVED		IV ALL RIGHTS RESERVED	
PAGE		PAGE	
77 OF 143		77 OF 143	
SHEET		SHEET	
72 OF 117		72 OF 117	





D

C

B

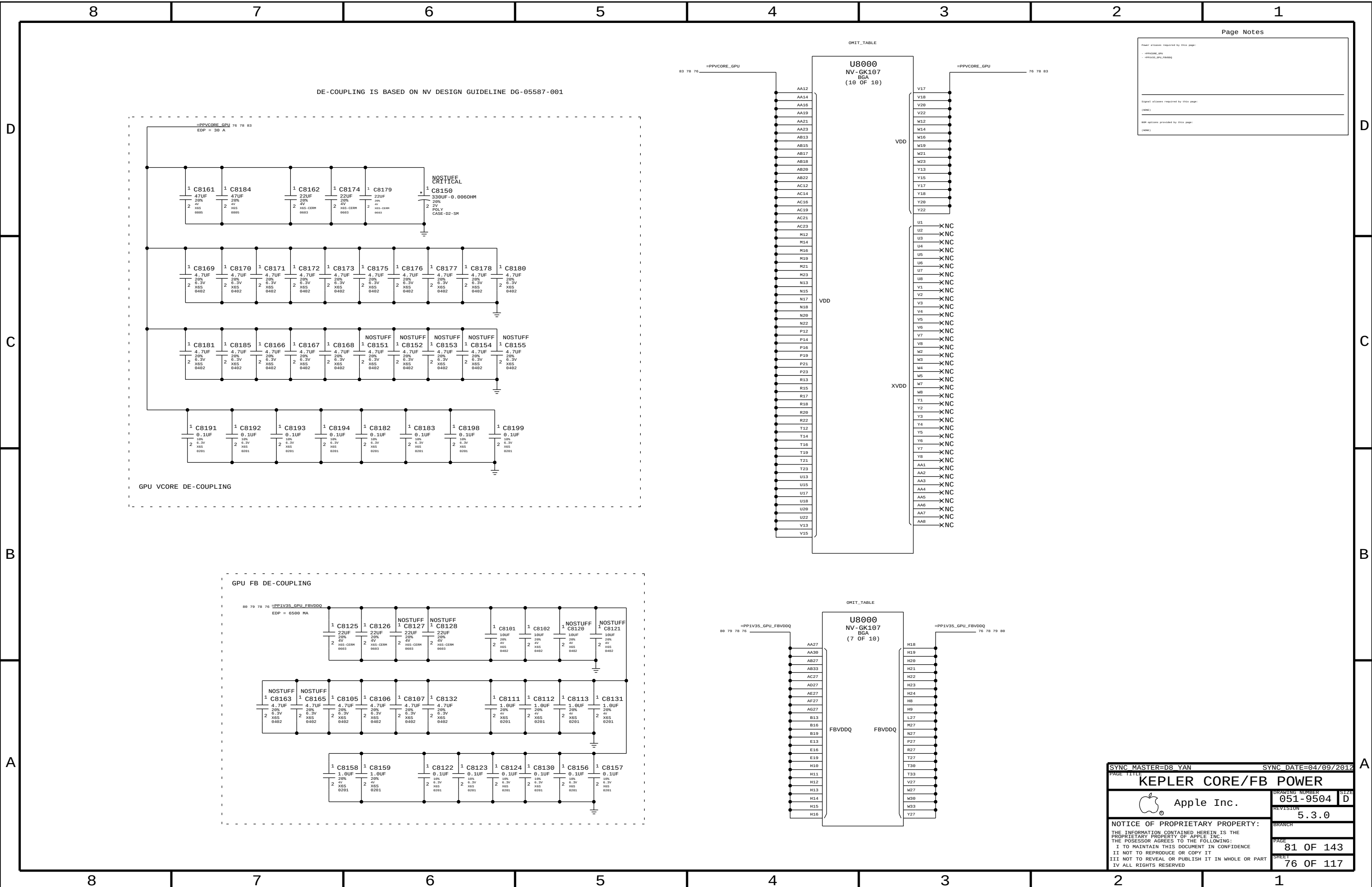
A

D

C

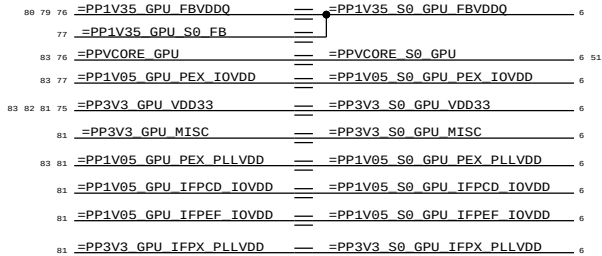
B

A

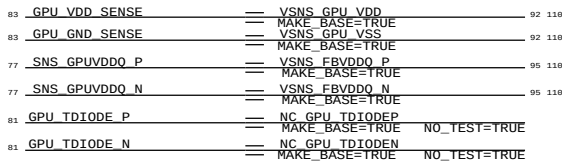


D

GPU POWER ALIAS

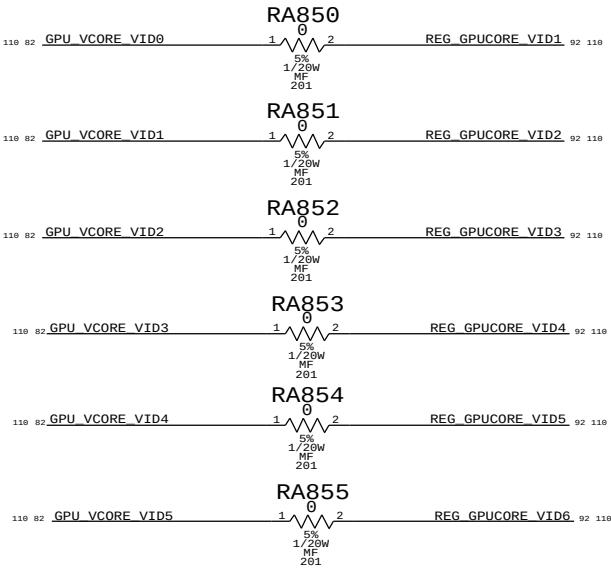


GPU SIGNAL & SENSE ALIAS



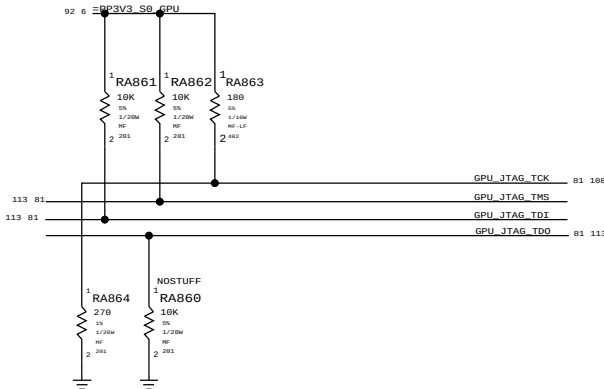
THESE POWER ALIASES ARE CRAETED TO MATCH D8 GK104

GPU VIDS ALIAS (VR VID0 IS TIED LOW)



B

PU/PD IS BASED ON RECOMMENDATION FROM NV FOR NVIDIA GPU JTAG DEBUGGER



A

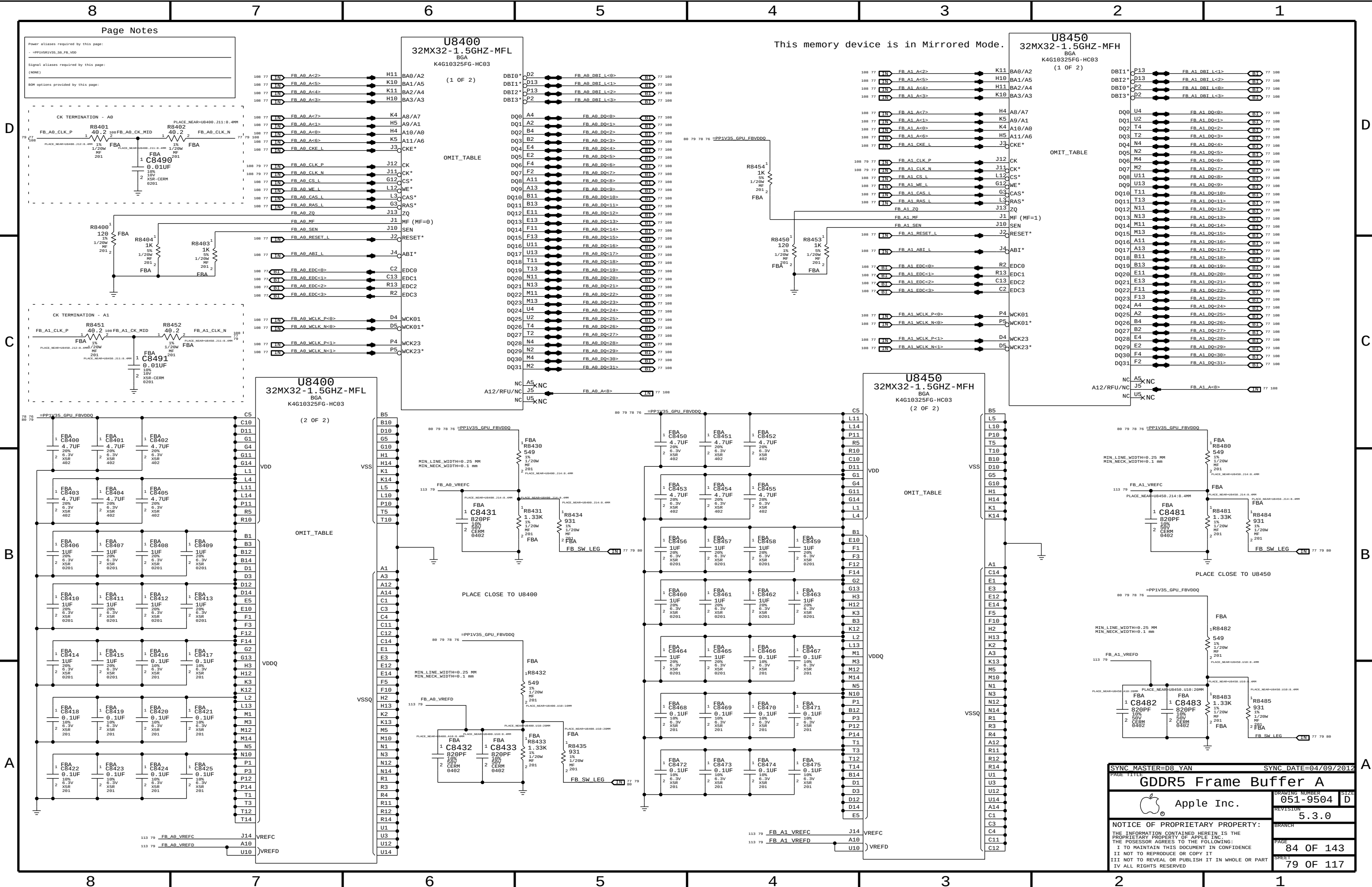
SYNC MASTER=D8 YAN		SYNC DATE=04/09/2012	
PAGE TITLE		GPU SIGNAL & POWER ALIASES	
		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	83 OF 143
		SHEET	78 OF 117

D

C

B

A



Page Notes

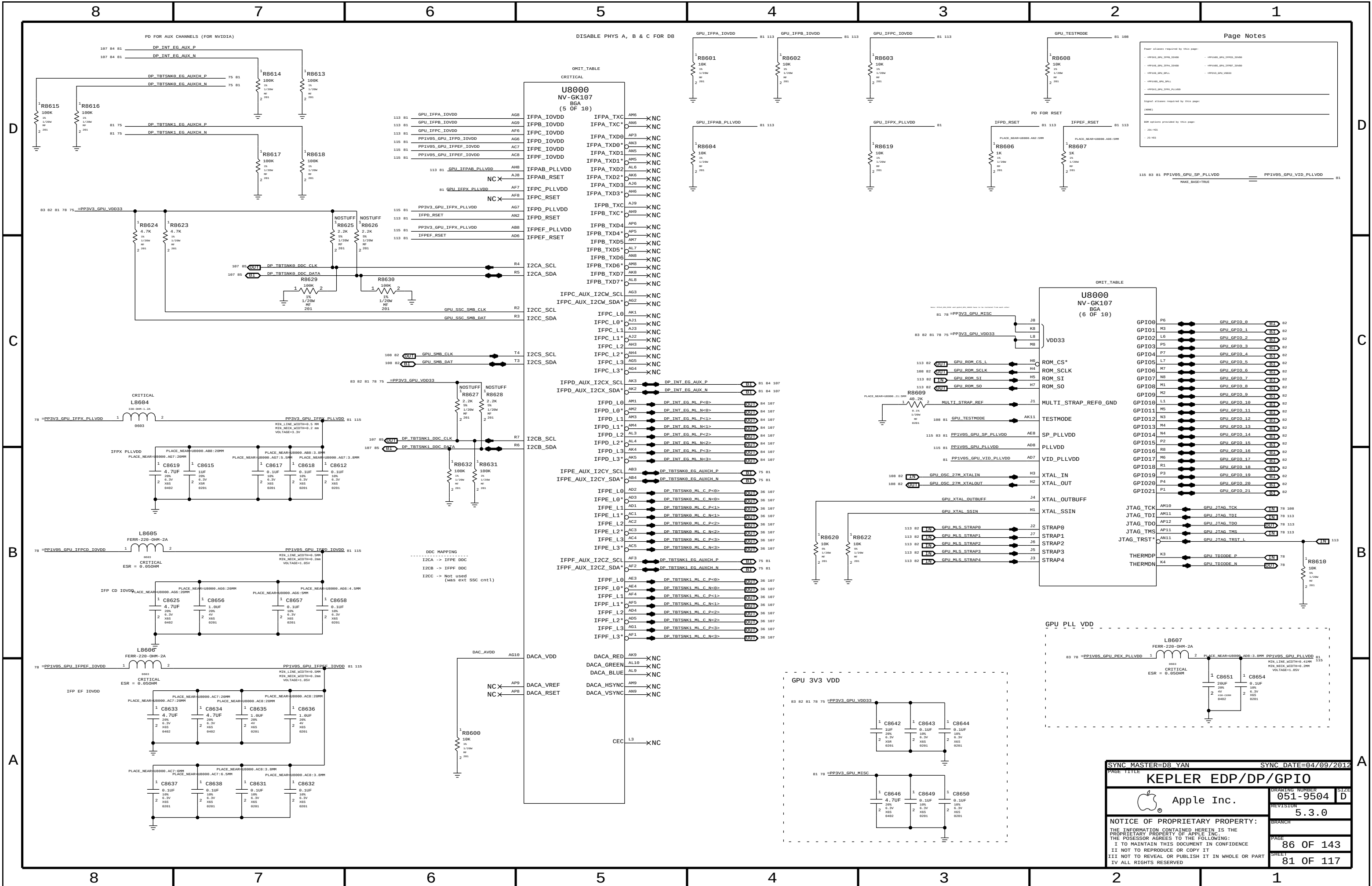
Power aliases required by this page:
- =PP1V35_V35_S8_FB_VDD

Signal aliases required by this page:
(NONE)

BOM options provided by this page:

This memory device is in Mirrored Mode.

SYNC MASTER=D8_YAN		SYNC DATE=04/09/2012		
PAGE TITLE				
GDDR5 Frame Buffer A		DRAWING NUMBER	051-9504	
 Apple Inc.		REVISION	5.3.0	
		BRANCH		
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE	84 OF 143	
		SHEET	79 OF 117	
		SIZE		D



Page Notes

Power planes required by this page:

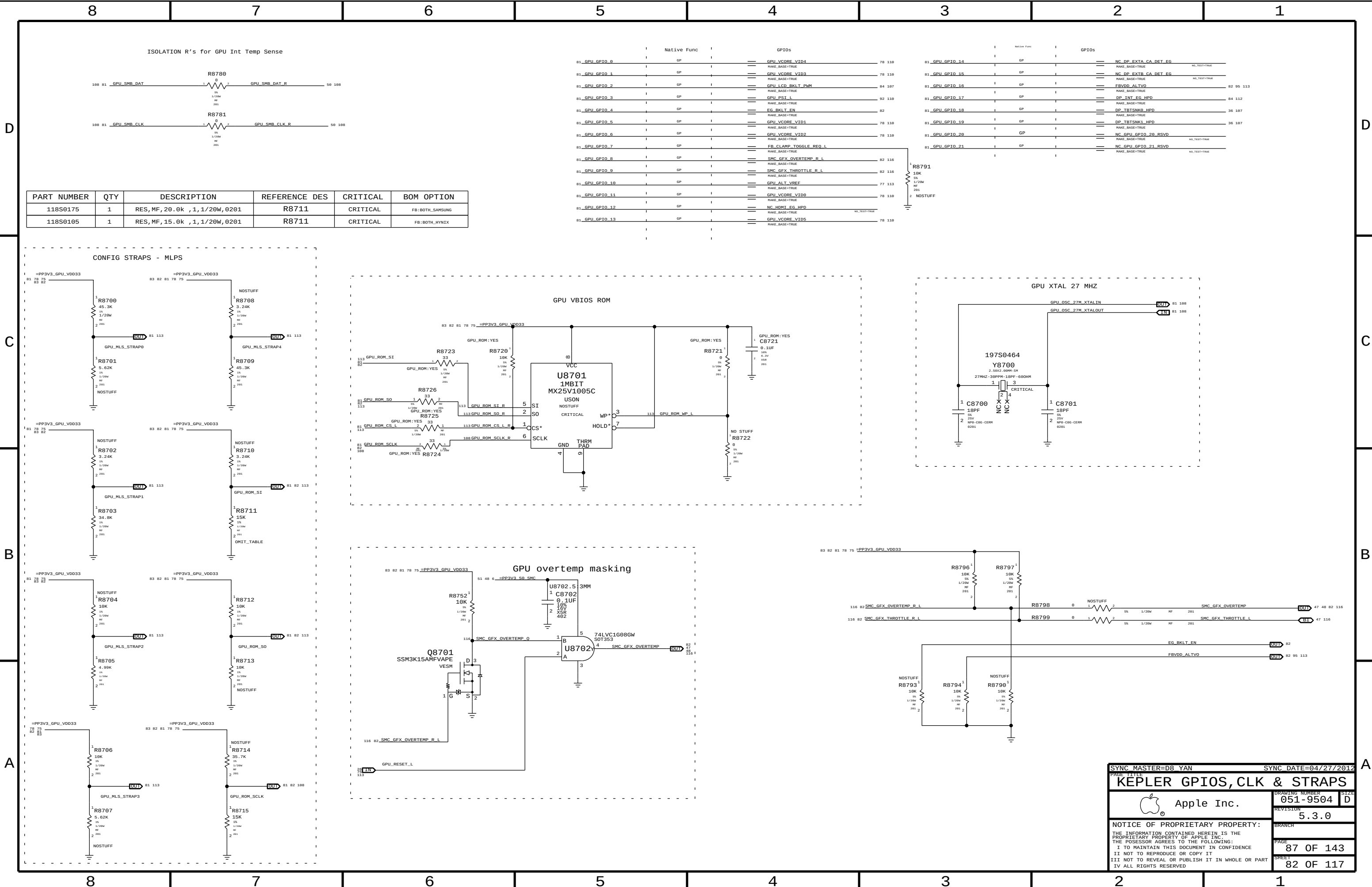
- PP3V3_GPU_SP_PLLVDD
- PP3V3_GPU_SP_PLLVDD
- PP3V3_GPU_SP_PLLVDD
- PP3V3_GPU_SP_PLLVDD
- PP3V3_GPU_SP_PLLVDD
- PP3V3_GPU_SP_PLLVDD

Signal planes required by this page:

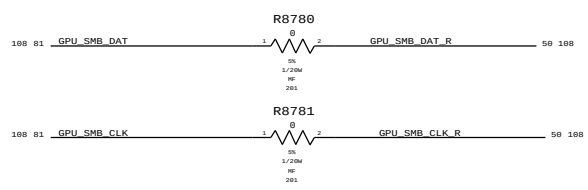
- PP3V3_GPU_SP_PLLVDD
- PP3V3_GPU_SP_PLLVDD

SW options provided by this page:

- PP3V3_GPU_SP_PLLVDD
- PP3V3_GPU_SP_PLLVDD



ISOLATION R's for GPU Int Temp Sense

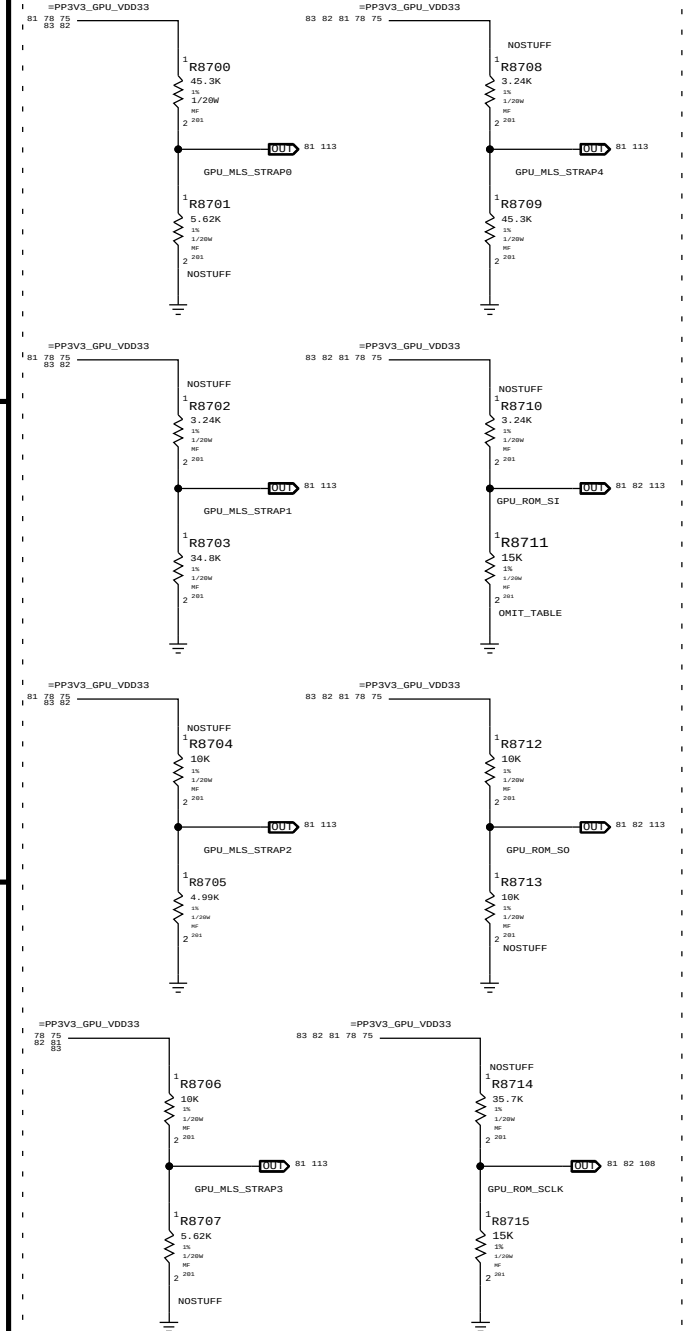


	Native Func		GPIOs	
81_GPU_GPIO_0	GP		GPU_VCORE_VID4	78 110
81_GPU_GPIO_1	GP		GPU_VCORE_VID3	78 110
81_GPU_GPIO_2	GP		GPU_LCD_BKLT_PWM	84 107
81_GPU_GPIO_3	GP		GPU_PSI_L	92 110
81_GPU_GPIO_4	GP		EG_BKLT_EN	82
81_GPU_GPIO_5	GP		GPU_VCORE_VID1	78 110
81_GPU_GPIO_6	GP		GPU_VCORE_VID2	78 110
81_GPU_GPIO_7	GP		FB_CLAMP_TOGGLE_REQ_L	82 116
81_GPU_GPIO_8	GP		SMC_GFX_OVERTEMP_R_L	82 116
81_GPU_GPIO_9	GP		SMC_GFX_THROTTLE_R_L	82 116
81_GPU_GPIO_10	GP		GPU_ALT_VREF	77 113
81_GPU_GPIO_11	GP		GPU_VCORE_VID0	78 110
81_GPU_GPIO_12	GP		NC_HDMI_EG_HPD	NO_TEST+TRUE
81_GPU_GPIO_13	GP		GPU_VCORE_VID5	78 110

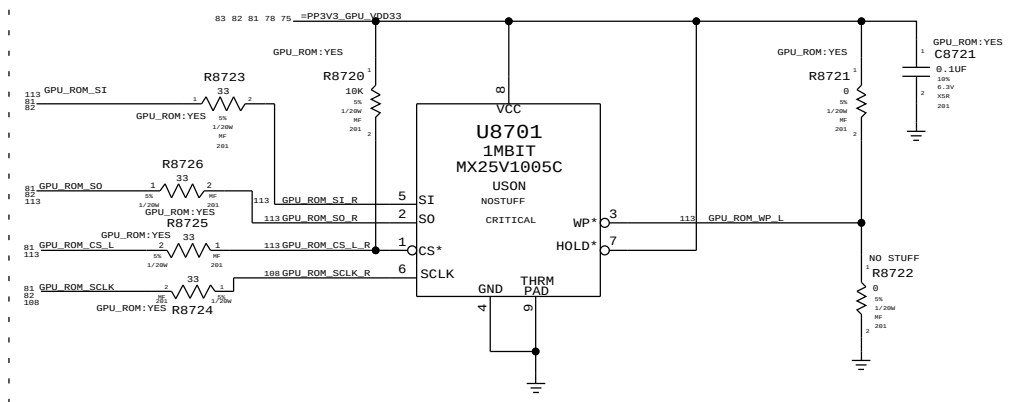
	Native Func		GPIOs	
81_GPU_GPIO_14	GP		NC_DP_EXTR_CA_DET_EG	NO_TEST+TRUE
81_GPU_GPIO_15	GP		NC_DP_EXTR_CA_DET_EG	NO_TEST+TRUE
81_GPU_GPIO_16	GP		FBVDD_ALTV0	82 95 113
81_GPU_GPIO_17	GP		DP_INT_EG_HPD	84 112
81_GPU_GPIO_18	GP		DP_TBTSNK0_HPD	36 107
81_GPU_GPIO_19	GP		DP_TBTSNK1_HPD	36 107
81_GPU_GPIO_20	GP		NC_GPU_GPIO_20_RSVD	NO_TEST+TRUE
81_GPU_GPIO_21	GP		NC_GPU_GPIO_21_RSVD	NO_TEST+TRUE

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0175	1	RES,MF,20.0k ,1,1/20W,0201	R8711	CRITICAL	FB:BOTH_SAMSUNG
118S0105	1	RES,MF,15.0k ,1,1/20W,0201	R8711	CRITICAL	FB:BOTH_HYWIX

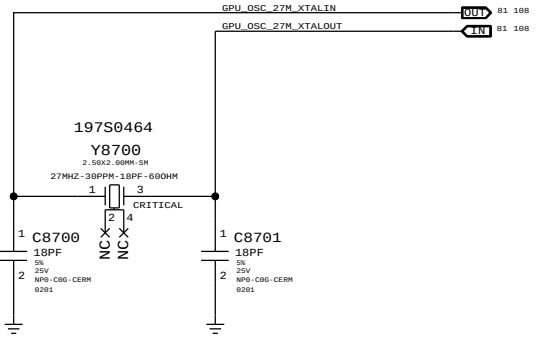
CONFIG STRAPS - MLPS



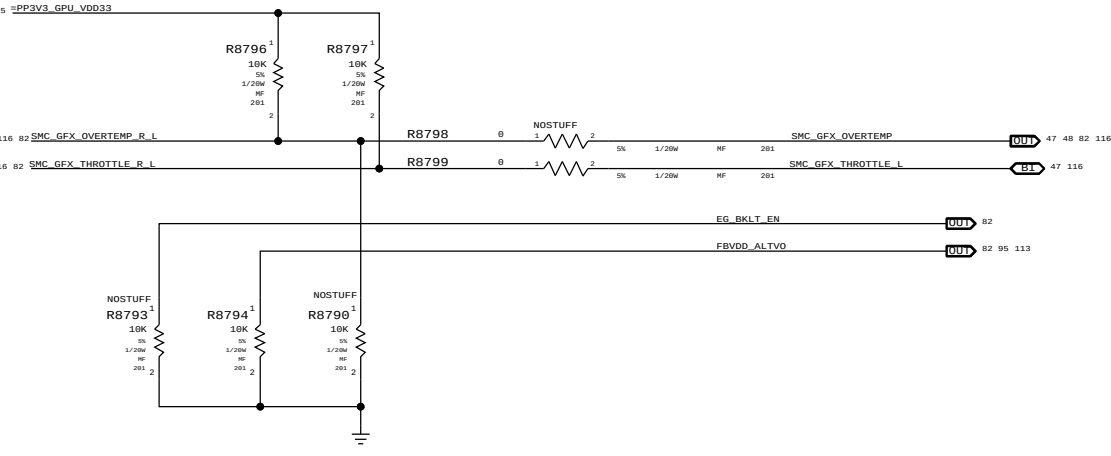
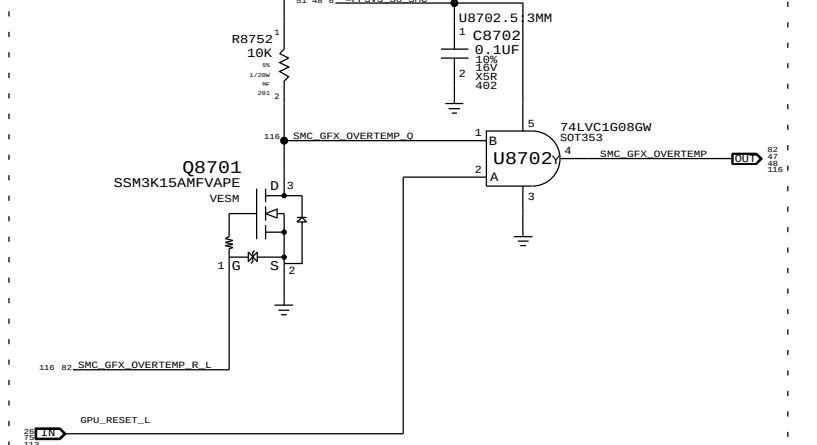
GPU VBIOS ROM



GPU XTAL 27 MHZ



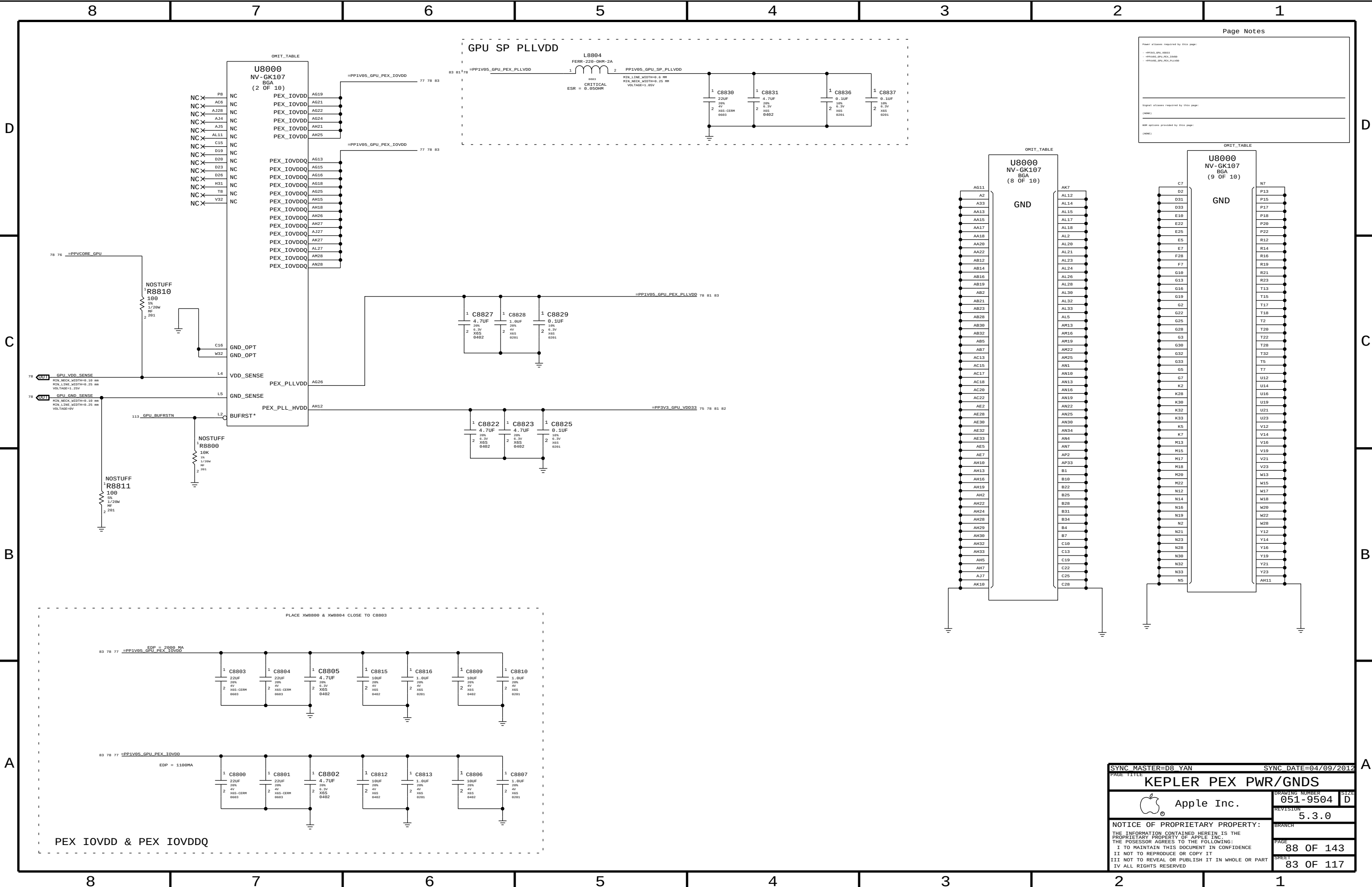
GPU overtemp masking



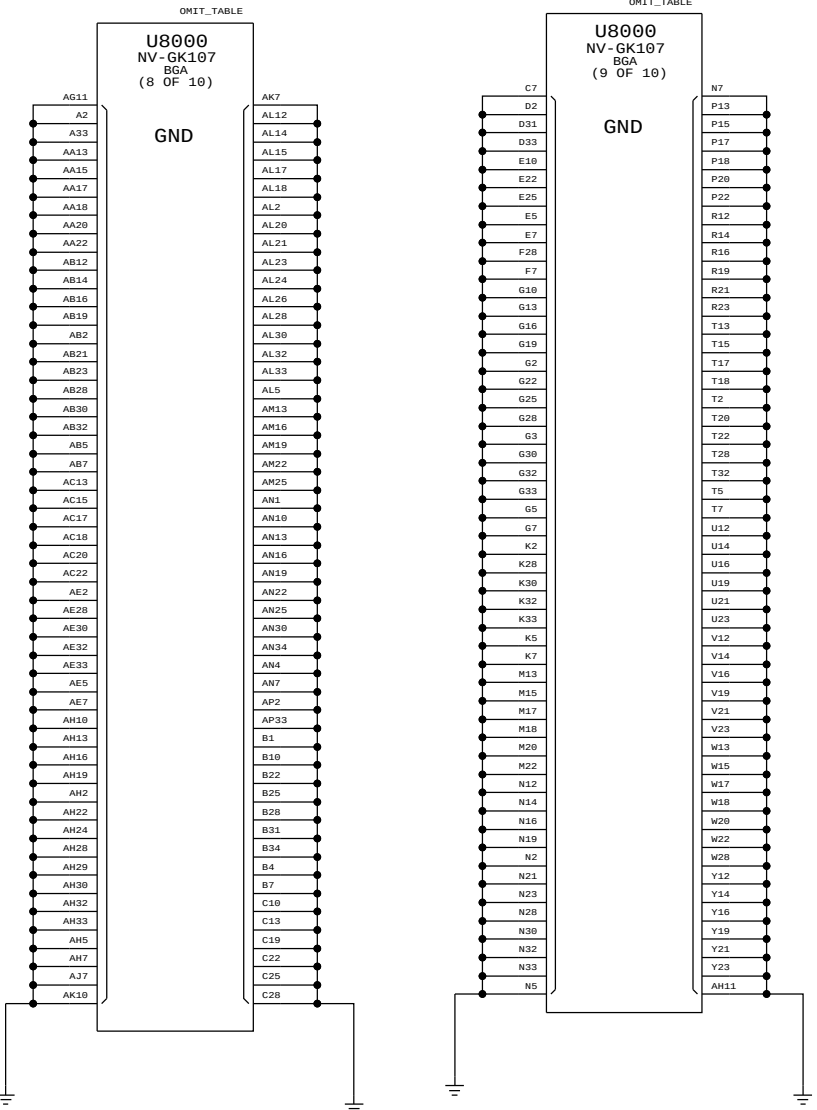
SYNC MASTER=D8_YAN

SYNC DATE=04/27/2012

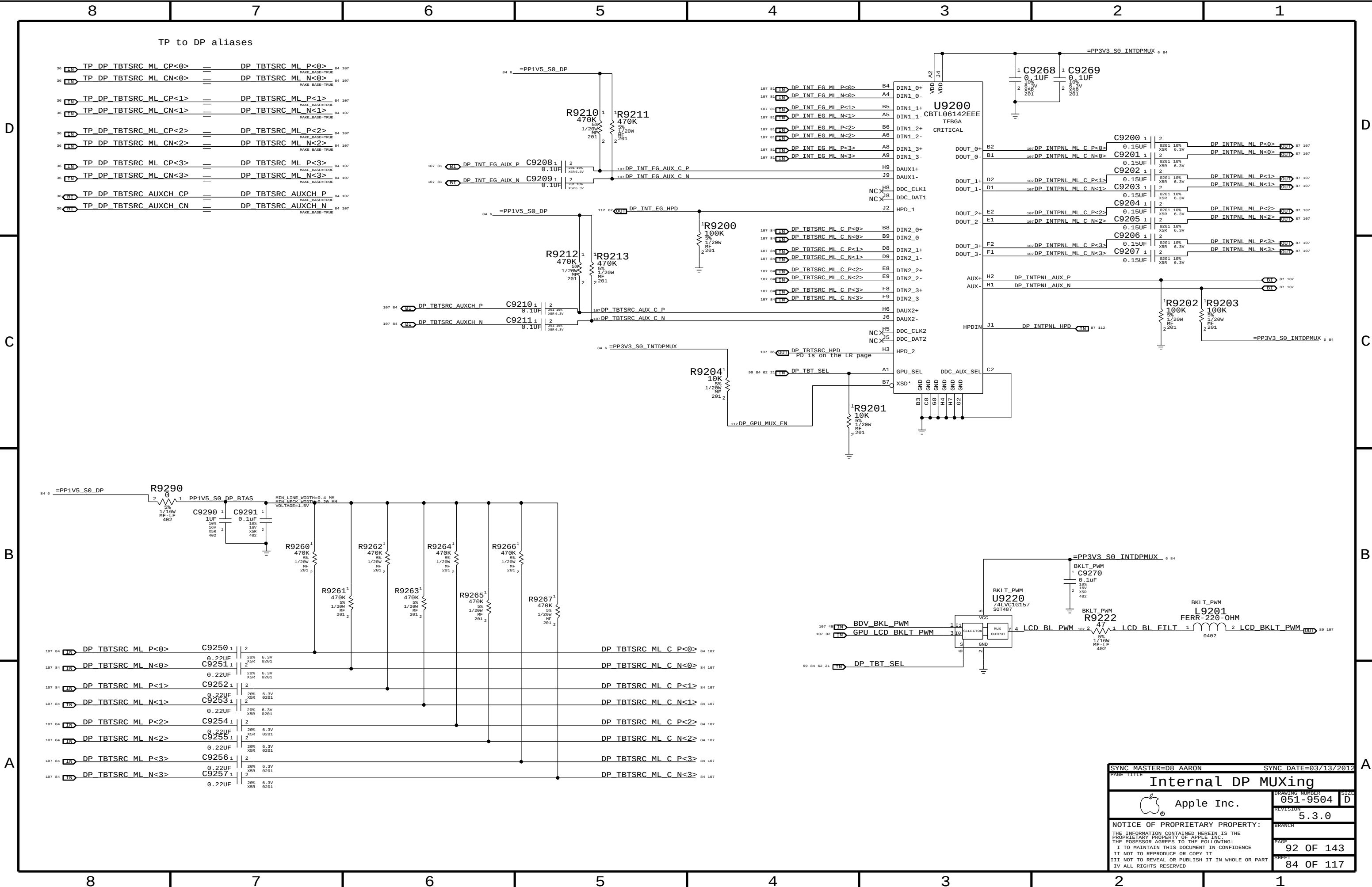
PAGE TITLE		KEPLER GPIOs, CLK & STRAPS	
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	87 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	82 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

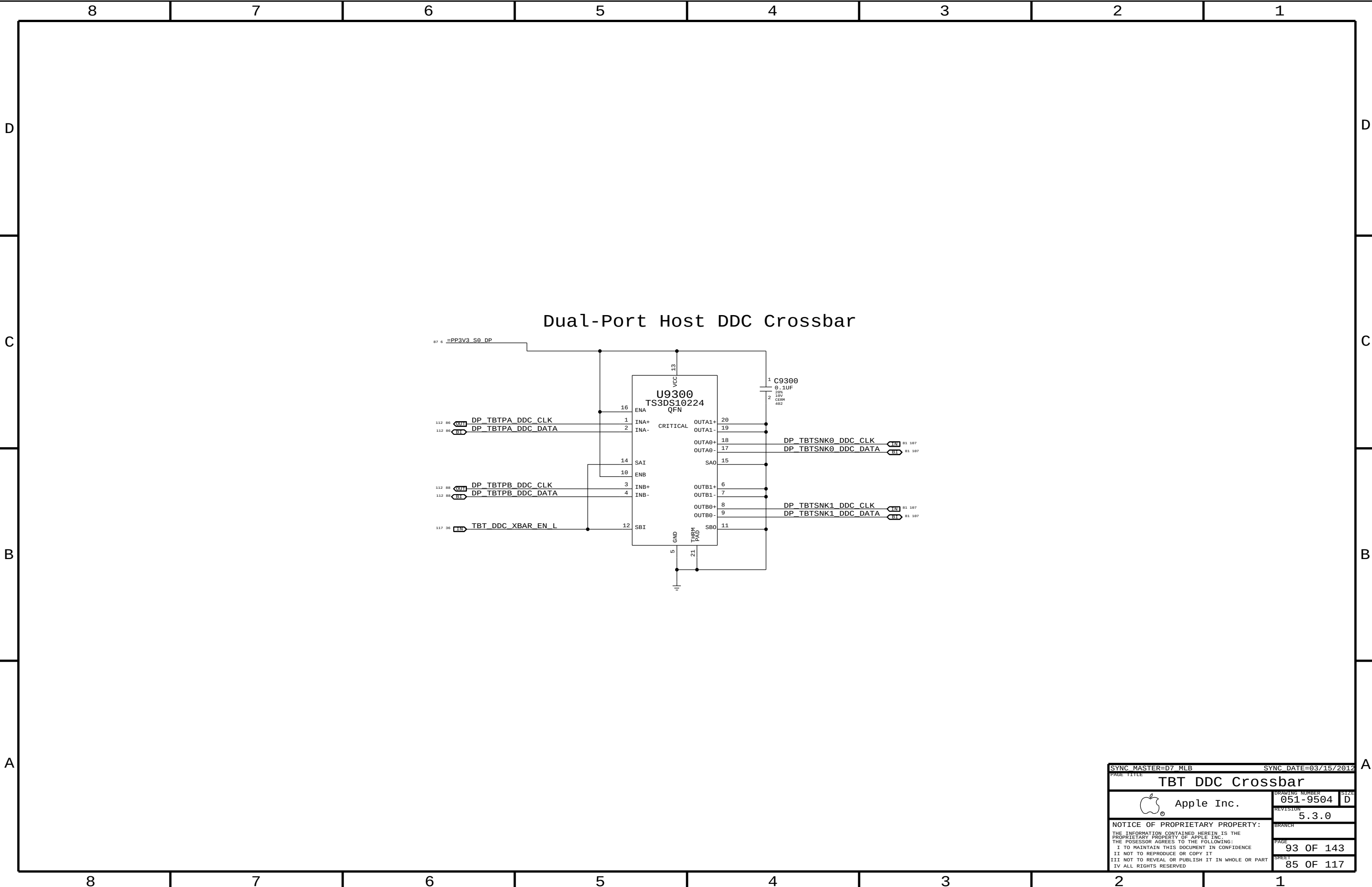


Page Notes	
Power aliases required by this page:	
- PP1V05_GPU_PEX_PLLVDD	
- PP1V05_GPU_PEX_IOVDD	
Signal aliases required by this page:	
(NONE)	
SOP options provided by this page:	
(NONE)	



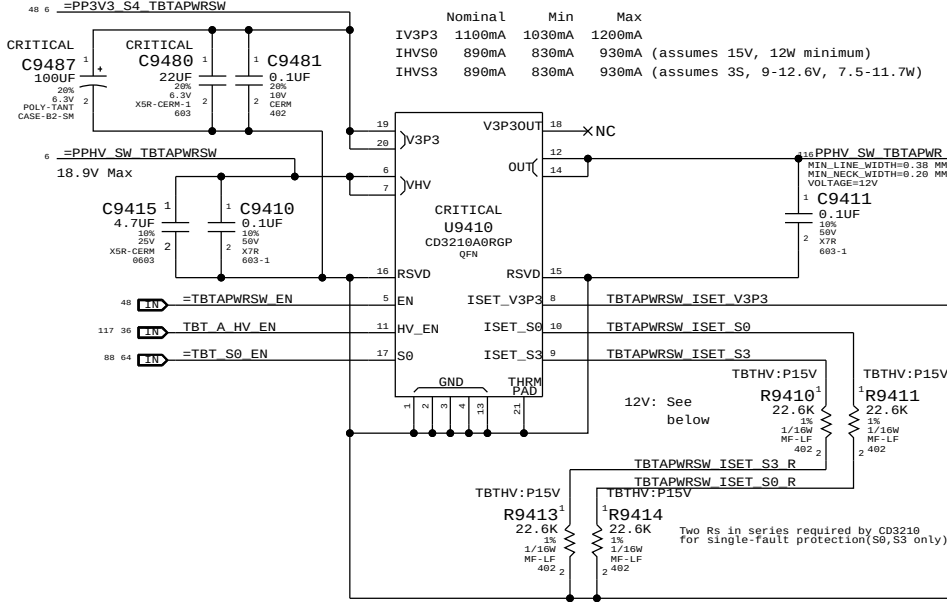
PAGE TITLE		SYNC DATE=04/09/2012	
KEPLER PEX PWR/GNDS		DRAWING NUMBER 051-9504	
Apple Inc.		REVISION 5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE 88 OF 143	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET 83 OF 117	
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			





3.3V/HV Power MUX

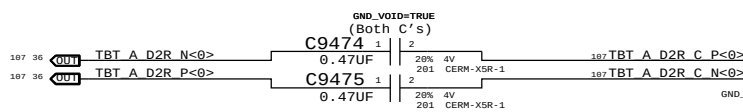
V3P3 must be S4 to support wake from Thunderbolt devices.



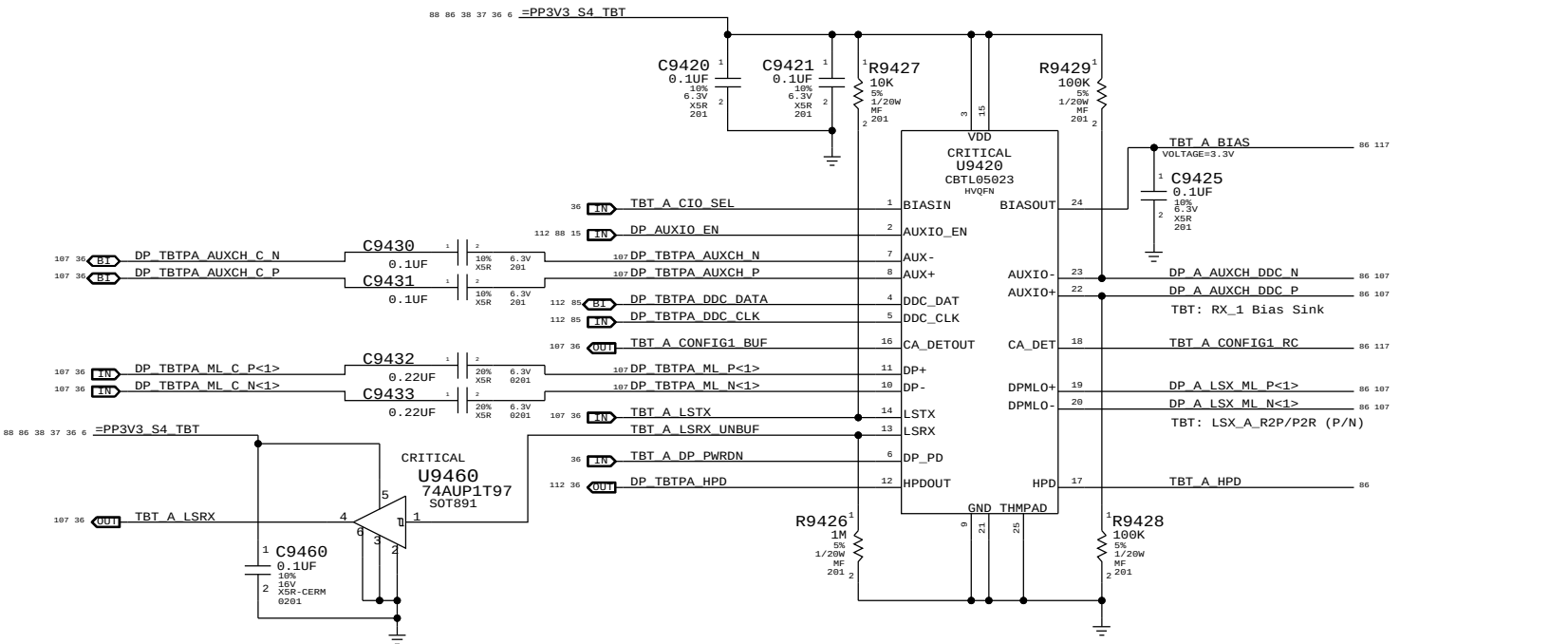
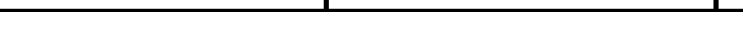
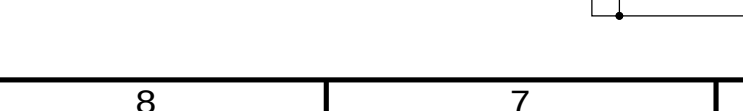
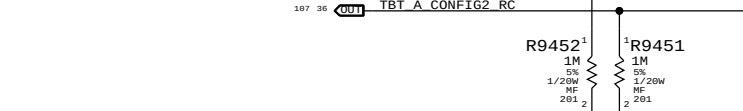
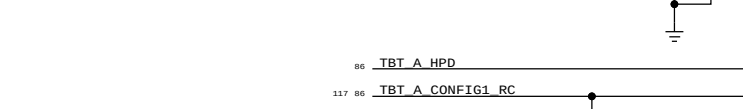
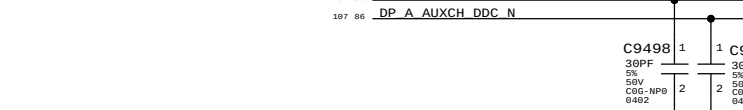
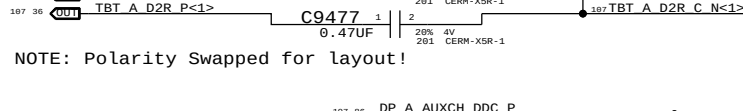
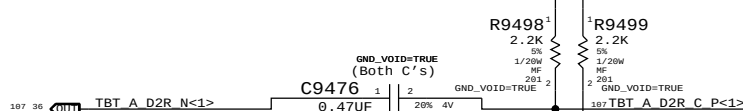
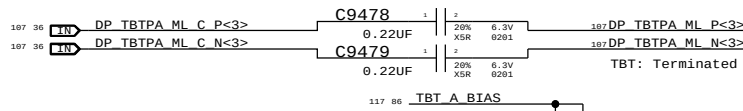
For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES, MTL FILM, 1/16W, 17.8K, 1, 0402, SMD, LF	R9410, R9413		TBTHV:P12V
114S0338	2	RES, MTL FILM, 1/16W, 17.8K, 1, 0402, SMD, LF	R9411, R9414		TBTHV:P12V

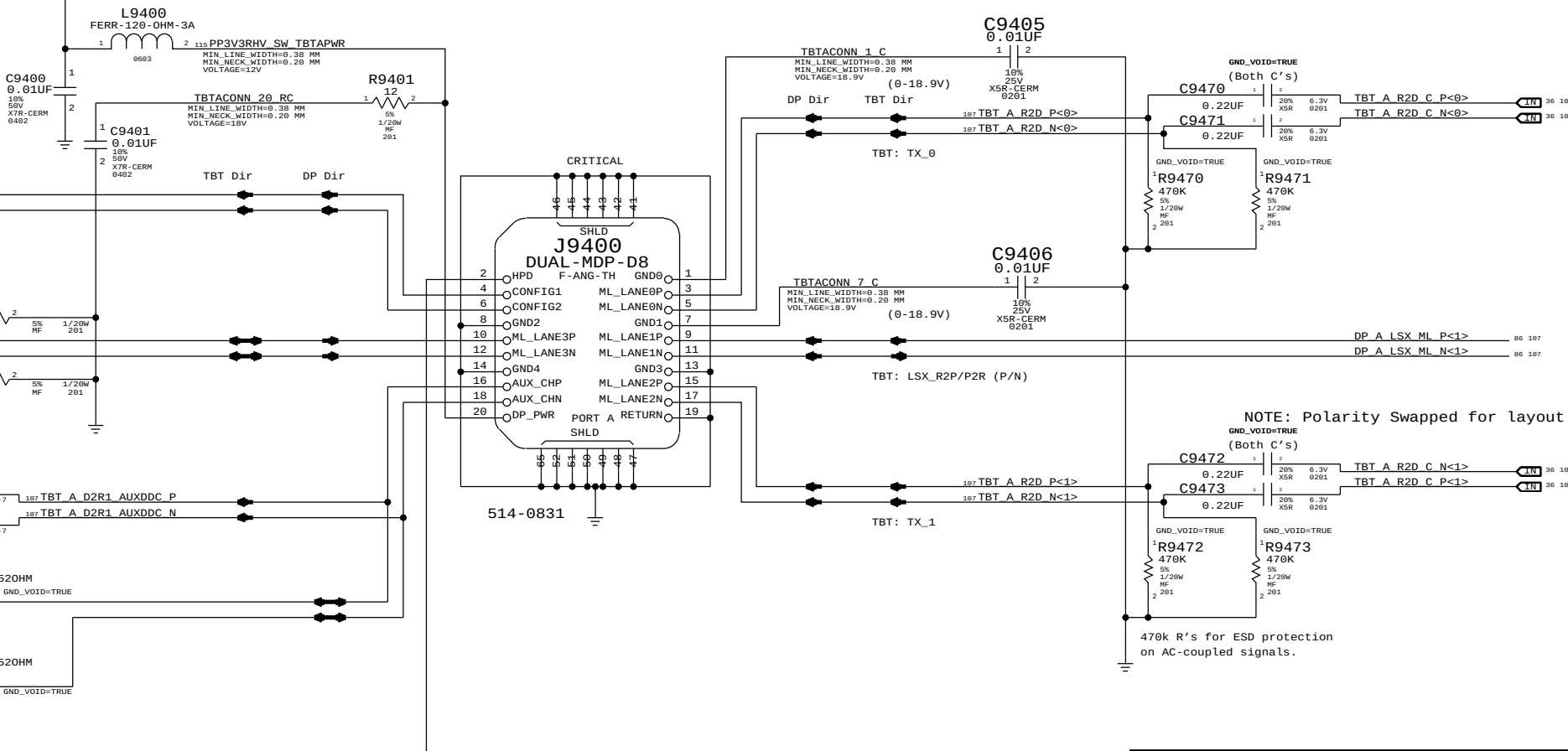
Nominal Min Max
IHVS0/S3 1120mA 1090mA 1170mA (12W minimum)



NOTE: Polarity Swapped for layout!



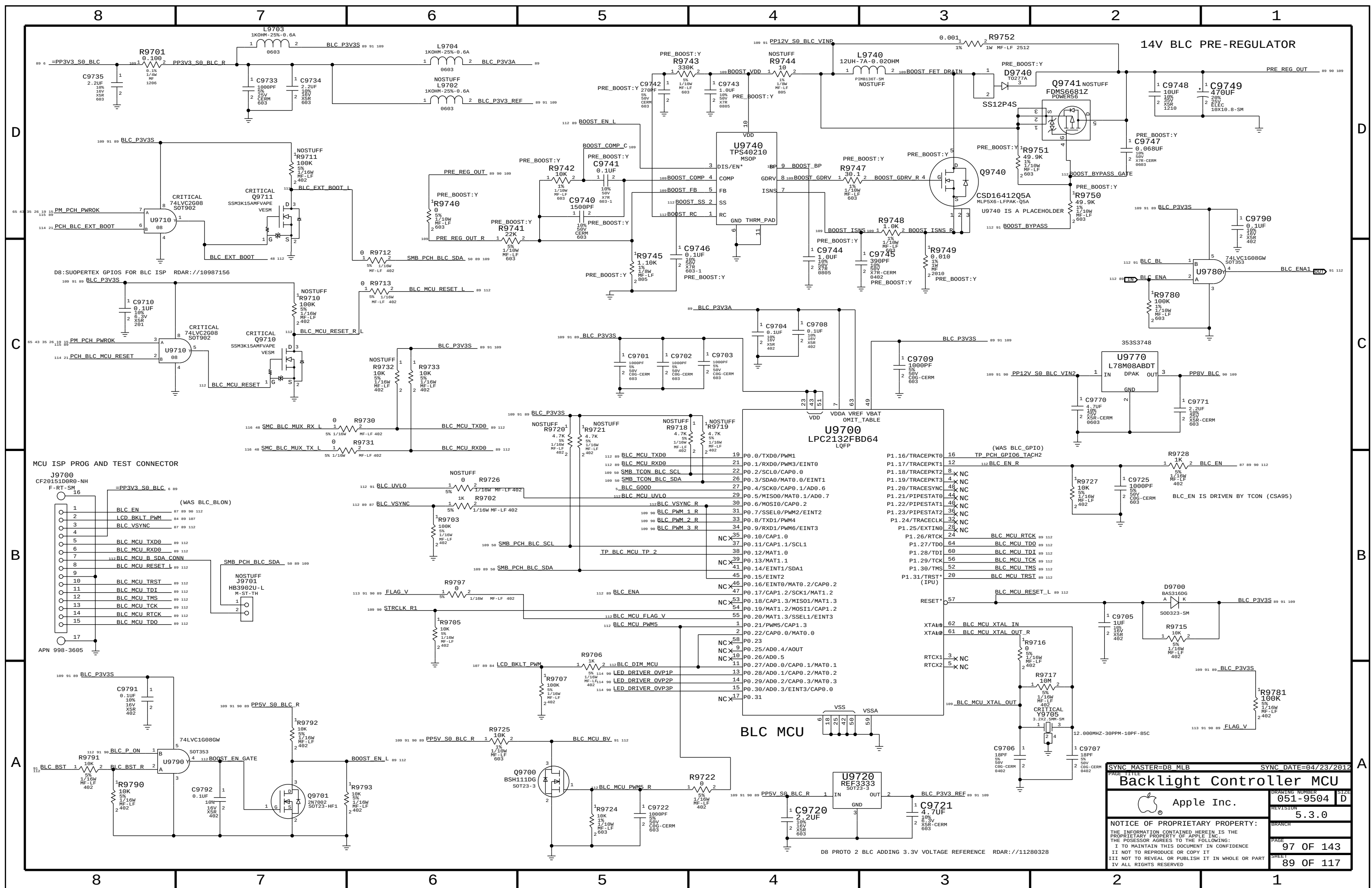
Thunderbolt Connector A

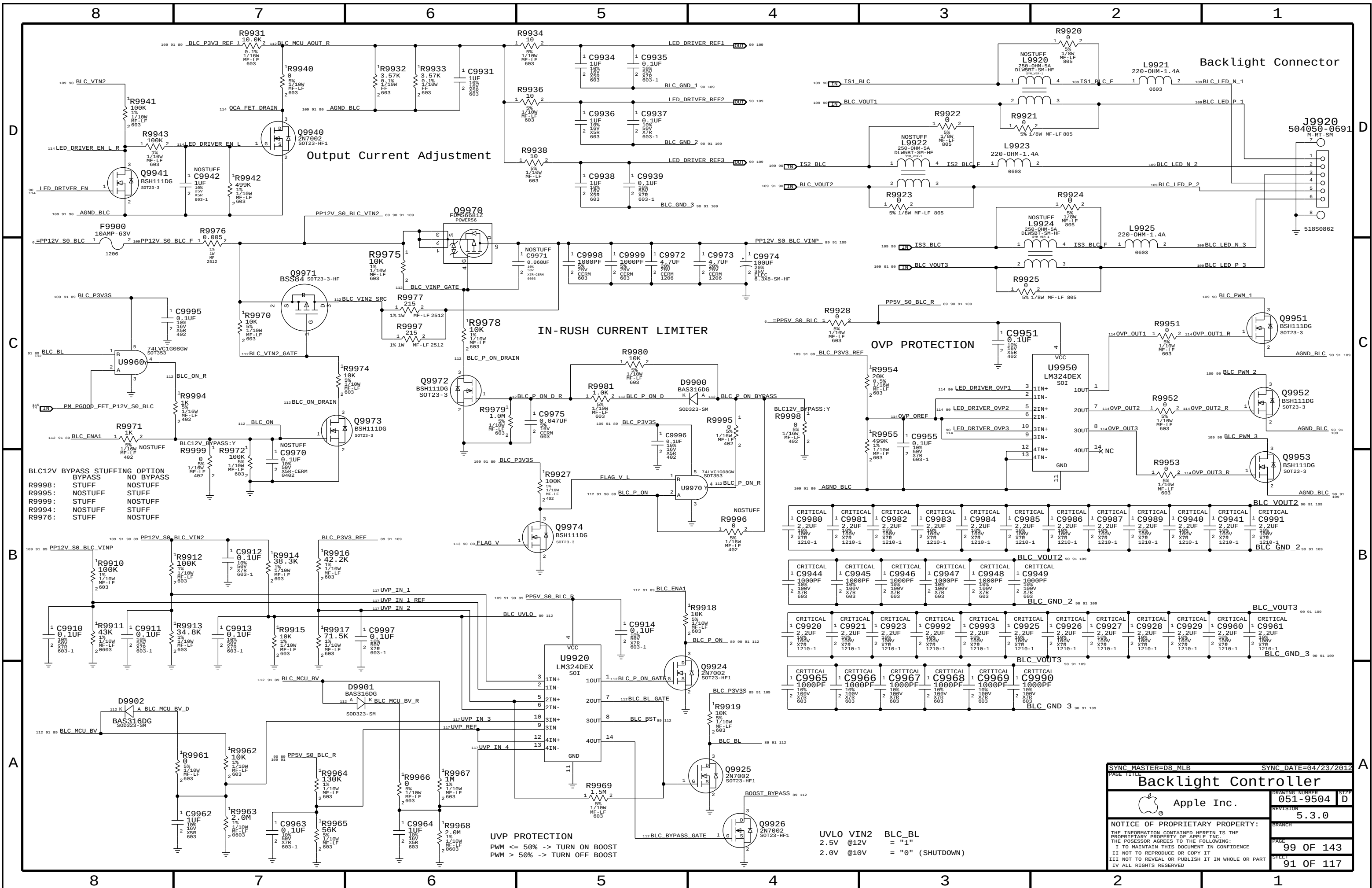


DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

SYNC MASTER=D8 AARON		SYNC DATE=03/13/2012	
PAGE TITLE		Thunderbolt Connector A	
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	94 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	86 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			





Output Current Adjustment

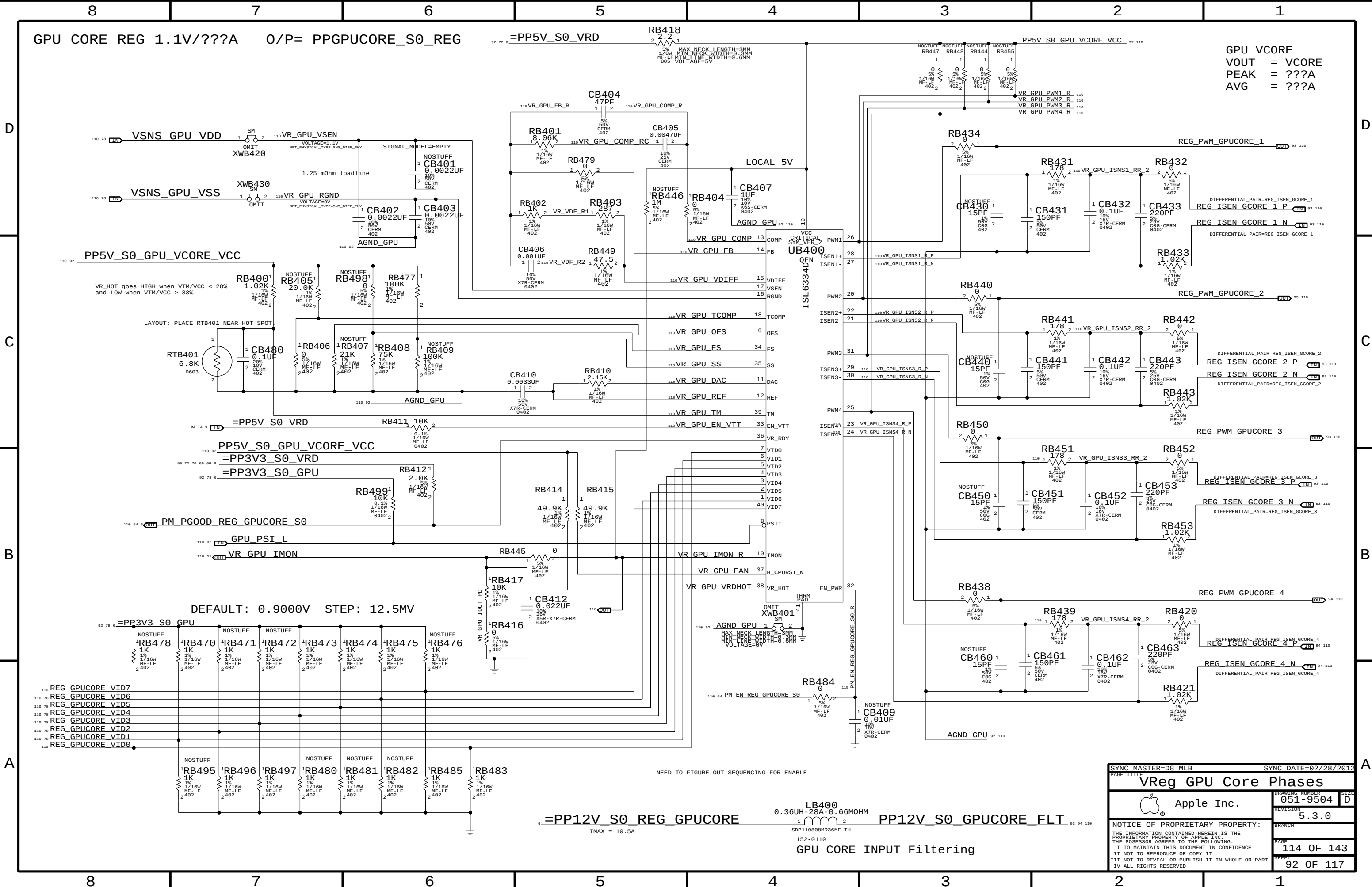
IN-RUSH CURRENT LIMITER

OVP PROTECTION

Backlight Connector

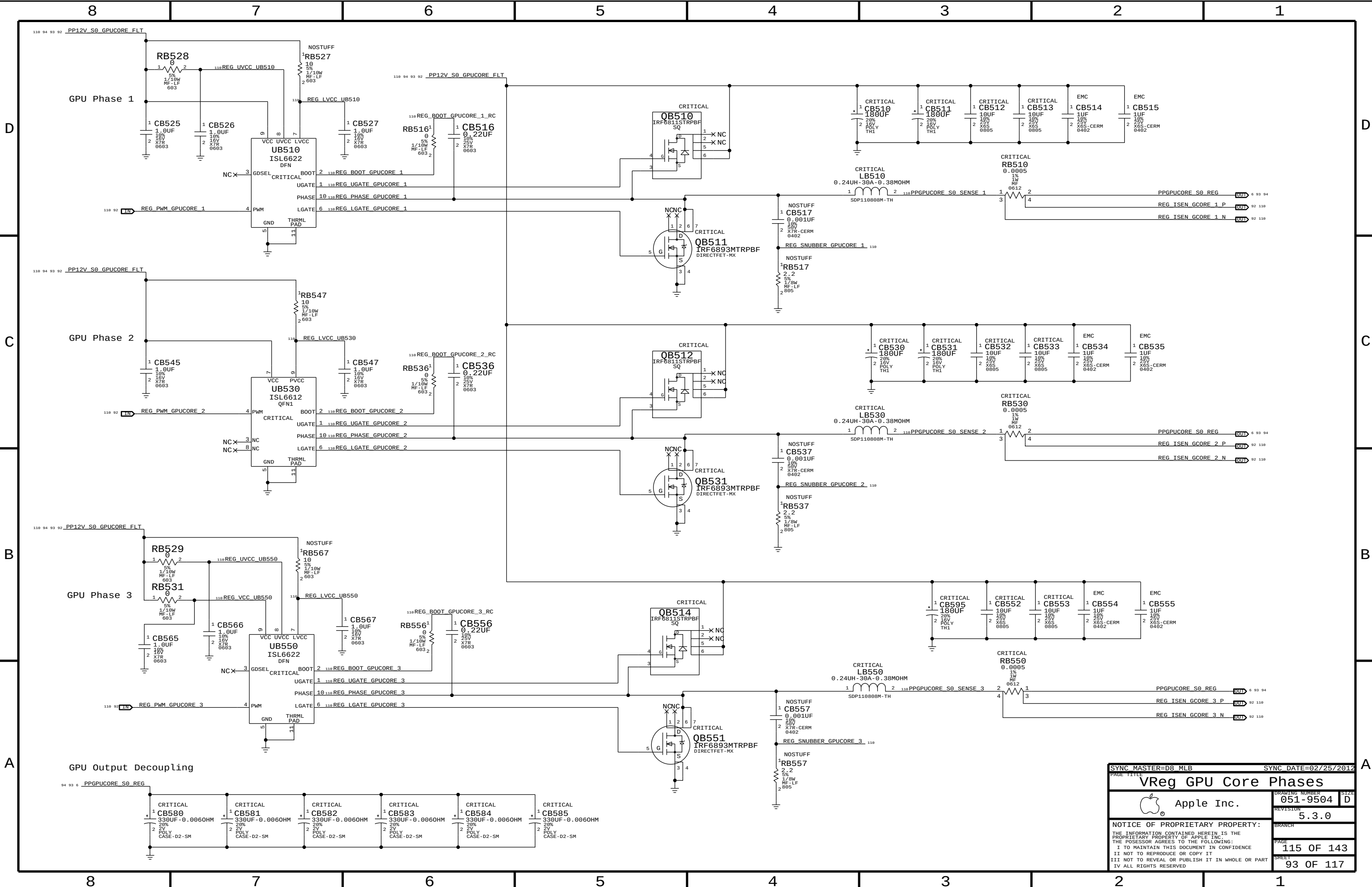
UVP PROTECTION
PWM <= 50% -> TURN ON BOOST
PWM > 50% -> TURN OFF BOOST

SYNC MASTER=D8 MLB		SYNC DATE=04/23/2012	
PAGE TITLE		Backlight Controller	
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	99 OF 143
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	91 OF 117
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

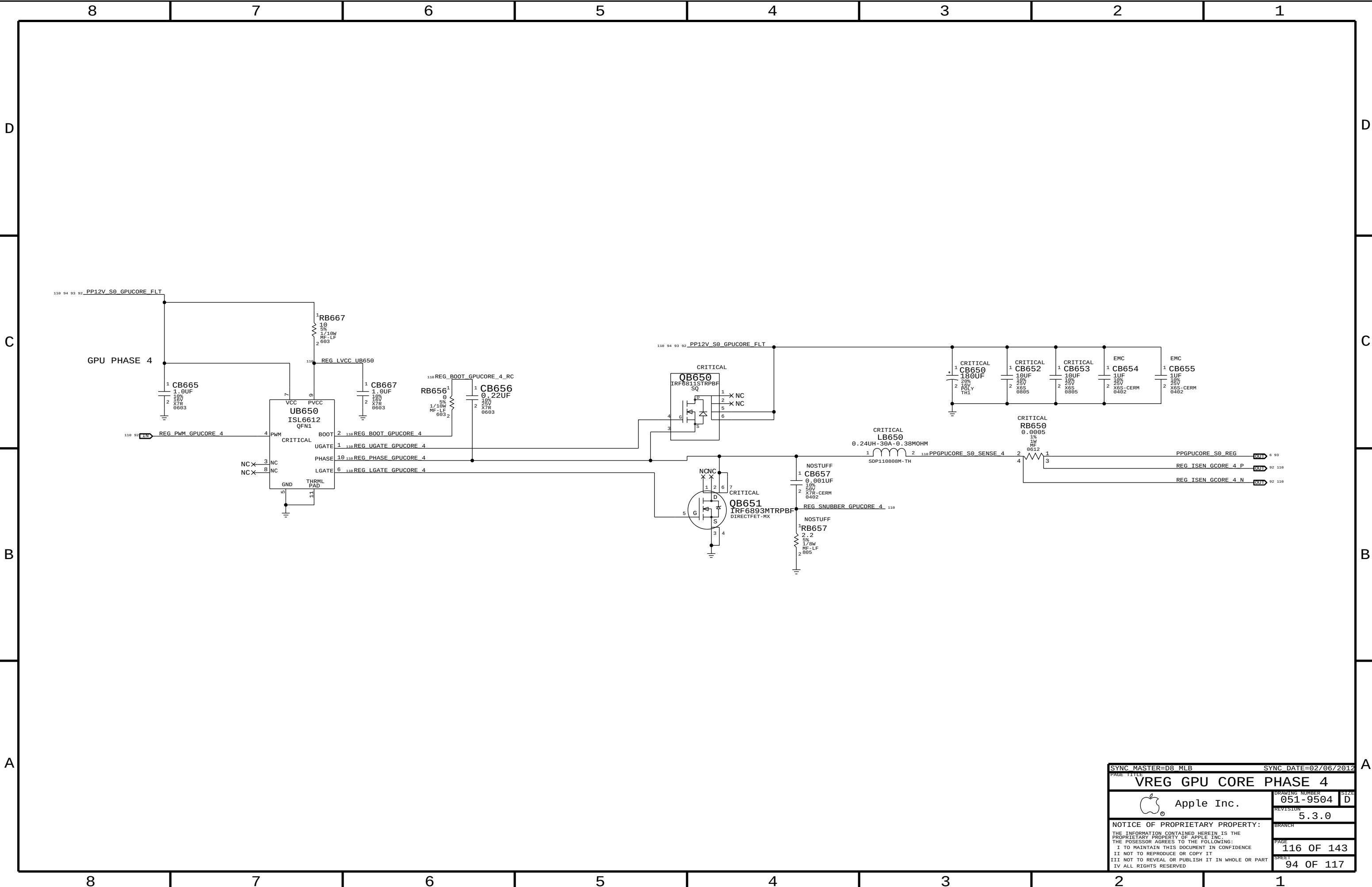


GPU VCORE
VOUT = VCORE
PEAK = ???A
AVG = ???A

PAGE TITLE		PAGE 1	
VReg GPU Core Phases		DRAWING NUMBER 051-9504	
Apple Inc.		REVISION 5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE 114 OF 143	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET 92 OF 117	
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



SYNC MASTER=D8_MLB		SYNC DATE=02/25/2012	
PAGE TITLE			
VReg GPU Core Phases		DRAWING NUMBER	051-9504
Apple Inc.		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	115 OF 143
		SHEET	93 OF 117



SYNC MASTER=D8_MLB

SYNC DATE=02/06/2012

VREG GPU CORE PHASE 4



Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

051-9504

SIZE

D

REVISION

5.3.0

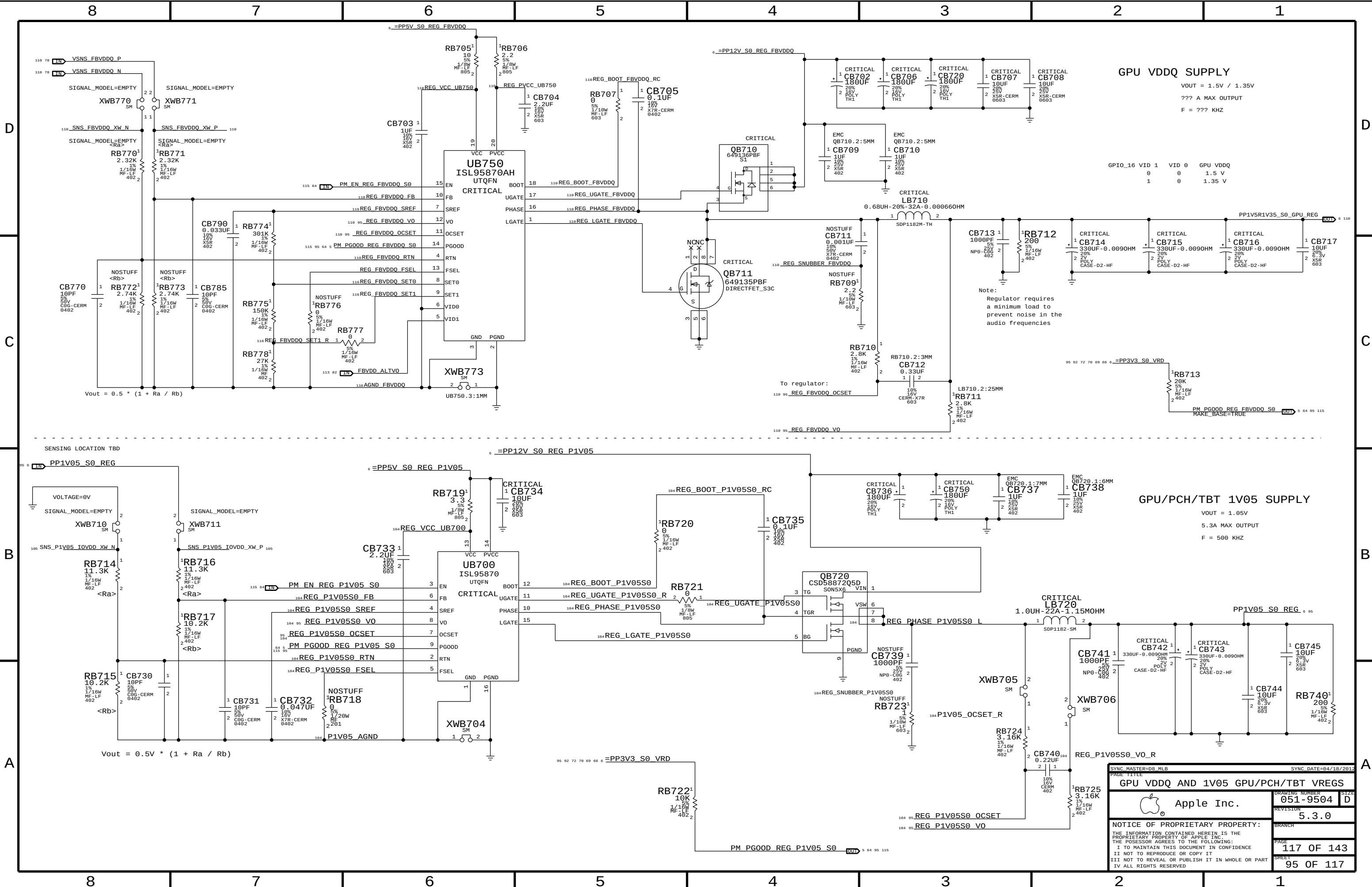
BRANCH

PAGE

116 OF 143

SHEET

94 OF 117



GPU VDDQ SUPPLY

VOUT = 1.5V / 1.35V
??? A MAX OUTPUT
F = ??? KHZ

GPIO_16 VID 1 VID 0 GPU VDDQ
0 0 1.5 V
1 0 1.35 V

GPU/PCH/TBT 1V05 SUPPLY

VOUT = 1.05V
5.3A MAX OUTPUT
F = 500 KHZ

GPU VDDQ AND 1V05 GPU/PCH/TBT VREGS	
Apple Inc.	
051-9504	SIZE D
5.3.0	
117 OF 143	
95 OF 117	

D8 BOARD SPECIFIC PHYSICAL AND SPACING CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM	NO_TYPE, BGA, BGA_TBT	MM	16.2

General Physical Rule Definitions

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.1 MM	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	*	Y	0.185 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	ISL5, ISL8	Y	0.205 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	TOP,BOTTOM	Y	0.220 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	*	Y	0.150 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	ISL5, ISL8	Y	0.165 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	TOP, BOTTOM	Y	0.175 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	*	Y	0.130 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	ISL5, ISL8	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	TOP, BOTTOM	Y	0.155 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	*	Y	0.115 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	ISL5, ISL8	Y	0.126 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	TOP, BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	0.090 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	ISL5, ISL8	Y	0.100 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP, BOTTOM	Y	0.105 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	*	Y	0.075 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	ISL5, ISL8	Y	0.080 MM	0.080 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	TOP, BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	Y	0.171 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.185 MM	0.085 MM	=STANDARD	0.150 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.136 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.141 MM	0.085 MM	=STANDARD	0.185 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.121 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.109 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.086 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.090 MM	0.085 MM	=STANDARD	0.230 MM	0.1 MM

Compensation Physical Rule Definition

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
COMP_SE	*	Y	0.305 MM	0.105 MM	3 MM	=STANDARD	=STANDARD

NOTE: line width based on 12 mil recommendation
NOTE: neck width based on 4 mil recommendation

General Spacing Definitions

Default

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

Fixed and Dielectric

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1X_DIELECTRIC	TOP, BOTTOM	0.071 MM	?
1X_DIELECTRIC	ISL3, ISL10	0.101 MM	?
1X_DIELECTRIC	*	0.076 MM	?

Board Stack-up

FINISHED BOARD THICKNESS: 1.94 MM

Layer	Material	Thickness
Top	Signal	0.5 oz (Cu plated)
	Prepreg	0.071 MM
2	Plane	1 oz
	Core	0.101 MM
3	Signal	0.5 oz
	Prepreg	0.115 MM
4	Plane	1 oz
	Core	0.076 MM
5	Signal	0.5 oz
	Prepreg	0.380 MM
6	Plane	1 oz
	Core	0.076 MM
7	Plane	1 oz
	Prepreg	0.380 MM
8	Signal	0.5 oz
	Core	0.076 MM
9	Plane	1 oz
	Prepreg	0.115 MM
10	Signal	0.5 oz
	Core	0.101 MM
11	Plane	1 oz
	Prepreg	0.071 MM
Btm	Signal	0.5 oz (Cu plated)

BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=STANDARD	?

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_ISO	*	=STANDARD	8000
GND_P2MM	*	=2:1_SPACING	1000
PWR_P2MM	*	=2:1_SPACING	1100

GENERIC

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GENERIC_ISO	*	=1:1_SPACING	?
PM_ISO	*	=1:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PM	*	*	PM_ISO
PM	GND	*	DEFAULT

BGA Area Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM

SYNCH MASTER=D8 KOSECOFF		SYNCH DATE=03/19/2012	
PAGE TITLE			
D8 RULE DEFINITIONS			
 Apple Inc.		DRAWING 051-9504	SIZE D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I. TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II. NOT TO REPRODUCE OR COPY IT III. NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV. ALL RIGHTS RESERVED		REVISION 5.3.0	
		BRANCH	
		PAGE 120 OF 143	
		SHEET 96 OF 117	

PCI EXPRESS

PCIe-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_80D
CLK_PCIE_PHY	*	PCIE_90D
COMP_PCIE_PHY	*	COMP_SE

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_ISO	TOP, BOTTOM	=5X_DIELECTRIC	?
PCIE_ISO	*	=4X_DIELECTRIC	?
CLK_PCIE_ISO	*	=5:1_SPACING	?
COMP_PCIE_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO
CLK_PCIE	*	*	CLK_PCIE_ISO
COMP_PCIE	*	*	COMP_PCIE_ISO

PCIe (PCH)

Electrical Constraint Set		Physical	Spacing	
x1 AirPort				
H599	PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE	PCIE AP R2D P NO TEST TRUE 35
H600	PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE	PCIE AP R2D N NO TEST TRUE 35
H601		PCIE_PHY	PCIE	PCIE AP R2D C P NO TEST TRUE 18 35
H602		PCIE_PHY	PCIE	PCIE AP R2D C N NO TEST TRUE 18 35
H603	PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE	PCIE AP D2R P NO TEST TRUE 18 35
H604	PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE	PCIE AP D2R N NO TEST TRUE 18 35
x1 Thunderbolt 3				
H605	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE CLK100M_AP P 18 35
H606	PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	PCIE CLK100M_AP N 18 35
x1 Caesar IV				
H607	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE ENET_R2D P NO TEST TRUE 39
H608	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE ENET_R2D N NO TEST TRUE 39
H609		PCIE_PHY	PCIE	PCIE ENET_R2D C P NO TEST TRUE 18 39
H610		PCIE_PHY	PCIE	PCIE ENET_R2D C N NO TEST TRUE 18 39
H611	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE ENET_D2R P NO TEST TRUE 18 39
H612	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE ENET_D2R N NO TEST TRUE 18 39
H613		PCIE_PHY	PCIE	PCIE ENET_D2R C P NO TEST TRUE 39
H614		PCIE_PHY	PCIE	PCIE ENET_D2R C N NO TEST TRUE 39
H615	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE CLK100M_ENET P 18 39
H616	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE CLK100M_ENET N 18 39

PCIE (PCH - TBT)

Electrical Constraint Set		Physical	Spacing	
x4 Thunderbolt				
[597]	PCIE_GEN2_R2D_P1NV	PCIE_PHY	PCIE	PCIE TBT R2D P<2..0> NO TEST TRUE 36
[597]	PCIE_GEN2_R2D_P1NV	PCIE_PHY	PCIE	PCIE TBT R2D N<2..0> NO TEST TRUE 36
[600]	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE TBT R2D P<3> NO TEST TRUE 36
[600]	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE TBT R2D N<3> NO TEST TRUE 36
[597]		PCIE_PHY	PCIE	PCIE TBT R2D C P<3..0> NO TEST TRUE 18 36
[597]		PCIE_PHY	PCIE	PCIE TBT R2D C N<3..0> NO TEST TRUE 18 36
[597]		PCIE_PHY	PCIE	PCIE TBT D2R P<3..0> NO TEST TRUE 18 36
[597]		PCIE_PHY	PCIE	PCIE TBT D2R N<3..0> NO TEST TRUE 18 36
[597]	PCIE_GEN2_D2R_P1NV	PCIE_PHY	PCIE	PCIE TBT D2R C P<1..0> NO TEST TRUE 36
[597]	PCIE_GEN2_D2R_P1NV	PCIE_PHY	PCIE	PCIE TBT D2R C N<1..0> NO TEST TRUE 36
[600]	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE TBT D2R C P<2> NO TEST TRUE 36
[600]	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE TBT D2R C N<2> NO TEST TRUE 36
[602]	PCIE_GEN2_D2R_P1NV	PCIE_PHY	PCIE	PCIE TBT D2R C P<3> NO TEST TRUE 36
[602]	PCIE_GEN2_D2R_P1NV	PCIE_PHY	PCIE	PCIE TBT D2R C N<3> NO TEST TRUE 36
[600]	PCIE_REF_CLK	CLK PCIE_PHY	CLK PCIE	PCIE CLK100M TBT P 18 36
[600]	PCIE_REF_CLK	CLK PCIE_PHY	CLK PCIE	PCIE CLK100M TBT N 18 36

PCIe (CPU)

Electrical Constraint Set	Physical	Spacing	
x16 Graphics			
E600 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D P<15> NO_TEST=TRUE 10 75
E601 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D N<15> NO_TEST=TRUE 10 75
E602	PCIE_PHY	PCIE	PEG R2D C P<15> NO_TEST=TRUE 75
E603	PCIE_PHY	PCIE	PEG R2D C N<15> NO_TEST=TRUE 75
E604 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R P<15> NO_TEST=TRUE 75
E605 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R N<15> NO_TEST=TRUE 75
E606	PCIE_PHY	PCIE	PEG D2R C P<15> NO_TEST=TRUE 10 75
E607	PCIE_PHY	PCIE	PEG D2R C N<15> NO_TEST=TRUE 10 75
E608 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D P<14> NO_TEST=TRUE 10 75
E609 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D N<14> NO_TEST=TRUE 10 75
E610	PCIE_PHY	PCIE	PEG R2D C P<14> NO_TEST=TRUE 75
E611	PCIE_PHY	PCIE	PEG R2D C N<14> NO_TEST=TRUE 75
E612 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R P<14> NO_TEST=TRUE 75
E613 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R N<14> NO_TEST=TRUE 75
E614	PCIE_PHY	PCIE	PEG D2R C P<14> NO_TEST=TRUE 10 75
E615	PCIE_PHY	PCIE	PEG D2R C N<14> NO_TEST=TRUE 10 75
E616 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D P<13> NO_TEST=TRUE 10 75
E617 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D N<13> NO_TEST=TRUE 10 75
E618	PCIE_PHY	PCIE	PEG R2D C P<13> NO_TEST=TRUE 75
E619	PCIE_PHY	PCIE	PEG R2D C N<13> NO_TEST=TRUE 75
E620 PCIE_GEN3_D2R_P1NV	PCIE_PHY	PCIE	PEG D2R P<13> NO_TEST=TRUE 75
E621 PCIE_GEN3_D2R_P1NV	PCIE_PHY	PCIE	PEG D2R N<13> NO_TEST=TRUE 75
E622	PCIE_PHY	PCIE	PEG D2R C P<13> NO_TEST=TRUE 10 75
E623	PCIE_PHY	PCIE	PEG D2R C N<13> NO_TEST=TRUE 10 75
E624 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D P<12> NO_TEST=TRUE 10 75
E625 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D N<12> NO_TEST=TRUE 10 75
E626	PCIE_PHY	PCIE	PEG R2D C P<12> NO_TEST=TRUE 75
E627	PCIE_PHY	PCIE	PEG R2D C N<12> NO_TEST=TRUE 75
E628 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R P<12> NO_TEST=TRUE 75
E629 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R N<12> NO_TEST=TRUE 75
E630	PCIE_PHY	PCIE	PEG D2R C P<12> NO_TEST=TRUE 10 75
E631	PCIE_PHY	PCIE	PEG D2R C N<12> NO_TEST=TRUE 10 75
E632 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D P<11> NO_TEST=TRUE 10 75
E633 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D N<11> NO_TEST=TRUE 10 75
E634	PCIE_PHY	PCIE	PEG R2D C P<11> NO_TEST=TRUE 75
E635	PCIE_PHY	PCIE	PEG R2D C N<11> NO_TEST=TRUE 75
E636 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R P<11> NO_TEST=TRUE 75
E637 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R N<11> NO_TEST=TRUE 75
E638	PCIE_PHY	PCIE	PEG D2R C P<11> NO_TEST=TRUE 10 75
E639	PCIE_PHY	PCIE	PEG D2R C N<11> NO_TEST=TRUE 10 75
E640 PCIE_GEN3_R2D_P1NV	PCIE_PHY	PCIE	PEG R2D P<10> NO_TEST=TRUE 10 75
E641 PCIE_GEN3_R2D_P1NV	PCIE_PHY	PCIE	PEG R2D N<10> NO_TEST=TRUE 10 75
E642	PCIE_PHY	PCIE	PEG R2D C P<10> NO_TEST=TRUE 75
E643	PCIE_PHY	PCIE	PEG R2D C N<10> NO_TEST=TRUE 75
E644 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R P<10> NO_TEST=TRUE 75
E645 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R N<10> NO_TEST=TRUE 75
E646	PCIE_PHY	PCIE	PEG D2R C P<10> NO_TEST=TRUE 10 75
E647	PCIE_PHY	PCIE	PEG D2R C N<10> NO_TEST=TRUE 10 75
E648 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D P<9> NO_TEST=TRUE 10 75
E649 PCIE_GEN3_R2D	PCIE_PHY	PCIE	PEG R2D N<9> NO_TEST=TRUE 10 75
E650	PCIE_PHY	PCIE	PEG R2D C P<9> NO_TEST=TRUE 75
E651	PCIE_PHY	PCIE	PEG R2D C N<9> NO_TEST=TRUE 75
E652 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R P<9> NO_TEST=TRUE 75
E653 PCIE_GEN3_D2R	PCIE_PHY	PCIE	PEG D2R N<9> NO_TEST=TRUE 75
E654	PCIE_PHY	PCIE	PEG D2R C P<9> NO_TEST=TRUE 10 75
E655	PCIE_PHY	PCIE	PEG D2R C N<9> NO_TEST=TRUE 10 75
E656 PCIE_GEN3_R2D_P1NV	PCIE_PHY	PCIE	PEG R2D P<8> NO_TEST=TRUE 10 75
E657 PCIE_GEN3_R2D_P1NV	PCIE_PHY	PCIE	PEG R2D N<8> NO_TEST=TRUE 10 75
E658	PCIE_PHY	PCIE	PEG R2D C P<8> NO_TEST=TRUE 75
E659	PCIE_PHY	PCIE	PEG R2D C N<8> NO_TEST=TRUE 75
E660 PCIE_GEN3_D2R_P1NV	PCIE_PHY	PCIE	PEG D2R P<8> NO_TEST=TRUE 75
E661 PCIE_GEN3_D2R_P1NV	PCIE_PHY	PCIE	PEG D2R N<8> NO_TEST=TRUE 75
E662	PCIE_PHY	PCIE	PEG D2R C P<8> NO_TEST=TRUE 10 75
E663	PCIE_PHY	PCIE	PEG D2R C N<8> NO_TEST=TRUE 10 75

PCIe (CPU)

[illegible]

SYNC MASTER=D8 AARON		SYNC DATE=03/13/2012	
PAGE TITLE			
CPU PCIe Constraints			
 Apple Inc.	DRAWING NUMBER	SIZE	
	051-9504	D	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	REVISION	5.3.0	
	BRANCH		
	PAGE	122 OF 143	
	SHEET	98 OF 117	

DMI

DMI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DMI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
DMI_PHY	*	DMI_85D

CPU Misc / FDI

CPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALL LOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
COMP_FDI_SE	*	Y	0.3 MM	0.25 MM	3 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
CPU_PHY	*	CPU_50S
COMP_FDI_PHY	*	COMP_FDI_SE

CPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_ISO	*	=3:1_SPACING	?
COMP_FDI_ISO	*	=4:1_SPACING	?

FDI Compensation Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Table	Trace	Design	Iso	Design	Comments
6-4	10	11.81	-	15.75	Using PCIe guidelines

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU	*	*	CPU_ISO
COMP_FDI	*	*	COMP_FDI_ISO

XDP

XDP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
XDP_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
XDP_PHY	*	XDP_50S

XDP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XDP_ISO	*	=2:1_SPACING	?
XDP_CLK_ISO	*	=4:1_SPACING	?

Desktop Debug Design Guide (Intel Doc# 430883)

Section	Imp	Design	Iso	Design	Comments
1.5	45-65	50	-	15.75	Isolation is for JTAG clocks. All signals default are 50 Ohm SE.

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XDP	*	*	XDP_ISO
CLK_JTAG	*	*	XDP_CLK_ISO
OBS_STR0BE	*	*	XDP_CLK_ISO

DMI

Electrical Constraint Set	Physical	Spacing	
DMI			
⚡ DMI N2S	DMI PHY	PCIE	DMI N2S P<3...0> NO TEST=TRUE 10 19
⚡ DMI N2S	DMI PHY	PCIE	DMI N2S N<3...0> NO TEST=TRUE 10 19
⚡ DMI S2N	DMI PHY	PCIE	DMI S2N P<3...0> NO TEST=TRUE 10 19
⚡ DMI S2N	DMI PHY	PCIE	DMI S2N N<3...0> NO TEST=TRUE 10 19
⚡ PCIE REF CLK	CLK PCIE PHY	CLK PCIE	DMI CLK100M CPU P 11 18
⚡ PCIE REF CLK	CLK PCIE PHY	CLK PCIE	DMI CLK100M CPU N 11 18
DMI Compensation			
⚡	COMP PCIE PHY	COMP PCIE	PCH_DMI_COMP 19
⚡	COMP PCIE PHY	COMP PCIE	PCH_DMI2RBIAS 19

CPU Misc.

Electrical Constraint Set	Physical	Spacing	
CPU Misc.			
PR20	CPU_PHY	CPU	CPU SKTOCC_L 11 64
PR25	CPU_PHY	CPU	CPU PROC SEL 11 19
PR27	CPU_PHY	CPU	CPU CATERR_L 11 48
PR35	CPU_PHY	CPU	CPU PECTI 11 21 47 48
PR36	CPU_PHY	CPU	CPU PROCHOT_L 11 47 48 66
PR39	CPU_PHY	CPU	CPU PROCHOT_R_L 11
PR40	CPU_PHY	CPU	CPU THRMTRIP_L 11 48
PR41	CPU_PHY	CPU	CPU RESET_L 11 26
PR47	CPU_PHY	CPU	PLT RESET LS1V05_L 11
PR50	CPU_PHY	CPU	PM SYNC 11 19
PR55	CPU_PHY	CPU	CPU PWRGD 11 21 25 28
PR56	CPU_PHY	CPU	PM MEM PWRGD 11 19 28
PR58	CPU_PHY	CPU	PM MEM PWRGD_R 11
PR59	CPU_PHY	CPU	CPU MEM RESET_L 11 28

FDI

Electrical Constraint Set	Physical	Spacing	
FDI Compensation			
12200	COMP_FDI_PHY	COMP_FDI	CPU_FDI_COMPIO

XDP

Electrical Constraint Set	Physical	Spacing	
XDP MISC.			
FE00 XDP_BPM_L	XDP_PHY	XDP	XDP BPM L<7..0> 11 25
FE01	XDP_PHY	XDP	CPU_CFG<17..16> 10 25
FE02	XDP_PHY	XDP	CPU_CFG<11..0> 10 15 25
FE03	XDP_PHY	XDP	XDP_DBRESET_L 11 25
FE04	XDP_PHY	XDP	XDP_CPU_PWRGD 25
FE05	XDP_PHY	XDP	XDP_CPU_PWRRTN_L 25
FE06	XDP_PHY	XDP	XDP_CPU_CFG<0> 25
FE07	XDP_PHY	XDP	XDP_VR_READY 25
FE08	XDP_PHY	XDP	XDP_CPU_PLTRST_L 25 26
FE09	XDP_PHY	XDP	XDP_PCH_PWRGD 25
FE0A	XDP_PHY	XDP	XDP_PCH_PWRBTN_L 25
FE0B	XDP_PHY	XDP	XDP_PCH_PLTRST_L 25 26
FE0C			
FE0D			
FE0E			
FE0F			
FE10	CLK_PCTE_PHY	CLK_PCTE	ITPCPU_CLK100M_P 11 15
FE11	CLK_PCTE_PHY	CLK_PCTE	ITPCPU_CLK100M_N 11 15
FE12 TTP_CLK_CONN	CLK_PCTE_PHY	CLK_PCTE	ITPXPDP_CLK100M_P 10 15 25
FE13 TTP_CLK_CONN	CLK_PCTE_PHY	CLK_PCTE	ITPXPDP_CLK100M_N 10 15 25
FE14	CLK_PCTE_PHY	CLK_PCTE	XDP_CPU_CLK100M_P 25
FE15	CLK_PCTE_PHY	CLK_PCTE	XDP_CPU_CLK100M_N 25
FE16			
FE17			
CPU JTAG			
FE18	XDP_PHY	CLK_JTAG	XDP_CPU_TCK 11 25
FE19	XDP_PHY	XDP	XDP_CPU_TMS 11 25
FE1A	XDP_PHY	XDP	XDP_CPU_TDI 11 25
FE1B	XDP_PHY	XDP	XDP_CPU_TDO 11 25
FE1C	XDP_PHY	XDP	XDP_CPU_TRST_L 11 25
FE1D	XDP_PHY	XDP	XDP_CPU_PRDY_L 11 25
FE1E	XDP_PHY	XDP	XDP_CPU_PREQ_L 11 25
FE1F			
PCH JTAG			
FE20	XDP_PHY	CLK_JTAG	XDP_PCH_TCK 10 25
FE21	XDP_PHY	XDP	XDP_PCH_TMS 10 25
FE22	XDP_PHY	XDP	XDP_PCH_TDI 10 25
FE23	XDP_PHY	XDP	XDP_PCH_TDO 10 25

Chipset Test Interface

Electrical Constraint Set	Physical	Spacing	
OBS PCH Side			
H193	XDP_PCH_OBS	XDP_PHY	XDP
H194	XDP_PCH_OBS	XDP_PHY	XDP
H195	XDP_PCH_OBS	XDP_PHY	XDP
H196	XDP_PCH_OBS	XDP_PHY	XDP
H197	XDP_PCH_OBS	XDP_PHY	XDP
H198	XDP_PCH_OBS	XDP_PHY	XDP
H199	XDP_PCH_OBS	XDP_PHY	XDP
H200	XDP_PCH_OBS	XDP_PHY	XDP
H201	XDP_PCH_OBS	XDP_PHY	XDP
H202	XDP_PCH_OBS	XDP_PHY	XDP
H203	XDP_PCH_OBS	XDP_PHY	XDP
H204	XDP_PCH_OBS	XDP_PHY	XDP
H205	XDP_PCH_OBS	XDP_PHY	XDP
H206	XDP_PCH_OBS	XDP_PHY	XDP
H207	XDP_PCH_OBS	XDP_PHY	XDP
H208	XDP_PCH_OBS	XDP_PHY	XDP
H209	XDP_PCH_OBS	XDP_PHY	XDP
H210	XDP_PCH_OBS	XDP_PHY	XDP
H211	XDP_PCH_OBS	XDP_PHY	XDP
H212	XDP_PCH_OBS	XDP_PHY	XDP
H213	XDP_PCH_OBS	XDP_PHY	XDP
H214	XDP_PCH_OBS	XDP_PHY	XDP
H215	XDP_PCH_OBS	XDP_PHY	XDP
H216	XDP_PCH_OBS	XDP_PHY	XDP
H217	XDP_PCH_OBS	XDP_PHY	XDP
H218	XDP_PCH_OBS	XDP_PHY	XDP
H219	XDP_PCH_OBS	XDP_PHY	XDP
H220	XDP_PCH_OBS_UNUSED	XDP_PHY	OBS_STROBE
OBS XDP Side			
H240	XDP_PHY	XDP	XDP
H241	XDP_PHY	XDP	XDP
H242	XDP_PHY	XDP	XDP
H243	XDP_PHY	XDP	XDP
H244	XDP_PHY	XDP	XDP
H245	XDP_PHY	XDP	XDP
H246	XDP_PHY	XDP	XDP
H247	XDP_PHY	XDP	XDP
H248	XDP_PHY	XDP	XDP
H249	XDP_PHY	XDP	XDP
H250	XDP_PHY	XDP	XDP
H251	XDP_PHY	XDP	XDP
H252	XDP_PHY	XDP	XDP
H253	XDP_PHY	XDP	XDP
H254	XDP_PHY	XDP	XDP
H255	XDP_PHY	XDP	XDP
H256	XDP_PHY	XDP	XDP
H257	XDP_PHY	XDP	XDP
H258	XDP_PHY	XDP	XDP
H259	XDP_PHY	XDP	XDP
H260	XDP_PHY	XDP	XDP
H261	XDP_PHY	XDP	XDP
H262	XDP_PHY	XDP	XDP
H263	XDP_PHY	XDP	XDP
H264	XDP_PHY	XDP	XDP
H265	XDP_PHY	XDP	XDP
H266	XDP_PHY	XDP	XDP
H267	XDP_PHY	XDP	XDP
H268	XDP_PHY	XDP	XDP
H269	XDP_PHY	XDP	XDP
H270	XDP_PHY	XDP	XDP
H271	XDP_PHY	XDP	XDP
H272	XDP_PHY	XDP	XDP
H273	XDP_PHY	XDP	XDP
H274	XDP_PHY	XDP	XDP
H275	XDP_PHY	XDP	XDP
H276	XDP_PHY	XDP	XDP
H277	XDP_PHY	XDP	XDP
H278	XDP_PHY	XDP	XDP
H279	XDP_PHY	XDP	XDP
H280	XDP_PHY	XDP	XDP
H281	XDP_PHY	XDP	XDP
H282	XDP_PHY	XDP	XDP
H283	XDP_PHY	XDP	XDP
H284	XDP_PHY	XDP	XDP
H285	XDP_PHY	XDP	XDP
H286	XDP_PHY	XDP	XDP
H287	XDP_PHY	XDP	XDP
H288	XDP_PHY	XDP	XDP
H289	XDP_PHY	XDP	XDP
H290	XDP_PHY	XDP	XDP
H291	XDP_PHY	XDP	XDP
H292	XDP_PHY	XDP	XDP
H293	XDP_PHY	XDP	XDP
H294	XDP_PHY	XDP	XDP
H295	XDP_PHY	XDP	XDP
H296	XDP_PHY	XDP	XDP
H297	XDP_PHY	XDP	XDP
H298	XDP_PHY	XDP	XDP
H299	XDP_PHY	XDP	XDP
H300	XDP_PHY	XDP	XDP
H301	XDP_PHY	XDP	XDP
H302	XDP_PHY	XDP	XDP
H303	XDP_PHY	XDP	XDP
H304	XDP_PHY	XDP	XDP
H305	XDP_PHY	XDP	XDP
H306	XDP_PHY	XDP	XDP
H307	XDP_PHY	XDP	XDP
H308	XDP_PHY	XDP	XDP
H309	XDP_PHY	XDP	XDP
H310	XDP_PHY	XDP	XDP
H311	XDP_PHY	XDP	XDP
H312	XDP_PHY	XDP	XDP
H313	XDP_PHY	XDP	XDP
H314	XDP_PHY	XDP	XDP
H315	XDP_PHY	XDP	XDP
H316	XDP_PHY	XDP	XDP
H317	XDP_PHY	XDP	XDP
H318	XDP_PHY	XDP	XDP
H319	XDP_PHY	XDP	XDP
H320	XDP_PHY	XDP	XDP
H321	XDP_PHY	XDP	XDP
H322	XDP_PHY	XDP	XDP
H323	XDP_PHY	XDP	XDP
H324	XDP_PHY	XDP	XDP
H325	XDP_PHY	XDP	XDP
H326	XDP_PHY	XDP	XDP
H327	XDP_PHY	XDP	XDP
H328	XDP_PHY	XDP	XDP
H329	XDP_PHY	XDP	XDP
H330	XDP_PHY	XDP	XDP
Standard usage branch off from OBS			
H227	XDP_PHY	XDP	USB_EXT_A_OC_R_L
H228	XDP_PHY	XDP	USB_EXT_B_OC_R_L
H229	XDP_PHY	XDP	USB_EXT_C_OC_R_L
H230	XDP_PHY	XDP	USB_EXT_D_OC_R_L
H231	XDP_PHY	XDP	USB_EXT_B_OC_EHCI_R_L
H232	XDP_PHY	XDP	USB_EXT_D_OC_EHCI_R_L
H233	XDP_PHY	XDP	AP_PWR_EN_R
H234	XDP_PHY	XDP	SDCONN_STATE_CHANGE_R
H235	XDP_PHY	OBS_STROBE	TBT_CIO_PLUG_EVENT_R
H236	XDP_PHY	XDP	ISOLATE_CPU_MEM_R_L
H237	XDP_PHY	XDP	GPU_GOOD_R
H238	XDP_PHY	XDP	DP_AUXCH_ISOL_R
H239	XDP_PHY	XDP	SATARDRV_R_EN_R
H240	XDP_PHY	XDP	DP_TBT_SEL_R
H241	XDP_PHY	XDP	JTAG_TBT_TCK_R
H242	XDP_PHY	XDP	AUD_IPHS_SWITCH_EN_PCH_R
H243	XDP_PHY	XDP	ENET_LOW_PWR_PCH_R
H244	XDP_PHY	OBS_STROBE	XDP_PIN03
Standard usage branch off from OBS			
H227	XDP_PHY	XDP	USB_EXT_A_OC_L
H228	XDP_PHY	XDP	USB_EXT_B_OC_L
H229	XDP_PHY	XDP	USB_EXT_C_OC_L
H230	XDP_PHY	XDP	USB_EXT_D_OC_L
H231	XDP_PHY	XDP	USB_EXT_B_OC_EHCI_L
H232	XDP_PHY	XDP	USB_EXT_D_OC_EHCI_L
H233	XDP_PHY	XDP	AP_PWR_EN
H234	XDP_PHY	XDP	SDCONN_STATE_CHANGE
H235	XDP_PHY	OBS_STROBE	TBT_CIO_PLUG_EVENT
H236	XDP_PHY	XDP	ISOLATE_CPU_MEM_L
H237	XDP_PHY	XDP	GPU_GOOD
H238	XDP_PHY	XDP	DP_AUXCH_ISOL
H239	XDP_PHY	XDP	SATARDRV_R_EN
H240	XDP_PHY	XDP	DP_TBT_SEL
H241	XDP_PHY	XDP	JTAG_TBT_TCK
H242	XDP_PHY	XDP	JTAG_TBT_TCK_ISOL
H243	XDP_PHY	XDP	AUD_IPHS_SWITCH_EN_PCH
H244	XDP_PHY	XDP	ENET_LOW_PWR_PCH
H245	XDP_PHY	XDP	DP_AUXCH_ISOL_EN

SYNCH MASTER=D8 ROSITA		SYNCH DATE=03/23/2012	
PAGE TITLE			
CPU MISC/DMI/FDI/XDP Constraints			
 Apple Inc.		DRAWING NUMBER 051-9504	SIZE D
		REVISION 5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE 123 OF 143	
		SHEET 99 OF 117	

SATA

SATA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SATA_PHY	*	SATA_90D
COMP_SATA_PHY	*	SATA_50S

SATA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ISO	*	=7.2X_DIELECTRIC	?
COMP_SATA_ISO	*	=5.4X_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA	*	*	SATA_ISO
COMP_SATA	*	*	COMP_SATA_ISO

SATA Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
15.2.1	90	95	20	23.62	SATA Gen2, SATA Gen3

SATA Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
15-3	50	50	15	15.75	SATA Gen2, SATA Gen3

SATA

Electrical Contrain Set	Physical	Spacing	
PCH SATA Port 0 (HDD)			
EB00 SATA_R2D	SATA_PHY	SATA	SATA_HDD_R2D_P 44
EB02 SATA_R2D	SATA_PHY	SATA	SATA_HDD_R2D_N 44
EB03 SATA_R2D	SATA_PHY	SATA	SATA_HDD_R2D_C_P 18 44
EB04 SATA_R2D	SATA_PHY	SATA	SATA_HDD_R2D_C_N 18 44
EB05 SATA_D2R	SATA_PHY	SATA	SATA_HDD_D2R_P 18 44
EB06 SATA_D2R	SATA_PHY	SATA	SATA_HDD_D2R_N 18 44
EB07 SATA_D2R	SATA_PHY	SATA	SATA_HDD_D2R_C_P 44
EB08 SATA_D2R	SATA_PHY	SATA	SATA_HDD_D2R_C_N 44
PCH SATA Port 1 (SSD)			
EB09 SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA_SSD_R2D_P 18 44
EB10 SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA_SSD_R2D_N 18 44
EB11 SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA_SSD_D2R_P 18 44
EB12 SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA_SSD_D2R_N 18 44
PCH SATA Compensation			
EB13A COMP_SATA_PHY	COMP_SATA_PHY	COMP_SATA	PCH_SATAICOMP 18
EB13B COMP_SATA_PHY	COMP_SATA_PHY	COMP_SATA	PCH_SATA3COMP 18
EB13C COMP_SATA_PHY	COMP_SATA_PHY	COMP_SATA	PCH_SATA3RBIAS 18

Unused

Electrical Constraint Set	Physical	Spacing

PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCH_ISO	*	=6.5X_DIELECTRIC	?	CLK_PCH	*	*	CLK_PCH_ISO
COMP_PCH_ISO	*	=2:1_SPACING	?	COMP_PCH	*	*	COMP_PCH_ISO
PCH_ISO	*	=3:1_SPACING	?	PCH	*	*	PCH_ISO

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCI_ISO	*	=3.6X_DIELECTRIC	?

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC_ISO	*	=1.5:1_SPACING	?
CLK_LPC_ISO	*	=3.6X_DIELECTRIC	?

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDA_ISO	*	=2X_DIELECTRIC	?	HDA	*	*	HDA_ISO

Crystal

Crystal-specific Physical Rules

[illegible]

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XTAL_ISO	*	=4X_DIELECTRIC	?	XTAL	*	*	XTAL_ISO

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SPI_ISO	*	=2X_DIELECTRIC	?	SPI	*	*	SPI_ISO

PCI

E14

Electrical Constraint Set	Physical	Spacing	
PCI Clock			
1600	CLK_PCT_50S	CLK_PCT	PCH_CLK33M_PCIIN 18 26
1600	CLK_PCT_50S	CLK_PCT	PCH_CLK33M_PCIOUT 20 26

LPC

Electrical Constraint Set	Physical	Spacing	
LPC			
	LPC 50S	LPC	LPC_AD<3..0> 18 47 49
	LPC 50S	LPC	LPC_R_AD<3..0> 18
	LPC 50S	LPC	LPC_FRAME_L 18 47 49
	LPC 50S	LPC	LFRAME_L 18
LPC Clocks			
	CLK LPC 50S	CLK LPC	LPC_CLK33M LPCPLUS 26 49
	CLK LPC 50S	CLK LPC	LPC_CLK33M LPCPLUS_R 20 26
	CLK LPC 50S	CLK LPC	LPC_CLK33M SMC 26 47
	CLK LPC 50S	CLK LPC	LPC_CLK33M SMC_R 20 26

PCH Clocks

Electrical Constraint Set	Physical	Spacing	
PCH Reference Clock			
 CLK_PCH_50S	CLK_PCH_50S	CLK_PCH	SYSClk_CLK25M_SB 26
 CLK_PCH_50S	CLK_PCH_50S	CLK_PCH	PCH_CLK25M_XTALIN 18 26
PCH Ref Clock Comp			
 PCH_50S	PCH_50S	COMP_PCH	PCH_XCLK_RCOMP 18
PCH RTC 32K			
 CLK_XTAL	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1 18 26
 CLK_XTAL	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2 18 26
 CLK_XTAL	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R 26
SMC 32K			
 CLK_PCH_50S	CLK_PCH_50S	CLK_PCH	PM_CLK32K_SUSCLK_R 19 48
 CLK_PCH_50S	CLK_PCH_50S	CLK_PCH	SMC_CLK32K 47 48

25 MHz Reference Clocks

Electrical Constraint Set	Physical	Spacing	
25M Reference Crystal			
	CLK_XTAL	XTAL	SYSClk CLK25M_X1 26
	CLK_XTAL	XTAL	SYSClk CLK25M_X2 26
	CLK_XTAL	XTAL	SYSClk CLK25M_X2_R 26
25M Reference Clocks			
	CLK_PCH_50S	CLK_PCH	SYSClk CLK25M_ENET 26 39
	CLK_PCH_50S	CLK_PCH	SYSClk CLK25M_ENET_R 26
	CLK_PCH_50S	CLK_PCH	SYSClk CLK25M_TBT 26 36
	CLK_PCH_50S	CLK_PCH	SYSClk CLK25M_TBT_R 36

HDA

E1

Electrical Constraints	Physical	Spacing	
HDA			
H605	HDA_50S	HDA	HDA_BIT_CLK 18 56
H606	HDA_50S	HDA	HDA_BIT_CLK_R 18
H605	HDA_50S	HDA	HDA_RST_L 18 56
H606	HDA_50S	HDA	HDA_RST_R_L 18
H606	HDA_50S	HDA	HDA_SDOUT 18 56
H607	HDA_50S	HDA	HDA_SDOUT_R 15 18
H606	HDA_50S	HDA	HDA_SYNC 15 18 56
H606	HDA_50S	HDA	HDA_SYNC_R 18
H606	HDA_50S	HDA	HDA_SDI0 18 56
H606	HDA_50S	HDA	AUD_SDI_R 56
H612	HDA_50S	HDA	SPI_DESCRIPTOR_OVERRIDE_R 15
SPDIF			
H630		HDA	AUD_SPDIF_CHIP 56
H632		HDA	AUD_SPDIF_OUT 56 60

SPI Bootrom

Electrical Constraint Set	Physical	Spacing	
SPI ROM			
H89D	SPT 56S	SPT	SPI CLK R 18 49
H89D	SPT 56S	SPT	SPI CLK 49
H89D	SPT 56S	SPT	SPI ALT CLK 49
H89D	SPT 56S	SPT	SPI SMC CLK 47 48
H89D	SPT 56S	SPT	SPI MLB CLK 48 49
H89D	SPT 56S	SPT	SPI CS0 R L 18 49
H89D	SPT 56S	SPT	SPI CS0 L 49
H89D	SPT 56S	SPT	SPI ALT CS L 49
H89D	SPT 56S	SPT	SPI SMC CS L 47 48
H89D	SPT 56S	SPT	SPI MLB CS L 48 49
H89D	SPT 56S	SPT	SPI MOSI R 18 49
H89D	SPT 56S	SPT	SPI MOSI 49
H89D	SPT 56S	SPT	SPI ALT MOSI 49
H89D	SPT 56S	SPT	SPI SMC MOSI 47 48
H89D	SPT 56S	SPT	SPI MLB MOSI 48 49
H89D	SPT 56S	SPT	SPI MISO 18 49
H89D	SPT 56S	SPT	SPI ALT MISO 49
H89D	SPT 56S	SPT	SPI SMC MISO 47 48
H89D	SPT 56S	SPT	SPI MLB MISO 48 49
H89D	SPT 56S	SPT	SPIROM USE MLB 21 49

USB

USB-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
USB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
USB2_PHY	*	USB_90D
USB3_PHY	*	USB_85D
USB_HUB_PHY	*	USB_50S

USB Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Comments
12.2.1	90	90	12/2.8 mils = 4.29:1	USB 2.0
13.3.1	85	85	20/2.8 mils = 7.14:1	USB 3.0
			50/2.8 mils = 17.9:1	USB 2.0/3.0

SMSC Hub Application Note 15.17

Single-ended impedance range from 45-80 ohm is acceptable

USB 2.0 Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB2_ISO	*	=4.4X_DIELECTRIC	?
USB2_ISO	TOP,BOTTOM	=4.4X_DIELECTRIC	?
USB2_CLK_ISO	*	=18X_DIELECTRIC	?
USB2_CLK_ISO	TOP,BOTTOM	=18X_DIELECTRIC	?
USB_HUB_ISO	*	=2:1_SPACING	?

USB 2.0 Spacing Rules

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB2	*	*	USB2_ISO
USB2	*CLK*	*	USB2_CLK_ISO
USB2	DISPLAYPORT	*	USB2_CLK_ISO
USB2	*TBT*	*	USB2_CLK_ISO
USB2	*ENET*	*	USB2_CLK_ISO
USB2	*SD*	*	USB2_CLK_ISO
USB3	PCIE	*	USB3_CLK_ISO
USB3	SATA	*	USB3_CLK_ISO
USB_HUB	*	*	USB_HUB_ISO

USB 3.0 Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_ISO	*	=7.3X_DIELECTRIC	?
USB3_ISO	TOP, BOTTOM	=7.3X_DIELECTRIC	?
USB3_CLK_ISO	*	=18X_DIELECTRIC	?
USB3_CLK_ISO	TOP, BOTTOM	=18X_DIELECTRIC	?
USB3_USB3_ISO	*	=7.3X_DIELECTRIC	?
USB3_USB3_ISO	TOP, BOTTOM	=7.3X_DIELECTRIC	?
USB3_USB3_ISO	ISL10	=5X_DIELECTRIC	?

USB 3.0 Spacing Rules

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3	*	*	USB3_ISO
USB3	*CLK*	*	USB3_CLK_ISO
USB3	DISPLAYPORT	*	USB3_CLK_ISO
USB3	*TBT*	*	USB3_CLK_ISO
USB3	*ENET*	*	USB3_CLK_ISO
USB3	*SD*	*	USB3_CLK_ISO
USB3	PCIE	*	USB3_CLK_ISO
USB3	SATA	*	USB3_CLK_ISO
USB3	USB3	*	USB3_USB3_ISO

CAMERA CONTROLLER

Camera Controller's SMIA Interface & MISC. Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMIA_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
CAM_SE	*	Y	0.2 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMIA_DIFF_PHY	*	SMIA_100D
CAM_PHY	*	CAM_SE

Camera Controller's SMIA Interface & MISC. Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMIA_DIFF_ISO	*	=6:1_SPACING	?
SMIA_DIFF2DIFF	*	=3:1_SPACING	?
CAM_ISO	*	=2:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMIA_DIFF	*	*	SMIA_DIFF_ISO
SMIA_DIFF	SMIA_DIFF	*	SMIA_DIFF2DIFF
CAM	*	*	CAM_ISO

USB 3.0 and USB 2.0 Trixies Muxing

Electrical Constraint Set		Physical	Spacing	
External Port A (J4600)				
H69P	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_A_RX_P 45
H69P	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_A_RX_N 45
H69P		USB3_PHY	USB3	USB3_EXT_A_RX_F_P 20 45
H69P		USB3_PHY	USB3	USB3_EXT_A_RX_F_N 20 45
H69P	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_A_TX_P 20 45
H69P	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_A_TX_N 20 45
H69P		USB3_PHY	USB3	USB3_EXT_A_TX_F_P 45
H69P		USB3_PHY	USB3	USB3_EXT_A_TX_F_N 45
H69P		USB3_PHY	USB3	USB3_EXT_A_TX_C_P 45
H69P		USB3_PHY	USB3	USB3_EXT_A_TX_C_N 45
H69P	USB2_MUXED_MUX0_CONN	USB2_PHY	USB2	USB_PCH_0_P 20 45
H69P	USB2_MUXED_MUX0_CONN	USB2_PHY	USB2	USB_PCH_0_N 20 45
H69P		USB2_PHY	USB2	USB2_EXT_A_MUXED_P 45
H69P		USB2_PHY	USB2	USB2_EXT_A_MUXED_N 45
H69P		USB2_PHY	USB2	USB2_EXT_A_MUXED_F_P 45
H69P		USB2_PHY	USB2	USB2_EXT_A_MUXED_F_N 45
External Port B (J4610)				
H69P	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_B_RX_P 45
H69P	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_B_RX_N 45
H69P		USB3_PHY	USB3	USB3_EXT_B_RX_F_P 20 45
H69P		USB3_PHY	USB3	USB3_EXT_B_RX_F_N 20 45
H69P	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_B_TX_P 20 45
H69P	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_B_TX_N 20 45
H69P		USB3_PHY	USB3	USB3_EXT_B_TX_F_P 45
H69P		USB3_PHY	USB3	USB3_EXT_B_TX_F_N 45
H69P		USB3_PHY	USB3	USB3_EXT_B_TX_C_P 45
H69P		USB3_PHY	USB3	USB3_EXT_B_TX_C_N 45
H69P	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_1_P 20 45
H69P	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_1_N 20 45
H69P		USB2_PHY	USB2	USB_PCH_9_P 20 45
H69P		USB2_PHY	USB2	USB_PCH_9_N 20 45
H69P		USB2_PHY	USB2	USB2_EXT_B_MUXED_P 45
H69P		USB2_PHY	USB2	USB2_EXT_B_MUXED_N 45
H69P		USB2_PHY	USB2	USB2_EXT_B_MUXED_F_P 45
H69P		USB2_PHY	USB2	USB2_EXT_B_MUXED_F_N 45
External Port C (J4700)				
H69P	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_C_RX_P 46
H69P	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_C_RX_N 46
H69P		USB3_PHY	USB3	USB3_EXT_C_RX_F_P 20 46
H69P		USB3_PHY	USB3	USB3_EXT_C_RX_F_N 20 46
H69P	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_C_TX_P 20 46
H69P	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_C_TX_N 20 46
H69P		USB3_PHY	USB3	USB3_EXT_C_TX_F_P 46
H69P		USB3_PHY	USB3	USB3_EXT_C_TX_F_N 46
H69P		USB3_PHY	USB3	USB3_EXT_C_TX_C_P 46
H69P		USB3_PHY	USB3	USB3_EXT_C_TX_C_N 46
H69P	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_P 20 46
H69P	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_N 20 46
H69P		USB2_PHY	USB2	USB2_EXT_C_F_P 46
H69P		USB2_PHY	USB2	USB2_EXT_C_F_N 46
External Port D (J4710)				
H69P	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_D_RX_P 46
H69P	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_D_RX_N 46
H69P		USB3_PHY	USB3	USB3_EXT_D_RX_F_P 20 46
H69P		USB3_PHY	USB3	USB3_EXT_D_RX_F_N 20 46
H69P	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_D_TX_P 20 46
H69P	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_D_TX_N 20 46
H69P		USB3_PHY	USB3	USB3_EXT_D_TX_F_P 46
H69P		USB3_PHY	USB3	USB3_EXT_D_TX_F_N 46
H69P		USB3_PHY	USB3	USB3_EXT_D_TX_C_P 46
H69P		USB3_PHY	USB3	USB3_EXT_D_TX_C_N 46
H69P	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_3_P 20 46
H69P	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_3_N 20 46
H69P		USB2_PHY	USB2	USB_PCH_10_P 20 46
H69P		USB2_PHY	USB2	USB_PCH_10_N 20 46
H69P		USB2_PHY	USB2	USB2_EXT_D_MUXED_P 46
H69P		USB2_PHY	USB2	USB2_EXT_D_MUXED_N 46
H69P		USB2_PHY	USB2	USB2_EXT_D_MUXED_F_P 46
H69P		USB2_PHY	USB2	USB2_EXT_D_MUXED_F_N 46
Camera (U4200)				
H69P	USB2_CONN_TNT	USB2_PHY	USB2	USB_CAMERA_P 42
H69P	USB2_CONN_TNT	USB2_PHY	USB2	USB_CAMERA_N 42
PCH USB Compensation		PCH_50S	COMP_PCH	PCH_USB_RBIA5 20

Electrical Constraint Set	Physical	Spacing	Voltage	
Camera Controller Local Ground				
IS27	GND_PHY	GND	0V	CAM_AGND 42
IS27	GND_PHY	GND	0V	CAM_PLLGND 42

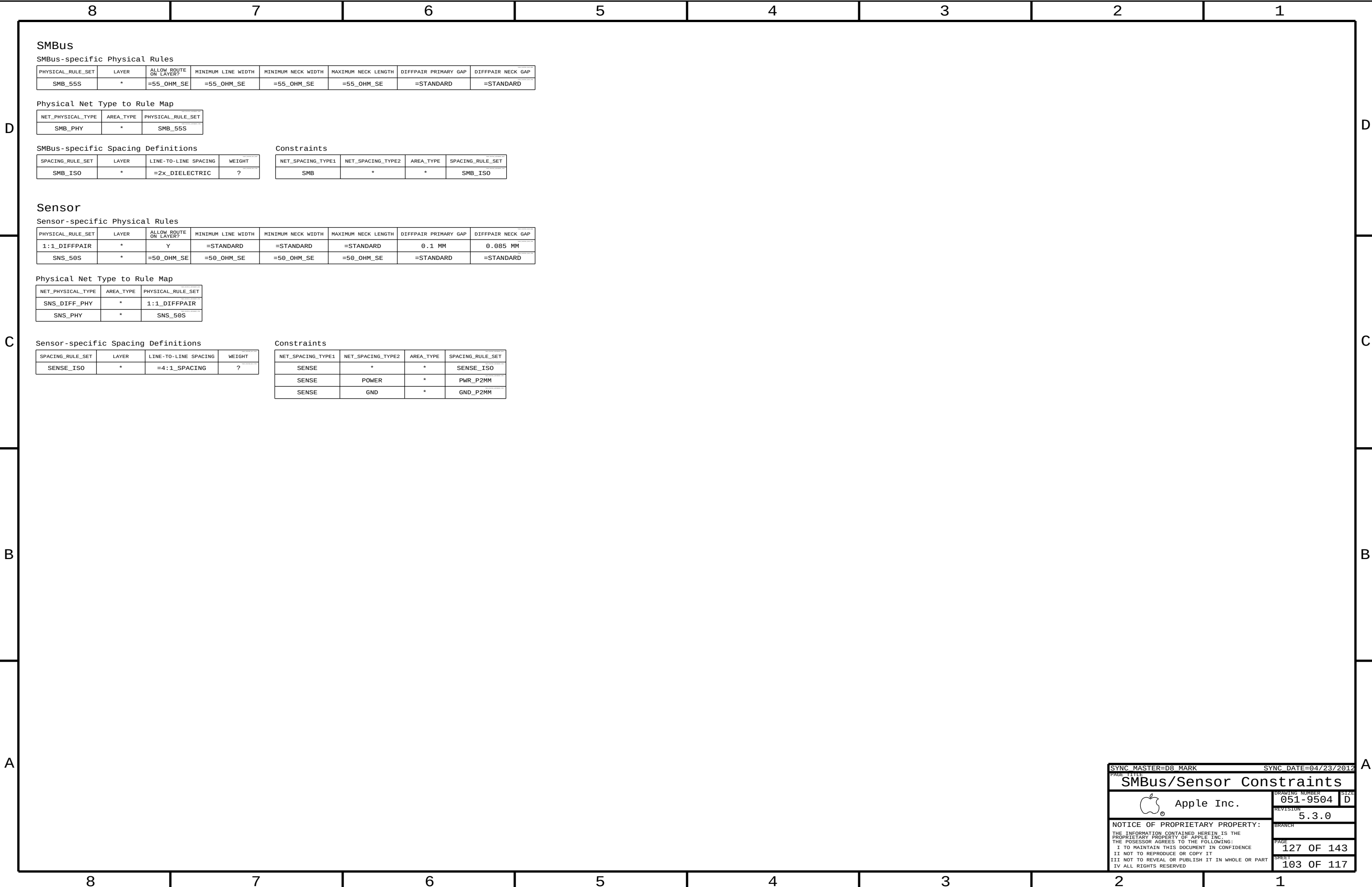
USB Hub

Electrical Constraint Set	Physical	Spacing	
USB 2.0 Hub			
 USB2_HUB_PCH	USB2_PHY	USB2	USB_PCH_7_P 29 27
 USB2_HUB_PCH	USB2_PHY	USB2	USB_PCH_7_N 29 27
 USB2_HUB_BT	USB2_PHY	USB2	USB_BT_P 27 38
 USB2_HUB_BT	USB2_PHY	USB2	USB_BT_N 27 38
 USB2_HUB_BT	USB2_PHY	USB2	USB_BT_MUX_P 35 27
 USB2_HUB_BT	USB2_PHY	USB2	USB_BT_MUX_N 35 27
	USB_HUB_PHY	USB_HUB	USB_HUB_2P 27 27
	USB_HUB_PHY	USB_HUB	USB_HUB_2N 27 27
USB 2.0 Hub Misc.			
	USB_HUB_PHY	USB_HUB	USB_HUB_RESET_L 27 27
	USB_HUB_PHY	USB_HUB	USB_HUB_VBUS_DET 27 27
	USB_HUB_PHY	USB_HUB	USB_HUB_NON_REM0 27 27
	USB_HUB_PHY	USB_HUB	USB_HUB_NON_REM1 27 27
	USB_HUB_PHY	USB_HUB	USB_HUB_HS_IND 27 27
USB 2.0 Hub Compensation			
	PCH_S0S	COMP_PCH	USB_HUB_RBIA5 27 27
USB 2.0 Hub Crystal			
	CLK_XTAL	XTAL	USB_HUB_XTAL1 27 27
	CLK_XTAL	XTAL	USB_HUB_XTAL2 27 27
	CLK_XTAL	XTAL	USB_HUB_XTAL2_R 27 27

Camera Controller

Electrical Constraint Set		Physical	Spacing	
SMIA				
HS02	SMIA_DATA	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA_P 42
HS02	SMIA_DATA	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA_N 42
HS04	SMIA_CLK	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK_P 42
HS20	SMIA_CLK	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK_N 42
MISC				
HS03		SPT_SBS	SPT	CAM_SF_CLK 42
HS03		SPT_SBS	SPT	CAM_SF_CLK_R 42
HS03		SPT_SBS	SPT	CAM_SF_DIN 42
HS03		SPT_SBS	SPT	CAM_SF_DIN_R 42
HS03		SPT_SBS	SPT	CAM_SF_CS_L 42
HS03		SPT_SBS	SPT	CAM_SF_WP_L 42
HS03		SPT_SBS	SPT	CAM_SF_DOUT 42
HS03		SPT_SBS	SPT	CAM_SF_DOUT_R 42
HS03		SPT_SBS	SPT	CAM_SF_HOLD_L 42
HS20		CAM_PHY	CAM	CAM_USB_VRES 42
HS03		CAM_PHY	CAM	MIPT_RESISTOR 42
HS03			PH	CAM_EXT_BOOT_L 43
HS03			PH	PCH_CAM_EXT_BOOT_R_L 23 43
HS03			PH	CAM_P1V2_RST_HOLDOFF 43
HS03			PH	CAM_P1V2_RST_HOLDOFF_L 43
I2C				
HS03		SMB_PHY	SMB	I2C_CAMSENSOR_SDA 42
HS03		SMB_PHY	SMB	I2C_CAMSENSOR_SCL 42
HS03		SMB_PHY	SMB	SMB_ALS_F_SDA 42
HS03		SMB_PHY	SMB	SMB_ALS_F_SCL 42
Camera Controller Crystal				
HS04		CLK_XTAL	XTAL	CAM_XTAL_IN 42
HS04		CLK_XTAL	XTAL	CAM_XTAL_OUT 42
HS20		CLK_XTAL	XTAL	CAM_XTAL_OUT_R 42

SYNC MASTER=D8 KOSECOFF		SYNC DATE=06/22/2012	
PAGE TITLE			
USB/Camera Constraints			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-9504	D
		REVISION	
		5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE			
PROPRIETARY PROPERTY OF APPLE INC.			
THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		126 OF 143	
I NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		102 OF 117	



D

D

C

C

B

B

A

A

SYNC_MASTER=D8_MARK

SYNC_DATE=04/23/2012

PAGE_TITLE

SMBus/Sensor Constraints

 Apple Inc.

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

051-9504

REVISION

5.3.0

BRANCH

PAGE

127 OF 143

SHEET

103 OF 117

DC - DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GND_P3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_CTL_PHY	*	POWER_P5MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S
POWER_PHY	*	POWER_P6MM
GND_PHY	*	GND_P5MM

Constraints

Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

Power-specific Spacing Definitions

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SWNODE_ISO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?
SWNODE_LG2SW	*	=3:1_SPACING	?

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?

CPU VccSA

Electrical Constraint Set	Physical	Spacing	Voltage	D1D2	NO_TEST	
Input Bus						
I24	POWER_PHY	POWER	5V			REG_VCC_U7500 70
I25	POWER_PHY	POWER	5V			REG_PVCC_U7500 70
Local Ground						
I26	GND_PHY	GND	0V			AGND_VCCSAS0 70
VCCIO						
I27	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VCCSAS0 70
I28	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_B00T_VCCSAS0 70
I29	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_B00T_VCCSAS0_RC 70
I30	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VCCSAS0 70
I31	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_VCCSAS0 70
I32	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_VCCSAS0 70
I33	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_OCSET 70
I34	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_VO 70
I35		SENSE				SNS_CPU_VCCSA 13 70
I36 VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE				SNS_VCCSAS0_XW_P 70
I37 VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE				SNS_VCCSAS0_XW_N 70
I38		SENSE				REG_VCCSAS0_FB 70
I39		SENSE				REG_VCCSAS0_RTN 70
I40	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_SREF 70
I41	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_FSEL 70
Output Bus						
I42	POWER_PHY	POWER	0_925V			PPVCCSA_S0 6

```
CPU VccIO PCH 1.05V S0
```

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus						
 P0	POWER_PHY	POWER	5V			REG_VCC_U7400 69
 P2	POWER_PHY	POWER	5V			REG_PVCC_U7400 69
Local Ground						
 A0	GND_PHY	GND	0V			AGND_CPU_P1V05S0 69
1.05V S0						
 P5	POWER_PHY	VB_SWITCH	12V	TRUE		REG_CPU_PHASE_P1V05S0 69
 P7	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_BOOT_P1V05S0 69
 P0	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_CPU_BOOT_P1V05S0_RC 69
 P0	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_UGATE_P1V05S0 69
 P0	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_CPU_UGATE_P1V05S0_R 69
 P0P0	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_CPU_LGATE_P1V05S0 69
 P0P2	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_CPU_SNUBBER_P1V05S0 69
 P0P0	VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_OCSET 69
 P0P0	VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_VD 69
 P0P0	VSNS_CPU_VCCIO	SENSE				SNS_CPU_VCCIO_P 13 69
 P0P0	VSNS_CPU_VCCIO	SENSE				SNS_CPU_VCCIO_N 13 69
 P0P0	SNS_DIEF_PHY	SENSE				SNS_CPU_P1V05S0_XW_P 69
 P0P0	SNS_DIEF_PHY	SENSE				SNS_CPU_P1V05S0_XW_N 69
 P0P0		SENSE				REG_CPU_P1V05S0_FB 69
 P0P0		SENSE				REG_CPU_P1V05S0_RTN 69
 P0P0	VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_SREF 69
 P0P0	VR_CTL_PHY	VR_CTL				REG_CPU_P1V05S0_FSEL 69
Output Bus						
 P0P0	POWER_PHY	POWER	1.05V			PP1V05_S0_CPU 6

PCH/GPU/TBT 1.05V S0

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus						
FE00	POWER_PHY	POWER	5V			REG_VCC_UB700 95
Local Ground						
LG00	GND_PHY	GND	0V			P1V05_AGND 95
1.05V S0						
PG00	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0 95
FE00	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0_L 95
FE00	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P1V05S0 95
FE00	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P1V05S0_RC 95
FE00	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0 95
FE00	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0_R 95
LG00	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_P1V05S0 95
LG00	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P1V05S0 95
FE00	VR_CTL_PHY	VR_CTL			TRUE	REG_P1V05S0_OCSET 95
FE00	VR_CTL_PHY	VR_CTL				REG_P1V05S0_VO 95
FE00	POWER_PHY	VR_SWITCH	12V			REG_P1V05S0_VO_R 95
FE00	POWER_PHY	VR_SWITCH	12V			P1V05_OCSET_R 95
FE00		SENSE				REG_P1V05S0_FB 95
LG00		SENSE				REG_P1V05S0_RTN 95
FE00	VR_CTL_PHY	VR_CTL				REG_P1V05S0_SREF 95
FE00	VR_CTL_PHY	VR_CTL				REG_P1V05S0_FSEL 95
Output Bus						
LG00	POWER_PHY	POWER	1.05V			PP1V05_S0_PCH 6
FE00	POWER_PHY	POWER	1.05V			PP1V05_S0 6
FE00	POWER_PHY	POWER	1.05V			PP1V05_TBTL 6 3H
FE00	POWER_PHY	POWER	1.05V			PP1V05_TRTCIO 6

SYNC MASTER=D8 MARK		SYNC DATE=02/10/2012	
PAGE TITLE			
VReg Constraints			
 Apple Inc.	DRAWING NUMBER	SIZE	
	051-9504	D	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	REVISION	5.3.0	
	BRANCH		
	PAGE	128 OF 143	
	SHEET	104 OF 117	

3.3V S5/5V S4

	Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus						
H12P	POWER_PHY	POWER	12V			REG_VIN_U7600
H12S	POWER_PHY	POWER	5V			REG_VCC1_U7600
H12S	POWER_PHY	POWER	5V			REG_VCC2_U7600
3.3V S5						
H12S	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P3V3S5
H12P	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_B00T_P3V3S5
H12S	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_B00T_P3V3S5_RC
H12S	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P3V3S5
H12S	VR_CTL_PHY	VR_IGATE	12V	TRUE		REG_LGATE_P3V3S5
H12S	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P3V3S5
H10P	VR_CTL_PHY	VR_CTL				REG_P3V3S5_ISEN
H10P	VR_CTL_PHY	VR_CTL				REG_P3V3S5_OCSET
H10S	VR_CTL_PHY	VR_CTL				REG_P3V3S5_FSET
H10S	VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT
H10P	VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT_R
H10P	VR_CTL_PHY	VR_CTL				REG_P3V3S5_FB
5V S3						
H12S	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P5V54
H12S	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_B00T_P5V54
H12S	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_B00T_P5V54_RC
H12S	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P5V54
H12S	VR_CTL_PHY	VR_IGATE	12V	TRUE		REG_LGATE_P5V54
H12S	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P5V54
H10S	VR_CTL_PHY	VR_CTL				REG_P5V54_ISEN
H10S	VR_CTL_PHY	VR_CTL				REG_P5V54_OCSET
H10S	VR_CTL_PHY	VR_CTL				REG_P5V54_FSET
H10S	VR_CTL_PHY	VR_CTL				REG_P5V54_VOUT
H10P	VR_CTL_PHY	VR_CTL				REG_P5V54_VOUT_R
H10P	VR_CTL_PHY	VR_CTL				REG_P5V54_FB
Output Bus						
H12S	POWER_PHY	POWER	5V			PP5V_S5
H12S	POWER_PHY	POWER	5V			PP5V_S4
H12S	POWER_PHY	POWER	3.3V			PP3V3_S5
FET Switched						
H12P	POWER_PHY	POWER	5V			PP5V_S0
H12P	POWER_PHY	POWER	3.3V			PP3V3_S4
H12S	POWER_PHY	POWER	3.3V			PP3V3_S0
H12P	POWER_PHY	POWER	3.3V			PP3V3_S4_ENET
H12S	POWER_PHY	POWER	3.3V			PP3V3_TBTLCL
H12S	POWER_PHY	POWER	3.3V			PP3V3_S4_AP
H12S	POWER_PHY	POWER	3.3V			PP3V3_S0_SSD

VDDQ S3 (1.5V)/VTT S0

Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus					
H139 POWER_PHY	POWER	5V			REG_V5IN_U7700 72
Local Ground					
H139 GND_PHY	GND	0V			AGND_VDDQS3 73
VDDQ S3					
H139 POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3 72
H139 POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3_L 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_VDDQS3 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_VDDQS3_RC 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3_R 72
H139 VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_VDDQS3 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_VDDQS3 72
H139 POWER_PHY	POWER	1.5V			PPVDDQ_S3_SENSE 72
H134 VR_CTL_PHY	VR_CTL				REG_VDDQS3_VDDQSN3 72
H139 VR_CTL_PHY	VR_CTL				REG_VDDQS3_VREF 72
H139 VR_CTL_PHY	VR_CTL				REG_VDDQS3_REFIN 72
H139 VR_CTL_PHY	VR_CTL				REG_VDDQS3_MODE 72
H139 VR_CTL_PHY	VR_CTL				REG_VDDQS3_TRIP 72
H139 VR_CTL_PHY	VR_CTL				LDO_DDRVITS0_SNS 72
Output Bus					
H139 POWER_PHY	POWER	1.5V			PPVDDQ_S3 6
H139 POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVTTS0_S0 6
FET Switched					
H139 POWER_PHY	POWER	1.5V			PP1V5_S0 6
Sensed					
H139 POWER_PHY	POWER	1.5V			PPVDDQ_S3_DDR 6
H139 POWER_PHY	POWER	1.5V			PP1V5_S0_CPU_MEM 6
H139 POWER_PHY	POWER	1.5V			PPFBVDDQ_S0_GPU 6

DDR3 Vref

	Physical	Spacing	Voltage	D1D2	NO_TEST	
Memory Vref						
HE35	POWER_PHY	POWER	3.3V			PP3V3_S4_VREFMRGN_DAC
HE36	POWER_PHY	POWER	3.3V			PP3V3_S4_VREFMRGN_CTRL
HE38B	POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVRFE_D0_MEM_A_S3
HE38C	POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVRFE_D0_MEM_B_S3
HE43	POWER_DDR_PHY	POWER_DDR	0.75V			CPU_DIMM_VREF_DAC_A
HE43	POWER_DDR_PHY	POWER_DDR	0.75V			CPU_DIMM_VREF_DAC_B
HE45B	POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVRFE_CA_MEM_A_S3
HE45C	POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVRFE_CA_MEM_B_S3
HE46B	POWER_DDR_PHY	POWER_DDR	0.75V			CPU_DDR_VREF
HE47	POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVRFE_S3

1.8V S0

Physical	Spacing	Voltage	DIDT	NO_TEST	
1.8V S0					
 POWER_PHY	VR_SWITCH	5V	TRUE		REG_PHASE_P1V8S0
 VR_CTL_PHY	VR_CTL				REG_P1V8S0_VFB
 VR_CTL_PHY	VR_CTL				REG_P1V8S0_SYNCH
Output Bus					
 POWER_PHY	POWER	1.8V			PP1V8_S0

HDD S0

Physical	Spacing	Voltage	DI0T	NO_TEST
HDD S0 POWER_PHY	POWER	5V		PP5V_S0_HDD

12V S5

Physical	Spacing	Voltage	DIOT	NO_TEST	
Input Bus					
 POWER_PHY	POWER	12V			PP12V ACDC 6
FET Switched					
 POWER_PHY	POWER	12V			PP12V S0 6
Sensed					
 POWER_PHY	POWER	12V			PP12V S5 6
 POWER_PHY	POWER	12V			PP12V G3H 6
 POWER_PHY	POWER	12V			PP12V G3H P3V42 75
 POWER_PHY	POWER	12V			PP12V S0 FBVDDQ 6
 POWER_PHY	POWER	12V			PP12V S0 CPU P1V05 6
 POWER_PHY	POWER	12V			PP12V S0 VCCSA 6
 POWER_PHY	POWER	12V			PP12V S0 P1V05 6
 POWER_PHY	POWER	12V			PP12V S0 HDD 6
 POWER_PHY	POWER	12V			PP12V S0 BLC 6

Ground/Common

Physical	Spacing	Voltage	D10T	NO_TEST
Common				
TEST GND_PHY	GND	6V		GND

3V42 S0

Physical		Spacing	Voltage	DIDT	NO_TEST	
3V42 S0						
	POWER_PHY	VR_SWITCH	5V	TRUE		P3V42G3H_SW 73
	VR_CTL_PHY	VR_CTL				P3V42G3H_FB 73
	VR_CTL_PHY	VR_CTL				P3V42G3H_SHDN_L 73
	POWER_PHY	POWER	3.3V			PP3V3_G3 6
	POWER_PHY	POWER	3.3V			PP3V42_G3H 6

SYNC MASTER=D8 MARK		SYNC DATE=02/10/2012	
PAGE TITLE			
Platform VReg Constraints			
	Apple Inc.	DRAWING NUMBER	051-9504
		SIZE	D
		REVISION	5.3.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE 130 OF 143	
		SHEET 106 OF 117	

GDDR5

GDDR5-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12 MM	=STANDARD	=STANDARD
GDDR_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	*	GDDR_45S
GDDR_ADBI_PHY	*	GDDR_45S
GDDR_CTRL_PHY	*	GDDR_45S
GDDR_CLK_PHY	*	GDDR_80D
GDDR_DQ_PHY	*	GDDR_45S
GDDR_EDC_PHY	*	GDDR_45S
GDDR_DBI_PHY	*	GDDR_45S
GDDR_WCK_PHY	*	GDDR_80D

Main Segment Min Spacing Rules for 4.5 Gbps or Less (AMD Doc# 49919)

[illegible]

GDDR5-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_ISO	*	=3X_DIELECTRIC	?
GDDR_ISO	TOP, BOTTOM	=5x_DIELECTRIC	?
GDDR_MA2MA	*	=2x_DIELECTRIC	?
GDDR_MA2MA	TOP, BOTTOM	=3X_DIELECTRIC	?
GDDR_ADBI2ADBI	*	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	*	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CLK2CLK	*	=3X_DIELECTRIC	?
GDDR_CLK2CLK	TOP, BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_DQ2DQ	*	=3x_DIELECTRIC	?
GDDR_DQ2DQ	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_EDC_ISO	*	=3X_DIELECTRIC	?
GDDR_EDC_ISO	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_EDC2EDC	*	=3X_DIELECTRIC	?
GDDR_EDC2EDC	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_DBI2DBI	*	=3x_DIELECTRIC	?
GDDR_DBI2DBI	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_WCK2WCK	*	=3X_DIELECTRIC	?
GDDR_WCK2WCK	TOP, BOTTOM	=5X_DIELECTRIC	?

Constraints ($x \in \{A, B\}$, $y \in \{0, 1\}$)

Memory Address: M_{Axy}[8:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_MA	*	*	GDDR_ISO
GDDR_*_*_MA	=SAME	*	GDDR_MA2MA

Address Dynamic Bus Inversion: ADBIx

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_ADBI	*	*	GDDR_ISO
GDDR_*_*_ADBI	=SAME	*	GDDR_ADBI2ADBI

Control: Reset, CKExy, CSxy, WExy, RASxy, CASxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	=SAME	*	GDDR_CTRL2CTRL

Clock: CKxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_CLK	*	*	GDDR_ISO
GDDR_*_*_CLK	=SAME	*	GDDR_CLK2CLK

GPU

GPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_GPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

GPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_GPU_ISO	*	=4:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_GPU	*	*	CLK_GPU_ISO

GDDR5 Frame Buffer A

Electrical Constraint Set		Physical	Spacing		
Memory Address					
MEM0	GDDR A0 MA	GDDR NA PHY	GDDR A 0 MA	FB A0 A<8..0>	NO TEST=TRUE 77 79
MEM1	GDDR A1 MA	GDDR NA PHY	GDDR A 1 MA	FB A1 A<8..0>	NO TEST=TRUE 77 79
Address Dynamic Bus Inv					
FE00	GDDR A0 ADBT	GDDR ADBT PHY	GDDR A 0 ADBT	FB A0 ABT L	NO TEST=TRUE 77 79
FE01	GDDR A1 ADBT	GDDR ADBT PHY	GDDR A 1 ADBT	FB A1 ABT L	NO TEST=TRUE 77 79
Control					
FE04	GDDR A0 CKE	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CKE L	NO TEST=TRUE 77 79
FE05	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CS L	NO TEST=TRUE 77 79
FE06	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 WE L	NO TEST=TRUE 77 79
FE07	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CAS L	NO TEST=TRUE 77 79
FE08	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 RAS L	NO TEST=TRUE 77 79
FE09	GDDR A1 CKE	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CKE L	NO TEST=TRUE 77 79
FE0A	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CS L	NO TEST=TRUE 77 79
FE0B	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 WE L	NO TEST=TRUE 77 79
FE0C	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CAS L	NO TEST=TRUE 77 79
FE0D	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 RAS L	NO TEST=TRUE 77 79
Clock					
FE0E	GDDR A0 CLK	GDDR CLK PHY	GDDR A 0 CLK	FB A0 CLK P	NO TEST=TRUE 77 79
FE0F	GDDR A0 CLK	GDDR CLK PHY	GDDR A 0 CLK	FB A0 CLK N	NO TEST=TRUE 77 79
FE10	GDDR A1 CLK	GDDR CLK PHY	GDDR A 1 CLK	FB A1 CLK P	NO TEST=TRUE 77 79
FE11	GDDR A1 CLK	GDDR CLK PHY	GDDR A 1 CLK	FB A1 CLK N	NO TEST=TRUE 77 79
Data					
FE12	GDDR A0 DQ BYTE0	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<7..0>	NO TEST=TRUE 77 79
FE13	GDDR A0 DQ BYTE1	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<15..8>	NO TEST=TRUE 77 79
FE14	GDDR A0 DQ BYTE2	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<23..16>	NO TEST=TRUE 77 79
FE15	GDDR A0 DQ BYTE3	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<31..24>	NO TEST=TRUE 77 79
FE16	GDDR A1 DQ BYTE0	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<7..0>	NO TEST=TRUE 77 79
FE17	GDDR A1 DQ BYTE1	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<15..8>	NO TEST=TRUE 77 79
FE18	GDDR A1 DQ BYTE2	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<23..16>	NO TEST=TRUE 77 79
FE19	GDDR A1 DQ BYTE3	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<31..24>	NO TEST=TRUE 77 79
Error Detection					
FE20	GDDR A0 EDC0	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<0>	NO TEST=TRUE 77 79
FE21	GDDR A0 EDC1	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<1>	NO TEST=TRUE 77 79
FE22	GDDR A0 EDC2	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<2>	NO TEST=TRUE 77 79
FE23	GDDR A0 EDC3	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<3>	NO TEST=TRUE 77 79
FE24	GDDR A1 EDC0	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<0>	NO TEST=TRUE 77 79
FE25	GDDR A1 EDC1	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<1>	NO TEST=TRUE 77 79
FE26	GDDR A1 EDC2	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<2>	NO TEST=TRUE 77 79
FE27	GDDR A1 EDC3	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<3>	NO TEST=TRUE 77 79
Data Dynamic Bus Inv					
FE28	GDDR A0 DBT0	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<0>	NO TEST=TRUE 77 79
FE29	GDDR A0 DBT1	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<1>	NO TEST=TRUE 77 79
FE2A	GDDR A0 DBT2	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<2>	NO TEST=TRUE 77 79
FE2B	GDDR A0 DBT3	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<3>	NO TEST=TRUE 77 79
FE2C	GDDR A1 DBT0	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBT L<0>	NO TEST=TRUE 77 79
FE2D	GDDR A1 DBT1	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBT L<1>	NO TEST=TRUE 77 79
FE2E	GDDR A1 DBT2	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBT L<2>	NO TEST=TRUE 77 79
FE2F	GDDR A1 DBT3	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBT L<3>	NO TEST=TRUE 77 79
Forwarded Clock					
FE30	GDDR A0 WCK0	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK P<0>	NO TEST=TRUE 77 79
FE31	GDDR A0 WCK0	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK N<0>	NO TEST=TRUE 77 79
FE32	GDDR A0 WCK1	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK P<1>	NO TEST=TRUE 77 79
FE33	GDDR A0 WCK1	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK N<1>	NO TEST=TRUE 77 79
FE34	GDDR A1 WCK0	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK P<0>	NO TEST=TRUE 77 79
FE35	GDDR A1 WCK0	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK N<0>	NO TEST=TRUE 77 79
FE36	GDDR A1 WCK1	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK P<1>	NO TEST=TRUE 77

GPU

Electrical Constraint Set	Physical	Spacing		
Clocks				
PEX0	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_PL	75
PEX1	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_NG	75
GPU0	CLK_PCTE_PHY	CLK_PCTE	GPU_TESTMODE	81
GPU0	CLK_PCTE_PHY	CLK_PCTE	GPU_PEX_TERM	75
GPU0	CLK_GPU_55S	CLK_GPU	FB_A0_CK_MID	NO_TESTTRUE 79
GPU0	CLK_GPU_55S	CLK_GPU	FB_A1_CK_MID	NO_TESTTRUE 79
GPU0	CLK_GPU_55S	CLK_GPU	FB_B0_CK_MID	NO_TESTTRUE 80
GPU0	CLK_GPU_55S	CLK_GPU	FB_B1_CK_MID	NO_TESTTRUE 80
GPU0	CLK_GPU_55S	CLK_GPU	FB_C0_CK_MID	NO_TESTTRUE 80
GPU0	CLK_GPU_55S	CLK_GPU	FB_C1_CK_MID	NO_TESTTRUE 80
GPU0	CLK_GPU_55S	CLK_GPU	FB_D0_CK_MID	NO_TESTTRUE 80
GPU0	CLK_GPU_55S	CLK_GPU	FB_D1_CK_MID	NO_TESTTRUE 80
GPU0	CLK_GPU_55S	CLK_GPU	GPU_JTAG_TCK	78 81
GPU0	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK	81 82
GPU0	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK_R	82
GPU0	CLK_GPU_55S	CLK_GPU	GPU_XTALOUT	
GPU0	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTALOUT	81 82
GPU0	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTALIN	81 82
SMB				
ESB0	SMB_PHY	SMB	GPU_SMB_CLK	81 82
ESB0	SMB_PHY	SMB	GPU_SMB_DAT	81 82
GPU0	SMB_PHY	SMB	GPU_SMB_CLK_R	50 82
GPU0	SMB_PHY	SMB	GPU_SMB_DAT_R	50 82
PCIe Compensation				
ESB0	PCIE_S0S	COMP_PCIE	FB_CAL_PD_VDDQ	77
ESB0	PCIE_S0S	COMP_PCIE	FB_CAL_PU_GND	77
ESB0	PCIE_S0S	COMP_PCIE	FB_CAL_TERM_GND	77

GDDR5 Frame Buffer B

Electrical Constraint Set		Physical	Spacing		
Memory Address					
H009	GDDR_B0_MA	GDDR_MA_PHY	GDDR_B_0_MA	FB_B0_A<8..0>	NO TEST-TRUE 77 80
H009	GDDR_B1_MA	GDDR_MA_PHY	GDDR_B_1_MA	FB_B1_A<8..0>	NO TEST-TRUE 77 80
Address Dynamic Bus Inv					
H009	GDDR_B0_ADBT	GDDR_ADBT_PHY	GDDR_B_0_ADBT	FB_B0_ABT_L	NO TEST-TRUE 77 80
H009	GDDR_B1_ADBT	GDDR_ADBT_PHY	GDDR_B_1_ADBT	FB_B1_ABT_L	NO TEST-TRUE 77 80
Control					
H009	GDDR_B0_CKE	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CKE_L	NO TEST-TRUE 77 80
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CS_L	NO TEST-TRUE 77 80
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_WE_L	NO TEST-TRUE 77 80
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CAS_L	NO TEST-TRUE 77 80
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_RAS_L	NO TEST-TRUE 77 80
H009	GDDR_B1_CKE	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CKE_L	NO TEST-TRUE 77 80
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CS_L	NO TEST-TRUE 77 80
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_WE_L	NO TEST-TRUE 77 80
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CAS_L	NO TEST-TRUE 77 80
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_RAS_L	NO TEST-TRUE 77 80
Clock					
H009	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_P	NO TEST-TRUE 77 80
H009	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_N	NO TEST-TRUE 77 80
H009	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_P	NO TEST-TRUE 77 80
H009	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_N	NO TEST-TRUE 77 80
Data					
H009	GDDR_B0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<7..0>	NO TEST-TRUE 77 80
H009	GDDR_B0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<15..8>	NO TEST-TRUE 77 80
H009	GDDR_B0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<23..16>	NO TEST-TRUE 77 80
H009	GDDR_B0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<31..24>	NO TEST-TRUE 77 80
H009	GDDR_B1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<7..0>	NO TEST-TRUE 77 80
H009	GDDR_B1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<15..8>	NO TEST-TRUE 77 80
H009	GDDR_B1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<23..16>	NO TEST-TRUE 77 80
H009	GDDR_B1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<31..24>	NO TEST-TRUE 77 80
Error Detection					
H009	GDDR_B0_EDC0	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<0>	NO TEST-TRUE 77 80
H009	GDDR_B0_EDC1	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<1>	NO TEST-TRUE 77 80
H009	GDDR_B0_EDC2	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<2>	NO TEST-TRUE 77 80
H009	GDDR_B0_EDC3	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<3>	NO TEST-TRUE 77 80
H009	GDDR_B1_EDC0	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<0>	NO TEST-TRUE 77 80
H009	GDDR_B1_EDC1	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<1>	NO TEST-TRUE 77 80
H009	GDDR_B1_EDC2	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<2>	NO TEST-TRUE 77 80
H009	GDDR_B1_EDC3	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<3>	NO TEST-TRUE 77 80
Data Dynamic Bus Inv					
H009	GDDR_B0_DBT0	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<0>	NO TEST-TRUE 77 80
H009	GDDR_B0_DBT1	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<1>	NO TEST-TRUE 77 80
H009	GDDR_B0_DBT2	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<2>	NO TEST-TRUE 77 80
H009	GDDR_B0_DBT3	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<3>	NO TEST-TRUE 77 80
H009	GDDR_B1_DBT0	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<0>	NO TEST-TRUE 77 80
H009	GDDR_B1_DBT1	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<1>	NO TEST-TRUE 77 80
H009	GDDR_B1_DBT2	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<2>	NO TEST-TRUE 77 80
H009	GDDR_B1_DBT3	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<3>	NO TEST-TRUE 77 80
Forwarded Clock					
H009	GDDR_B0_WCLK	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<0>	NO TEST-TRUE 77 80
H009	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<0>	NO TEST-TRUE 77 80
H009	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<1>	NO TEST-TRUE 77 80
H009	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<1>	NO TEST-TRUE 77 80
H009	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<0>	NO TEST-TRUE 77

Frame Buffer Reset

Electrical Constraint Set	Physical	Spacing	
Reset			
H4A5 GDDR_A0_RESET	GDDR_S0S	GDDR_CTLB1	FB_A0 RESET L 77.78
H4A6 GDDR_A1_RESET	GDDR_S0S	GDDR_CTLB1	FB_A1 RESET L 77.78
H4A9 GDDR_B0_RESET	GDDR_S0S	GDDR_CTLB1	FB_B0 RESET L 77.80
H4A9 GDDR_B1_RESET	GDDR_S0S	GDDR_CTLB1	FB_B1 RESET L 77.80
H4A9 GDDR_C0_RESET	GDDR_S0S	GDDR_CTLB1	FB_C0 RESET L 77.80
H4A9 GDDR_C1_RESET	GDDR_S0S	GDDR_CTLB1	FB_C1 RESET L 77.80
H4B9 GDDR_D0_RESET	GDDR_S0S	GDDR_CTLB1	FB_D0 RESET L 77.80
H4B9 GDDR_D1_RESET	GDDR_S0S	GDDR_CTLB1	FB_D1 RESET L 77.80

SYNC MASTER=D8 AARON		SYNC DATE=03/13/2012	
PAGE TITLE			
GDDR5/GPU Constraints			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-9504	D
		REVISION	
		5.3.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE 132 OF 143 SHEET 108 OF 117	

D

C

B

A

Constraints

BLC High Voltage Output

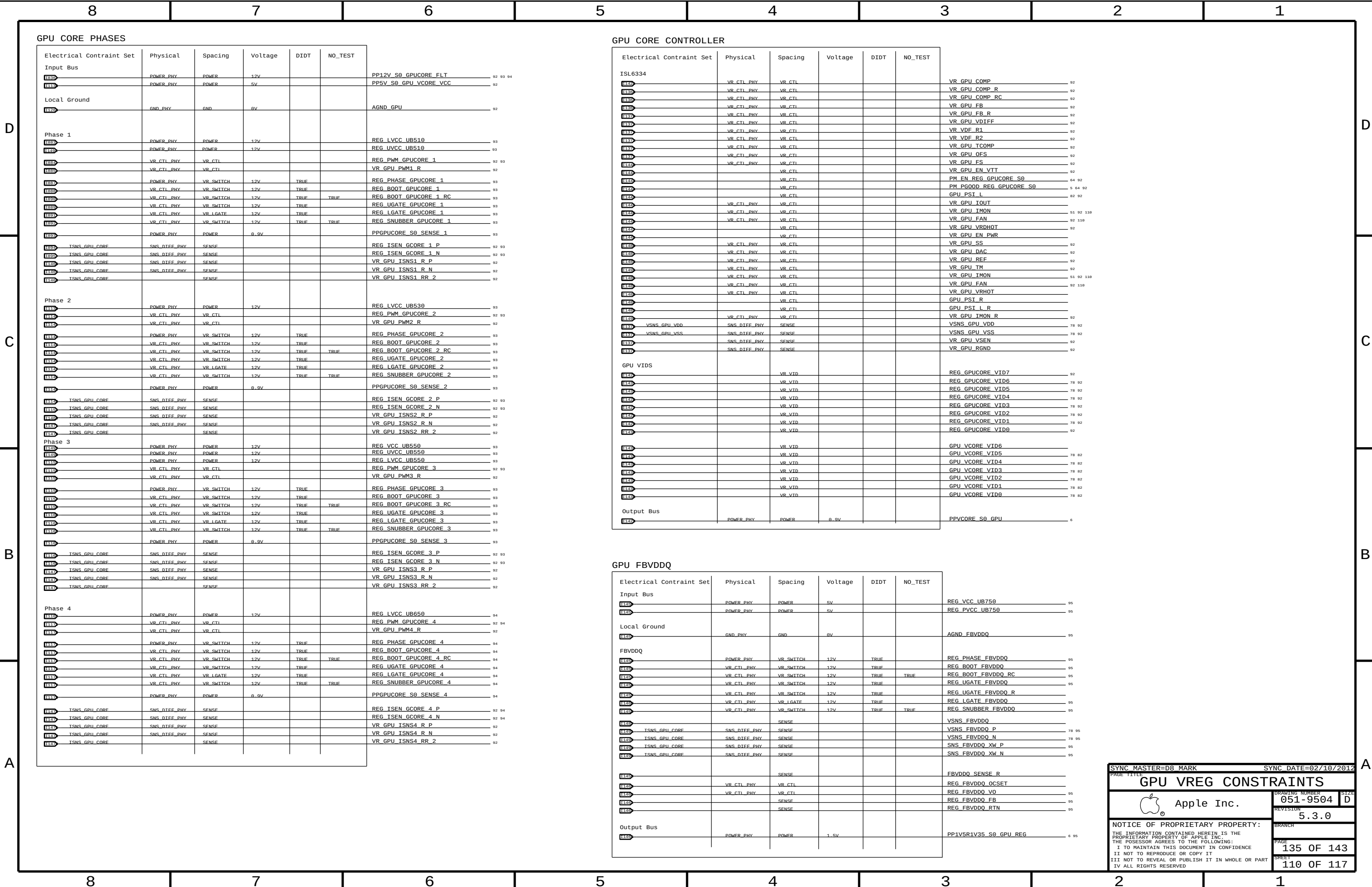
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	POWER	*	PHASE_SW2PWR
BLC_PHASE	GND	*	PHASE_SW2GND

BLC Control			
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

Physical		Spacing	Voltage	DI0T	NO_TEST	
Input	Bus					
H460	POWER_PHY	POWER	12V			PP12V_S0_BLC_VIN2
H477	POWER_PHY	POWER	12V			PP12V_S0_BLC_VINP
H480	POWER_PHY	POWER	14V			PRE_REG_OUT
H481	POWER_PHY	POWER	3_3V			BLC_P3V3S
H482	POWER_PHY	POWER	3_3V			BLC_P3V3_REF
H483	POWER_PHY	POWER	3_3V			BLC_P3V3
H484	POWER_PHY	POWER	3_3V			PP3V3_S0_BLC_R
H484	POWER_PHY	POWER	8V			SPTX_VIN
H485	POWER_PHY	POWER	8V			PP8V_BLC
H486	POWER_PHY	POWER	8V			BLC_VIN2
H487	POWER_PHY	POWER	14V			BOOST_FET_DRAIN
H488	POWER_PHY	POWER	12V			BOOST_VDD
H489	POWER_PHY	POWER	5V			PP5V_S0_BLC_R
H490	POWER_PHY	POWER	12V			PP12V_S0_BLC_F
Local Ground						
H491	BLC_CTL_PHY	BLC_PHASE	8V			BLC_GND_1
H492	BLC_CTL_PHY	BLC_PHASE	8V			BLC_GND_2
H493	BLC_CTL_PHY	BLC_PHASE	8V			BLC_GND_3
H494	GND_PHY	GND	8V			AGND_BLC
Backlight						
H495	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1
H496	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1_R
H497	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2
H498	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2_R
H499	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3
H500	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3_R
H501	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_RC_1
H502	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_RC_2
H503	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_RC_3
H504	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS1
H505	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS2
H506	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS3
H507	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_C1
H508	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_C2
H509	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_C3
H510	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_1
H511	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_2
H512	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_3
H513	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK1
H514	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK2
H515	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK3
H516	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1_R
H517	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2_R
H518	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3_R
H519	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1
H520	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2
H521	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3
H522	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF1
H523	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF2
H524	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF3
H525	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP1
H526	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP2
H527	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP3
H528	BLC_CTL_PHY	BLC_CTL				BCOMP1
H529	BLC_CTL_PHY	BLC_CTL				BCOMP2
H530	BLC_CTL_PHY	BLC_CTL				BCOMP3
H531	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT1
H532	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT2
H533	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT3
H534	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_1
H535	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_2
H536	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_3
H537	BLC_CTL_PHY	BLC_CTL				PRE_REG_OUT_R
H538	BLC_CTL_PHY	BLC_CTL				BOOST_FB
H539	BLC_CTL_PHY	BLC_CTL				BOOST_COMP
H540	BLC_CTL_PHY	BLC_CTL				BOOST_COMP_C
H541	BLC_CTL_PHY	BLC_CTL			TRUE	BOOST_GDRV
H542	BLC_CTL_PHY	BLC_CTL			TRUE	BOOST_GDRV_R
H543	BLC_CTL_PHY	BLC_CTL				BOOST_ISNS
H544	BLC_CTL_PHY	BLC_CTL				BOOST_ISNS_R
H545	BLC_CTL_PHY	BLC_HV	80V			LED_DRVR_DRAIN_1
H546	BLC_CTL_PHY	BLC_HV	80V			LED_DRVR_DRAIN_2
H547	BLC_CTL_PHY	BLC_HV	80V			LED_DRVR_DRAIN_3
OUTPUT BUS						
H548	POWER_BLC_RET	BLC_HV	80V			IS1_BLC_F
H549	POWER_BLC_RET	BLC_HV	80V			IS2_BLC_F
H550	POWER_BLC_RET	BLC_HV	80V			IS3_BLC_F
H551	POWER_BLC_RET	BLC_HV	80V			IS1_BLC
H552	POWER_BLC_RET	BLC_HV	80V			IS2_BLC
H553	POWER_BLC_RET	BLC_HV	80V			IS3_BLC
H554	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_1
H555	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_1
H556	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_2
H557	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_2
H558	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_3
H559	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_3
H560	POWER_BLC	BLC_HV	80V			BLC_VOUT1
H561	POWER_BLC	BLC_HV	80V			BLC_VOUT2
H562	POWER_BLC	BLC_HV	80V			BLC_VOUT3

D



8

7

6

5

4

3

2

1

GPU VREG CONSTRAINTS

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

051-9504

5.3.0

135 OF 143

110 OF 117

8

7

6

5

4

3

2

1

Caesar IV (Ethernet/SD)

CIV-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
ENET_COMP_PHY	*	ENET_50S
ENET_DIFF_PHY	*	ENET_100D
SD_PHY	*	SD_50S

CIV-specific Spacing Definitions
Ethernet

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF_ISO	*	=6:1_SPACING	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_TRANS_ISO	*	1.27 MM	?
COMP_ENET_ISO	*	=4:1_SPACING	?

Constraints
Ethernet

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_DIFF	*	*	ENET_DIFF_ISO
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF
ENET_TRANS	*	*	ENET_TRANS_ISO
COMP_ENET	*	*	COMP_ENET_ISO
ENET_TRANS	ENET_TRANS	*	ENET_DIFF2DIFF

2 kV isolation

SD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_ISO	*	=3:1_SPACING	?

SD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SD	*	*	SD_ISO

VENI, VIDI, VICI!

Electrical Constraint Set	Physical	Spacing	
ENET_MDI_NORM	ENET_DIFF_PHY	ENET_DIFF	ENETCONN MDI P<3..0>
ENET_MDI_NORM	ENET_DIFF_PHY	ENET_DIFF	ENETCONN MDI N<3..0>
	ENET_DIFF_PHY	ENET_TRANS	ENETCONN MDI T P<3..0>
	ENET_DIFF_PHY	ENET_TRANS	ENETCONN MDI T N<3..0>
		ENET_TRANS	ENETCONN MCT3
		ENET_TRANS	ENETCONN MCT2
		ENET_TRANS	ENETCONN MCT1
		ENET_TRANS	ENETCONN MCT0
ENET_COMP_PHY	ENET_COMP_PHY	COMP_ENET	ENET_RDAC
SD_DATA	SD_PHY	SD	ENET_CR_DATA<7..0> NO_TEST=TRUE
	SD_PHY	SD	SDCONN_DATA<7..0>
	SD_PHY	SD	SDCONN_DATA R<7..0> NO_TEST=TRUE
SD_CMD	SD_PHY	SD	ENET_SD_CMD NO_TEST=TRUE
	SD_PHY	SD	SDCONN_CMD NO_TEST=TRUE
	SD_PHY	SD	SDCONN_CMD R NO_TEST=TRUE
SD_CLK	SD_PHY	SD	ENET_SD_CLK NO_TEST=TRUE
	SD_PHY	SD	SDCONN_CLK NO_TEST=TRUE
	SD_PHY	SD	SDCONN_CLK R NO_TEST=TRUE
	SD_PHY	SD	ENET_MEDIA_SENSE
	SD_PHY	SD	ENET_SD_DETECT L
	SD_PHY	SD	SDCONN_WP
CIV SPI	SPI_50S	SPI	ENET_SCLK
	SPI_50S	SPI	ENET_MISO
	SPI_50S	SPI	ENET_MOSI
	SPI_50S	SPI	ENET_CS_L

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

DCBABA

BOOST_*		
	Spacing	Netname
1600	GENERIC ISO	BOOST BP
1600	GENERIC ISO	BOOST BYPASS
1600	GENERIC ISO	BOOST BYPASS_GATE
1600	PM	BOOST_EN_GATE
1600	PM	BOOST_EN_L
1600	GENERIC ISO	BOOST_RC
1600	GENERIC ISO	BOOST_SS

DP_*			
	Physical	Spacing	Netname
FE80		PM	DP_AUXIO_EN
FE80		PM	DP_GPU_MUX_EN
FE80		GENERIC_ISO	DP_INTPNL_HPD
FE80		GENERIC_ISO	DP_INT_EG_HPD
FE80	TBT_GEN_55S	TBT_GEN	DP_TBTPA_DDC_CLK
FE80	TBT_GEN_55S	TBT_GEN	DP_TBTPA_DDC_DATA
FE80		GENERIC_ISO	DP_TBTPA_HPD
FE80	TBT_GEN_55S	TBT_GEN	DP_TBTPB_DDC_CLK
FE80	TBT_GEN_55S	TBT_GEN	DP_TBTPB_DDC_DATA
FE80		GENERIC_ISO	DP_TBTPB_HPD

AUTO-CONSTRAINTS PG 3

ISNS_*

Physical	Spacing	Netname
SNS_PHY	SENSE	ISNS_CPUXAG_FB
SNS_PHY	SENSE	ISNS_CPUXAG_N
SNS_PHY	SENSE	ISNS_CPUXAG_P
SNS_PHY	SENSE	ISNS_CPUCORE_FB
SNS_PHY	SENSE	ISNS_CPUCORE_N
SNS_PHY	SENSE	ISNS_CPUCORE_P
SNS_PHY	SENSE	ISNS_GPUCORE_FB
SNS_PHY	SENSE	ISNS_GPUCORE_N
SNS_PHY	SENSE	ISNS_GPUCORE_P
SNS_PHY	SENSE	ISNS_P12VG3H_R
SNS_PHY	SENSE	ISNS_P12VS0_CPU_P1V05_R
SNS_PHY	SENSE	ISNS_P12VS0_CPU_VCCSA_R
SNS_PHY	SENSE	ISNS_P12VS0_FBDVDQ_R
SNS_PHY	SENSE	ISNS_P12VS0_HDD_R
SNS_PHY	SENSE	ISNS_P12VS0_P1V05_R
SNS_PHY	SENSE	ISNS_P1V05S0_PCH_R
SNS_PHY	SENSE	ISNS_P1V5S0_CPU_MEM_R
SNS_PHY	SENSE	ISNS_P3V3S0_SSD_R
SNS_PHY	SENSE	ISNS_P3V3S4_AP_R
SNS_PHY	SENSE	ISNS_P5VS0_HDD_R
SNS_PHY	SENSE	ISNS_PVDDQS3_DDR_R

ISOLATE_*

Spacing	Netname
PM	ISOLATE_CPU_MEM_5V
PM	ISOLATE_CPU_MEM_5V_L

JTAG_*

Physical	Spacing	Netname
XDP_PHY	XDP	JTAG_GMUX_TMS
XDP_PHY	XDP	JTAG_TBT_TDI
XDP_PHY	XDP	JTAG_TBT_TDI_ISOL
XDP_PHY	XDP	JTAG_TBT_TDO
XDP_PHY	XDP	JTAG_TBT_TDO_ISOL
XDP_PHY	XDP	JTAG_TBT_TMS
XDP_PHY	XDP	JTAG_TBT_TMS_ISOL

LED_*

Spacing	Netname
PM	LED_DRIVER_EN
PM	LED_DRIVER_EN_L
PM	LED_DRIVER_EN_L_R
GENERIC_ISO	LED_DRIVER_OVP1
GENERIC_ISO	LED_DRIVER_OVP1P
GENERIC_ISO	LED_DRIVER_OVP1_OUT
GENERIC_ISO	LED_DRIVER_OVP2
GENERIC_ISO	LED_DRIVER_OVP2P
GENERIC_ISO	LED_DRIVER_OVP2_OUT
GENERIC_ISO	LED_DRIVER_OVP3
GENERIC_ISO	LED_DRIVER_OVP3P
GENERIC_ISO	LED_DRIVER_OVP3_OUT

LPCPLUS_*

Spacing	Netname
GENERIC_ISO	LPCPLUS_GPIO

LPC_*

Spacing	Netname
GENERIC_ISO	LPC_PWRDWN_L
GENERIC_ISO	LPC_SERIRQ

MEMVTT_*

Spacing	Netname
PM	MEMVTT_EN
PM	MEMVTT_EN_L

MEM_*

Spacing	Netname
GENERIC_ISO	MEM_EVENT_L

MOJO_*

Physical	Spacing	Netname
XDP_PHY	XDP	MOJO_RX_L
XDP_PHY	XDP	MOJO_TX_L

OCA_*

Spacing	Netname
GENERIC_ISO	OCA_FET_DRAIN

OVP_*

Spacing	Netname
GENERIC_ISO	OVP_OREF
GENERIC_ISO	OVP_OUT1
GENERIC_ISO	OVP_OUT1_R
GENERIC_ISO	OVP_OUT2
GENERIC_ISO	OVP_OUT2_R
GENERIC_ISO	OVP_OUT3
GENERIC_ISO	OVP_OUT3_R

P1V2_*

Spacing	Netname
PM	P1V2_S4_EN

P1V8_*

Spacing	Netname
PM	P1V8_S4_EN

P3V3AP_*

Spacing	Netname
GENERIC_ISO	P3V3AP_VM0N

P3V3_*

Spacing	Netname
GENERIC_ISO	P3V3_S0_EN_G
GENERIC_ISO	P3V3_S3_EN_G

P3V42G3H_*

Spacing	Netname
GENERIC_ISO	P3V42G3H_B00ST

P5V_*

Spacing	Netname
GENERIC_ISO	P5V_S0_EN_G

PCA9557D_*

Spacing	Netname
GENERIC_ISO	PCA9557D_RESET_L

PCH_*

Physical	Spacing	Netname
	PM	PCH_BLC_EXT_BOOT
	PM	PCH_BLC_EXT_BOOT_R
	PM	PCH_BLC_MCU_RESET
	PM	PCH_BLC_MCU_RESET_R
	PM	PCH_CAM_RESET
	PM	PCH_CAM_RESET_R
	GENERIC_ISO	PCH_DSWMEN
CPU_PHY	CPU	PCH_PECT
CPU_PHY	CPU	PCH_PROCPWRGD
	GENERIC_ISO	PCH_RCIN_L
	GENERIC_ISO	PCH_RT_L
	GENERIC_ISO	PCH_SMBALERT_L
	GENERIC_ISO	PCH_SRTCST_L
	GENERIC_ISO	PCH_STRP_TOPBLK_SWP_L
	GENERIC_ISO	PCH_SUSWARN_L

PCIE_*

Spacing	Netname
GENERIC_ISO	PCIE_CLKREQ5_GPIO44_L
PM	PCIE_WAKE_L

PEG_*

Spacing	Netname
GENERIC_ISO	PEG_CLKREQ_L

PGOOD_*

Spacing	Netname
PM	PGOOD_P1V5_S0_DLY

PLT_*

Spacing	Netname
PCH	PLT_RESET_L
PCH	PLT_RST_BUF_L

PM_*

Spacing	Netname
PM	PM_CLKRUN_L
PM	PM_DSW_PWRGD
PM	PM_EN_FET_P12V_S0
PM	PM_EN_FET_P12V_S0_R
PM	PM_EN_FET_P3V3_S0
PM	PM_EN_FET_P3V3_S4
PM	PM_EN_FET_P5V_S0
PM	PM_EN_FET_VDDQ_S0
PM	PM_EN_LD0_DDRVTT_S0
PM	PM_EN_REG_CPUCORE_S0

AUTO-CONSTRAINTS GENERATED Fri 04/06/12 07:24:10 PM

AUTO-CONSTRAINTS 3

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER
051-9504

REVISION
5.3.0

BRANCH

PAGE
140 OF 143

SHEET
114 OF 117

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

DCB

A

I72

I74
I75

I74

I75

182

187
188

I89190190I92I92I92I95I95

Y415

I116

I117

I128

PAGE TITLE		DRAWING NUMBER		SIZE
AUTO-CONSTRAINTS 4		051-9504		D
 Apple Inc.		REVISION		
		5.3.0		
NOTICE OF PROPRIETARY PROPERTY:				
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:				
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE				
II NOT TO REPRODUCE OR COPY IT				
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART				
IV ALL RIGHTS RESERVED				
BRANCH		PAGE		
		141 OF 143		
SHEET		115 OF 117		

2	1
---	---

2	1
---	---

AUTO-CONSTRAINTS PG 6

SMC_*

Physical	Spacing	Netname
16	GENERIC ISO	SMC ROMBOOT 48 49
16	GENERIC ISO	SMC RUNTIME SCI L 15 21 47
16	GENERIC ISO	SMC RX L 47 48 49
16	PM	SMC S4 WAKESRC_EN 35 47 48
16	GENERIC ISO	SMC TCK 47 48 49
16	GENERIC ISO	SMC TDI 47 48 49
17	GENERIC ISO	SMC TDO 47 48 49
16	PM	SMC THRMTRIP 47 48
16	GENERIC ISO	SMC TMS 47 48 49
16	GENERIC ISO	SMC TO BLC_RX_L 48
16	GENERIC ISO	SMC TO BLC_TX_L 48
16	GENERIC ISO	SMC TX_L 47 48 49
16	PM	SMC WAKE_L 47
16	GENERIC ISO	SMC WAKE_SCI_L 15 21 47
16	CLK XTAL	SMC XTAL 47 48
16	CLK XTAL	SMC XTAL_R 48

SML_*

Physical	Spacing	Netname
16	SMB_PHY	SMB 18 50
16	SMB_PHY	SMB 18 50
16	SMB_PHY	SMB 18 50
16	SMB_PHY	SMB 18 50

SPI_*

Spacing	Netname
16	GENERIC ISO SPI_DESCRIPTOR_OVERRIDE_L 15 47

SSD_*

Spacing	Netname
16	GENERIC ISO SSD_CLKREQ_L 15 18

SYS_*

Spacing	Netname
16	PM SYS_PWROK_R 65

TBT_*

Physical	Spacing	Voltage	Netname
12	GENERIC ISO	3.3V	TBT_A_BIAS 66
12	TBT_GEN_55S	TBT_GEN	TBT_A_CONFIG1_RC 66
12	PM		TBT_A_HV_EN 36 66
12	GENERIC ISO	3.3V	TBT_B_BIAS 68
12	TBT_GEN_55S	TBT_GEN	TBT_B_CONFIG1_RC 68
12	PM		TBT_B_HV_EN 36 68
12	GENERIC ISO		TBT_CLKREQ_ISOL_L 38
12	GENERIC ISO		TBT_CLKREQ_L 15 38
12	GENERIC ISO		TBT_DDC_XBAR_EN_L 36 85
12	GENERIC ISO		TBT_EN_CIO_PWR 38
12	PM		TBT_EN_CIO_PWR_L 36 38
12	GENERIC ISO		TBT_EN_LC_ISOL 38
12	GENERIC ISO		TBT_EN_LC_PWR 36 38
12	GENERIC ISO		TBT_PCH_CLKREQ_L 15 21
12	PM		TBT_PWR_EN 15 26 36
12	PM		TBT_PWR_EN_PCH 18 26
12	PM		TBT_PWR_ON_POC_RST_L 36 38
12	PM		TBT_PWR_REQ_L 15 20 36
12	PM		TBT_S0_EN 64
12	PM		TBT_SW_RESET_L 21 38

TSNS_*

Electrical	Physical	Spacing	Netname
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_1_1_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_1_1_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_1_2_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_1_2_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_1_3_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_1_3_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_1_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_1_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_2_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_2_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_3_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_3_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_4_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_4_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_5_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_5_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_6_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_6_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_7_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_2_7_P 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_ACDC_N 6 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_ACDC_P 6 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_SKIN_N 53
12	SNS_TEMP	SNS_DIFF_PHY	SENSE TSNS_SKIN_P 53

UVP_*

Spacing	Netname
16	PM UVP_IN_1 91
16	PM UVP_IN_1_REF 91
16	PM UVP_IN_2 91
16	PM UVP_IN_3 91
16	PM UVP_IN_4 91
16	PM UVP_REF 91

VREFMRGN_*

Spacing	Netname
12	GENERIC ISO VREFMRGN_CA_SODIMMA_EN 34
12	GENERIC ISO VREFMRGN_CA_SODIMMB_EN 34

VTTCLAMP_*

Spacing	Netname
12	GENERIC ISO VTTCLAMP_EN 28
12	GENERIC ISO VTTCLAMP_L 28

WOL_*

Spacing	Netname
12	GENERIC ISO WOL_EN 15 21 48